

**AN INVESTIGATION OF THE FACTORS THAT INFLUENCE PIPE
BURSTS IN THE WATER DISTRIBUTION NETWORK IN
BARBADOS AND THE IMPLICATIONS OF CLIMATE CHANGE**

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ABSTRACT

An Investigation of the Factors that Influence Pipe Bursts in the Water Distribution Network in Barbados and the Implications of Climate Change

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Understanding pipe burst behaviour in a water distribution network is fundamental for informing how this infrastructural asset which transports potable water can be managed more efficiently and prudently. For this research, a data-driven approach which was enhanced by domain knowledge was adopted to identify the factors that contribute to pipe burst occurrences in a country-wide case study for Barbados. Additionally, the potential impacts of climate change on pipe bursts were assessed. A research of this scope has not been carried out before as there is lack of pipe burst studies conducted in sub-tropical and tropical regions, and an even greater gap in the evaluation of climate change on pipe failure worldwide.

In this study, three discrete datasets that recorded historical pipe burst incidents at different temporal and spatial scales were compiled for analysis. Two methods (self-organising maps and evolutionary polynomial regression) based on machine learning (i.e., a branch of artificial intelligence) were used for data mining to detect patterns among pipe bursts and eight possible explanatory factors. The derived knowledge informed the development of a pipe burst risk model created with fuzzy logic. That model was also used to incorporate downscaled projections from a global climate model to assess the distribution of pipe bursts across four climate change scenarios via hotspot mapping.

It was concluded that the main drivers of local pipe bursts were rainfall, pressure, pipe diameter, soil type, and pipe material. Also, the climate change assessment indicated that a significant reduction in rainfall has the potential to cause an increase in pipe burst rates particularly in places that contain clays that shrink and swell due to soil moisture fluctuations. Overall, the research showed that despite data challenges, a strategic implementation of methods can produce useful outputs that may be used to support current and future decision-making

pertaining to pipe replacement. This can assist in addressing water losses, energy losses, compromised water quality, and service disruptions which are all challenges that can result from frequent pipe bursts.

Keywords: Water distribution network (WDN); pipe failure; pipe burst modelling; data-driven; self-organising map (SOM); evolutionary polynomial regression (EPR); fuzzy logic; climate change; geographic information system (GIS); hotspot mapping; Barbados.

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DEDICATION

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LIST OF ACRONYMS

ANN	Artificial Neural Network
BWA	Barbados Water Authority
CAP	Condition Assessment and Prioritisation
CIP	Cast Iron Pipe
COD	Coefficient of Determination
COF	Consequence of Failure
CSV	Comma-separated Values
DEM	Digital Elevation Model
DIP	Ductile Iron Pipe
EPR	Evolutionary Polynomial Regression
FCA	Failure Cluster Analysis
FFA	Failure Factor Analysis
FIS	Fuzzy Inference System
GCM	Global Climate Model
GHG	Greenhouse Gases
GI	Galvanised Iron
GIS	Geographic Information System
HDPE	High Density Polyethylene
IPCC	Intergovernmental Panel on Climate Change
LOF	Likelihood of Failure
PE	Polyethylene
PI	Physical Inspection
PVC	Polyvinyl Chloride
RCM	Regional Climate Model
RCP	Representative Concentration Pathway
RQ	Research Question
SIDS	Small Island Developing States
SOM	Self-organising Map
UWI	The University of the West Indies
WDN	Water Distribution Network

NOMENCLATURE

Symbol	Unit Name	Quantity
mm	Millimetre	Length
m	Metre	Length
km	Kilometre	Length
km ²	Kilometres Squared	Area
m ³	Metres Cubed	Volume
L	Litres	Volume
°C	Degrees Celsius	Temperature
PSI	Pounds Per Square Inch	Pressure
USD	United States Dollar	Currency

1 INTRODUCTION

Day Zero, a term coined by the South African Government to convey to residents that the legislative capital, Cape Town, was on the brink of running out of water. In 2017, that impending reality was grim as a population of 4 million-plus (Western Cape Government 2018) were facing a life without water in a matter of months. The cause of *Day Zero* was attributed to a multi-year drought that steered the regulating authorities to implement measures which mostly focused on the consumption, conservation, and allocation of water (Sarni 2020; Trivedi and Chertock 2019). According to Alexander (2019), that drought-induced crisis was looming to be the biggest modern-day catastrophe in water resources management. Fortunately, the situation was averted through extensive water-saving efforts and the return of average rainfall. Nevertheless, the local water utility was cautioned that its water supply system was still not resilient and remained vulnerable since one of the underlying issues behind that crisis was climate change (Millington and Scheba 2021; Alexander 2019).

Climate change is defined as ‘a change in the state of the climate that can be identified by changes in the mean and/or the variability of its properties and that persists for an extended period, typically decades or longer’ (IPCC 2018). For the crisis in Cape Town, the drought from 2015 to 2018 explained the deficits in dam water levels. However, Ziervogel (2019) found that the drought event was extreme, prolonged, and up to six times more likely to have occurred within the past century because of climate change. In fact, Pascale et al. (2020) not only quantified the contribution of anthropogenic climate change to that multi-year drought but also found substantial increases in the probability of such an event reoccurring under intermediate and high emission scenarios by the year 2100. These studies showed that if drivers of events that impact water resources management are climate-related then a wider lens that takes climate change into account should be used to gain a more comprehensive, accurate, and prudent outlook.

Though some may consider that situation in Cape Town was extreme and uncommon, the fact remains, many countries in the world are water stressed.

Globally, the residents of Cape Town represent a miniscule fraction (0.2%) of the two billion people living in countries under high water stress (i.e., water scarce) – that is, less than 1,000 m³ per capita per year (United Nations 2018). Dr Peter Gleick, President Emeritus of Pacific Institute, stated that “what happened in Cape Town could happen anywhere” (Leahy 2018). From that perspective, every country should evaluate the threats imposed by climate change on water supply systems especially if considered water scarce.

Day Zero increased global attention to the important linkages between water resources management and climate change. However, studies that have investigated both subject areas often focused on water sources and water consumption – as in the example of Cape Town (Brühl and Visser 2021; Pascale et al. 2020; Ziervogel 2019). The impacts of climate change on a water distribution network (WDN), which is used to transport water from a source to consumers, is rarely considered despite it being an essential part in a water supply system.

In a WDN, pipes are subjected to inherent deterioration processes in which climatic factors such as precipitation and temperature can play a major role. Such processes, which can result in pipe bursts, can be sped up by climate change (Wols and van Thienen 2016). According to the Government of the City of Cape Town (CCT 2018), total water losses were approximately 16% which included potable water lost from the WDN via pipe bursts. However, the effect of climate change on the efficiency of the WDN has not been assessed. For a holistic and sustainable approach to water resources planning and management, all components of a water supply system should be analysed to determine patterns at the local level with consideration to the effects of climate change (Ziervogel 2019; Brekke et al. 2009).

The efficiency of the WDN in Cape Town far exceeds those in other countries. In the Caribbean, water losses can range between 24% and 63% (Burdescu et al. 2020). This research uses the country of Barbados – a water scarce, small island developing state located in the Caribbean – as a case study to assess the possible impact of climate change on pipe bursts in the local WDN, which has

an average water loss of approximately 44% (Burdescu et al. 2020, Halcrow Inc. 2012). However, before a climate change assessment, this research will first seek to identify and understand the drivers of pipe burst occurrences in the WDN. An evaluation of significant factors and patterns at the local level are especially critical (Ziervogel 2019; Brekke et al. 2009) as it is clear from the example of Day Zero that what was perceived as the cause of the problem influenced the response.

The aim of this chapter is to introduce the main topic of this research – investigating factors that contribute to pipe bursts in WDNs and the impact of climate change – and to situate the relevance of this study within that context. In this chapter, the following was presented:

1. An overview of WDNs and challenges in managing these systems.
2. The motivation for this research.
3. A background of the study area.
4. The scope of this research.
5. An outline the structure of this thesis.

The differences in how WDNs are managed worldwide was given to show some of the disproportionate problems that are being faced by water utilities across the globe. This comparison highlights the need for this research in a country where the implications of water management and climate change are particularly acute. Existing gaps and challenges that prevent most water utilities from building additional resiliency to future shocks into the infrastructural management of the WDN are also highlighted in this chapter.

1.1 Water Distribution Networks

Centuries ago, civilisations had to be built near to water sources (e.g., rivers, lakes, springs, and groundwater aquifers). However, in modern times, engineering has evolved to mechanically pump and transport water from the location of the source to a plant for treatment and storage before it is distributed directly to the individual consumers. The entire scheme is referred to as a water

supply system which consists of three main components: the source (supply), the consumers (demand), and the WDN (infrastructure).

The basic components of a WDN are storage tanks, pipes, pumps, and valves (Nyende-Byakika 2018). In the typical design of a WDN, components are hydraulically linked to form an interconnected system that can efficiently deliver pressurised water to consumers. Water is conveyed through pipes which usually comprise the lengthiest and most expensive part of a WDN. Pipes are hollowed cylindrical objects that are manufactured from various materials. Essentially all pipes deteriorate with time ultimately leading to cracks, leaks, and bursts. Besides the loss of water, pipe deterioration can result in wasted energy, compromised water quality, service disruptions, and challenges with pressure management (Jafar, Shahrou, and Juran 2010; Mohamed and Zayed 2008; Boxall et al. 2007; Mamlook and Al-Jayyousi 2003). Understanding the dynamics of pipe deterioration is critical in the management of WDNs.

The networks of pipes that run beneath most communities are ageing assets which deteriorate as initially stable conditions are altered with time. Determining the condition of pipes can be a challenge, especially for utilities with limited resources, as these pipes are usually buried underground and run for hundreds to thousands of kilometres in total length. Furthermore, the factors that influence pipe deterioration are many, and the physical processes of pipe deterioration are complex and non-linear (Kleiner, Adams, and Rogers 2001). Thus, it is extremely difficult to model pipe deterioration, locate leaks, and predict pipe breaks. However, many attempts have been made to analyse and model the deterioration behaviour of pipes in an effort to understand the physical dynamics that take place beneath the surface.

Investigating pipe deterioration usually requires that substantial data of relatively good quality be collected for the factors that contribute to pipe deterioration. The data would normally comprise of pipe asset data, network information, and other environmental data. However, water utilities usually lack the data that is necessary to build complex models found in the literature (Zhang et al. 2018; Pelletier, Mailhot, and Villeneuve 2003). While this remains true to

some extent, in recent years, water utilities have started to collect more data than ever before. The increase in data is owed to digitisation of information, automated control systems, remote sensing technologies, accessibility and affordability of software designed for water utilities, and enhanced capacities of data storage platforms. However, collecting more data does not necessarily result in better information or knowledge. Furthermore, it is estimated that less than 0.5% of data is ever used and analysed worldwide (Gantz and Reinsel 2012) which, in the water sector, can be compounded by issues with control over proprietary data. Therefore, water utilities should correspondingly explore methodologies and methods that can utilise and process large datasets and discover patterns in complex or sparse data.

All water utilities contend with sparse and incomplete data, aged and deteriorating assets, and limited and limiting resources. However, in most developed countries (also referred to as the Global North), water utilities make large investments towards data gathering and research on water losses from pipe networks. Such efforts may be driven by standards and incentives imposed by regulating authorities. For example, in England (where most water utilities are privately owned), penalties are imposed on water utilities that are not found in compliance of leakage standards set by the Water Services Regulation Authority, OFWAT (Tang, Parsons, and Jude 2019). Additionally, most of the research found in the literature on pipe failure in water distribution systems is carried out in developed countries. All of the above may account for lower water losses in countries such as Netherlands - 4%, United States of America - 16%, and England and Wales - 20%. Whereas, in developing countries (also referred to as the Global South) where most water utilities are state-owned, operated, and regulated by the government, the average rate of non-revenue water loss is 44% (Burdescu et al. 2020). These utilities are typically characterised as having weak regulatory and legal frameworks in water governance, in addition to a strong lack in capacity and research.

In contrasting the overall status and approach to managing WDNs in the Global North against the Global South, it was evident that serious consideration needs

to be given to improving in the latter the capacity to gather, analyse, and utilise data to inform management practices. However, having established that climate change poses a triple threat on each component of a water supply system, water utilities in the Global South must go beyond increasing the efficiency of WDNs. Developing countries in the Global South, particularly Small Island Developing States (SIDS), are on the frontline of climate change. Water utilities in those countries should also seek to promote resiliency by understanding the potential implications of climate change on WDNs.

1.2 Research Problem and Motivation

All water utilities across the globe face challenges in efficiently managing WDNs; however, the underperformance of these systems is much higher in SIDS, including those in the Caribbean. However, there is a lack of pipe failure studies that have been carried out in sub-tropical and tropical countries (Shi, Zhang, and Ho 2013) and more specifically in SIDS. Conducting such studies would assist water utilities in those countries to manage WDNs more efficiently. Studies should first be conducted to analyse and characterise local pipe failure. But for many water utilities, especially those in SIDS, the real-world application of the pipe failure models found in the literature would be a considerable challenge due to limitations in the quality of data available, technical and human capacity, and financial resources. Nonetheless, attempts to carry out such studies should be made since the main drivers of pipe failure identified in the Global North may not be the same due to differences in operational and environmental factors.

Climate change poses a considerable threat to the sustainable management of a water supply system. However, the impacts of climate change on water supply systems are not wholly understood as more emphasis is placed on the *source* and the *consumer* than on the *WDN* (i.e., the infrastructure used to transport water from the source directly to the consumer). Given this major gap in knowledge and research, this study also focused on investigating the implications of climate change on a WDN.

In consideration of the above, the WDN in Barbados – a Caribbean SID – was selected as the case study for this research. In this way, the research was carried out in a place where it is critically needed and findings from this research that used real data would have wider theoretical and practical relevance due to the similarities in water management issues of Caribbean SIDS.

Similar to other Caribbean SIDS, the state-owned water utility in Barbados struggles with operational efficiency, finances, and quality of service (Burdescu et al. 2020). The local water utility was reported as having water losses between 43% and 48% (Halcrow Inc. 2012), an average of 700 pipe bursts per month (Downes 2018), challenges with accessing and sustaining finances (BGIS 2020), and an array of service issues such as low pressure and frequent water shutoffs (Forde-Craigg 2020). Furthermore, Barbados is a water scarce country which makes the implications of climate change on its water supply system more serious. Thus, Barbados was considered an apt choice as a case study for this research.

The motivation for this research was not only due to its theoretical and practical relevance but also owed to personal interest. The author has experienced first-hand the shortcomings and consequences of having inadequate water supply systems after having visited and lived in several Caribbean-SIDS such as Barbados, the Bahamas, Antigua and Barbuda, Jamaica, and Trinidad and Tobago. With the impacts of climate change already being observed (e.g., more droughts and increased frequency and intensity of tropical storms), this research can have a significant bearing on decision-making in the management of water resources within the region.

1.3 Background of Study Area (Barbados)

Barbados is the most eastern island in the Caribbean (see Figure 1.1). It has a tropical climate with monthly averages for temperature and rainfall at 27°C and 106 mm, respectively (Barbados Meteorological Services 2020). With a total of 287,025 residents (World Bank 2019) in its 430 km² land area, it is one of the most densely populated countries in the world and highest in the Caribbean.



Figure 1.1 - The location of Barbados within the wider Caribbean [Source: http://lloydcripe.com/boqueteweather/map_caribbean.htm].

Barbados is divided into 11 parishes (see Figure 1.2). The capital city, Bridgetown, is located in the southwest of the parish of St Michael – the highest populated parish. The country has two main urban corridors; along the western and southern coasts. The local economy is largely sustained by service-based industries such as tourism, finance, and government services. Primary industries such as agriculture and fisheries have seen a major decline over the past decades leading to a higher dependency on tourism which has also made the local economy more vulnerable to external economic shocks. The existing high public debt increased after the global recession in 2008-2009. In fact, the ratio of public debt to the gross domestic product was 157% in 2017 and 147% in 2020 (CIA 2021). This debt, along with other factors, prompted a sovereign debt restructuring in 2018-2019. However, it was projected that Barbados will experience its third consecutive year of recession, prolonged because of the ongoing global Covid-19 pandemic (World Bank 2020). Despite these economic challenges, Barbados is classified by the World Bank as being a high-income country.



Figure 1.2 - A satellite image of Barbados showing parish boundaries.

Barbados gained its independence from Great Britain in 1966. The government system in Barbados is based on a constitutional monarchy with parliamentary democracy. The government is headed by a Prime Minister, and parliament comprises 30 members elected to the House of Assembly and 21 members appointed to the Senate. The Prime Minister chooses members from either the House of Assembly or Senate to comprise the cabinet. The existing 25 members of cabinet take a lead on specific policy areas such as health, education, tourism, foreign trade, energy, and water resources and the like.

1.3.1 Water in Barbados

In Barbados, the available sources of water are groundwater (79%), surface water (12%), spring water (6%), wastewater (2%), and runoff (1%) (Inniss 2001). The geological makeup of the island is primarily of karstic coral limestone which acts as a natural filter for the rain-fed groundwater aquifers. Therefore, rainfall is a very important factor for the natural replenishment of the groundwater supply in Barbados.

Total annual rainfall, on average, is approximately 1,500 mm in Barbados. However, there is a distinct variation in rainfall over a calendar year. From June to November is the wet season and from December to May is the dry season. Normally 70% of total annual rainfall occurs during the wet season while the remainder falls during the dry season. In addition, the spatial distribution of rainfall also varies across the island with greater amounts received in areas of higher elevation than in those that are low lying (see Figure 1.3).

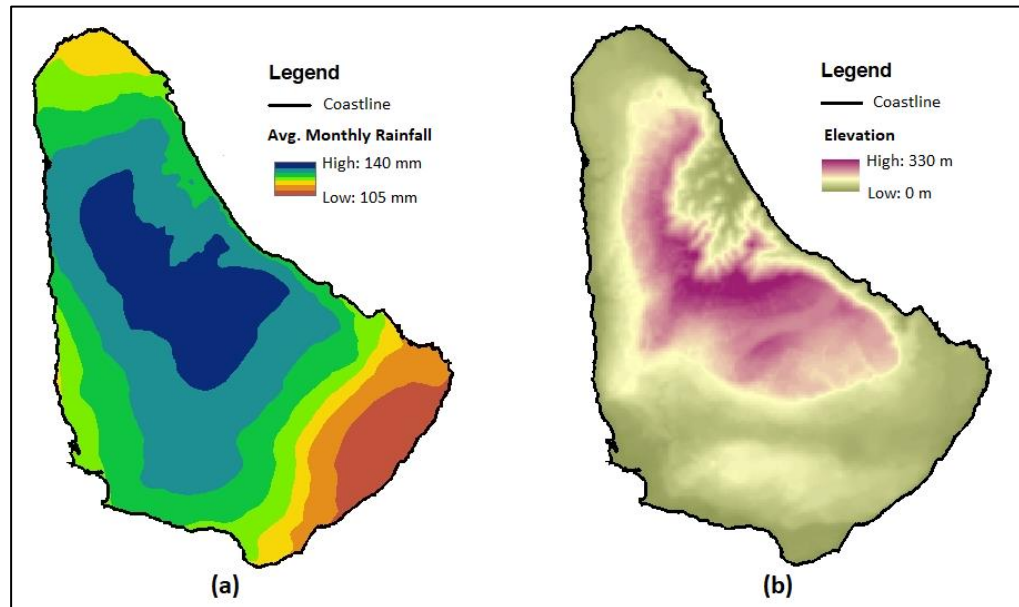


Figure 1.3 - A map of average monthly rainfall in Barbados in (a) and an elevation map of Barbados in (b).

It was estimated that only 20% of rainfall percolates into the ground to contribute to the supply of freshwater from aquifers in Barbados (Hutchinson 2010). The remaining 80% is lost to evaporation, evapotranspiration, and run-off. Furthermore, higher rainfall intensities are experienced during the wet season which causes decreased infiltration throughout the wettest period (Lowe et al. 2018) due to increased run-off. The impact of climate change can result in serious implications for a rainfall-dependent country such as Barbados since it could alter rainfall patterns by decreasing actual rainfall amounts or increasing intensities (Hoegh-Guldberg et al. 2018). Thus, it is imperative that every drop of groundwater is accounted for and managed as efficiently as possible.

1.3.2 Water Management in Barbados

In Barbados, the distribution of potable water is the sole responsibility of the Barbados Water Authority (BWA) – a statutory, state-owned entity. The commitment of BWA to monitor, assess, control, and protect local water resources can be framed against several international conventions and environmental agreements which the country has ratified or adopted. There are also national level policies and legislation to promote the sustainable use of water resources and regulate the provision of water services. Although the governance framework of BWA is considered as neutral being neither well-developed nor underdeveloped (Burdescu et al. 2020), organisational, human resource capacity, data, and financial constraints persist (GOB 2008). Key players in the oversight of the BWA are the Minister with responsibility for Water Resources (who falls under the Ministry of Transport, Works, and Water Resources), and a Board of Directors, while the day-to-day operations fall under the purview of a General Manager.

From an operational standpoint, the BWA reports a 100% coverage of potable water service to its 106,580 customers, of whom 95% are residential (Burdescu et al. 2020). Water is primarily sourced from aquifers via wells and boreholes. However, this supply is augmented with water from two natural springs and a privately owned desalination plant. Collectively, BWA pumps about 130 million litres of water each day into its system (Innis 2001). Within the WDN there is approximately 2,000 km of interconnected pipes (see Figure 1.4). The pipe network, described by the General Manager of BWA as a century-old network, has continued to exhibit signs of extensive deterioration (Downes 2018).

Total non-revenue water for the BWA, which includes water losses and unbilled authorised consumption, was reported at 55% (Burdescu et al. 2020). This figure has increased since Halcrow Inc. (2012) estimated water losses near 45% with real losses (i.e., bursts and leaks) accounting for approximately 36%. According to the system grading standards developed by McDonald, Daigle, and Fe'lio (1994), the pipe network in Barbados is considered as being in poor

condition since it averages over 40 pipe bursts per 100 km. Such statistics are quite alarming for a water scarce country with high public debt that augments its natural supply of freshwater with desalinated water at additional costs to the state-owned water utility that has financial challenges.

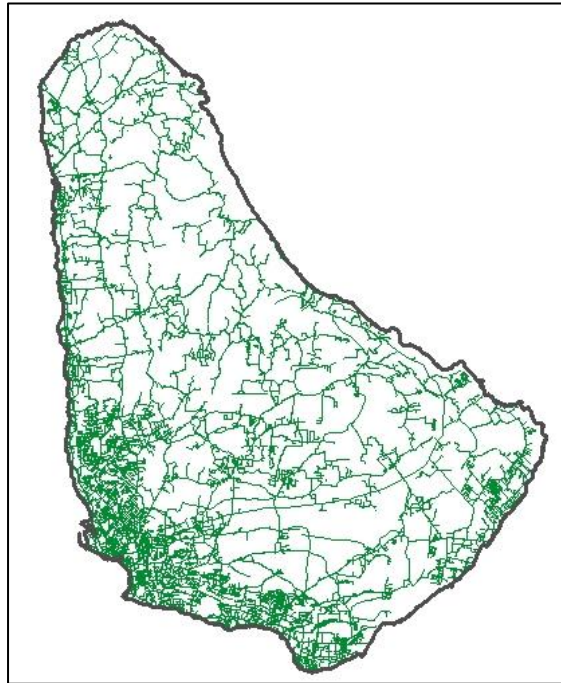


Figure 1.4 - Pipes in the WDN in Barbados.

1.3.3 Addressing Water Loss in Barbados

Over the past decade, BWA has undertaken a few projects that aimed to replace aged pipes and reduce non-revenue water and water losses. Such projects are typically financed through loans obtained by the Government on behalf of BWA since the revenue generated from water services to customers is mostly used to fund the day-to-day operations. Some of those projects are listed below:

- The ‘Water and Sanitation Systems Upgrade Project’ from 2011 to 2016 was estimated around USD\$55 million which was received as a loan from the Inter-American Development Bank (BWA 2014). About USD\$35.6 million was allocated towards the rehabilitation of the potable water supply system that included a pipe replacement component. An international consultancy company, Halcrow Inc., produced a pipe replacement plan for BWA that targeted the most critical pipes (Halcrow Inc. 2012).

- In 2015, the Caribbean Development Bank approved a USD\$39.5 million loan to assist in upgrading the water supply network to address operational inefficiencies (CDB 2015).
- In 2020, the Development Bank of Latin America provided USD\$10 million towards the ‘Barbados Water Infrastructure Rehabilitation Project’ for the modernisation of the water and wastewater treatment infrastructure and mitigating the adverse effects of climate change (CAF 2020).
- In 2020, the European Investment Bank also agreed to supply a USD\$12 million loan towards the ‘Barbados Water Infrastructure Rehabilitation Project’ (EIB 2020).

Recently, the government admitted to the mismanagement of funds, some of which should have been allocated to rehabilitating the pipe network. In fact, there is an on-going investigation of at least 20 active contracts summing to over USD\$750 million which does not proportionately reflect improvement to the service or infrastructure of BWA (BGIS 2020). Water losses have remained high and the utility has resorted to measures such as water shut-offs, trucking water, water restrictions, and revision of tariff structure. As a solution to water shortages, BWA has considered building two desalination plants to augment the water supply. However, an increased water supply does not infer better performance from the WDN. Thus, the issue of rehabilitating the pipe network cannot be circumvented if improvements are to be achieved.

Besides the rehabilitation and replacement of pipes under the project initiatives given above, the BWA also carries out *ad hoc* repairs when pipe bursts are reported. As a solution to the high burst rate of the pipe network, a former Minister of Water Resources expressed that the solution was simple – change all of the old pipes. This statement has serious connotations not only because it may be inaccurate, but it was also the stance of the chairperson of the Board and likely reflected a shared mentality of personnel within the organisation. While pipe age is certainly a factor as some pipes in the local WDN are more than 100 years old, frequent bursts are not necessarily a function of age (Boxall et al. 2007). In fact, many factors can contribute to pipe burst occurrences (see

Section 2.3). Thus, there is a need for evidence-based decision-making in the planning of pipe repair and replacement interventions in order to maximise the limited resources of the BWA and increase the resiliency of the WDN.

1.4 Research Aim

In Barbados, the most reported form of pipe failure in the local WDN is pipe bursts. The aim of this research was to identify and understand which factors contribute to pipe bursts in the WDN in Barbados, and thus better inform the planning of pipe replacement interventions taking into consideration the implications of climate change. The influence of multiple factors on pipe failure was investigated to determine which combination of factors contributed to frequent pipe burst occurrences. By doing this, areas of the pipe network most vulnerable to those factors could be ascertained and the impact of climate change on pipe burst incidents could be assessed. The effects of climate change on pipe failure in a WDN have not been widely studied, especially in tropical regions.

1.5 Research Questions

In order to achieve the research aim (see Section 1.4), several questions were posed to sequentially guide the research process:

1. What are the main drivers of pipe burst occurrences in the WDN in Barbados?
2. What patterns can be identified in the behaviour of pipe bursts in the WDN in Barbados?
3. In Barbados, which areas of the pipeline in the WDN are most susceptible to frequent pipe bursts based on the main drivers identified?
4. How can the effect of climate change impact the occurrences of pipe bursts in the WDN in Barbados?
5. What considerations should be given by the Barbados Water Authority in planning pipe replacement interventions with respect to climate change?

The subject of each research question was used to narrow the focus of the literature review carried out for this study (see Chapter 2). After the literature

review was conducted, research objectives were appended to the research questions which were jointly used as a basis to configure the research approach and strategy (see Chapter 3).

1.6 Research Scope

For this study, the main area of research was pipe burst occurrences in a WDN. The scope of this research only investigated reported pipe bursts. Some pipe bursts that have not been reported could possibly be located by performing extensive field observations. However, this was not feasible due to constraints in resources, technology, and time. Nonetheless, the scope encompassed the entire WDN in Barbados. A country-wide scale research was conducted for several reasons:

- Due to the challenges usually experienced with pipe failure data from water utilities (Zhang et al. 2018), an island-wide scale increased the amount of data available for analysis. Additional data could enhance model performance and hence improve insights.
- Previous work carried out by Mackey, Cashman, and Cumberbatch (2014) investigated pipe failure in only 2 (out of 16) of the districts in Barbados. Whereas this study included all pipes in the WDN.
- Analysis could be carried out at different resolutions (i.e., scales) depending on how the data are collected.
- A network-wide analysis could complement and lend the greatest support to on-going and future initiatives of the BWA.
- Case studies that investigated pipe failure are not usually conducted at a country-wide scale. Thus, the example and findings from this study could prove valuable for other SIDS, and the wider body of similar works.

1.7 Thesis Structure

This thesis is prepared as a traditional monograph in accordance with the regulations for higher degrees of the University of the West Indies. The main body of the thesis contains nine chapters – each started with an introduction and ended with a summary. The introduction sets the aim, content, and layout of the

chapter, while the summary captures the key points in the chapter and shows how the next chapter is linked. The thesis structure, by chapter, is shown below in Table 1.1.

Table 1.1 - The structure of the thesis by chapter.

Thesis Structure	
Chapter	Title and Description
1	<i>Introduction</i> This chapter provides a background to the status and challenges of managing WDNs across the world and the threats imposed by climate change. This background is given in order to provide an understanding of the main topics of this research - pipe failure in a WDN and climate change. The research problem, aim, and questions are stated and an overview of the arrangement of the thesis is given.
2	<i>Literature Review</i> This chapter is a literature review on pipe failure in WDNs. Focus is placed on the factors that contribute to pipe failure, approaches and methods for modelling pipe failure, the effects of climate change on pipes in a WDN, limitations and gaps in the body of knowledge, and implications for the research.
3	<i>Research Methodology</i> The research, in and of itself, is the sole focus of this chapter. Attention is placed on describing key elements of the research such as the objectives, approach, strategy, modelling framework, and ethics.
4	<i>Creation of the Database</i> This chapter is about the data used in this research. As this is a data-driven study, the issues around the obtaining and the integrity of the data are of importance and have a direct bearing on the validity of the research. The particulars of the techniques used to clean, create, fill-in, compile, and validate data are detailed. Ultimately, three datasets are derived.
5	<i>Implementation of Methods</i> This chapter presents the analytical process and details the implementation of the three methods that were applied for data analysis.
6	<i>Results – Data Mining and Pipe Burst Modelling</i> This chapter shows the results from the data mining analysis carried out on each dataset. The synthesis of the outputs and creation of a final pipe burst risk model are detailed, and a hotspot model of pipe bursts in the WDN of Barbados is produced.
7	<i>Results – Climate Change Assessment on Pipe Bursts</i> This chapter presents and analyses rainfall projections for Barbados under various climate change scenarios. The projections are used to facilitate an assessment on the effect of future rainfall on the medium and long-term distribution of pipe burst hotspots in the WDN in Barbados.
8	<i>Discussion</i> This chapter presents an interpretation of the major findings from the analyses carried out in the previous two chapters. The theoretical, methodological, and practical implications of each major finding are elaborated on. The limitations of the research are discussed and recommendations are given.

9	<i>Conclusion</i> This chapter revisits the research aim and research questions. Key findings are summarised and the contributions of the research are acknowledged.
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1.8 Chapter Summary

Most water utilities provide potable water using a water supply system that broadly consists of three main components: water sources (supply); consumers (demand); and WDNs (infrastructure). Although climate change poses a triple threat on a water supply system, the potential impacts of climate change on WDNs has not been widely studied – especially in SIDS where the implications are particularly acute. Thus, this research aims *to identify and understand which factors contribute to pipe bursts in the WDN in Barbados, and thus better inform the planning of pipe replacement interventions that considers the implications of climate change*. In the next chapter, a literature review of the fundamental aspects of pipe failure analysis in a WDN was conducted and used as a basis to inform and guide the research process.

2 LITERATURE REVIEW

2.1 Chapter Introduction

The aim of this chapter is to review works in the literature that would assist in the identification of: (a) factors that can contribute to pipe failure in WDNs; (b) methods that can be used to model pipe failure and analyse factors; and (c) possible impacts of climate change on pipe failure. This threefold literature review provides a basis used to inform how this research was designed and implemented. The topics covered in the sections of this chapter are:

- (1) An overview of WDNs and pipe failure.
- (2) A characterisation of the influence of factors that are commonly used in pipe failure studies from established findings.
- (3) Approaches and methods that can be used for investigating factors that contribute to pipe failure.
- (4) An evaluation of other works that examined the effects of climate change on pipe failure.
- (5) Limitations and gaps found in the various works included in this literature review.
- (6) Implications from this literature review on the research.

This chapter is laid out in such a way to show a progression of working through the literature from concepts and theories, to observations and findings, to methods and techniques, and to wider applications and implications of pipe failure analyses.

2.2 Water Distribution Networks and Pipe Failure

A WDN is an infrastructure of assets consisting of interconnected components used to transport water from a source to consumers. Typically, untreated water is extracted from two of the main sources that contain a natural water supply - groundwater or surface water. From the source, water is pumped to a plant for treatment where it can be stored in a reservoir or tank. Large diameter pipes, known as trunk mains, are used for the bulk transport of potable water from the treatment plant to the distribution area (Trifunovic 2020). Secondary mains and

distribution mains, which are medium diameter pipes, form the skeletal layout of the pipe network and are usually laid adjacent to roads, respectively (Trifunovic 2020). Smaller diameter pipes, or service mains, are connected to distribution mains to deliver water directly to private or public connections (Trifunovic 2020). Besides pipes, pumps, and reservoirs, other common components and devices in the network are joints, valves, and water meters.

According to Trifunovic (2020), the main objectives of a WDN are to maintain an adequate quantity and acceptable quality of water. One of the ways this is achieved is by keeping the potable water in the network pressurised. In doing so, water retention times are reduced, stagnation is prevented, chances of contamination are decreased, and water is delivered to consumers at sufficient pressure levels. Hence, it is common practice to divide a WDN into pressure or distribution zones or geographical districts to facilitate better management. However, managing a network of buried pipe has many challenges.

Pipes are traditionally laid underground instead of aboveground in most WDNs. The laying of pipes beneath the ground can provide protection, add bedding support, and enhance efficiency of distributing water directly to points of consumption due to less obstructions as compared to the surface. However, these buried assets become weakened over time due to different mechanisms such as internal and external loadings that can cause pipe joints to bend or longitudinal or circumferential cracks in pipes. Bends, cracks, and splits in pipes, which are evidence that structural integrity has been compromised, consequentially lead to leaks. Complete failure is signalled when pipes burst. The issue of bursts is important since it represents a loss of resources (i.e., water and energy), contributes to non-revenue water, becomes a potential point of contamination, and may require service disruptions, traffic interruptions, and road excavations during repair (Giraldo-González and Rodríguez 2020; Wu and Liu 2017; Berardi et al. 2008). Essentially, all pipes have a limiting life cycle that ends in inevitable failure (Singh and Adachi 2013).

It is fundamental to understand the life cycle of buried pipes for anticipating pipe failure. In theory, the operational life cycle of a pipe is depicted as a bathtub

curve (see Figure 2.1). The curve is partitioned into three phases: (1) early life, (2) useful life, and (3) wear out life (Singh and Adachi 2013; Andreou, Marks, and Clark 1987). The first phase is characterised by early pipe failure that occurs immediately after installation due to poor construction procedures or defects in pipe manufacturing. The second phase is distinguished by a relatively low and steady pipe failure rate which may be disrupted by random stresses such as abrupt changes in pressure or ground movement that promote failure. The third phase is characterised by a pronounced increase in the pipe failure rate with pipe age and extended exposure to factors that contribute to pipe deterioration. Although the general contributory factors may be inferred, in order to properly understand pipe failure, rigorous investigation of the factors needs to be undertaken to gain a better understanding of the underlying causes and present the possibility of being able to predict failures.

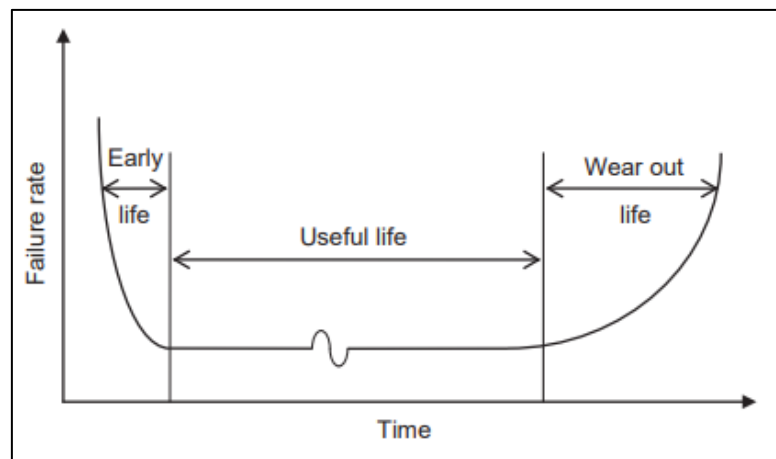


Figure 2.1 - The bathtub curve showing the theoretical rate of pipe failure for a typical pipe.

2.3 Factors that Contribute to Pipe Failure

Factors that contribute to pipe failure are typically grouped into three categories: physical, operational, and environmental. Physical factors are pipe-intrinsic and pertain to the actual attributes of a pipe. Operational factors describe or directly affect network parameters, conditions, and practices. Environmental factors refer to or affect the surrounding conditions of the pipe network. A list of these factors was compiled and shown in Table 2.1.

Table 2.1 - Factors that contribute to pipe failure. (Farmani et al 2017, Wood and Lence 2009, Savic and Walters 1999.

Factors that Contribute to Pipe Failure		
Pipe-Intrinsic	Operational	Environmental
• Pipe age • Pipe diameter • Pipe material • Pipe length • Pipe installation era • Pipe depth • Pipe wall thickness • Pipe manufacturing procedures • Pipe lining • Pipe protection (wrapped) • Pipe condition (remaining life)	• Pressure • Corrosion • Pipe burst history • Leakage • Connection density • Water quality • Flow velocity • Operational and maintenance practices	• Temperature • Rainfall • Soil type • Frost loading • Land-use • Traffic classification and loading • Under roadway • Groundwater • Ground movement • Pipe bedding or backfill material • Surface type

Many studies have investigated pipe failure by examining the relationships that exists between pipe failure and various combinations of the factors listed in Table 2.1. However, there are disagreements in the literature about which factors should be considered for analysis, the level of importance of factors, and the understanding of the mechanisms and interactions among explanatory factors. This debate is due to the fact that factors which contribute to pipe failure are dependent on the varying conditions and circumstances in different WDNs (Rajani and Tesfamariam 2007). Thus, findings will vary by case study.

Case studies that analysed pipe failure have reported findings about the significance and contribution of explanatory factors. These findings have been collated numerous times in the literature (Barton et al. 2019; Farmani et al. 2017; Wood and Lence 2009). For this literature review, insights from the findings of other case studies were used to form a knowledge base. This knowledge base was compiled to assist in gaining an understanding of some of the root causes of pipe failure. According to Tang, Parsons, and Jude (2019), this can be achieved by identifying risk factors and understanding the linkages among those factors. Therefore, in this section, this information was compiled for several factors identified from the literature. The factors included were either identified as significant, frequently included in analyses, considered relevant to the case study area of this study, or data were available. Additionally,

consideration was also given to how factors were modelled when included in analyses. In this way, gaps pertaining to data could be identified which in turn determines the type of approaches and methods that can be utilised for analysis.

2.3.1 Pipe Age

Pipe age is usually defined as the length of time (in years) since a pipe has been laid in a WDN. Older pipes tend to be more prone to pipe failure than younger pipes (Arsenio et al. 2015, Boxall et al. 2007) since longer exposure times to adverse operating conditions increase the likelihood of failure. Given that, pipe age is a variable that is frequently included in pipe failure studies (Tang, Parsons, and Jude 2019). Berardi et al. (2008) considered pipe age as the most important time-dependent factor for pipe deterioration analysis due to higher failure rates and burst frequencies of aged pipes. Many other studies have also found pipe age to be a significant factor of pipe failure (Wols et al. 2019, Li et al. 2015, Cooper et al. 2000). However, some studies debated that an increase in failure in older pipes should not automatically be attributed solely as a function of pipe age (Barton et al. 2019, Boxall et al. 2007, O'Day 1982) but other factors such as diameter, material or, corrosiveness of soil could be the underlying cause (Goulter and Kazemi 1988).

Some older pipes can be more resistant to deterioration due to thicker pipe walls that allow for greater resistance to corrosion, loading, and other forces (Barton et al. 2019). In fact, Ho, Lin, and Lo (2010) found that the performance of prediction models for pipe failure had improved when pipe age was not used as an explanatory variable.

For analysis, pipes are usually sorted into cohorts based on intervals of pipe age. An example of pipe cohorts by pipe age by Tang, Parsons, and Jude (2019) is given below:

- 0 (less than 1 year).
- Between 1 and 20 years.
- Between 20 and 40 years.
- Between 40 and 60 years.

- Between 60 and 80 years.
- More than 80 years.

Often, the age of all pipes in a distribution is not known. Thus, proxy variables for pipe age are used in pipe failure studies. Examples of common proxy variables for pipe age are year of manufacture, installation era, pipe material, and urban development age (Barton et al. 2019, Cooper et al. 2000).

2.3.2 Pipe Diameter

Pipe diameter is the measured length of a two-dimensional straight-line segment having both end points touching the circumference of the pipe passing through the centre. The diameter of imperial sized pipes (in inches) is measured by the inner diameter of a pipe, whereas, the diameter of metric sized pipes (in millimetres) is measured by the outer diameter of a pipe. For this study, pipe diameter is measured in imperial units.

Pipe diameter is considered as one of the most important factors in the analysis of pipe failure given its function in determining the ability of a pipe to resist loading forces (Boxall et al. 2007). Thus, pipe diameter has been included as an input variable in most pipe failure analyses. The general consensus in the literature is that there is an inverse relationship between pipe diameter and pipe failure; that is, smaller diameter pipes are more prone to failure than larger diameter pipes (Rezaei, Ryan, and Stoianov 2015, Shi, Zhang, and Ho 2013).

2.3.3 Pipe Material

Pipes are built using different types of materials. Properties such as strength, impact resistance, and ductility vary by material (Tang, Parsons, and Jude 2019). Therefore, the type of material of a pipe has a major impact on its structural integrity. Usually, the type of pipe material is chosen based on the function of a pipe. According to Tang, Parsons, and Jude (2019), trunk mains that convey large volumes of water at higher flows and pressures are usually made of materials known for strength (such as iron) whereas smaller service mains are usually made of materials known for elasticity (such as plastic).

Pipes in a WDN are commonly manufactured from the following materials: iron, steel, cement, and plastic. Iron pipes, such as cast iron and ductile iron, are known for their rigidity. Cast iron pipes, the oldest set of pipes, can be dated back to the mid-1800s. Ductile iron pipes, which came about a century after cast iron pipes, are characterised as being more resilient (Barton et al. 2019). Therefore, ductile iron has become the preferred choice to cast iron. Despite the known strength of iron pipes, iron pipes are susceptible cold temperatures, cold internal water temperatures, frost, and corrosive soils (Barton et al. 2019). Steel pipes came about near the same era as ductile iron pipes. In contrast to ductile iron pipes, steel pipes can withstand higher water pressures but are more susceptible to corrosion (Barton et al. 2019). Asbestos cement pipes were typically installed between the early and mid-1900s. However, due to health concerns, they were phased out of being manufactured and installed worldwide (Barton et al. 2019; Williams and Aspern 2011).

Plastic pipes have a higher resistance to corrosion than metal pipes and due to elasticity and ease of installation, there has been a substantial increase in the use of plastic pipes since its manufacturing began in the 1950s (Tang, Parsons, and Jude 2019). Common types of plastic pipes are polyvinyl chloride (PVC), polyethylene (PE), medium density polyethylene (MDPE), and high density polyethylene (HDPE). While plastic pipes have several advantages over other pipe materials, PVC pipes are susceptible to warm temperatures, shrink and swell action of soils, and exposure to sunlight (Barton et al. 2019).

2.3.4 Temperature

Temperature is a physical quantity that is used to describe the degree of heat or cold of air, soils, water, ground surface, or any other surface. The most commonly used standard unit for measuring temperature is the Celsius scale (°C). In the majority of pipe failure studies, ‘temperature’ referred to air temperature unless explicitly stated otherwise. Thus, any reference to temperature in this literature review also refers to air temperature.

Temperature is often used as an explanatory variable in pipe failure studies (Farmani et al. 2017; Laucelli et al. 2013; Rajani, Kleiner, and Sink 2012; Hu

and Hubble 2007) and is the most frequently used environmental factor. In cold regions, frost has shown to promote pipe failures due to external loading stress and ground movements caused by freezing and thawing of the ground (Barton et al. 2019; Rezaei, Ryan, and Stoianov 2015). Colder temperatures can also cause pipes to become more brittle and thus more susceptible to breaking (Rezaei, Ryan, and Stoianov 2015).

The temperature of the water found in pipes can also influence pipe failure. Cold water temperatures can cause pipes to contract which can result in circumferential breaks (Barton et al. 2019). According to Farrow et al. (2017), such failures are more common than failures due to frost. Pipes are also vulnerable to thermal expansion due to the temperature of the soil in hot climates or the freezing of water in pipes in cold climates. Pipes made from plastic material, particularly PVC and PE pipes, are less vulnerable to thermal expansion and contraction.

Various descriptors of temperature have been used in the literature in pipe failure studies: average minimum air temperature, average maximum air temperature, and changes in temperature. Some studies also included proxy variables for temperature such as freezing index and freezing air temperature (Farmani et al. 2017, Wols and van Thienen 2016).

2.3.5 Soil Type

A typical installation of a pipe involves the following steps: (1) soil excavation, (2) laying of the pipe, and (3) backfilling and covering the pipe. Hence, the nature of the soil material in which pipes are laid, soil moisture levels, the presence or absence of groundwater, and the way pipe trenches are backfilled can all have an influence on pipe failure.

The type of soil that pipes are laid in is an important factor. Soil types such as clays or peat-rich are known to shrink and swell depending on soil moisture content (Barton et al. 2019; Wood and Lence 2009). Changes in the levels of soil moisture can not only result in soil movement which can cause pipes to bend. Soil moisture is mostly influenced by rainfall but can also be impacted by

the presence of ground water if the water table is high. The presence of water in soils enables oxidation which can promote corrosion. Other properties of the soil such as chemical content, dissolved salt content, and temperature can have an effect on the corrosion of pipes – particularly pipes of metal materials. Fluctuations in soil moisture can also affect external loading on pipes and differential soil settlement. Changing levels in groundwater can also induce differential soil settlement and place a bending stress on pipes. Soil deformation is crucial in the early stages of pipe failure (Arsénio et al. 2015).

The backfilling of trenches provide support for pipes. Despite this fact, water utilities sometimes use native soils as backfill which may result in rocks, debris, and other objects being found in the soil that causes unlevelled bedding and induced loads on pipes (Wood and Lence 2009). Poor installation practices and unsuitable choices of soil types for backfill can result in failures that could have otherwise been slowed or prevented. Lengthy pipes can span multiple soil types (Mackey, Cashman, and Cumberbatch 2014) which can make these pipes more vulnerable.

Common explanatory variables pertaining to soil type that are commonly used in pipe failure studies are corrosive indicators and the shrink-swell capacity of soils (Tang, Parsons, and Jude 2019). Proxy variables for soil type are physical loading, level of pipe protection, drainage capacity, construction practices, bedding or backfill material (Wood and Lence 2009), ground movement (Savic and Walters 1999), and soil moisture deficit (Wols and van Thienen 2016).

2.3.6 Rainfall

Rainfall is the amount of rain that occurs within a specified area and period. The effect of rainfall on pipe failure can be physical or chemical. An example of a physical effect is founded on the basis that rainfall is the predominant factor that determines the moisture content of soils. The variation in soil moisture can promote soil movement which in turn can contribute to pipe failure (see Sections 2.3.5). In this way, rainfall has a physical effect and direct impact on pipe failure (Savic and Walters 1999). Another effect of rainfall on pipe deterioration is the electrochemical reaction with iron-based pipes that can

promote corrosion. Li et al. (2015) observed increased corrosion of galvanised iron pipes during periods of high rainfall. In the instance where rainfall induces soil erosion, removing soil from around the pipe and possibly exposing the pipe to the open environment, both a physical and chemical effects can take place.

A correlation between pipe failure and the effect of local seasonality in rainfall patterns has been recorded in the literature (Barton et al. 2019; Fuchs-Hanusch et al. 2013). Lower pipe failure rates are observed when soils are wet and saturated than when soils are dry (Barton et al. 2019). It was also established that climate-based phenomena such as droughts can cause an increase in pipe failure and correlations have been observed with indices such as rain deficit and antecedent precipitation index (Wols and van Thienen (2016). Several studies such as Wols and Van Thienen (2014), Hu and Hubble (2007) have included drought or rain-based indices as explanatory variables in the statistical analyses and prediction of pipe failure.

2.3.7 Water Pressure

Water pressure (commonly referred to as pressure) is the force used to move water through a pipe. The standard unit of measuring pressure in a WDN is pounds per square inch (PSI), an Imperial unit of measure. As the flow of water in a pipe is increased, the pressure exerted on the walls of the pipe is also amplified. Thus, pressure plays a major role in determining the internal force and load inside of pipes (Marzouk, Hamid, and El-Said 2015, Savic and Walters 1999) – increasing the likelihood of failure if there are weaknesses or defects (Wols et al. 2019). If the internal force exceeds the tensile strength of a pipe, then that pipe can fail and burst. Sudden changes in pressure (i.e., pressure transients) caused by rapid changes in flow in pipes including negative pressures, which results in an effect known as water hammer, can also result in pipe bursts (Punurai and Davis 2017, Rogers and Calvo (2015), Savic and Walters 1999). Additionally, cyclical pressure has also been reported to promote early pipe failure (Barton et al. 2019, Rezaei, Ryan, and Stoianov 2015).

Traditionally, hydraulic models of the pipe network are created in software to simulate pressure at various points in a WDN. However, most hydraulic models

found in water utilities are not well calibrated especially given the maintenance that is required to keep the model up to date (Wu and Liu 2017). Nevertheless, hydraulic models can provide an estimation of the distribution of pressure in pipes in the WDN. Alternatively, sensors can be used to monitor in pressure across the WDN in real-time. Such technology is useful for detecting and recording changes in pressure that can abruptly occur like a water hammer – a pressure surge or shock in a pipe that happens when flow changes suddenly. Without deployed sensors, events like water hammer are difficult to monitor.

The general consensus in the literature is that pressure is a significant factor in the occurrence of pipe bursts in a WDN (Barton et al. 2019, Ghorbanian, Karney, and Guo 2016, Marzouk, Hamid, and El-Said 2015). Pressure transients and cyclic pressure loadings can cause pipe failure in both small and large diameter pipes (Barton et al. 2019, Punurai and Davis 2017). In spite of operational standards of minimum and maximum pressure values to promote a safe, consistent, and economic service to customers (Ghorbanian, Karney, and Guo 2016), issues with pressure management can still occur due to incorrectly sized pipes, inadequate booster pumps, significant pipe deterioration in WDN, and challenges with customers located at higher elevations.

2.3.8 Traffic Loading

Traffic loading is the external load that is placed on pipes that are under or adjacent to roads because of vehicular traffic. Traffic loading can increase the external beam loading caused from the physical loading of roads, wheel loads, and traffic volumes (Wood and Lence 2009). This results in higher internal strains that can accelerate pipe deterioration and make pipes more susceptible to bursts – especially small diameter pipes (Moerman, Wols, and Diemel 2016). Traffic loading is cyclic, and it is usually higher in urbanised areas with paved roadways where drivers of motorised vehicles typically commute. Pipes that are openly exposed or laid without proper bedding support are most susceptible to traffic loading.

The quantification of traffic loading has not been standardised in pipe failure studies. Given that, it is a variable that is usually not directly included in the

analysis of pipe failure and other variables, such as land-use, are identified as proxy variables (Mounce et al. 2014; Savic and Walters 1999). Thus, the correlation between pipe failure and traffic loading has not been well established (Moerman, Wols, and Diemel 2016). Mancuso et al. (2016) used a method based on expert judgment to determine that traffic loading was a significant factor; whereas Mackey, Cashman, and Cumberbatch (2014) used a combination of expert judgment and geostatistical methods. Cooper et al. (2000) used the number of buses per hour to estimate the probability of failure for trunk mains. Moerman, Wols, and Diemel (2016) employed geostatistical methods (using road crossings, other road sections, and speed bumps to determine the frequency of pipe failure at those locations with potential peak traffic loads) and concluded that the overall contribution of traffic-loading to pipe failure was small. Although the frequency of pipe failure at road crossings was statistically significantly higher than average, the reason for this observation was not given.

2.3.9 Connection Density

Connection density can be defined as the number of service connections within a specified area. To an extent, connection density can indicate the degree of urbanisation (Skipworth et al. 2002). Areas of high connection density are characterised by increased concentrations of pipes, valves, and joints. Pipes in areas with high connection densities are considered more susceptible to bursts since joints are viewed as potential points of weakness (Boxall et al. 2007). With more connections, indicating an area of high water consumption, it is likely that water flow and pressure would be increased or comparably higher than other parts of the network in order to meet the water demand. Imposing a greater demand on the WDN can place additional stress on joints or other weak points.

Despite the implications of connection density on pipe burst occurrences in a WDN, it has not been widely used in pipe failure studies. Cooper et al. (2000) used a GIS to calculate pipe density as an input variable in a probability failure model. Also using a GIS, Mackey, Cashman, and Cumberbatch (2014) combined the spatial densities of pipeline and building rooftops contained in specified study areas to estimate connection density. Other works noted that

variables such as length (Boxall et al. 2007) and frequency of pressure transients (Skipworth et al. 2002) can be used as surrogate variables for connection density. Diameter can also be related to connection density since service connections are found on small diameter pipes.

2.4 Assessing Pipe Failure

Pipe failure studies can be grouped by the type of approach that is adopted. In general, four approaches are used to evaluate pipe failure. In this section, these approaches are described in addition to the different types of pipe failure models found in the literature.

2.4.1 Approaches for Evaluating Pipe Failure

The most common research questions investigated in pipe failure studies are listed in Table 2.2. The type of approach that is usually adopted to address each question is also denoted in Table 2.2.

Table 2.2 - Common research questions and approaches in the evaluation of pipe failure.

Evaluation of Pipe Failure		
Research Question	Main Approach	Strategy
Why do pipes fail?	Failure factor analysis (FFA)	To quantitatively assess the impact of factors that contribute to pipe failure
What is driving pipe failure?		
When is failure likely to happen?		
How do pipes fail?	Physical inspection (PI)	To excavate pipes for direct physical inspection
Where is pipe failure most prevalent?	Failure cluster analysis (FCA)	To spatially measure, assess, and map areas where pipe failure is frequent
Which pipes are frequently bursting?	Condition assessment and prioritisation (CAP)	To rate the condition of pipes and prioritise pipes by risk for replacement

In the literature, Failure Factor Analysis (FFA) is the most frequently applied approach in the evaluation of pipe failure. The (supposed) ability of FFA to answer the *why*, *what*, and *when* questions around pipe failure is one of the main reasons for its widespread use. Not only can FFA be used to directly address these key questions, but the knowledge generated can provide a basis and give

insight into the remaining questions that majority of water utilities seek to answer (see Table 2.2). Therefore, the adoption of an FFA approach is a preferred and logical starting point for evaluating pipe failure in any WDN.

Physical Inspection (PI) is an expensive and extensive approach since it involves the actual excavation of buried pipes or in-situ assessment (Giraldo-González and Rodríguez 2020). Furthermore, deriving the mathematical expressions to model the dynamics of the physical mechanisms is a complex process. Thus, PI is an approach that is seldom used. The other approaches, Failure Cluster Analysis (FCA) and Condition Assessment and Prioritisation (CAP), are more commonly used than PI. There has been a surge in the adoption of the FCA approach largely owed to the increased applications of geographic information systems (GIS) in pipe failure studies (see Section 2.7.4). Also, pipe failure studies that use a CAP approach continue to build on standard industry methods such as estimating the Likelihood of Failure (LOF) and Consequence of Failure (COF). The implications of the four general approaches on this study were evaluated in Section 2.7.1.

2.4.2 Pipe Failure Models and Methods

The physical inspection of an underground pipeline can pose considerable logistical, technical, and financial challenges, as stated before. As an alternative, a widely used technique is pipe failure modelling. The types of pipe failure models and common methods which are used were described in this section. This study draws on some of the methods presented.

Physical Models

Traditionally, pipe failure modelling was divided into physical and statistical models. In a physical model, the capacity of a pipe to withstand loading and corrosion are analysed (Giraldo-González and Rodríguez 2020). These physical mechanisms that can compromise the structural integrity of a pipe are simulated to inform pipe failure predictions (El-Abbasy et al. 2019, Wood and Lence 2009). Physical models can be quite accurate (Giraldo-González and Rodríguez 2020). However, the input data required for physical models are extremely

difficult to obtain due to time, costs, and constraints in data measurements (El-Abbasy et al. 2019, Xu et al. 2018). Due to these challenges, statistical models are typically preferred to physical models since the input data required for the former are considered more cost-effective and easier to obtain (El-Abbasy et al. 2019).

Statistical Models

In a statistical model, observed data (i.e., historical pipe failure as the dependent variable and quantifiable explanatory factors as the independent variables) are used to develop a tool that can predict pipe failure or pipe condition (Giraldo-González and Rodríguez 2020; St. Clair and Sinha 2012; Wood and Lence 2009). The concept on which statistical models for pipe failure are based was denoted by Skipworth et al. (2002) who expressed pipe bursts as a function of explanatory factors (see Equation 1). In the equation, ‘X’ represents an explanatory factor and ‘n’ represents the number of factors included.

Equation 1 - Pipe bursts as a function of its factors (Skipworth et al. 2002).

$$\text{Pipe Bursts} = f(X_1, X_2, X_3, \dots, X_n)$$

Statistical models are also used to assess relationships among input variables. The models assume that past behaviour can be used to predict future behaviour. This is a reasonable assumption though applicability becomes limited when observed data are deficient or unseen data are considered. Additionally, metrics are used to measure the goodness-of-fit or performance of statistical models. A frequently used metric is the coefficient of determination (COD or R^2) – a value between 0 and 1 which represents how much the variance in the independent variable can be explained by the independent variables included in the model. Examples of other metrics include the mean absolute error and mean square error.

Many studies have employed statistical methods such as regression (e.g. linear, exponential, logistic, and Poisson) and survival models (e.g. proportional hazard models) to analyse pipe failure (Alizadeh et al. 2019; Carlyle 2019; Motiee and Ghasemnejad 2018; St. Clair and Sinha 2012; Wang et al. 2010;

Alvisi and Franchini 2010). A deterministic approach was taken by Shamir and Howard (1979) in which exponential relationships of pipe failure rates were found on homogeneous pipes with similar age, diameter, and material using regression analysis. Similarly, Kettler and Goulter (1985) established a simple linear model that related pipe failure with pipe age while Su et al. (1987) determined a relationship between the rate of pipe bursts to pipe diameter size. A probabilistic approach adopted by Andreou, Marks, and Clark (1987) calculated the probability of the time between consecutive failures with the development of a proportional hazards model. These earlier studies were fundamental in establishing the knowledge base on the behaviour of pipe bursts and explanatory factors.

Statistical models are still being used in recent works. Bruaset and Saegrov (2018) examined the correlation between pipe failure rates and temperature data using a linear regression model. Xu et al. (2018) used lognormal, exponential, and Poisson functions to model the number of and variations in pipe failure rates. The models depicted failure over the entire life cycle of pipes and uncertainty analysis was used to determine the reliability of the model for different lifetime periods (Xu et al. 2018).

Data-driven Models

For nearly three decades, statistical methods were primarily used to model and predict pipe failure. More recently, there has been a growing interest and increased application of data-driven methods in pipe failure studies. While data-driven methods are built on statistical theory, data-driven methods can be considered as a sophisticated evolution of statistical methods that can detect non-linear and complex relationships even in large multivariate datasets. These methods utilise advanced algorithms based on artificial intelligence that have proven successful for mining useful information in a timely manner from compounded data that are too difficult for humans to handle and analyse (Wu and Liu 2017; Romano, Kapelan, and Savic 2014).

Data mining is a process that extracts knowledge by identifying patterns and relationships, which were previously unknown, in unstructured

multidimensional data. This process uses tools such as machine learning algorithms to move from data to patterns. There are multiple steps in the data mining process. These steps usually include data cleaning, feature selection, pre-processing, automating and performing statistical machine learning tasks, and the creation of visual outputs. Given the complexities of pipe failure behaviour and the upsurge in the volume and range of data collected from automated sensors and large customer databases, data mining techniques are increasingly being used to support data-driven approaches to pipe failure analysis (Giraldo-González and Rodríguez 2020; Zhang et al. 2018; Mounce and Machell 2006; Savic and Walters 1999). Hence, data-driven models have been recognised as a third type of pipe failure model (Giraldo-González and Rodríguez 2020, Xu et al. 2018, Wu and Liu 2017).

Several types of data-driven methods have been commonly used in studies that investigated different aspects of WDNs including pipe failure: artificial neural networks (ANNs), self-organising maps (SOMs), K-means clustering, evolutionary polynomial regression (EPR), decision trees, and fuzzy logic. In a review of data-driven approaches in pipe failure modelling, Wu and Liu (2017) found that the most frequently adopted approach was to combine methods – particularly one for prediction and the other for classification. This approach was shown effective and a major advantage was the better handling of uncertainty (Wu and Liu 2017). Some of the combinations of data-driven methods use in pipe failure studies were: ANNs and Fuzzy Logic – Christodoulou and Deligianni (2010), SOMs and decision trees – Ellis et al. (2015), Aksela, Aksela, and Vahala (2009), SOMs and K-means Clustering – Juntunen et al. (2013), EPR and K-Means Clustering – Kakoudakis et al. (2017), and SOMs and EPR – Mounce et al. (2016). Several of these methods are explored further below.

Artificial Neural Networks

An artificial neural network (ANN) is a model that mimics the workings of biological neural systems. In an ANN, the unit that carries out the nonlinear processing is called a neuron. ANNs consist of many interconnected neurons

that can carry out processes in parallel and pass signals (or inputs) to other neurons. As input is passed to the ANN, a neuron performs mathematical operations which transforms the input into a set of outputs. The function that determines the transformation is based on how strong the connections (or weights) are between neurons which is learnt from training examples. The training of an ANN using labelled data (or known targets) is called supervised machine learning. The goal of supervised machine learning is to predict outcomes for new data based on observed (or seen) data. Such methods are typically used to solve problems of classification or regression. The other type of machine learning, unsupervised learning, is when data are unlabelled and patterns are determined from computational algorithms without human intervention or knowledge of the clustering classes.

ANNs are considered as one of the most proficient methods in forecasting and predicting phenomena pertaining to engineering and infrastructure (Alizadeh et al. 2019). In pipe failure studies, ANNs have been used to model the rates of pipe deterioration, predict pipe bursts, and estimate the probability of pipe failure (Giraldo-González and Rodríguez 2020; Alizadeh et al. 2019; Van der Walt, Heyns, and Wilke 2018; Romano, Kapelan, and Savic 2014; Asnaashari et al. 2013; St. Clair and Sinha 2012; Christodoulou and Deligianni 2010; Mounce and Machell 2006).

Mounce and Machell (2006) used an ANN to detect pipe bursts from the analysis of water flow and pressure data. The ANNs successfully learned the patterns of pipe bursts and was useful for detecting relationships in the data (Mounce and Machell 2006). Romano, Kapelan, and Savic (2014) used an ANN in a similar application as Mounce and Machell (2006). The results from the ANN to predict signal values of pressure and flow was used to develop a Bayesian Inference System that predicted the probability of a pipe burst event (Romano, Kapelan, and Savic 2014). Combining the use of an ANN with an inference system has proven effective (Romano, Kapelan, and Savic 2014; Christodoulou and Deligianni 2010). Asnaashari et al. (2013) successfully used an ANN to predict pipe failure and concluded that ANNs are more capable of

solving nonlinear problems than classical statistical modelling methods such as multiple linear regression. Asnaashari et al. (2013) also found that the ANN was able to incorporate all input variables.

Romano, Kapelan, and Savic (2014) successfully employed several data-driven methods to automatically detect the occurrence of pipe bursts using data retrieved from a supervisory control and data acquisition (SCADA) system. An ANN was used to predict signal values of pressure and flow, statistical process control was used to analyse changes in pressure and flow for a pipe burst event, and the probability of a pipe burst event was predicted by a Bayesian Inference System (Romano, Kapelan, and Savic 2014).

Self-Organising Maps (SOMs)

A self-organising map (SOM) is an unsupervised ANN developed by Teuvo Kohonen (1990). As an unsupervised ANN, a SOM constructs its own clustering of the input data that is passed to the network so that patterns in the data can be detected without any prior knowledge. A SOM network comprises of two layers: an input layer and an output (competitive) layer. An input variable is represented as a regular low-dimensional grid (i.e., a 2-D plane) which is also called a component plane. Each grid contains neurons (or nodes) which are the mathematical linkages between the input and output layers. In the computational process, as each data-point (or observation) is iteratively analysed, neurons move closer to the best matching data-point to indicate clusters in the data. This process is repeated until the positions of neurons are stabilised. The output of low-dimensional grids makes SOMs a useful technique for visualising multidimensional datasets. The network of a SOM is depicted as a schematic model in Figure 2.2.

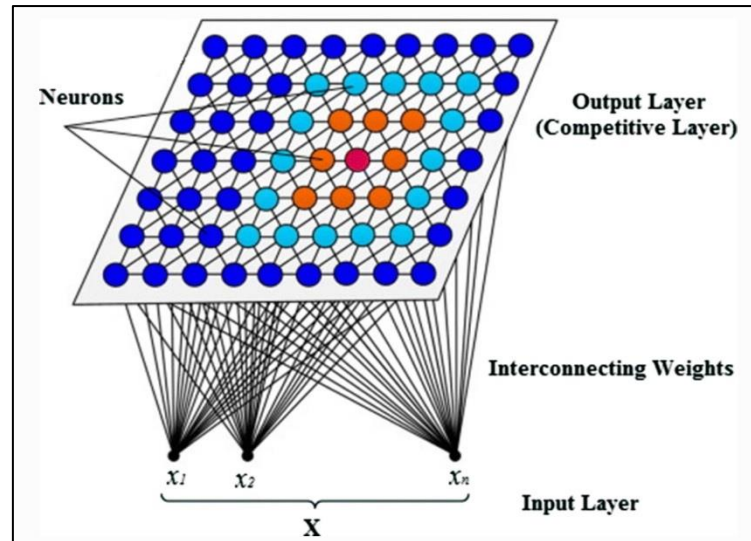


Figure 2.2 - A schematic model of a SOM network produced by Faradonbeh et al. (2020)

Figure 2.2 shows that neurons are typically laid out in a spatially ordered fashion. Neurons are represented as colour-coded hexagons in the component plane. The colouring or shading corresponds to actual numerical values of an input variable which may be referenced to a scale bar adjacent to the component plane. Usually, shades of blue and red corresponds to low and high values of said variable, respectively. This type of output presents an intuitive representation of a complex dataset which improves visualisation and facilitates interpretation (Ellis et al. 2015). Through visual inspection of component planes, patterns across variables can be assessed while the original topology of the dataset is maintained.

SOMs are noise tolerant; that is, it is a method that can handle anomalies, irregularities, and weak observations in data. SOMs can be used with sparse data (i.e., when many gaps exist in the data) and non-numeric (i.e., categorical) variables. These characteristics of SOMs, including its capabilities to process data, makes it a useful method for data mining, clustering, and classification, (Ellis et al. 2015). Thus, SOMs are considered a suitable data-driven tool for various applications in the water industry.

A review of the use of SOMs in water resources was presented in Kalteh, Hjorth and Berndtsson (2008). In the field of hydrology, SOMs were used to group data into homogeneous classes or areas (Lin and Chen 2006; Hall and Minns

1999); model stream flow (Parasuraman, Elshorbagy, and Carey 2006; Hsu et al. 2002); and decompose hydrographs and rainfall-runoff processes (Jain and Srinivasulu 2006; Furundzic 1998). The application of SOMs to promote knowledge discovery and for modelling and analysing potential factors affecting drinking water quality appear in several studies (Speight, Mounce, and Boxall 2019; Blokker et al. 2016; Mounce et al. 2016; Juntunen et al. 2013; Mounce et al. 2012). However, fewer studies have used SOMs in the analysis of pipe failure.

Aksela, Aksela, and Vahala (2009) created a leak detection model by using SOMs to predict values for a leak function using water flow data. Ellis et al. (2015) used SOMs for data exploration to detect patterns that indicated factors that were contributing to failure in unlined iron pipes. The findings from the data exploration were used to inform the development of a model to predict iron failures and to identify relevant input variables (Ellis et al. 2015). Despite its wider use in works that investigated water quality in WDNs, no other pipe failure studies were found employing SOMs.

Evolutionary Polynomial Regression

Evolutionary polynomial regression (EPR) is a hybrid technique that combines linear regression and genetic algorithms to generate mathematical models. A multi-objective algorithm is used to optimise the balance between the law of parsimony (i.e., models that provide the simplest explanation) and goodness of fit (i.e., a measure of how accurate the model fits the data). From one EPR run, multiple models are generated representing that trade-off between the number of terms in an expression and accuracy (Giraldo-González and Rodríguez 2020; Berardi et al. 2008). The models produced are usually polynomial expressions with terms that can consist of a combination of input variables with assigned exponents (Lauccelli et al. 2013). The EPR technique was developed by Giustolisi and Savic (2006) and further enhanced by Giustolisi and Savic (2009).

EPR has been used for modelling in several fields including water resources management, engineering, environment, and agriculture. As it pertains to pipe

failure in WDNs, EPR has been used to model and predict failure rates (Giraldo-González and Rodríguez 2020; Kakoudakis et al. 2017; Laucelli et al. 2013; Xu et al. 2011; Berardi et al. 2008). Key findings from these studies that used EPR to analyse pipe failure are discussed below.

Regarding the accuracy of EPR models, Berardi et al. (2008) found that it improved when pipes were homogeneously grouped. Similarly, Giraldo-González and Rodríguez (2020) determined that EPR models had sufficient accuracy for pipes grouped by pipe age and pipe diameter although model performance declined for pipes with smaller diameters. The performance of models produced by EPR was compared to another data-driven technique, genetic programming, by Xu et al. (2011). It was found that the mathematical expressions produced by EPR showed uniformity and good parameter estimation; however, genetic programming was better at detecting complex relationships (Xu et al. 2011).

Laucelli et al. (2013) used EPR to investigate the relationships between pipe bursts and climatic variables. The data were partitioned by warm and cold seasons. Laucelli et al. (2013) found that EPR models generated for the cold season had a better performance than for the warm season. The reason for the difference in model performances was that the most significant factor, freezing index, was captured by proxy variables.

Kakoudakis et al. (2017) complemented the use of EPR with K-means clustering. K-means clustering is an unsupervised machine learning method that can partition data into a predetermined number of clusters. Kakoudakis et al. (2017) used EPR to generate prediction models for pipe failure using pipe age, pipe diameter, and pipe length as explanatory variables, while employing K-means clustering to split the cohorted data into clusters of training and testing data. Pipe cohorts were grouped according to pipe age, pipe diameter, and soil type. Kakoudakis et al. (2017) only used this approach for the cast iron pipes in the WDN. However, this approach of combined methods was found to improve model accuracy when failures were low and high (Kakoudakis et al. 2017).

Fuzzy Logic

In classical logic, a concept is defined as either false or true. The values of zero and one are theoretically assigned to the classes of true and false, respectively. However, in fuzzy logic it is posited that a continuous scale exists between true and false; that is, the set of real numbers between 0 and 1 (Zadeh 1965). Therefore, in fuzzy logic a concept is defined based on a degree of truth. The theory of fuzzy logic, developed by Zadeh (1965), uses multi-valued logic to represent concepts that are ambiguous or inexact. Fuzzy logic has been used in different fields to resemble human reasoning in decision-making systems that can be expert or knowledge based.

In some studies that have analysed pipes in a WDN, fuzzy logic was used to predict pipe failure and deterioration rates (El-Abbasy et al. 2019; Rajani and Tesfamariam 2007; Najjaran, Sadiq, and Rajani 2006; Kleiner, Sadiq, and Rajani 2004), to model the risk of pipe failure (Christodoulou and Deligianni 2010; Fares and Zayed 2010), and to develop a pipe replacement or rehabilitation plan (Salehi, Ghazizadeh, and Tabesh 2018; Ho, Lin, and Lo 2010; Makropoulos and Butler 2005). St. Clair and Sinha (2012) showed how most of these works effectively paired fuzzy logic in combination with other methods such as artificial neural networks (Christodoulou and Deligianni 2010; Ho, Lin, and Lo 2010).

Fuzzy logic was also found useful when shortcomings in the data existed or the performance of prediction models were poor (Salehi, Ghazizadeh, and Tabesh 2018; Rajani and Tesfamariam 2007; Kleiner, Sadiq, and Rajani 2004). In a study that estimated pipe failure, Rajani and Tesfamariam (2007) applied fuzzy logic after not being able to successfully established reliable probability distributions due to limitations in available data. Rajani and Tesfamariam (2007) found that fuzzy logic was able to deal with the uncertainties in the data. Likewise, Salehi, Ghazizadeh, and Tabesh (2018) recognised fuzzy logic as a preferred method to handle data received from water utilities which are typically imprecise and vague.

2.5 Evaluating the Impact of Climate Change on Pipe Failure

In 1979, a global appeal was made at the First World Climate Conference to expand the knowledge that existed on climate by determining the impacts that a changing climate might impose on humanity in the future (Zillman 2009). Since that time, there has been a substantial increase in the understanding of those impacts largely owing to the state-of-the-art advancements in climate modelling. Examples of those advancements included downscaling global climate models (GCMs) to regional climate models (RCMs), and disseminating short- and medium-term projections to analysts and practitioners which were more useful than 100-year forecast horizons. Such achievements have allowed climate model outputs to be used in multiple research fields (Heaphy 2015; Vohland, Rannow, and Stagl 2014), inclusive of pipe failure studies.

As shown earlier in this chapter, climatic factors such as temperature and rainfall can contribute to pipe failure in a WDN (see Sections 2.3.4 and 2.3.6). The effect of climate change and climate variability can increase or decrease the baseline intensity and distribution of such climatic factors which may accelerate or decelerate pipe deterioration processes (Wols and van Thienen 2016, 2014). In the instance of accelerated pipe deterioration, the structural integrity of a pipeline can be further compromised by climate change and result in eventual pipe failure (Wols and van Thienen 2016). For example, the effect of freezing and thawing of the ground can place additional loading stress on buried pipes (see Section 2.3.5). If climate change causes winters to become harsher in such locations, then the effect of freezing and thawing can be exacerbated and likely result in increased failures. Nonetheless, only a few studies have analysed the impact of climate change on pipe failure.

Using linear regression, Wols and van Thienen (2013) conducted a case study in the Netherlands establishing correlations between pipe failure and weather-related factors such as temperature and drought. Building on that study, Wols and van Thienen (2016, 2014) developed a model that used ambient temperature as a dependent variable to predict pipe failure rates. To assess the impact of climate change on pipe failure, future values of daily mean temperature were

extracted from a climate explorer tool to be analysed and incorporated into the model. Wols and van Thienen (2016) used those climate outputs to derive a function for daily mean temperature for 4 climate scenarios for the years 2050 (a time slice between 2036 and 2065) and 2100 (a time slice between 2086 and 2115). In the analysis, pipe age and pipe material were also considered. However, Wols and van Thienen (2016) had not included drought since temperature had a greater influence on pipe failure in that network and a correlation was detected between temperature and drought. The results of that study found the effects of climate change on pipe failure varied by type of pipe material. Wols and van Thienen (2016) concluded that WDNs which predominantly contained asbestos cement pipes will experience higher frequencies of pipe failure than WDNs with an equivalent amount of grey cast iron pipes. Additionally, WDNs that comprise mostly of PVC pipes are expected to incur the lowest frequencies of pipe failure in light of climate change (Wols and van Thienen 2016).

Similar to Wols and van Thienen (2016), Bruaset and Sægrov (2018) also used linear regression and future projections of temperature to evaluate the impacts of climate change on pipe failure in WDNs across Norwegian territories. Temperature projections were based on the climate change scenarios, Representative Concentration Pathways (RCPs), adopted by the International Panel of Climate Change (IPCC). The projections were downscaled, and values were generated to the year 2070. Bruaset and Sægrov (2018) showed that an increase in temperature resulted in a decrease in pipe failure by 2.7% to 7.2% as rates were usually highest during cold months. It was also shown that a substantial number of failures were caused by interactions between pipes and soil, such as frost loading or changes in soil temperatures, that induced bending (Bruaset and Sægrov 2018).

In a more recent study, Wols et al. (2019) included explanatory variables such temperature, wind gust, precipitation, and evapotranspiration to assess the effects of weather on pipe failure in the Netherlands and the implications of climate change. Non-linear methods based on machine learning were used for

data mining to detect correlations between pipe failure rates, weather conditions, and soil characteristics. Ultimately, Wols et al. (2019) found soil settlements to be the biggest threat on a WDN posed by climate change. However, the impacts of climate change were less when compared to the effects of pipe age and pipe replacements (Wols et al. 2019).

2.6 Challenges and Gaps in Pipe Failure Studies

From this literature review, a common challenge encountered in conducting pipe failure analyses was identified along with a major research gap. These shortcomings are discussed in the sections below.

2.6.1 Data Quantity and Quality

Pipe failure models are limited by the deficiencies in the quality and quantity of available data regarding pipe failure history and pipe asset data from water utilities (Zhang et al. 2018, Boxall et al. 2007, Wood and Lence 2009). Moreover, many factors which are known to contribute to pipe failure are not directly quantifiable (Boxall et al. 2007). These factors include connection density, traffic loading, improper pipe choice, electrical corrosion, overload and vibration, water hammer, ground collapse, and construction damage (Ho, Lin, and Lo 2010). The difficulties in quantifying such factors, in addition to shortcomings in data quality and quantity, has led to the reliance on a limited set of perceived contributing variables (i.e., pipe age, pipe material, and pipe diameter) and the use of cohorts (i.e., grouped data) for analysis. The robustness of a methodology can be negatively impacted when pipe failure models do not take operational and environmental factors into consideration (El-Abbasy et al. 2019). Therefore, collating data from various sources and findings new ways to quantify possible contributing factors may prove beneficial in advancing and enhancing pipe failure studies (Van Daal and Raterman 2013).

2.6.2 Assessing the Impacts of Climate Change

Despite the increased availability of climate model outputs and recognition of some possible effects of future climate on pipes in a WDN, there is still a gap in the knowledge of how climate change will impact pipe failure rates globally

(Bruaset and Sægrov 2018). Relative to the total number of pipe failure studies carried out, only a small percentage of those works evaluated the possible effects of climate change (see Section 2.5). Additionally, these works have mostly examined the effects of changes in temperature and frost and have been predominantly conducted using case studies from Europe. In fact, most pipe failure studies are carried out in Europe and North America (Shi, Zhang, and Ho 2013). Such cold regions are expected to experience different impacts from climate change than warmer regions (Bruaset and Sægrov 2018). Therefore, more studies are needed in warmer regions in general (Shi, Zhang, and Ho 2013) and in instances where there is evidence that climate-related factors contribute to pipe failures. In doing so, the influence of climate change on these factors can be investigated to analyse what changes in the amount and frequency would cause a significant impact on pipe failure. This was also identified as a gap since studies that have considered the effects of climate change did not provide enough details in explaining the dynamics or quantifying the impacts (Bruaset and Sægrov 2018; Wols and van Thienen 2016; UKWIR 2012). However, before carrying out such an investigation, the scale of the resolution required for downscaled outputs from GCMs and RCMs to capture such variations would need to be determined.

2.7 Implications for the Research

In this literature review, various works that analysed pipe failure in WDNs were examined and critiqued. The existing research was generally based on the concept that pipe failure can be expressed as function of a set of explanatory factors. This was likewise used as the theoretical basis for this research. Although this foundational concept of pipe failure analysis is widely accepted and applied, the study design of different works varied. Therefore, this literature review was also used to inform the selection of approaches and methods for analysis, factor selection and data-handling techniques, and considerations for assessing climate change – all of which were critical aspects of this study. The implications of the literature review on each of the abovementioned aspects are discussed in the sections below.

2.7.1 Selection of Approaches

The approaches used to evaluate pipe failure are failure factor analysis (FFA), physical inspection (PI), failure cluster analysis (FCA), and condition assessment and prioritisation (CAP) – as presented in Table 2.2. FFA, FCA, and CAP are approaches that are more feasible to carry out than PI. However, FFA can provide more fundamental insight than FCA and CAP since it can address several key aspects of pipe failure. PI was deemed not applicable and out of scope since it required pipes to be excavated. In consideration of the pros and cons, the approaches were matched to the research questions of this study (see Section 1.5) based on how suitable the approach was for addressing the research questions (Table 2.3).

Table 2.3 – Matching approaches for evaluating pipe failure to the research questions.

Applicable Approaches for Evaluating Pipe Failure	
Research Question	Approach
1. What are the main drivers of pipe burst occurrences in the WDN in Barbados?	Failure factor analysis
2. What patterns can be identified in the behaviour of pipe bursts in the WDN in Barbados?	Failure factor analysis; Failure cluster analysis
3. In Barbados, which areas of the pipeline in the WDN are most susceptible to frequent pipe bursts?	Failure cluster analysis
4. How can the effect of climate change on rainfall impact the occurrences of pipe bursts in the WDN in Barbados?	Failure factor analysis; Failure cluster analysis
5. What considerations should be given by the Barbados Water Authority in planning pipe replacement interventions with respect to climate change?	Condition assessment and prioritisation

From Table 2.3, the FFA approach matched with most of the research questions and thus, it was selected as the main approach to evaluate pipe failure. FCA and CAP were also identified as being most suitable for some of the research questions (see Table 2.3). Thus, aspects of these approaches would also be adopted.

2.7.2 Selection of Methods

The three types of methods used to analyse pipe failure were physical, statistical, and data-driven (see Section 2.4.2). Physical models usually

produced very accurate models; however, it was not feasible to collect the extensive amount of data required about the physical mechanisms and interactions that occur beneath the surface. Statistical methods have been the predominant method for pipe failure studies for decades as these were shown effective for pipe failure modelling. However, statistical methods were not found to be as robust as data-driven methods especially for the handling of data that is sparse, complex, and contains a mixture of numeric and non-numeric variables. Therefore, data-driven methods were for the analyses carried out in this study.

It was observed that combining data-driven methods generated better modelling results and also handled the level of uncertainty associated with pipe failure data (Wu and Liu 2017). Giraldo-González and Rodríguez (2020) found that few studies have incorporated more than two methods and case studies in developing nations are even fewer. Thus, this study used the methods of SOMs and EPR for data mining and Fuzzy Logic for developing a model to depict pipe burst risk.

SOM was selected since it was an unsupervised machine learning method that did not require *a priori* knowledge for discerning patterns. Although SOMs have not been used as frequently in pipe failure studies when compared to water quality studies, it has been successfully implemented in many studies and showed to possibly generate better results when combined with predictive models. Therefore, EPR was chosen as a secondary, complementary method since its advanced regression techniques generated mathematical expressions that could be used to assess the strength of relationships between variables. Fuzzy Logic was selected since it was an inference system, which is also capable of handling vagueness in data, that could use the findings from the data mining analysis and expert knowledge to inform the development of a pipe burst risk model. Furthermore, other works have shown that Fuzzy Logic has not only been coupled with other data-driven methods to evaluate pipe failure but has also found useful when models performed poorly.

2.7.3 Selection of Factors

From the literature, an extensive list of factors that can contribute to pipe failure was compiled (see Table 2.1). A review was carried out on nine factors for which local data were available or for its surrogate factors (see Section 2.3). From that set of factors, it was determined some were frequently used in pipe failure studies due to significance (i.e., age, diameter, material, temperature, soil type, and pressure). However, some factors had theoretical significance but were often not included because of challenges with quantification (i.e., connection density and traffic loading) or greater use of proxy variables and indices (i.e., rainfall). Given that this study was exploratory, eight of the factors (or proxy variables) were incorporated. Temperature was not included in the research since it was relatively stable and did not show significant variance in Barbados even beyond annual seasonality. Although temperature was a significant variable in pipe failure studies that assessed climate change, it was the effect of temperature on the surrounding ground and soil that was the mechanism promoting failure. However, data for local soil temperatures were not available.

The review was also used to advise on grouping parameters by value (i.e., cohorts), assigning risk values to parameters of categorical variables such as material and soil type based on a consensus of findings, creating and quantifying factors (i.e., connection density and traffic loading), and providing a foundation for understanding the interactions between pipe failure and its explanatory factors that were used in the interpretation of pipe failure model outputs.

2.7.4 Using a Geographic Information System

A geographic information system (GIS) is the most widely used platform built with geospatial tools to assist analysts in visually representing data. In the early 1990s, GIS was being used as a tool in water utilities for mapping and illustrating management plans (Aburawe 2019). Since that time, geospatial tools have evolved to include functions such as geocoding, georeferencing, and digitisation which have led to major advancements in digital mapping and spatial analysis.

An observation from the literature was that pipe failure studies that incorporated variables with spatial characteristics, which were primarily environmental factors, used GIS as a multi-functional tool for data-linking, spatial analysis, and visualisation of outputs (Aburawe 2019, Van Daal and Raterman 2013, Shi, Zhang, and Ho 2013). Given that this study included variables such as rainfall and soil type, a GIS was used to carry out interpolations, perform spatial calculations, and link variables in datasets. It was also used for creating and quantifying new variables and compiling the final datasets (see Chapter 4). Furthermore, FCA and CAP were two of the approaches adopted for this study (see Section 2.7.1). For both approaches, GIS is a suitable tool for conducting spatial and non-destructive condition assessments (Arsenio et al. 2015) and generating hotspot maps to visualise the most critical pipes (see Section 5.5).

2.8 Chapter Summary

In this chapter, three sets of literature were reviewed: (1) factors that can influence pipe failure in a WDN, (2) methods that can be used to analyse pipe failure factors, and (3) the possible effects of climate change on pipe failure. It was found that a range of factors can contribute to pipe failure. Findings from various pipe failure studies on a subset of factors that are commonly used in pipe failure analysis or of particular interest to this research were detailed. Additionally, it was discerned that while statistical methods have been traditionally used to analyse pipe failure, there has been an uptake and increase in applying methods based on machine learning which are proving to be accurate, flexible, and more robust. The literature review also highlighted a major gap in the literature – the limited number of pipe failure studies that considered the implications of climate change. Given all the above, this literature review steered the selection of factors that were included in the analyses carried out in this study, the methods that were adopted, and the approach used to assess the effects climate change on pipe failure. In the next chapter, all critical components used to structure, frame, and guide the research were detailed.

3 RESEARCH METHODOLOGY

3.1 Chapter Introduction

The aim of this chapter is to define, illustrate, explain, and substantiate key components that guided how this research was conducted. Here, the following is presented about the research: objectives, approach, strategy, modelling framework, and ethics. This chapter is laid out to logically present the research structure and the rationale of the combination of approaches and methods chosen.

3.2 Research Objectives

A research methodology is a set of logical steps used to investigate a research problem and to evaluate the methods adopted (Patel and Patel 2019; Creswell 2014). For this study, the problem was identified as the need for evidence-based research on pipe failure occurrences in the WDN in Barbados (see Section 1.2). Thus, the aim of this research was *to identify and understand which factors contribute to pipe bursts in the WDN in Barbados, and thus better inform the planning of pipe replacement interventions that considers the implications of climate change* (see Section 1.4). In order to achieve the overall aim, a series of interlinked research questions were framed (see Section 1.5). These questions were expanded out to research objectives in Table 3.1.

Table 3.1 - Research questions and objectives.

Research Questions	Research Objectives
1. What are the main drivers of pipe burst occurrences in the WDN in Barbados?	1a. To make efficient use of various types of available data to build a database of the characteristics of the pipe network and historical pipe bursts.
	1b. To create a detailed geo-spatial database containing GIS layers of explanatory factors that contribute to pipe burst occurrences.
	1c. To evaluate the importance of explanatory factors by modelling pipe bursts.
2. What patterns can be identified in the behaviour of pipe bursts in the WDN in Barbados?	2a. To determine temporal and spatial trends in historical pipe burst occurrences.
	2b. To discover relationships that may exist between pipe bursts and explanatory factors using quantitative methods.

Research Questions	Research Objectives
	2c. To assess which methods of analysis can derive information that reflect on observed pipe burst occurrences and promote knowledge discovery.
3. In Barbados, which areas of the pipeline in the WDN are most susceptible to frequent pipe bursts based on the main drivers identified?	3a. To develop a risk ranking system to that can be used to inform the planning of pipe replacement interventions that target peak pipe bursts.
	3b. To generate a hotspot map of the pipe network that depicts risk to burst occurrences.
4. How can the effect of climate change impact the occurrences of pipe bursts in the WDN in Barbados?	4. To assess the effects of climate change on the location and frequency of pipe burst hotspots in the future.
5. What considerations should be given, if any, by the Barbados Water Authority in planning pipe replacement interventions with respect to climate change?	5. To provide insight that can assist in the medium and long-term planning of pipe replacement interventions.

3.3 Research Approach

Research approaches are broadly classified as deductive or inductive. In a deductive approach, hypotheses are postulated from theories and tested using empirical data. In an inductive approach, patterns are detected from empirical data and used to derive theories. For this study, although the widely accepted concept that pipe bursts are a function of its explanatory factors (Skipworth et al. 2002) was used as the basis, an inductive approach was essentially taken.

An inductive approach was adopted in order to facilitate an exploratory research. This approach was considered appropriate since the methodologies that were proposed in the literature were not found suitable and therefore this study would start at the beginning. Additionally, most pipe failure studies in the literature had not been carried out in subtropical and tropical areas, and in Barbados, the few instances of previous works used subjective methods to evaluate pipe failure (Mackey, Cashman, and Cumberbatch 2014; Halcrow Inc. 2012). Due to this lack of comparative studies, exploration was carried out before modelling. Thus, this research was designed similar to a data-driven case study – exploring and detecting patterns from observational data that used a comprehensive range of factors to reduce biases instead of overreliance on just a few variables. In this way, knowledge about pipe burst behaviour was

established for Barbados. This approach also offered the possibility of generalising the findings in such a way as to inform further studies and water utilities in other locales in the region.

Based on the approach, research methods were chosen from the following categories: quantitative, qualitative, and mixed methods (Creswell 2014). Quantitative data and methods were mostly used for this research. As evident from the literature, quantitative data and methods have been predominantly used to analyse pipe failure (see Section 2.4.2). However, qualitative data such as ordinal variables can enhance the performance of pipe failure models (Wu and Liu 2017) and promote knowledge discovery with the application of data-driven methods. Thus, non-numeric (or categorical) variables were included in this research. The categorical variables that were included required ordering at the point in analysis that assessed risk (see Section 4.5). This ordering required qualitative judgements to be made which were informed by the literature (see Section 2.3). In addition, the analysis of this research was not purely data-driven since:

- (1) Methods, such as SOMs, required qualitative judgements to be made in the interpretation of outputs (see Section 6.2).
- (2) Results from the data mining analysis were compared and synthesised (see Section 6.4).
- (3) Membership functions in the pipe burst risk model developed with fuzzy logic were user-defined (see Section 6.5.1).

Therefore, critical aspects of this research were qualitative in nature. Nevertheless, steps were taken to reduce subjectivity and increase validity and reliability. These steps were presented and detailed in Chapters 4 and 8.

3.4 Research Strategy

A research strategy was developed as a plan to achieve the aim and objectives of this study. The structure of the strategy was based on seven stages. In this section, the strategy is explained through the purpose and steps taken at each stage. The layout of the strategy is depicted as a schematic diagram in Figure

3.1. In the schematic diagram, research objectives were integrated into the strategy flow and the steps taken were colour-coded to indicate the part of the research process being carried out.

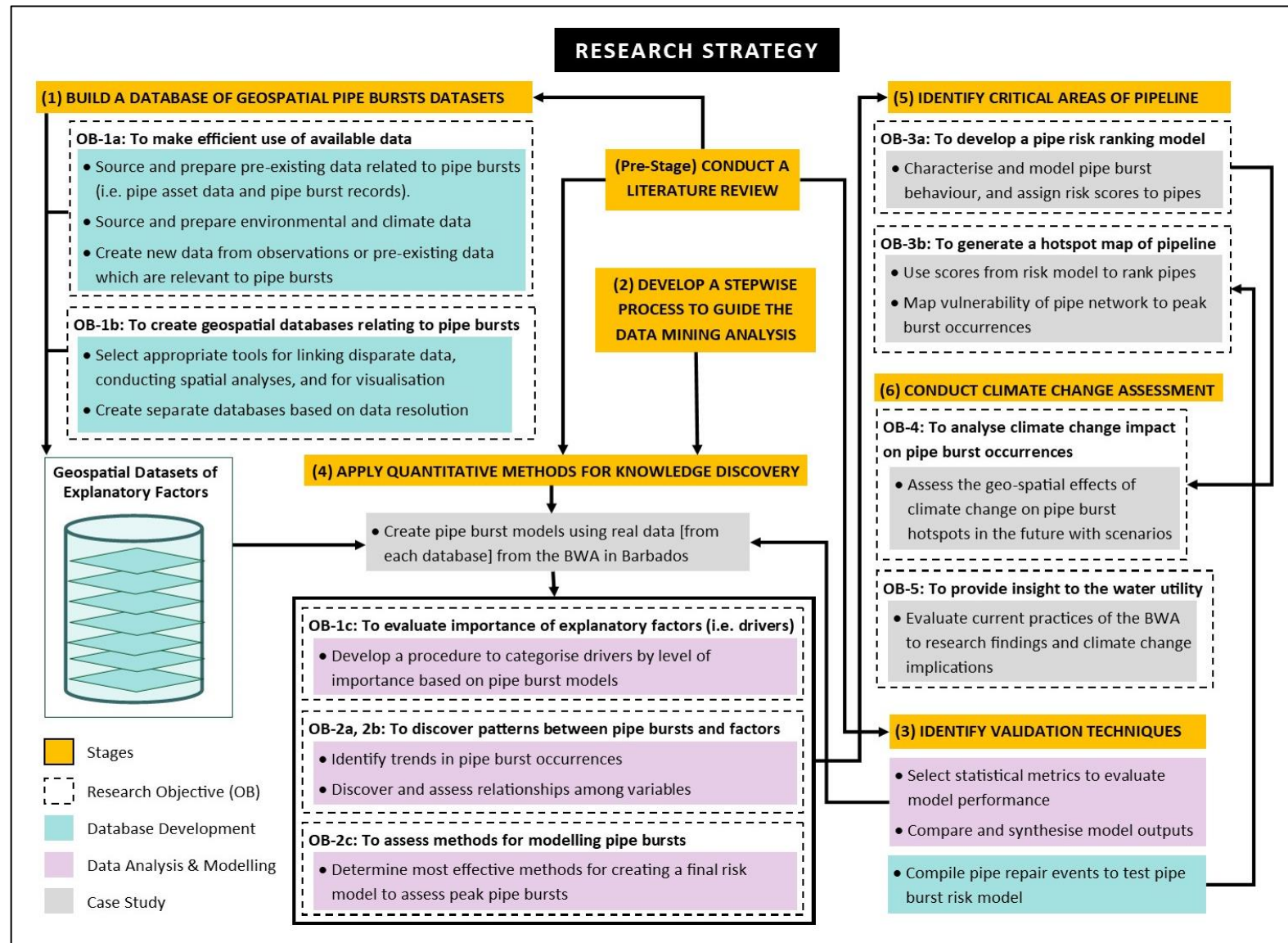


Figure 3.1 - A schematic diagram of the research strategy.

Pre-Stage

This research strategy followed from a literature review conducted on pipe failure in a WDN. Information was sourced from peer-reviewed journal articles, books, reports (i.e., consultancy, academic, and organisational), newspaper and online articles, and conference proceedings (see References). From the review, it was determined that modelling pipe bursts as a function of explanatory factors was the best approach for achieving the first part of the research aim (i.e., identifying and understanding which factors contribute to pipe bursts). The set of possible explanatory factors that could be included in the pipe bursts analysis were identified from the findings of other pipe failure studies carried out across the globe, though predominantly in the Global North. An understanding of the physical and dynamic interactions between factors was gained. The review was also used to evaluate the pros and cons of methods that were available for supporting an exploratory analysis. The lessons learnt were used to inform which methods were selected and how the methods were applied. This stage of the research strategy is captured in Chapter 2.

Stage 1

Informed by the literature review, a database that contained geospatial datasets of reported pipe bursts and explanatory factors was built. Data were sourced from multiple entities, however, the BWA was the primary source (see Section 4.2). Based on the factors identified from the review and data that were available, the final set of variables included in data mining analysis were (1) reported pipe bursts, (2) pipe material, (3) pipe diameter, (4) pipe installation era (i.e., a proxy for pipe age), (5) water pressure, (6) rainfall, (7) soil type, (8) connection density, and (9) land-use (i.e. a proxy for traffic loading – see Section 4.2). Reported pipe bursts were recorded by pipe diameter, district, and subdistrict in three distinct datasets. These datasets could not be merged, and thus, all were incorporated into the analysis to make efficient use of available data (i.e., research objective 1a) and to provide more data to facilitate data mining. A GIS was selected as an appropriate tool to transform all data into geospatial data so that data could be linked, analysed, and visualised. Thus, the

final database comprised of three distinct geospatial pipe burst datasets (i.e., research objective 1b). This stage of the research strategy is captured in Chapter 4 of this thesis.

Stage 2

An analytical process to guide the data mining analysis was developed. This was a necessary step for this study since two methods were identified from the literature review for data mining and pipe burst modelling: SOMS and EPR. Given that, along with the three datasets that were to be incorporated into the analysis, a set of stepwise rules were developed in an iterative manner to systematically guide how the factors in each dataset would be analysed with each method. This stage of the research strategy is captured in Chapter 5 of this thesis.

Stage 3

Having selected the methods and developed an analytical process, it was important to identify parts of the analysis that required validation. Therefore, metrics and techniques that would be used to evaluate and validate pipe burst models were identified. Additionally, all findings from modelling pipe bursts were compared and synthesised to enhance the validity of the knowledge derived from data mining. This knowledge was used to inform the development of a single pipe burst risk model that depicted burst behaviour of the pipe network. The validity of the pipe burst risk model was tested using pipe repair events of the BWA. This stage of the research strategy is captured in Chapter 5 of this thesis.

Stage 4

A data mining analysis was carried out to promote knowledge discovery. The analysis was informed from the literature review and guided by the stepwise analytical rules. The analysis incorporated the three geospatial datasets created for this study which were analysed using SOMS and EPR. This resulted in multiple pipe failure models, the outputs of which were assessed to identify trends and discover relationships between pipe bursts and explanatory factors

(i.e., research objectives 2a and 2b). This also allowed for an evaluation of the effectiveness methods employed (i.e., research objective 2c) using validation techniques. The findings from the data mining analysis were synthesised and used to identify the most significant factors (i.e., research objective 1c) and characterise pipe burst behaviour. By doing this, the first part of the research aim could be achieved. This stage of the research strategy is captured in Chapter 6 of this thesis.

Stage 5

In this stage, the most critical parts of the pipe network which were susceptible to peak pipe bursts (or the highest recorded burst frequency) were identified. This was accomplished by utilising the knowledge derived from the data mining analysis and pipe failure modelling to inform the development a single pipe burst risk model (i.e., research objective 3a). This pipe burst risk model was based on fuzzy logic and created with a fuzzy inference system. The model assigned risk scores to pipes based on its status in relation to factors deemed as most significant. The model was tested and validated using pipe repair events of the BWA. Ultimately, the model was used to assess the pipe network under current conditions. The outputs were used to generate a hotspot map which is research objective 3b. This stage of the research strategy is captured in Chapter 6 of this thesis.

Stage 6

For the final stage, an assessment on the impact of climate change on pipe bursts in Barbados was conducted. In the previous stage, a pipe failure model based on the factors identified as significantly contributing to peak pipe bursts in Barbados was developed. Future rainfall for Barbados was incorporated into the abovementioned model by transforming climate change projections for rainfall into geospatial data that could then be linked to the input dataset using a GIS. The outputs from the pipe failure model that included rainfall projections produced a ranking that was used to identify critical pipes and a hotspot map of the pipe network. This procedure was carried out for several climate change scenarios (i.e., research objective 4). These assessments were used to provide

insight to the BWA about the implications of climate change on pipe bursts (i.e., research objective 5). This stage of the research strategy is captured in Chapter 7 of this thesis.

3.5 Modelling Framework

A modelling framework was designed for this research (see Figure 3.2). This framework, comparable to Winkler et al. (2018), used machine learning to model pipe failure in a WDN for a real-life case study. While Winkler et al. (2018) employed a supervised machine learning method which required models to be trained using specified outputs, this study used an unsupervised machine learning method which does not specify outputs but detects patterns in the data. In unsupervised learning, the outputs are not labelled since the goal is to determine the natural structure and clusters in the data. This form of machine learning is most appropriate for an exploratory analysis. The framework for this research was also designed to incorporate a means to evaluate the impact of climate change on pipe failure. In this section, the structure of the framework for this research is described along with a justification of the selected methods and how they were applied.

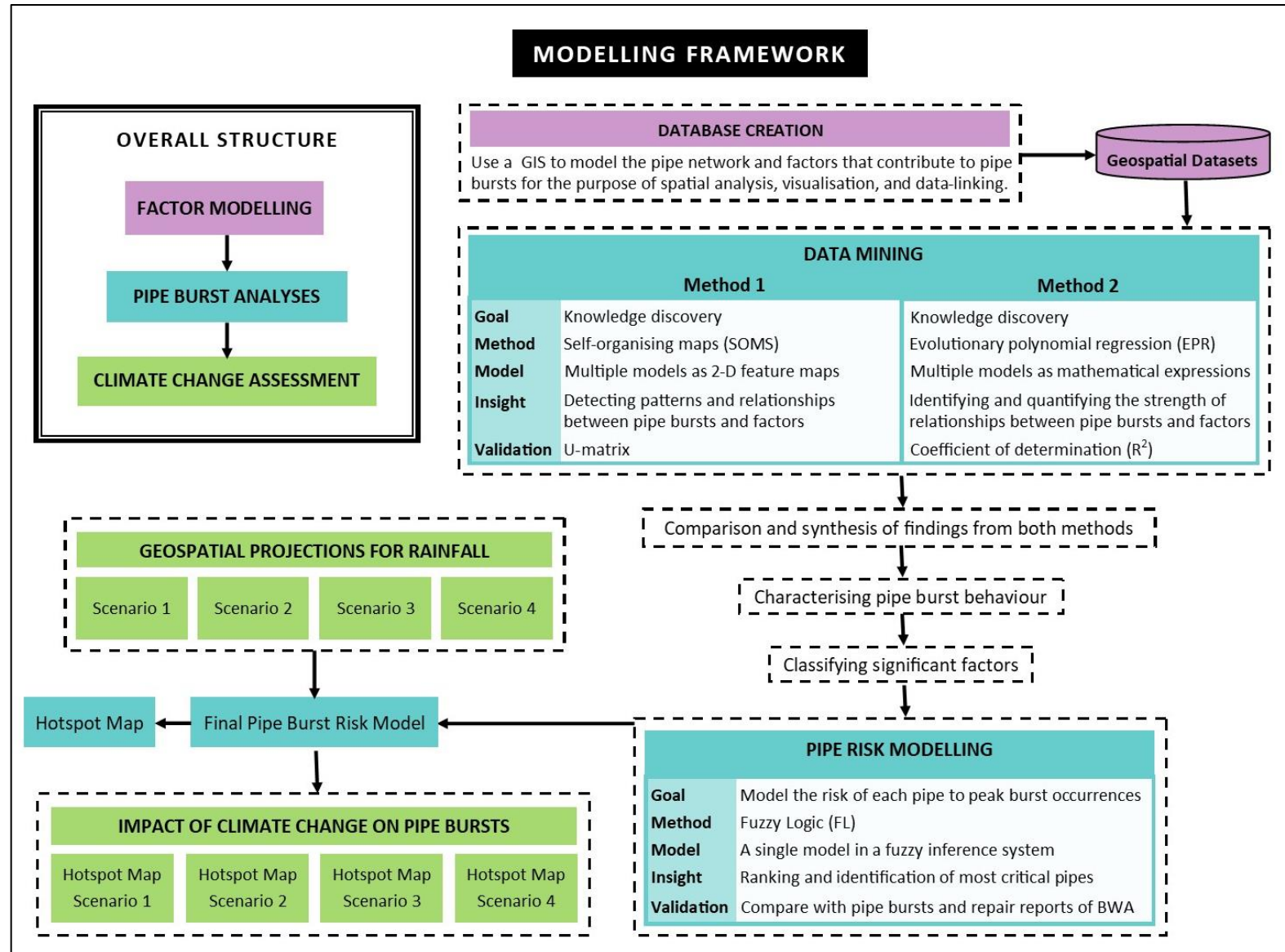


Figure 3.2 - A schematic diagram of the research modelling framework.

The overall structure of the modelling framework is shown as an inset in Figure 3.2. For the factor modelling component, a GIS was used to depict the pipe network and geospatially model eight possible explanatory factors (see Section 4.4) that were selected based on data availability and implications from the literature review (Section 2.7). This geospatial data was linked to three datasets that reported pipe bursts at different spatial and temporal scales. Other pipe failure studies that have included environmental variables also used a GIS for spatial analysis and data-linking (see Section 2.7.4). The finalised geospatial datasets facilitated the widely accepted concept that pipe failure can be assessed by using methods to quantitatively model pipe bursts as a function of its explanatory factors (Skipworth et al. 2002).

The literature review was also used to inform the selection of the methods that were applied in this study. SOMs, an unsupervised machine learning method, was chosen as the primary method to model pipe bursts based on the pros given in Section 2.7.2. Although the use of SOMs for that specific purpose had not been reported in the literature, it has been successfully applied to investigate water quality in WDNs (Speight, Mounce, and Boxall 2019; Blokker et al. 2016; Mounce et al. 2016; Mounce et al. 2014). While K-means clustering, another unsupervised machine learning method, has been used to model pipe failure (Kakoudakis et al. 2017), it does not allow for an ease of comparison of the attributes of each individual factor like SOMs. Therefore, SOMs was used as a data mining tool to detect patterns in pipe burst and among explanatory factors.

EPR was another data-driven technique which had shown high accuracy in predicting pipe failure and parameter estimation (Giraldo-González and Rodríguez 2020; Kakoudakis et al. 2017; Xu et al. 2011; Berardi et al. 2008). However, EPR was reported as not being able to detect complex relationships when compared to other methods (Xu et al. 2011). Nonetheless, the mathematical expressions generated from EPR could be used to identify and quantify the strength of relationships between factors. Therefore, EPR was also used for data mining as the secondary method in complement with SOMs. The goal of applying both methods was knowledge discovery as the outputs were

subsequently encoded in a risk rating system to evaluate the impacts of climate change. The types of models, nature of the insight gained, and validation techniques of SOMS and EPR were shown in Figure 3.2. The findings from the multiple models generated from both methods were compared and synthesised in order to characterise pipe burst behaviour and identify the main drivers of frequent pipe burst occurrences – which was the first part of the research aim.

To achieve the second part of the research aim – assessing the implications of climate change – a model that assigned a risk value to each pipe was created based on the outputs from the data mining analysis. These outputs consisted of numeric and non-numeric data that were not precise (i.e., estimated ranges from synthesised findings). Fuzzy logic was chosen as the method to create the pipe burst risk model since it could handle imprecise data and be driven by expert knowledge – that is, the knowledge discovered from the data mining analysis. Fuzzy logic has been shown as an effective method for modelling pipe failure and deterioration rates in several studies (St. Clair and Sinha 2012; Ho, Lin, and Lo 2010; Mounce, Boxall, and Machell 2010; Faiz and Edirisinghe 2009). Decision trees, another expert driven system used to model pipe failure in WDNs (Winkler et al. 2018), were considered but since it was a supervised machine learning method with known disadvantages such as overfitting and challenges with continuous numerical variables it was not selected.

The output from the fuzzy model was visualised as a hotspot map and validated against reported bursts and burst repairs of the BWA. The fuzzy model was used to incorporate the projections for rainfall from four climate change scenarios established by the IPCC – RCP2.6 in 2035, RCP2.6 in 2050, RCP8.5 in 2035, and RCP8.5 in 2050. In this way, the medium and long-term impacts were evaluated. The outputs for each scenario were mapped to visualise and spatially assess the changes in the distribution of hotspots in the future.

3.6 Research Ethics

This research was carried out in full compliance with the Policy on Research Ethics set by the University of the West Indies (UWI) and conducted with due consideration of other relevant internal policies. This research did not involve

human participants or any interviews. No ethical clearance was required from the UWI to conduct this research.

A written request was sent to the Barbados Water Authority (BWA) on 1 March 2016 for access and use of pipe asset and other relevant data, and any other technical assistance that may be needed to facilitate the research. In a follow-up meeting to that request, the General Manager of the BWA agreed that the research would be supported by the water utility. In a showing of reciprocity, the BWA would be provided the following:

- Assurance of confidential handling and use of data.
- Acknowledgment of the BWA's support of the research.
- Access to the final geospatial database created for this study and the report on the research findings.
- Invitations to academic seminars where the status of the research at that point in time and any findings are formally presented.

3.7 Chapter Summary

In this chapter, it is explained how an inductive approach was adopted to facilitate an exploratory analysis since a study of this scope for Barbados had never been carried out prior to this research. It is outlined in the research methodology that three discrete datasets would be created using reported pipe burst data and a combination of eight explanatory variables. These datasets would be included in a data mining analysis that applied two methods – SOMs and EPR – in a systematic manner to generate multiple pipe burst models. The knowledge derived from data mining was used to inform the development of a hotspot model of the network based on fuzzy logic that would also facilitate a climate change assessment. Schematic diagrams were created to depict the research strategy and modelling framework, illustrating the order and flow in which research processes were conducted and methods were implemented. The next chapter details the creation of the three geospatial pipe burst datasets which were critical for data mining and pipe burst modelling.

4 CREATION OF THE DATABASE

4.1 Chapter Introduction

The aim of this chapter is to detail the steps taken to create the geospatial datasets used to facilitate the data mining analysis carried out in this study. In this chapter the following is presented:

- (1) Preliminary steps carried out before data processing.
- (2) Characteristics of the three base datasets used for pipe burst modelling.
- (3) Processes used to clean, model, and explore the set of explanatory factors with statistical and geospatial techniques.
- (4) Procedures used to code non-numeric explanatory factors.
- (5) The final structure of the three pipe burst datasets.
- (6) Data challenges and limitations.
- (7) Tools and actions used to enhance validity and reliability of the data.

A highlight of this chapter is the innovative techniques used to create, quantify, and geospatially model the explanatory factors using a GIS – particularly, land-use and connection density.

4.2 Preliminary Steps and Data Acquisition

For this study, the approach conceptualised by Skipworth et al. (2002) where pipe bursts are modelled as a function of a set of explanatory variables was adopted. Thus, the first step in creating the database was factor identification. The following factors were chosen to be included in the analysis: pipe diameter, pipe material, pipe installation era, soil type, rainfall, pressure, land-use, and connection density. These factors were selected based on the relevance and use of factors in other studies (see Section 2.3), implications from the literature review (see Section 2.7), and the availability of data.

This list of identified factors builds on existing knowledge. Other factors such as traffic loading, pipe depth, and water quality (see Table 2.1) were taken into consideration but ultimately not included. These factors were not incorporated into this study for the following reasons:

- Traffic loading was considered but after acquiring, cleaning, and plotting the data in a GIS, it was found that the data were too limited. Data for traffic volumes were collected on a seemingly *ad hoc* basis by the local Ministry of Transport of Works and thus there were many gaps and inconsistencies. Additionally, the data-points for traffic volumes which were identified by a general address and not an exact location were clustered near the city of Bridgetown. This biased spatial distribution of data-points would result in a skewed data interpolation. Therefore, land-use was used as a proxy variable for traffic loading (see Section 2.3.8).
- Pipe depth was considered but only 5 burst repair records that estimated depths for 5 out of 18,618 individual pipes in the network were able to be sourced.
- Water quality data were requested from the BWA but not given because of data sensitivity.

As set out in Table 3.1, an objective of this study was to gather as much of the available and relevant data as possible from various sources and make efficient use of it. The creation of a robust database is critical to the process of data mining. Thus, the second step was to ascertain the type of data that were needed, identify possible entities from which the data could be sourced, and gather the data. The list of data that were sourced was shown in Table 4.1.

Table 4.1 - List of sourced data to facilitate database creation.

Data	Data Format	Source Entity
BWA districts	GIS shapefile	*BWA
Bursts by diameter	Spreadsheet	BWA
Bursts by district	Spreadsheet	BWA
Bursts by subdistrict and pipe diameter	GIS shapefile	BWA
Pipeline layout and pipe asset data (i.e., material, diameter, installation era, and length)	GIS shapefile	BWA
Water meter locations	GIS shapefile	BWA
Pressure	Hydraulic model	BWA
Soil type	Paper map	*CERMES
Rainfall	Spreadsheet	CERMES, *MOA

Rain days	Spreadsheet	*BMS
Land-use	GIS shapefile	CERMES
Traffic volume points	Spreadsheet	*MTW
Roads	GIS shapefile	CERMES, BWA
Rooftops	GIS shapefile	BWA
Population data	GIS shapefile	*BSS
Contours	GIS shapefile	CERMES
Barbados coastline and parishes	GIS shapefile	CERMES
* BWA – Barbados Water Authority CERMES – Centre for Resource Management and Environmental Studies MOA – Ministry of Agriculture BMS – Barbados Meteorological Services MTW – Ministry of Transport and Works BSS – Barbados Statistical Services		

The third step was to identify and acquire the various software that would be needed in creating the database. The software that were used in this study for building the datasets are described below:

- ESRI's ArcGIS ArcMap (Esri 2017): a geographic information system (GIS) with many built-in functions to conduct geospatial analyses, mapping, and visualisation. Given that data were sourced in different formats and factors were geospatially modelled, GIS was also used a tool for linking disparate data.
- Microsoft Excel (Microsoft Corporation 2018): a spreadsheet program with tools for carrying out a variety of statistical analyses and data visualisation.
- Bentley WaterGEMS (Bentley Systems 2017): a software designed for the building, simulating, and evaluating hydraulic models for a WDN.

These preliminary steps were carried out prior to the procedures used to clean and compile the data which were detailed in Sections 4.3 and 4.4.

4.3 Datasets

For this research, data on pipe burst occurrences in the water distribution system were sourced solely from the BWA. It was not feasible to carry out physical assessments as another means to collect data on pipe bursts since such approaches required extensive time and use of advanced technology that were

not available. In general, this type of assessment is beyond the capabilities of water utilities. A request was made to the BWA to receive all available data on pipe bursts occurrences in the WDN and thus, the following datasets were obtained:

1. A spreadsheet of total monthly reported pipe bursts, recorded by diameter size, between 2002 and 2012.
2. A spreadsheet of total monthly reported pipe bursts, recorded by districts, between 2002 and 2012.
3. A geospatial dataset, received as a GIS shapefile, of average annual reported pipe bursts, recorded by subdistrict, between 2000 and 2009.

These datasets could not be merged since pipe bursts were recorded by different variables (i.e., diameter, district, and subdistrict). Each of these distinct datasets were modelled separately in a GIS. In that same way, the set of possible explanatory factors (see Section 4.2) were also geospatially modelled so that the data for those variables could be linked to each pipe burst dataset. A detailed description of the characteristics of each dataset is given in addition to any data cleaning procedures that was carried out, a descriptive statistical summary of pipe bursts, and the explanatory factors that were linked. For ease of reference, datasets are named and referred to hereafter as:

1. Simple Dataset: pipe bursts by diameter.
2. Compound Dataset: pipe bursts by district.
3. Complex Dataset: pipe bursts by subdistrict.

4.3.1 Simple Dataset: Pipe Bursts by Diameter

The Simple Dataset was built on historical pipe burst data received from the BWA. Pipes had been grouped by diameter size and the total number of monthly reported bursts was recorded for each diameter group for the period between 2002 and 2012. The data was initially received as a spreadsheet that contained the following variables: pipe diameter size, total monthly reported pipe bursts, month, and year for all pipes in the local WDN. After inspection of the data, it was determined that no cleaning procedures had to be carried out since the

dataset did not contain any missing values, outliers, or errors. A summary of the data is shown in Table 4.2.

Table 4.2 - Summary of reported pipe bursts by diameter size in the Simple Dataset.

Summary of Bursts by Diameter (2002 - 2012)					
Diameter (inches)	Total Bursts	Avg. Annual Bursts	Avg. Monthly Bursts	Peak Season for Bursts	Peak Rainfall Season for Bursts
3	3,142	285.6	23.8	Spring	Dry
4	5,553	504.8	42.1	Winter	Dry
5	146	13.3	1.1	Spring	Dry
6	1,014	92.2	7.7	Summer	Wet
7	91	8.3	0.7	Autumn	Wet
8	423	38.4	3.2	Winter	Dry
9	110	10.0	0.8	Winter	Dry
10	230	20.9	1.7	Winter	Dry
12	105	9.5	0.8	Spring	Dry
14	60	5.5	0.5	Autumn	Wet
15	41	3.7	0.3	Winter	Dry
16	31	2.8	0.2	Autumn, Winter	Wet
18	3	0.3	0.0	-	-
20	0	0	0	-	-

From Table 4.2, it was seen that pipe bursts were most frequent in smaller diameter pipes, particularly in 3-inch and 4-inch pipes, rather than in larger diameter pipes. Overall, peak pipe bursts primarily took place during winter and spring which coincided with the dry season in Barbados. For this research, the Simple Dataset was modelled in GIS (see Figure 4.1). Since pipe bursts were grouped by diameter size, the extent of the data was an island-wide scale. Barbados is approximately 20 km wide by 30 km long.

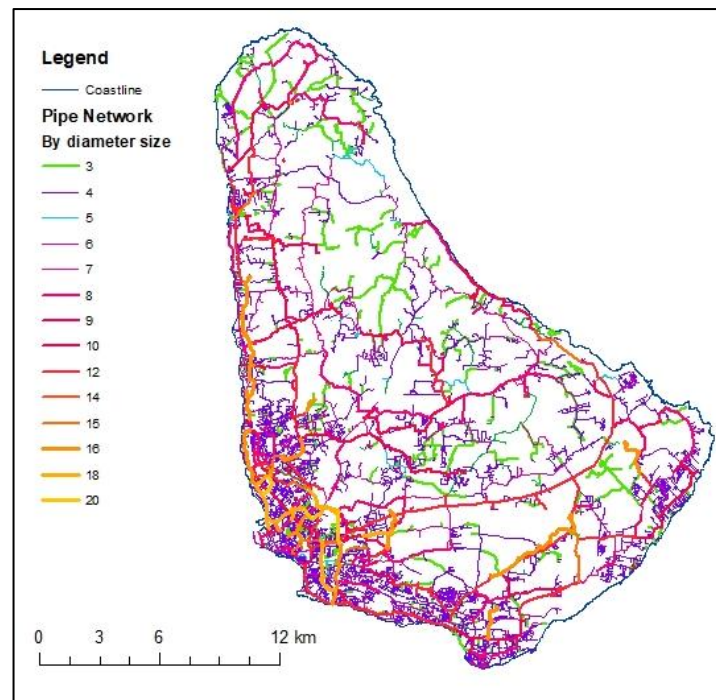


Figure 4.1 - A geospatial model of pipes grouped by diameter in the Simple Dataset.

4.3.2 Compound Dataset: Pipe Burst by District

The Compound Dataset was built on historical pipe burst data received from the BWA. Pipes had been grouped by 16 districts with boundaries defined by the BWA (see Figure 4.2) and the total number of monthly reported bursts was given likewise for the period between 2002 and 2012.

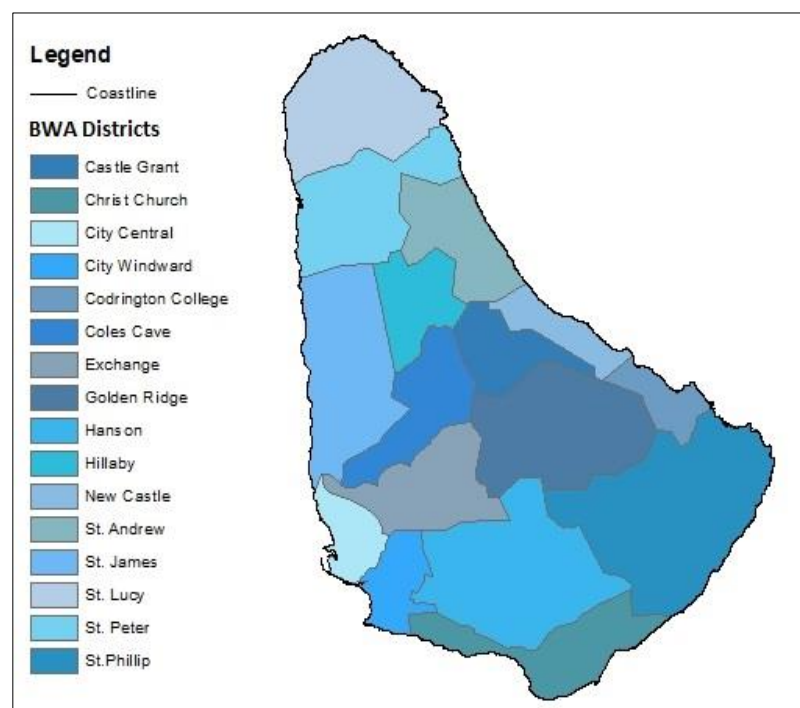


Figure 4.2 - Map of 16 districts used in the Compound Dataset.

Data for the Compound Dataset were initially received as a spreadsheet that contained the following variables: district, total monthly reported pipe bursts, month, and year for all pipes in the local WDN. The data were not shown to contain any missing values, outliers, or errors; and thus, no data cleaning procedures were required. A summary of the data is shown in Table 4.3.

Table 4.3 - Summary of reported pipe bursts by district in the Compound Dataset.

BWA Districts	Summary of Bursts by BWA District				
	Dry Season		Wet Season		TOTAL
	Winter	Spring	Summer	Autumn	
EAST	465	415	372	468	1,720
Castle Grant	68	86	61	87	302
Codrington College	91	82	52	101	326
Golden Ridge	166	120	139	144	569
New Castle	140	127	120	136	523
NORTH	507	397	446	486	1,836
Hillaby	59	55	61	81	256
St. Andrew	90	70	64	95	319
St. Lucy	214	175	177	172	738
St. Peter	144	97	144	138	523
SOUTH	1,195	1,172	963	1,061	4,391
Christ Church	293	273	229	267	1,062
Hanson	372	391	334	389	1,486
St. Philip	530	508	400	405	1,843
WEST	798	782	740	831	3,151
City Central	227	208	166	218	819
City Windward	69	73	67	54	263
Coles Cave	69	69	52	100	290
Exchange	165	172	167	175	679
St. James	268	260	288	284	1,100
TOTAL	2,965	2,766	2,521	2,846	11,098

In Table 4.3, districts were grouped by location according to the relative geographic area (i.e., north, east, south, and west). Pipe bursts were higher for districts in the south and west which were known as the urban corridors of the island having larger population counts than districts in the north and east. It can also be discerned that bursts were slightly higher in the dry season than wet season.

4.3.3 Complex Dataset: Pipe Bursts by Subdistrict

The Complex Dataset was built on historical pipe burst data received from the BWA in which pipes had been grouped by subdistrict and by diameter size. Likewise, average annual reported pipe bursts were recorded by subdistrict and by diameter between 2002 and 2012. For example, *‘no pipe bursts were recorded between 2002 and 2012 for all 6” pipes in the Airy Cot subdistrict’*. The data was initially received in geospatial format as a GIS shapefile with polylines that represented the actual pipe network. For each pipe, the following attributes were given: subdistrict name, pipe diameter, average annual reported pipe bursts, pipe material, and pipe installation era.

The name of the subdistrict in which each pipe was located was given. However, the land area bounded within each subdistrict was not drawn. In fact, the boundaries of the subdistricts had not been geographically delineated by the BWA. Having the boundaries defined as polygons in a GIS would facilitate geospatial analysis and data-linking since all data in this dataset would be compiled in relation to subdistricts. Thus, steps were carried out to draw the geographic boundaries of subdistricts.

Boundaries for subdistricts were generated using the Minimum Boundary Geometry tool in Esri’s ArcGIS. The tool created rectangular polygons around pipes having the same subdistrict name. Before applying the tool, some data cleaning of the names of subdistricts were required since a boundary would only be drawn around pipes having an exact name match as the tool was sensitive to case (i.e., uppercase and lowercase letters), spaces, and symbols. Application of the tool resulted in the creation of 834 subdistricts (see ‘A’ in Figure 4.3). However, it can be seen that some boundaries were oversized. By investigation, it was observed that the oversized boundaries contained pipes that had the same subdistrict name though the pipes were substantially distanced. This issue was resolved by identifying and renaming groups of pipes that were not contiguous within a subdistrict since, in theory, all pipes within a subdistrict should be linked to hydraulically deliver water. For example, two groups of pipes that were not contiguous in Melvin’s Hill subdistrict (see ‘B’ in Figure 4.3) were

renamed as Melvin's Hill North and Melvin's Hill South. The cleaning of oversized subdistricts resulted in a final count of 851 subdistricts (see 'C' in Figure 4.3).

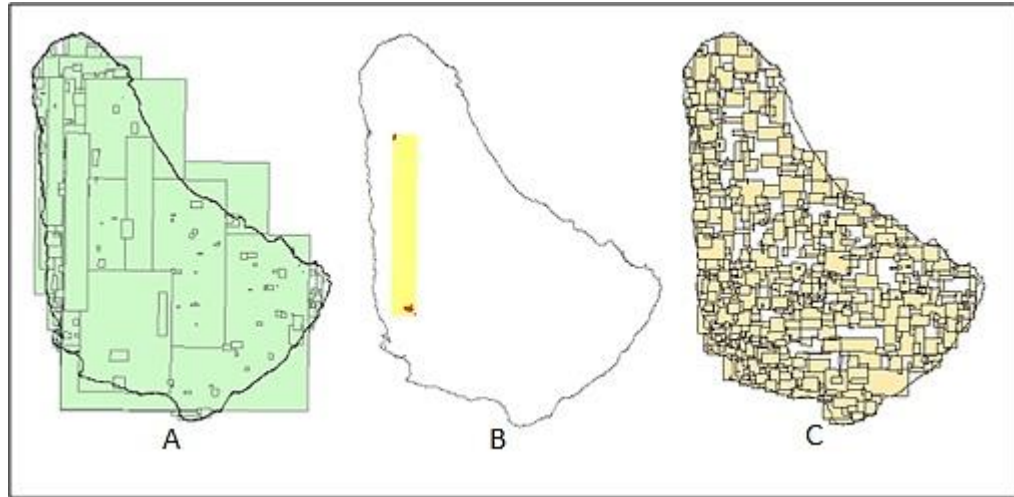


Figure 4.3 - Oversized subdistrict boundaries in 'A'; a boundary that formed due to 2 sets of pipes having the same subdistrict name in 'B'; and finalised subdistrict boundaries in 'C'.

The splitting of subdistricts was a necessary action since an oversized boundary far exceeded the scale of the pipes within it. It was imperative that the size of a subdistrict was comparable to the scale of the pipes within it so that other factors calculated in relation to subdistricts accurately represented the conditions experienced by said pipes. For example, if average rainfall was calculated for an oversized boundary, it would not provide the most accurate representation of the amount of rainfall that pipes in that subdistrict experienced. However, splitting of subdistricts resulted in the splitting of bursts. For example, if all 3-inch pipes in Melvin's Hill had an average of 35 bursts, then splitting the subdistrict into Melvin's Hill North and Melvin's Hill South resulted in the group of 3-inch pipes in both new subdistricts being attributed to having 35 bursts each. In such an instance, it could not be determined how bursts were actually distributed since burst data were not given at the pipe-level (i.e., for each individual pipe). As a workaround, bursts were proportioned according to the total number of pipes found in each newly created subdistrict. Other alternatives were considered but not chosen: (i) assigning the number of reported bursts to each pipe group but this could be considered as duplication

and overestimation; (ii) assigning the total number of reported bursts to one group but zero to another but this would be a major assumption with no justifiable or logical basis; or (iii) dividing the number of reported burst according to the number of newly created subdistricts but some districts may contain substantially less number and length of pipes than others. In total, 59 subdistricts were split with 51% having bursts redistributed.

The base geospatial model of the Complex Dataset was shown in Figure 4.4. The average spatial scale of the subdistricts was less than 2.5 km. Thus, this was the highest resolution dataset in this study which allowed all of the possible explanatory variables to be linked:

- Pipe diameter was the diameter size of the pipes in each cohort.
- Pipe material was the predominant type of material in each pipe cohort.
- Pipe installation era was the predominant era category in each pipe cohort.
- Soil type was the predominant soil type within each subdistrict.
- Rainfall was the average annual rainfall in each subdistrict over the period 2002 and 2012.
- Pressure was the average static pressure of the pipes in each pipe cohort.
- Land-use was the predominant type of land-use of each subdistrict.
- Connection density was the average connection density of the pipes in each pipe cohort.

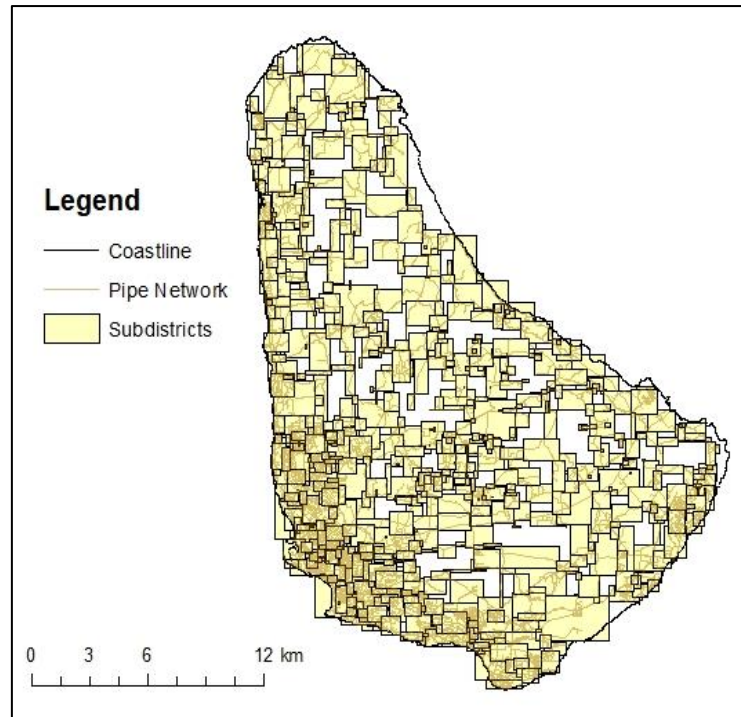


Figure 4.4 - A geospatial model of the pipe network and subdistricts in the Complex Dataset.

4.4 Factor Modelling

The set of possible explanatory factors of pipe bursts occurrences that were selected for analysis in this study were geospatially represented using a GIS. In geospatial format, data for the explanatory factors could be linked to the three pipe burst datasets (i.e., Simple Dataset, Compound Dataset, and Complex Dataset) that were used for data mining and pipe burst modelling. In this section, data cleaning procedures and the steps taken to model each factor to the scale of each dataset are detailed.

4.4.1 Pipe Diameter

Data for pipe diameter were available at the pipe-level in the Complex Dataset. After an inspection of the data, the only inconsistency found was an entry error of “999” for the diameter of a pipe in the subdistrict of Heywoods. Based on the location and length, it was determined that the pipe was a service main. Thus, the error entry was changed to 4-inch – the standard size of a service main. A summary of the distribution of pipe diameter is given below in Table 4.4.

Table 4.4 - Summary of pipe diameter in the Complex Dataset.

Pipe Diameter (inches)	Pipe Length	
	km	%
0.8	0.8	0.04
1	6.7	0.3
1.3	1.9	0.1
1.5	14.1	0.7
2	15.7	0.8
3	246.6	12.2
4	992.9	49.0
5	32.0	1.6
6	235.8	11.6
7	55.5	2.7
8	104.9	5.2
9	39.1	1.9
10	103.8	5.1
11	8.9	0.4
12	92.8	4.6
14	2.4	0.1
15	15.3	0.8
16	38.6	1.9
18	15.6	0.8
20	4.0	0.2
21	0.2	0.01

Pipe diameter were only available for Simple and Complex Datasets. Pipe diameter could not be joined to the Compound Dataset.

4.4.2 Pipe Material

Data on pipe material for each pipe in the network was provided by the BWA. A summary of this data was given in Table 4.5. For approximately 67% of the total length of the pipeline, the pipe material was not known (see Table 4.5). Some descriptive statistical analysis methods used to analyse pipe burst data were adopted from Zhao, Li, and Zhou (2014) in an attempt to fill missing data in unknown pipe material.

Table 4.5 - Summary of initial pipe material data in the Complex Dataset.

Pipe Material	Pipe Length	
	km	%
Asbestos Cement	4.2	0.2
Cast Iron (CIP)	450.8	22.3
Ductile Iron (DIP)	122.2	6.0
Galvanised Iron	25.7	1.3
High-density Polyethylene (HDPE)	0.2	0.01
Polyvinyl chloride (PVC)	56.9	2.8
Steel	2.7	0.13
Unknown	1,363.9	67.3

The first method was to create bivariate tables to compare material to other pipe asset data such as diameter, installation era, and length. However, no relationships were observed. The second method was to map pipe material to determine if spatial clusters could be detected from visual inspection. Again, no patterns were found. Assistance was requested from the BWA and a set of formalised rules from a report (BWA 1997) was received. These rules were internally standardised to guide the design of the WDN. Given that, there was confidence in using the rules listed below:

- All odd diameter pipes (e.g., 3-inch, 5-inch, 7-inch, etc.) were cast iron.
- Unknown 4-inch, 12-inch, 18-inch, and 20-inch pipes were ductile iron.
- Unknown 2-inch pipes were galvanised iron.

The results of applying the formalised rules were shown in Table 4.6. The percentage of unknown pipe material decreased from 67% to 17%. This data could not be improved any further. Thus, the percentage of pipes with unknown material remained unspecified.

Table 4.6 - Summary of pipe material after applying formalised rules.

Pipe Material	Revised Pipe Length	
	km	%
Asbestos Cement	4.2	0.2
Cast Iron (CIP)	454.5	22.4
Ductile Iron (DIP)	1,116.8	55.1
Galvanised Iron	39.0	1.9
High-density Polyethylene (HDPE)	0.2	0.01
Polyvinyl chloride (PVC)	56.9	2.8
Steel	2.7	0.13
Unknown	352.4	17.4

The types of pipe material were grouped after a preliminary analysis using self-organising maps (see Section 5.4.1) was performed. It was determined that a higher-level grouping of pipe material would better facilitate detecting patterns in the data. Thus, pipe materials were grouped as:

- Asbestos Cement
- Iron (CIP, DIP, and galvanised iron)
- Plastic (HDPE and PVC)
- Steel
- Unknown

After the procedures for cleaning and regrouping of categories were performed, the data for pipe material were finalised for the Complex Dataset. The predominant pipe material of each pipe cohort (i.e., grouped by subdistrict and by diameter) was determined. Data for pipe material were available only for the Complex Dataset.

4.4.3 Pipe Installation Era

For this study, pipe installation era was used as a proxy variable for pipe age (see Section 2.3.1). Data on the installation era for each pipe in the WDN was obtained from the BWA. A summary of that data is shown in Table 4.7. It can be seen that the installation era of 94% of the total length of pipe in the network was not known.

Table 4.7 – Summary of initial pipe installation era data in the Complex Dataset.

Pipe Installation Era	Pipe Length	
	km	%
Pre-1970	22.5	1.11
1970 - 1980	29.9	1.48
1980 - 1990	11.4	0.56
1990 - Present	56.8	2.80
Unknown	1,905.8	94.05

The same descriptive statistical analysis methods used in attempt to fill missing data for pipe material were also carried out for pipe installation era. However,

these attempts were unsuccessful. Ultimately, the formalised rules for network design that were obtained from the BWA (1997) were applied accordingly:

- Cast iron pipes were laid between 1860 and 1969.
- Asbestos cement pipes were laid in 1964.
- Ductile iron pipes were laid between 1970 and 1996.
- PVC pipes were laid between 1990 and 1996.

The results of applying the formalised rules as well as reclassifying the categories for pipe installation era were shown in Table 4.8. The percentage of unknown pipe installation era improved from 94% to 19%. This data could not be improved any further, so pipes with unknown installation era remained unspecified.

Table 4.8 - Summary of pipe installation era after applying formalised rules.

Pipe Installation Era	Revised Pipe Length	
	km	%
1860 - 1969	454.46	22.4
1964	4.23	0.2
1970 - 1996	1,116.82	55.1
1990 - Present	56.90	2.8
Unknown	394.13	19.4

After the cleaning procedures for pipe installation were carried out, the data were joined to the Complex Dataset in the GIS. The predominant installation era of each pipe cohort (i.e., grouped by subdistrict and by diameter) was determined and assigned. Data for pipe installation could only be joined to the Complex Dataset.

4.4.4 Soil Type

Soil type is an environmental factor that was shown in various works found in the literature to contribute to pipe burst occurrences due its role in soil movement, external loading, and pipe corrosion (see Section 2.3.5). Despite these observations, soil type was not recorded in the datasets that were obtained from the BWA. In fact, geospatial data of soil types in Barbados could not be sourced from any of the local data sources used for this research (see Table 4.1).

Thus, geospatial data for soil type had to be created so that it to be included as an explanatory factor in analyses conducted in this research.

A soil map of Barbados drawn by Vernon and Carroll (1966) was sourced from the Centre for Resource Management and Environmental Studies (CERMES) at the University of the West Indies. The map by Vernon and Carroll (1966) is the most detailed and frequently referenced map of local soils in Barbados. The map was scanned, imported into a GIS, and digitised using tracing tools in ESRI's ArcGIS Editor toolbox. The final digitised map of the six main soil types identified by Vernon and Carroll (1966) is shown in Figure 4.5.

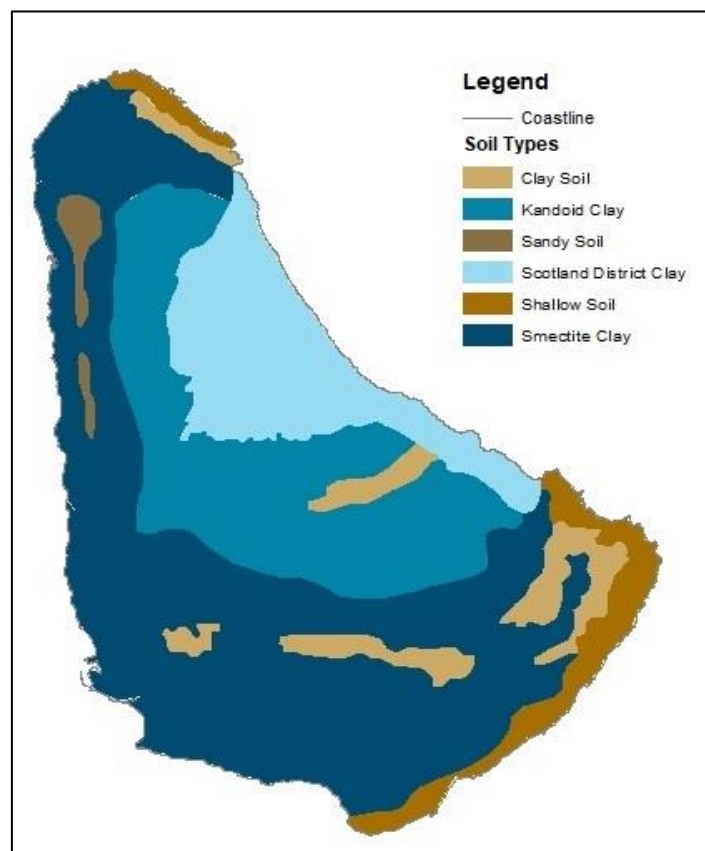


Figure 4.5 - Digitised soil map of Barbados (adapted from Vernon and Carroll 1966).

For each of the main soil types, its characteristics and potential to promote bursts in a WDN are given below:

- Scotland District clays are impervious and impermeable which results in high rainfall runoff (Vernon and Carroll 1966). This soil type is considered immature due to topography of the Scotland District area which is very

unstable because of erosion, soil creep, and large landslides (Vernon and Carroll 1966). The instability associated with this soil type and area can expose pipes to the elements, promote bending and external stress, or cause bursts due to landslides.

- Smectite clays are known as swelling clays. Soils with a high content of smectite clay minerals have a tendency to shrink in very dry conditions and swell with increased moisture (Abbey, Eyo, and Ng'ambi 2020). In climates with pronounced wet and dry seasons, the shrinking and swelling action of smectite clays are greatest (Krenz, Lee, and Owens 2006). Since Barbados has pronounced wet and dry seasons, the movement and bending stress caused by cyclic shrink-swell actions of this soil type can considerably promote pipe bursts.
- Kandoid clays contained high contents of clay and silt materials, similar to smectite clays. However, a key difference between the two soil types was that Kandoid clays were very well-drained and had a low moisture supplying capacity (Vernon and Carroll 1966). Hence, Kandoid clays were not characterised by the shrink-swell action observed in smectite clays. In this way, Kandoid clays did not pose a relative risk to the occurrence of pipe bursts.
- Shallow soils were found along the windward coasts of the parishes of St Lucy and St Philip which are characterised by a dry climate and strong winds. Shallow soils are considered immature with a weak structure and very low moisture supplying capacity. Due to proximity to the coast, these soils are known to experience salt blasts that may encourage the corrosion of iron pipes. Therefore, shallow soils posed some risk in promoting pipe bursts.
- Sandy soils were developed from beach deposits of sand derived from corals. These soils, which were found along a strip of the western coast of Barbados, were composed mostly of coarse and fine sand. Due the porous nature of this soil type, it was moderately permeable, well-drained, and had a low moisture capacity (Vernon and Carroll 1966). Compared to the other

soil types, Sandy soils did not pose a substantial risk to the occurrence of pipe bursts.

- Clay soils were developed from sea, lagoonal, or other mixed clay deposits. Overall, these soils had poor drainage properties. However, the moisture capacity of these soils was between moderate and high. During the wet season, the water table can be found at relatively high levels compared to other soils (Vernon and Carroll 1966). Such characteristics caused these clay soils to be more stable when compared to other local clays (i.e., Scotland District clay or Smectite clay). Thus, this group of clay soils posed a relatively lower threat to the occurrence of pipe bursts to the other local clays.

It was only possible to join soil data to the Compound and Complex Datasets. Therefore, the predominant soil type for each district and subdistrict had to be calculated. Spatial analysis was used to perform the calculations.

To find the predominant soil type in each district, the polygon features (i.e., enclosed shapes with boundaries) that were used to represent the distribution of soil type in a GIS (see Figure 4.5) had to be converted to raster data (i.e., discrete pixel data). The conversion to raster data allowed for the application of the Zonal Calculation tool in ESRI's ArcGIS ArcMap (see Section 4.2). The function of this tool allowed raster calculations to be conducted. Using two data inputs – soil raster data and district boundaries – the Zonal Calculation tool totalled the number of pixels of each soil type in each district. This procedure was also carried out using the soil raster data and subdistrict boundaries as input data. However, due to the small size and overlap of some subdistricts, calculations were missing. To work around this shortcoming, larger subdistricts for which calculations were performed were removed from the layer and the Zonal Calculation tool was run again. This was carried out several times until calculations were completed for all subdistricts. The results exported from the GIS into Microsoft Excel to efficiently tally the data to determine the predominant soil types for districts and subdistricts. The results were reimported

into the GIS, joined to the Compound and Complex Datasets, and mapped for visualisation (see Figure 4.6 and Figure 4.7).

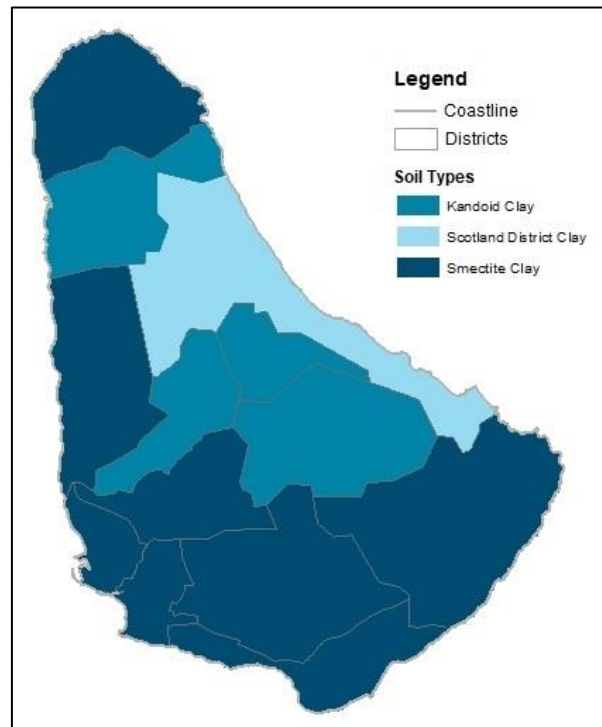


Figure 4.6 - Map of predominant soil types in each district for the Compound dataset.

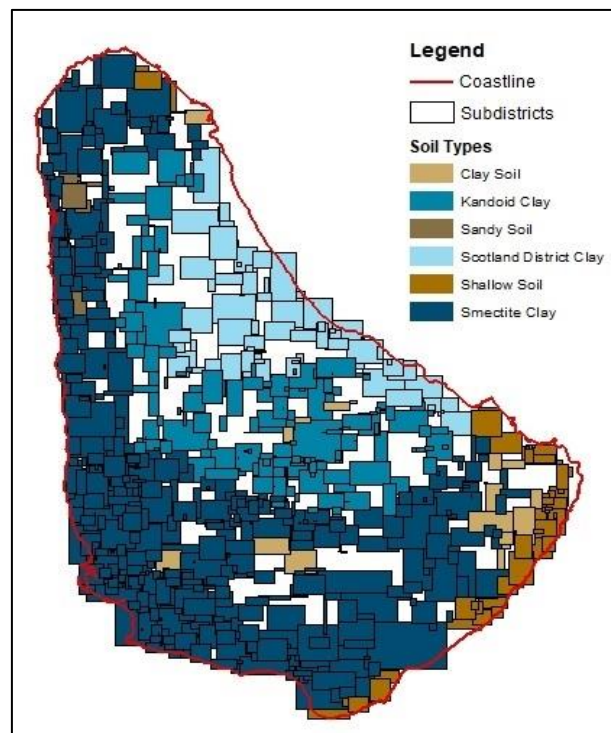


Figure 4.7 - Map of predominant soil types in each subdistrict for the Complex dataset.

4.4.5 Rainfall

The seasonality effect of rainfall was shown to correlate with the frequency of pipe burst occurrences mainly due to the influence of rainfall on soil moisture, soil movement, and soil stability (Barton et al. 2019). For this reason, and its role in the electrochemical corrosion of pipes (Li et al. 2015), rainfall has been included as a variable in several pipe failure studies in the literature (see Section 2.3.6). For this study, although historical rainfall data for Barbados were available for a few decades, monthly rainfall data from 2002 to 2012 were used in the creation of the datasets since this span was the same period and frequency of the data acquired for historical pipe bursts. Rainfall data were available for the 24 stations across the island (see Figure 4.8). Given that most of these rainfall stations were being managed by the Ministry of Agriculture, most of the stations were located in rural agricultural areas rather than in the densely populated urban corridors on the west and south coasts.



*Figure 4.8 - Location of 24 rainfall stations in Barbados.
(Source: Base map imagery from ESRI ArcGIS)*

The sourced rainfall data did not require any cleaning, as there were no missing values, outliers, or any other detectable errors in the data. In exploring the data, the average monthly rainfall was plotted in Figure 4.9. The pronounced wet

season (June to November) and dry season (December to May) can be observed in the graphic (see Figure 4.9). Additionally, a tabular contrast of the total amount of rainfall in a typical wet year, dry year, and baseline period was shown in Table 4.9. The observations from this statistical exploration of rainfall data prompted the splitting of the datasets by month, meteorological season (winter, spring, summer, and autumn), and rainy season (wet and dry) in order to investigate any correlation between pipe bursts and rainfall in the data mining analysis.

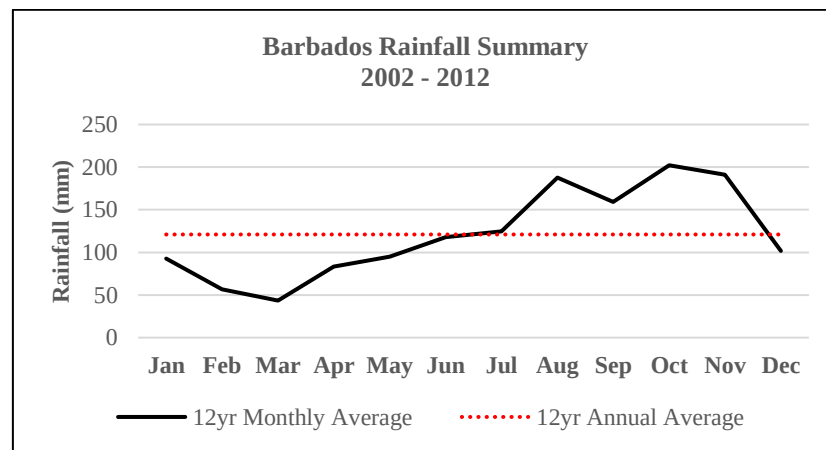


Figure 4.9 - Average monthly rainfall from 2002 - 2012 in Barbados.

Table 4.9 - A contrast of the distribution of rainfall during the wet and dry seasons in Barbados.

Rainfall Distribution in Wet and Dry Seasons		
Periods	Total & Average Rainfall (mm)	
	Wet (Jun to Nov)	Dry (Dec to May)
2003 (dry year)	977 (79%)	261 (21%)
2004 (wet year)	1,136 (62%)	706 (38%)
2000 to 2015	15,726 (68%)	7,568 (32%)

A spatial exploration of the data was also carried out to investigate the best method for interpolation. Rainfall data had to be interpolated in order to calculate values for average rainfall by district and subdistrict. This exploration was prompted by the observation of similarities between the distribution of historical rainfall and elevation (see Figure 4.10). The correlation between

rainfall and elevation was likely due to the prevailing easterly wind and assisted orographic uplift with heights peaking near the centre of the island. In other studies, the inclusion of a digital elevation model (DEM) significantly improved the accuracy of interpolation (Ahmed, Shahid, and Harun 2014; Delbari, Afrasiab, and Jahani 2013). Thus, while Kriging was one of the most widely used methods for rainfall interpolation, Co-Kriging (a variation of the method) which allowed other variables to be included into the analysis was also considered. Using the Geostatistical Analyst Toolbox in ESRI's ArcGIS ArcMap (see Section 4.2), these two methods were tested. Ultimately, the Co-kriging method which incorporated the DEM of Barbados was chosen as the method used for interpolating rainfall in this study since it was more accurate than the Kriging model.

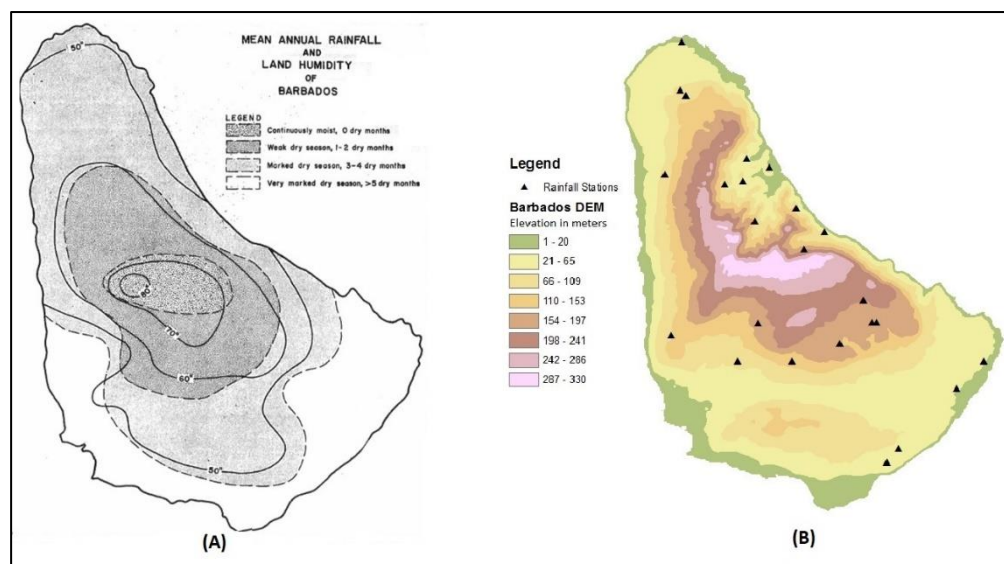


Figure 4.10 – Map of rainfall by Vernon and Carrol (1966) in (A) and a digital elevation map (DEM) of Barbados with location of rainfall station in (B).

In order to link interpolated rainfall data to the Compound and Complex Datasets, average rainfall had to be calculated by district and subdistrict, respectively. These spatial analyses were conducted with the Spatial Analyst Toolbox in ESRI's ArcGIS ArcMap. The Zonal Statistics function was used to perform the spatial calculations. For the Compound Dataset, 22 interpolation maps were produced to show the average monthly rainfall for each rain season between 2002 and 2012. For each of those maps, the Zonal Statistics function

was applied to compute the average rainfall by district. For the Complex Dataset, only 1 interpolation map was produced that depicted the annual average of monthly rainfall between 2002 and 2012. The Zonal function was executed several times in order to compute values in smaller subdistricts. Since the Simple Dataset was on an island wide scale, no spatial analyses were required for that dataset – only basic statistical calculations.

4.4.6 Pressure

Water pressure exerts an internal force inside pipes. When this force exceeds the tensile strength of a pipe, bursts can occur. In the literature it was found that higher, transient, and cyclical pressures promoted pipe burst occurrences (see Section 2.3.7). Hydraulic models are typically used to model pressure in a WDN (see Section 2.3.7). For this study, a hydraulic model of the local WDN was sourced directly from the BWA. This hydraulic model was fine-grained in detail as it considered all pipes in the network rather than a skeletonised model. A comparison of the actual pipes in the WDN (see Figure 1.4) and the hydraulic model showed that the number of pipes in the hydraulic model was nearly doubled even though the total lengths are similar, differing by only 2% (see Table 4.10). The difference in pipe count between the models was investigated and the methods used to reconcile inconsistencies were detailed.

Table 4.10 - Comparison of hydraulic model and pipe network model.

Properties	Pipe Network Model	Hydraulic Model
Developer(s)	BWA	CH2M Hill
	Halcrow Inc.	
Software	ESRI ArcGIS	Bentley Systems WaterGEMS
Year	2013	2016
Total number of pipes	18,618	30,911
Total length of pipes (in km)	2,027	2,067
Main features	Pipe network	Pipe network and junctions
Key parameters	Pipe material, pipe diameter, historical pipe bursts, pipe length, and installation era	Average daily pressure, pipe length, start node, and end node

In the WaterGEMS hydraulic modelling software (see Section 4.2), a *pipe* is a link that connects two nodes which could but does not necessarily reflect the actual length of a pipe in a network. The placement of a node is usually prompted by changes in diameter or material along a pipe, or at a point where several pipes are joined. Consequently, it is expected that a hydraulic model that attempts to model each pipe in a network should contain more *pipes* than the model of the actual pipe network. Moreover, a popular rule of thumb among users of WaterGEMS is having at least 1,000 pipes in the model for every 10,000 persons of the population. Application of this rule of thumb indicates that a hydraulic model for a population of Barbados, approximately 285,000 persons, should contain about 28,500 pipes. This estimate is near to the actual total number of pipes in the hydraulic model, 30,911 pipes. Closer inspection of the data in a GIS revealed that the hydraulic model had multiple pipes (i.e., segments) representing an actual pipe in the network model in most instances.

The spatial layout of the pipes of the two models were compared using ESRI's ArcGIS ArcMap. The *Select by Location* tool was used with the hydraulic pipes layer as the target variable and the pipe network layer as the source layer. The spatial selection method that was chosen was '*Share a line segment with the source layer feature*'. That is, if a pipe in the hydraulic model was perfectly aligned with a pipe from the network model, then the pipe in the hydraulic model would be selected and highlighted. Otherwise, if not perfectly aligned, it would not be selected (see Figure 4.11 and Figure 4.12). It was found that 97% of the pipes in the hydraulic model were perfectly aligned with the pipes in the network model. In remaining instances, edits were made to align pipes in the hydraulic model with the pipe network.

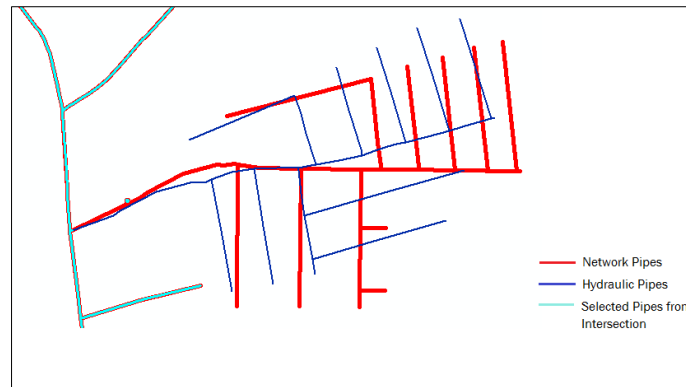


Figure 4.11 – Example of orientation error in alignment of the hydraulic model and network.

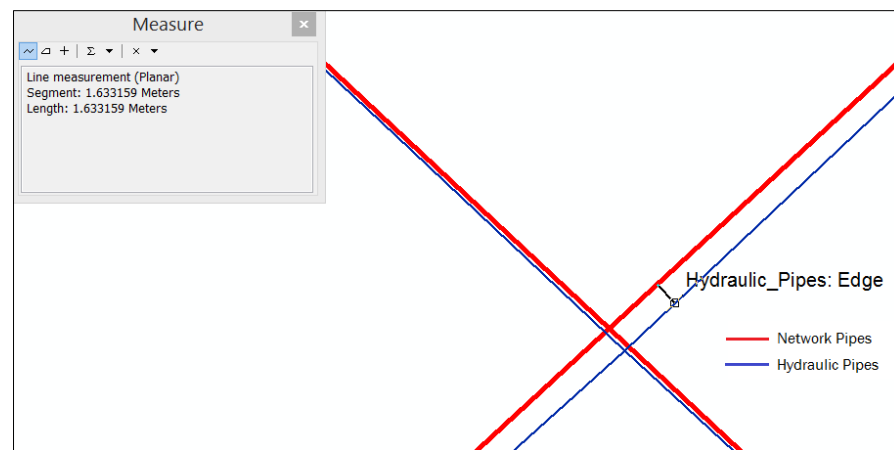


Figure 4.12 - Example of proximity error in alignment of the hydraulic model and network.

Modelled pressure values at the nodes of the hydraulic model were also examined. It was observed that the hydraulic model contained a total of 24,655 nodes with 5,557 nodes (23%) having negative values for average pressure. The BWA maintained that the model was not finalised and expected revisions to be made to the model. It was expected that with additional observations of real-time pressure data, the hydraulic model could be recalibrated to improve model accuracy as is the general procedure in hydraulic modelling. However, the collection of in-field measurements and the model revisions required went beyond the scope of this study. Despite this deficiency, the cleaning and exploration of this data continued as it was goal to produce and inspect a map of pressure distribution to see if it resembled the known pressure management issues – that is, higher pressures in higher elevated and remote areas.

Values for pressure could not be directly extracted from WaterGEMS due to the limitations of the academic version of the software (i.e., no runs for models that contain more than 2,000 pipes). Despite this limitation, existing layers in the hydraulic model could be extracted. Thus, the layers for pipes and junction nodes were extracted. Various steps were taken to calculate the average static pressure which was averaged over a series of 24-hour profiles taken during each month of the year in 2016.

- Step 1: The junction layer that contained pressure data for all nodes were spatially joined to the hydraulic pipe layer. This resulted in the linking of values of static pressure for the start and end nodes to each pipe in the hydraulic pipe layer.
- Step 2: A geometric intersection of the hydraulic pipe layer and the network pipe layer was carried out to associate the attributes (start nodes and end nodes) of pipes from the hydraulic model to actual pipes in the network model.
- Step 3: The attribute table resulting from the intersection in Step 2 was exported from GIS into a spreadsheet format for further manipulation to calculate the average pressure between nodes for each pipe.
- Step 4: A Microsoft Excel add-on, Combine Rows Wizard by Ablebits, was used to merge rows of multiple hydraulic pipes associated with each pipe and pipe cohort.
- Step 5: The final pressure values (see Figure 4.13) for each pipe in the network model was calculated by averaging the values for static pressure obtained in the previous steps.

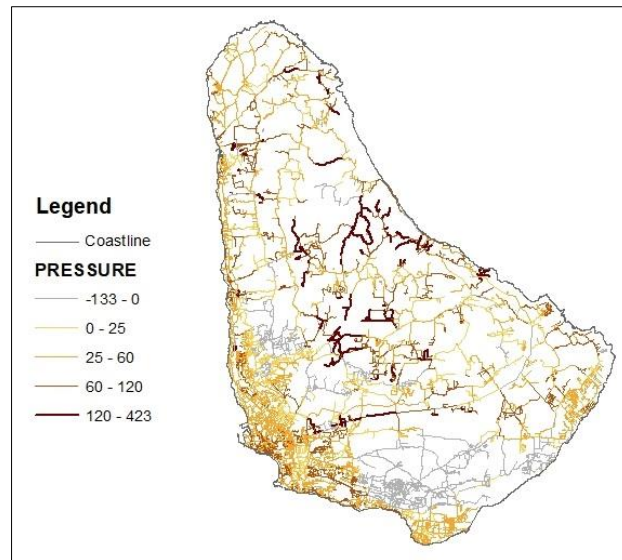


Figure 4.13 - GIS layer of average pressure values in pipe network.

From inspection of Figure 4.13, it was observed that higher values for pressure were in areas of higher elevation and remote areas – which reflects the current pressure management situation. This was a good indicator that the model had some accuracy. It was feasible to use this data only for the Complex Dataset. Thus, pressure would be computed as the average static pressure of the pipes in a cohort of that dataset. Taking the average would assist in reducing the extreme values in the hydraulic model. Due to the abovementioned, and since there were no other data on pressure available and it had already been reported by the General Manager of the BWA that pressure management was a major issue for the BWA (Forde-Craig 2020), this data was used in this study.

4.4.7 Land-use

For this study, land-use was included as a possible explanatory factor and used as a proxy for traffic loading (Savic and Walters 1999). Data to quantify traffic loading was too limited (see Section 4.2). Therefore, it was not feasible to include traffic loading as a factor for analysis. The type of land-use implied the level of activity in an area and hence the expected level of road traffic activity.

Data for land-use was originally sourced as a GIS shapefile that contained 19 categories for land-use: agriculture, beach, commercial, heavy industry, landfill, light industry, major retail, marina jetties, natural features, office, private golf

course, public institution, public open space, residential (high density), residential (low density), resource extraction, tourism, transportation, and vacant. Inspection of the data revealed discrepancies especially when compared to a satellite image of Barbados. For example, several known communities were classified as agriculture area instead of residential. Also, the percentage of total land area classified as high-density residential areas and low-density residential areas were 0.02% and 17.82%, respectively. Given that Barbados is one the most densely populated countries in the world, these classifications were not considered to be accurate. The land-use data appeared outdated and/or inaccurate, and without any meta data, information about the creator of this data, data sources, and definition of categories were unknown. Therefore, there was limited confidence in this data. Nonetheless, the data was used in a preliminary analysis using self-organising maps (see Section 5.4.1). From that analysis, it was difficult to detect patterns in the data likely due to the large number of categories which may have been inaccurately classified. Due to the gaps in the data and the results of the preliminary analysis, alternative methods for depicting land-use were explored.

The simplest classification of land-use is the distinction between urban and rural areas which was also adopted in other related studies (Mounce et al. 2016). An approach by Dijkstra and Poelman (2014) was to classify land-use by using a population grid of 1km x 1km to determine if an area was urban or rural. According to Dijkstra and Poelman (2014), a 1 km grid box that contained a density of at least 300 inhabitants was considered urban. Thus, population data for Barbados were obtained in GIS format. The thematic map in Figure 4.14 showed the distribution of population by enumeration district. However, this data could not be used because of the graphical distortion caused by the expansion of the size of some enumeration districts in less dense areas to contain higher counts of persons for the purpose of census taking.

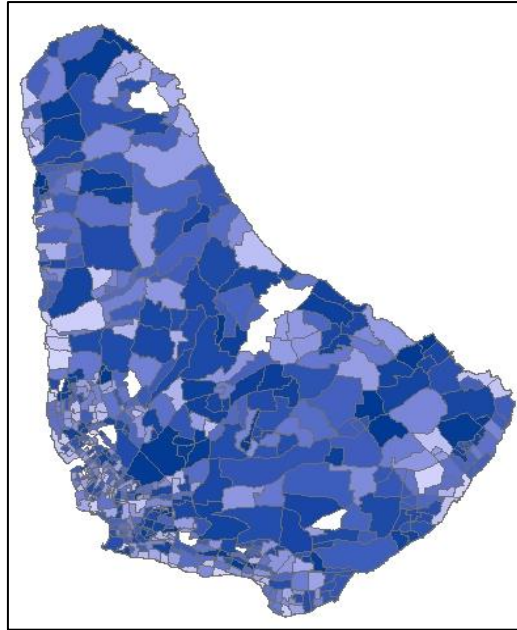


Figure 4.14 - Population by enumeration district with higher counts depicted in darker shades of blue.

Alternatively, rooftops were substituted as a proxy for population density in the approach of Dijkstra and Poelman (2014). In GIS, a polygon layer of rooftops was overlaid on a 1km x 1km grid. The total number of rooftops were 151,233. The number of households reported in the 2010 census was 78,936 which is approximately 52% of the total number of rooftops. Given that Barbados has an average household size of three (3), then one (1) grid box having more than 100 households would be classified as urban. Using these assumptions, 200 rooftops (50% of which are assumed to be residential households) was used as the determining threshold for land-use classification. This assumption was made since there were no data available to differentiate the type of building (i.e., residential, commercial, etc.). Thus, this was a shortcoming of using this approach as this only addresses the degree of urbanisation with no specificity to rural areas. Nonetheless, the results of the geo-spatial analysis were shown in Figure 4.15 which closely reflects on-the-ground observations.

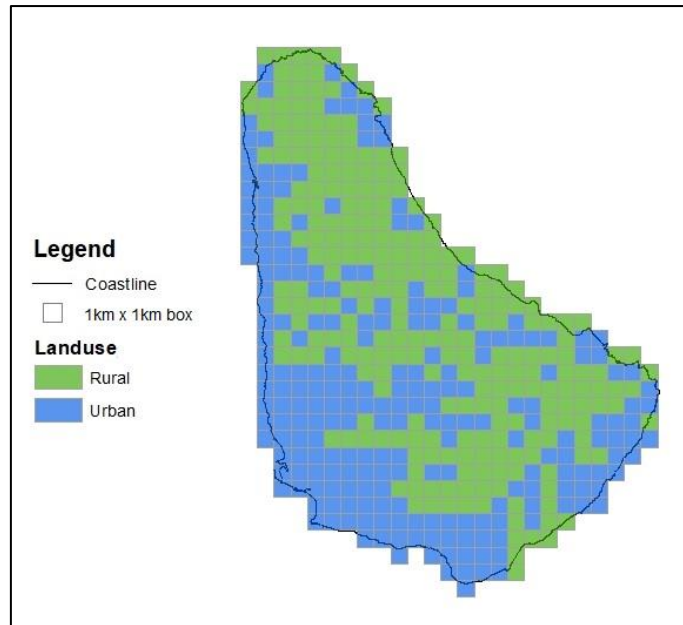


Figure 4.15 - 1km x 1km grid showing distribution of land-use.

The output in Figure 4.15 was used as a base layer for the classification of districts and subdistricts by land-use. The combinational use of the Zonal Calculation tool in ESRI's ArcGIS ArcMap and Microsoft Excel were used to determine the predominant land-use type of each district and subdistrict. The final land-use maps to depict the classification of districts and subdistricts by land-use were shown in Figure 4.16 and Figure 4.17, respectively. Both maps were compared to satellite images of Barbados for verification. Based on visual inspection and expert knowledge of the areas, both maps were considered accurate.

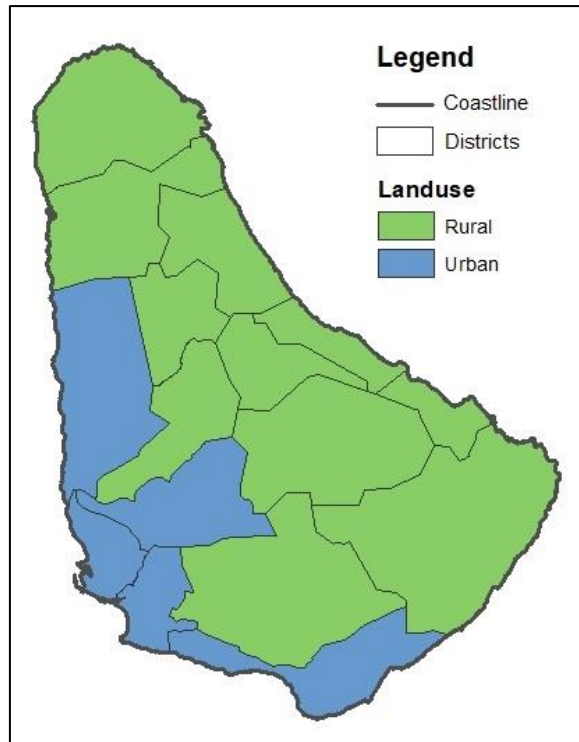


Figure 4.16 – Classification of districts by land-use for the Compound Dataset.

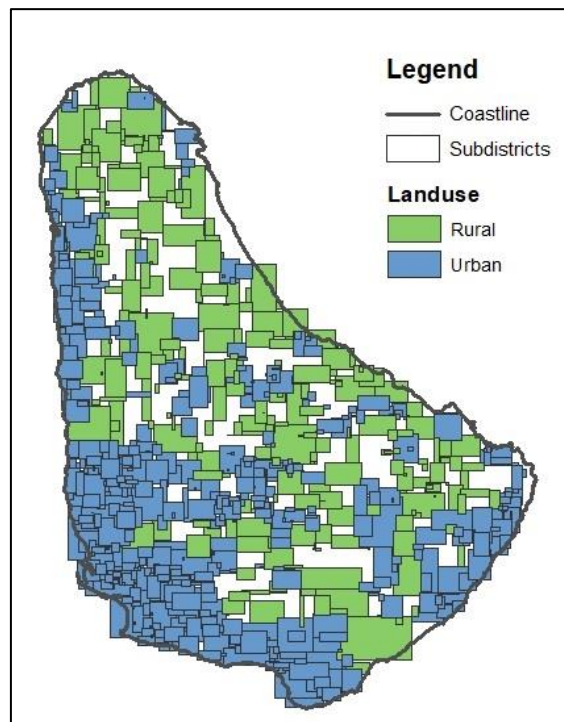


Figure 4.17 - Classification of subdistricts by land-use for the Complex Dataset.

4.4.8 Connection density

The parts along the network where pipes are connected (i.e., joints) are considered as points of weaknesses. Thus, areas of high connection density may indicate parts of the pipeline that are susceptible to pipe bursts (see Section 4.4.8). However, connection density was not often included in pipe failure studies likely because of a lack of a standardise means of quantification. For this study, connection density was the only factor that was GIS-derived.

Connection density was computed by dividing the number of connections along a pipe, represented by geo-referenced meters, by the length of that pipe. The Geoprocessing Tool in ESRI's ArcGIS ArcMap was used to determine the meters that were nearest to each pipe. The result of the calculations for connection density was shown in Figure 4.18.

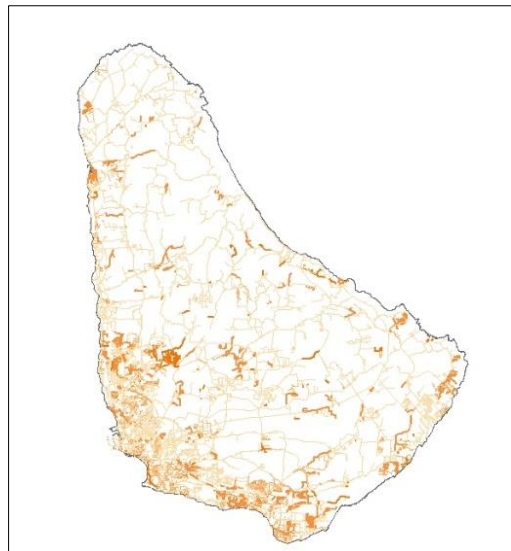


Figure 4.18 - GIS layer for connection density, with higher values depicted by darker colours.

It was expected that there would be some correlation with connection density and land-use (i.e., classification of urban and rural areas). However, possible autocorrelation of these variables was not introduced in the data mining analysis as these variables were not used together. Land-use was only used as a category to label SOM models and not as a numeric variable to drive the SOM and land-use as a binary variable was not incorporated in the EPR analysis of the Complex Dataset (see Sections 6.2 and 6.3).

4.5 Coding Non-Numeric (Categorical) Variables

For this study, eight possible explanatory factors were chosen to analyse pipe burst occurrences in the WDN in Barbados (see Section 4.2). From that set of factors, the following were non-numeric (i.e., variables with parameters that were categories rather than real numbers): pipe material, pipe installation era, soil type, and land-use. Non-numeric variables can be included in an analysis either in its original categorical form (i.e., qualitative data) or as numbers that represented an ordering or ranking of categories (i.e., quantitative data). Both forms of representing non-numeric variables were used in the analyses conducted in this study.

The inclusion of both forms of non-numeric variables in this study was done for several reasons. Firstly, from the methods chosen for the data mining analysis carried out in this study, only SOMS processed and analysed non-numeric input data while EPR could not. Given that, categories of non-numeric factors were assigned numerical values that represented a relative ordering of influence on pipe burst occurrences the WDN. In this way, non-numeric factors could be included in analyses using both methods. Secondly, in the SOMS analysis, non-numeric data did not drive the actual models but were treated as labels in the clustering process (see Section 5.4.1). By incorporating the quantitative form of non-numeric factors, the influence of those variables on the models could also be assessed. Thirdly, and lastly, the stepwise rules used to guide the data mining analysis allowed for a systematic comparison of model outputs using both forms of non-numeric variables. Given that this was an exploratory analysis with the objective of data mining, this approach would have provided useful insight in the handling of different data types in pipe failure analyses.

The coding of categories for non-numeric variables in this study was informed from literature. The coding was structured such that each category was numbered (starting from 1 and ending with n , where n was the total number of parameters for a variable) and ordered according to its relative influence on pipe bursts. Exceptions to this coding structure were made in instances where a variable had only two parameters. Lower ranking numbers implied a lesser

influence in contributing to pipe bursts while higher numbers implied a greater influence. The final coding and associated rationale for each non-numeric factor were given below.

Pipe Material

There were four main types of pipe materials identified in the database. These materials were ranked based on findings in the literature regarding the strength and resistance of these materials to pipe failure, the threat posed to water quality (particularly for asbestos cement) and the era during which these materials were typically manufactured and installed (see Section 2.3.3). The remaining category consisted of pipes for which the material was not known. This category was given the highest ranking and considered to pose the largest threat. It was assumed that these pipes likely represented older pipes in the system (for which the material was probably iron or asbestos cement) and not younger plastic pipes that were more recently installed. The coding for pipe material was shown in Table 4.11.

Table 4.11 - Coding of categories for pipe material.

Coding Pipe Material	
Categories	Ranking
Plastic	1
Steel	2
Iron	3
Asbestos Cement	4
Unknown	5

Pipe Installation Era

For installation era, pipes laid during older periods were considered to pose a higher risk than pipe laid in more recent periods. This was based on the limitations of the life span associated with all manufactured pipes (see Figure 2.1) and findings from the literature regarding pipe age being a significant factor of pipe burst occurrences (see Section 2.3.1). Pipes for which the installation was not known was assumed to be the older pipes in the network and given the highest risk ranking. The coding for installation era was shown in Table 4.12.

Table 4.12 - Coding of categories for pipe installation era.

Coding Pipe Installation Era	
Categories	Ranking
1990 – Present	1
1970 – 1996	2
1964	3
1860 – 1969	4
Unknown	5

Soil Type

There were six main soil types in Barbados (see Figure 4.5). The ranking of these soils according to the contribution of promoting pipe bursts was based on the soil characteristics such as the inclination to soil movement (i.e., shrinking, swelling, and land slippage) and drainage capacity (see Section 4.4.4). The ranking was also informed from literature (see Section 2.3.5). The coding for soil type was shown in Table 4.13.

Table 4.13 - Coding of categories for soil type.

Coding Soil Type	
Categories	Ranking
Sandy Soil	1
Shallow Soil	2
Kandoid Clay	3
Clay Soils	4
Scotland District Clay	5
Smectite Clay	6

Land-use

After a preliminary analysis using self-organising maps (see Section 5.4.1), land-use was ultimately grouped into two classes: urban and rural. Given that, the use of this factor as a quantitative variable was binary. That is, rural and urban were coded as 0 and 1, respectively.

4.6 The Final Database

For this study, the database that was created to facilitate the data mining analysis comprised of three datasets: Simple Dataset, Compound Dataset, and Complex Dataset. Each of these discrete datasets were based on reported pipe bursts that

were recorded at different scales (see Section 4.3). From the literature review, eight possible explanatory factors were identified (see Section 4.2), modelled (see Section 4.4), and linked to each dataset using GIS wherever it was feasible. Properties of the three finalised datasets were given in Table 4.14.

Table 4.14 - Summary of finalised datasets created for this study.

Properties	GEOSPATIAL DATASETS		
	Simple	Compound	Complex
Bursts Records	Bursts by diameter size (21 diameter groups)	Bursts by district (16 districts)	Bursts by subdistrict (851 subdistricts)
Bursts Period	Monthly	Monthly	12-year annual average
Years	2002 - 2012	2002 - 2012	2002 - 2012
Variables	1. Bursts 2. Diameter 3. Rainfall 4. Month 5. Meteorological Season 6. Rain Season 7. Year	1. Bursts 2. BWA district 3. Rainfall 4. Land-use 5. Soil type 6. Location 7. Month 8. Meteorological Season 9. Rain Season 10. Year	1. Bursts (per km) 2. Diameter 3. Installation Era 4. Pipe material 5. Pressure 6. Connection density 7. Rainfall 8. Soil type 9. Land-use 10. Location
Scale	20 km - 30 km	4 km – 10 km	Less than 2.5 km

Table 4.14 showed the list of variables contained in each dataset. The explanatory factors were defined as follows:

- Diameter: as is.
- Material: predominant material of pipe cohort.
- Installation era: predominant installation era of pipe cohort.
- Soil type: predominant soil type of district and subdistrict.
- Rainfall: average rainfall based on time and spatial scales.
- Pressure: average static pressure of pipe cohort.
- Land-use: predominant land-use type.
- Connection density: average connection density of pipe cohort.

For the Compound Dataset, location was added as an indicator variable to represent the general geographic position of a district (see Table 4.3). In the same way, location was also defined for the Complex Dataset. In the Complex Dataset, pipe bursts were normalised by pipe length. That is, the total number of bursts in each cohort was divided by the total length of pipe in that cohort. In this way, the burst frequency would be scaled and give a more accurate depiction of the occurrence of bursts across subdistricts. According to Alizadeh et al. (2019), the consideration of pipe length with pipe burst rates may result in better model performances. Additionally, the ranges of the spatial scale for each dataset were given in Table 4.14. The differences in scale were depicted for visualisation in Figure 4.19.

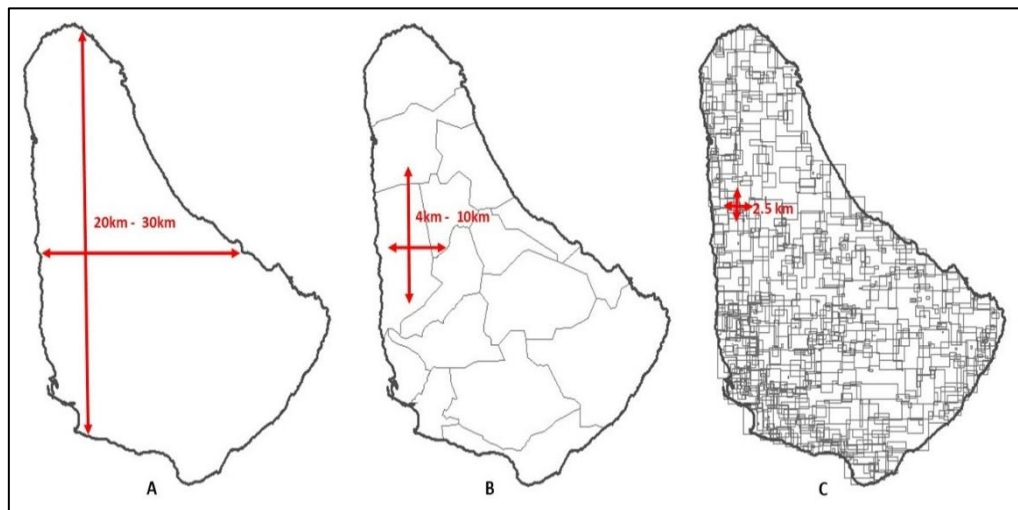


Figure 4.19 - Visualising the spatial scales of the Simple Dataset (A), Compound Dataset (B), and Complex Dataset (C).

4.7 Constraints of the Data and Other Challenges

The constraints and challenges encountered with the data were summarised below. If there was an associated implication, it was identified along with the steps taken to address it.

- The data collected on pipe bursts was carried out by the BWA in an ad-hoc manner except for the period (i.e., 2002 to 2012) for which data on reported pipe bursts were obtained. This is likely the reason behind the request that was made for additional data on historical pipe burst beyond this period not

being fulfilled by the BWA. This had implications on the amount of data available for data mining and for validation.

- Reported pipe bursts in a WDN represented a subset of the total number of bursts. Reported pipe bursts depended on the evidence of a burst incident being detected and brought to the attention of the BWA. Furthermore, given that pipes in the network were buried in the ground, bursts may have unknowingly occurred to the BWA. However, this limitation could not be circumvented.
- Pipe burst data were not recorded at the pipe-level – that is, for each individual pipe. This limitation could not be circumvented and any attempt to increase the resolution would prompt for major assumptions. Thus, all available data on historical pipe burst incidents were used at the scales that data were originally recorded.
- Pipe burst data were recorded at the sub-district level but the boundaries for subdistricts had not been delineated by the BWA. This was addressed by using geospatial techniques to draw boundaries around pipes with the same subdistrict name and extensively cleaning this data (see Section 4.3.3).
- There were major gaps in the data for pipe material and pipe installation era such that these variables would not have been useful in any analysis. This limitation was addressed by sourcing the operational procedures of the BWA to substantially fill-in missing and unknown data (see Sections 4.4.2 and 4.4.3).
- There were a limited number of rainfall stations in the urban corridors. This challenge was addressed by testing different interpolation methods and choosing the most accurate method.
- The hydraulic model that was sourced from the BWA contained nearly 25% of negative values for modelled pressure. Also, gaps in pipe material data which was a critical parameter for calculations were likely if shortcomings in this variable (see Section 4.4.2) were not addressed by the external consultancy company that created the hydraulic model (see Table 4.10). These inconsistencies could not be addressed as the additional information need to improve the hydraulic model was beyond the scope of this research.

Furthermore, the academic version of WaterGEMS could not run the hydraulic model to calculate the average static pressure due to restrictions on the size of the model. As a workaround, geostatistical techniques were used to calculate the average static pressure of each pipe in a GIS. Despite the shortcomings, the model exhibited consistency to a certain extent as modelled values of high pressure aligned with subdistricts known to have problems with pressure management.

- There were challenges and limitations with data for traffic loading, pipe depth, and water quality that could not be circumvented. Thus, these possible explanatory factors were not included in this study.
- The preliminary procedures (see Section 4.2) along with the techniques used to clean, model, and compile the final datasets (see Sections 4.3 and 4.4) took more than two years to complete – mostly owed to delays in receiving the data from local entities and the extensive cleaning and geospatial analyses that were required.

Although the above list was lengthy, most of the challenges were addressed to the furthest extent possible. The final database was the best quality which could be achieved within the major limitations of the base datasets.

4.8 Validity and Reliability of the Data

In creating the database, it was important to ensure that the factors that were modelled were accurate and reliable (i.e., what was intended to be measured was actually measured) and that the techniques used were consistent and replicable.

The validity of the data was enhanced by:

- Data cleaning and filling-in missing values for pipe asset data (i.e., pipe material and pipe installation era) with information sourced directly from the BWA.
- In modelling land-use in GIS, the final output was overlaid on a satellite image of Barbados to visually inspect if areas classified as urban or areas

were aligned with built-up or scarcely populated (or heavily covered with vegetation) areas, respectively.

- For rainfall, different interpolation methods (i.e., Kriging and Co-kriging) were tested. Based on the accuracy of prediction of the test models which used historical rainfall data for Barbados, the co-kriging method was chosen to interpolate rainfall.

The reliability of the data was enhanced by:

- Sequentially applying a set of geospatial analysis techniques in the modelling of each factor that would produce the same outputs if repeated.
- Depicting critical aspects of the factors that were GIS-derived by using key ‘fixed’ variables. For land-use, the density of rooftops was used as an indicator of the population density to determine if an area was urban or rural. ‘Rooftops’ was more or less a fixed variable that was a reliable proxy that is typically used for estimating population density. For connection density, the pipe network and location of water meters were used as input variables for calculation.

4.9 Chapter Summary

Data on historical pipe bursts in the local WDN were sourced from the BWA in the form of three discrete datasets. These datasets recorded pipe bursts on different scales, namely: Simple Dataset – bursts by diameter, Compound Dataset – bursts by district, and Complex Dataset – bursts by subdistrict. In this chapter, it was shown how these datasets and the chosen set of possible explanatory factors (i.e., diameter, material, installation era, soil type, rainfall, pressure, land-use, and connection density) were geospatially modelled and visually represented. Each dataset was used as a separate base, and factors were linked with a GIS wherever it was feasible. Challenges in data quality were addressed as far as possible within the scope of the research and actions were taken to ensure datasets were accurately depicting factors. Having prepared the data for analysis, the next chapter details the methodology and how the methods selected for analysis would be implemented.

5 IMPLEMENTATION OF METHODS

5.1 Chapter Introduction

The aim of this chapter is to explain how the methods that were selected for this study were implemented at each stage of the research methodology. The following is presented in this chapter:

- (1) An overview of the analytical process.
- (2) The stepwise rules that were developed to guide data mining analysis.
- (3) The implementation process of each research method.
- (4) The geospatial techniques used to create and analyse hotspot maps of the pipe network.
- (5) The procedures used to test the pipe burst risk model.
- (6) The procedure used to assess the impacts of climate change on pipe bursts.

The layout of this chapter follows the order that the steps in the analytical process were carried out.

5.2 The Analytical Process

An analytical process was developed to specify the flow and sequence of how the methods and techniques would be applied. The analytical process was based on the research strategy and modelling framework (see Sections 3.4 and 3.5). A depiction of the analytical process is shown in Figure 5.1.

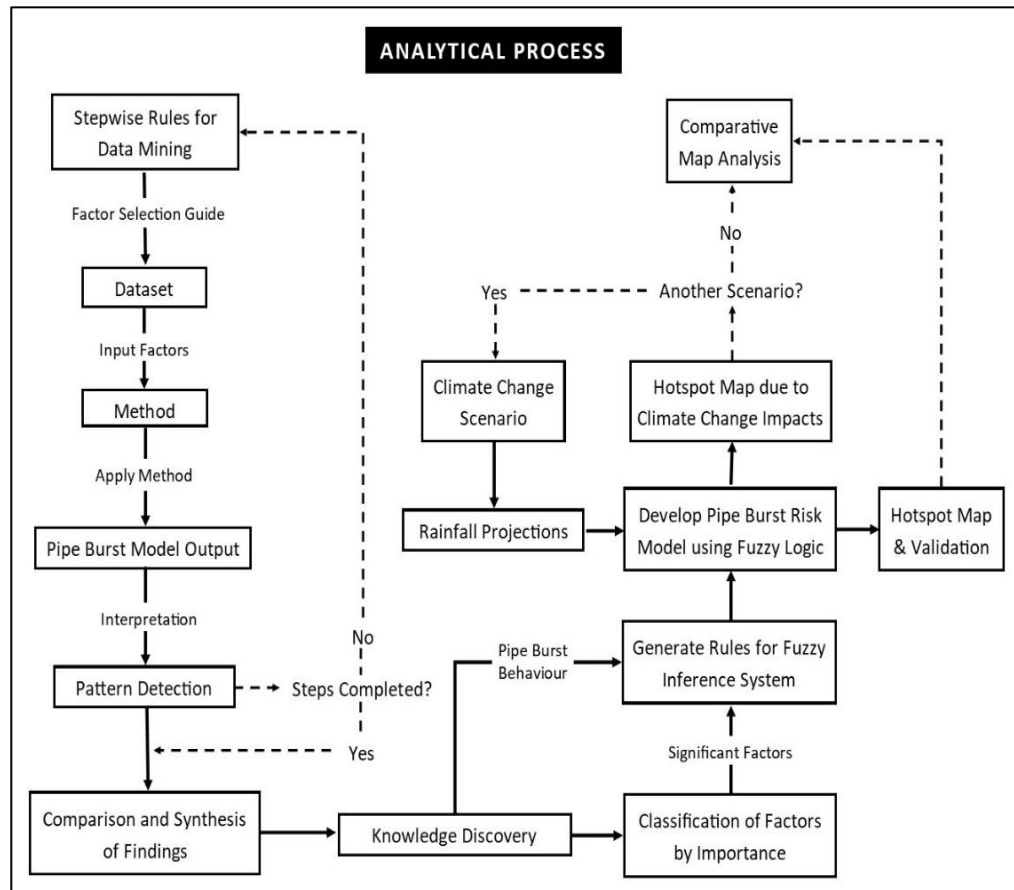


Figure 5.1 - The analytical process of the research.

The analytical process began with employing a set of stepwise rules (see Section 5.3) to guide the data mining analysis in alignment with the research aim and objectives (see Section 3.2). The datasets that had been compiled based on informed data gathering (see Section 4.6) were used for the data mining analysis which was conducted in a systematic manner. Beginning with the Simple Dataset, the stepwise rules allowed for the objective selection of variables rather than selecting all or an arbitrary combination of variables. The selection of variables would be used as inputs in the analysis of pipe bursts which would be initially carried out using SOMs.

Applying SOMs produced a pipe burst model in the form of a 2-D feature map that required interpretation of the outputs (see Section 5.4.1). This resulted in the detection of patterns that would be used to inform or signal proceeding to the next step. The iterative stepwise rules were applied until all steps had been carried out and each dataset had been analysed. Once completed, the process

began again using EPR (see Section 5.4.2). The findings from the SOMS and EPR analyses were compared and synthesised to promote knowledge discovery. This new knowledge was used to give insight into pipe burst behaviour and facilitated the classification of explanatory factors by level of importance.

In the classification, a factor was categorised as significant if it was found to contributed to the occurrence of peak pipe bursts (or the highest recorded burst frequency). The understanding of the pipe burst behaviour of the significant factors was used to define the rules that governed the pipe burst risk model developed with fuzzy logic (see Section 5.4.3). This model was used to rank and identify the most critical areas of the pipe network with a hotspot map (see Section 5.5). The pipe burst risk model was tested using reported pipe bursts and pipe repair data of the BWA (see Section 0).

In the same way that the pipe burst risk model was used to generate a hotspot map of the network, projections for rainfall were incorporated into the model to assess how climate change might affect the distribution of peak pipe bursts in the future. Various climate change scenarios were assessed, producing several hotspot maps (see Section 5.7). Ultimately, the hotspot maps under the different climate scenarios were compared to the hotspot map of the network under current conditions. The findings from the comparative analysis of hotspot maps could be used to provide information to assist in the planning of pipe replacement interventions of the local water utility.

5.3 Stepwise Rules to Guide the Data Mining Analysis

For this research, an inductive approach was taken to carry out an exploratory analysis. Due to the effectiveness of implementing data-driven methods in pipe failure studies shown in Section 2.4.2, data mining was chosen as the main type of analysis to detect patterns from pipe burst modelling. To facilitate the data mining analysis, three datasets (i.e., Simple Dataset, Compound Dataset, and Complex Dataset) were created (see Chapter 4) and two methods (SOMS and EPR) were chosen to model pipe bursts. To guide the data mining analysis in how each dataset would be analysed using each method, stepwise rules were

developed. In each step, the analytical technique produced an output from which insights were applied to the subsequent step. The steps were set out as follows:

Step 1. Using the Simple Dataset, assess the following:

- 1a. Temporal trends in the occurrences of bursts starting with the initial base variables of the dataset: bursts, diameter, and month.
- 1b. From the trend detected in 1a, identify and include other explanatory variables that exhibited a similar trend.
- 1c. The effect of different statistics (i.e., average and standard deviation) of the variable identified in 1b on the model.

Step 2. Using the Compound Dataset, assess the following:

- 2a. Temporal and spatial trends in the occurrences of bursts starting with the initial base variables of the dataset: bursts, location, month, and year.
- 2b. The contribution of the variables identified in 1b to the model.
- 2c. The contribution of other the remaining geospatial variables to the model.

Step 3. Using the Complex Dataset, assess the following:

- 3a. Key variables identified in Steps 1 and 2.
- 3b. The contribution of other physical and operational variables.
- 3c. The contribution of other environmental variables.
- 3d. The inclusion of the risk values assigned to non-numeric variables.

The rationale for developing these stepwise rules, including reasons for how steps were arranged, is listed below:

- To create a systematic approach for ordering the analysis of variables to assist in the detection of patterns and determining the type of explanatory factors (i.e., physical operational, and environmental) that improved model performance rather than an *ad hoc* or simultaneous approach.
- To let the data mining process be continuously informed as trends and relationships are detected.

- To standardise the application of each method to promote consistency and reliability.
- To adopt the law of parsimony (i.e., the methodological principle that the simplest explanation which the least assumptions is preferred) for the data mining process by beginning with and building on basic models.
- To include and evaluate the contribution of all explanatory variables.

The stepwise rules were carried out for each method - first using SOMS and then EPR. The details on the implementation of these methods were discussed in the next section.

5.4 Implementation of Research Methods

In the modelling framework that was developed for the analyses carried out in this study (see Figure 3.2), the main methods were SOMs, EPR, and Fuzzy Logic. The procedures used for implementing each method were described in the sections below.

5.4.1 Self-Organising Maps (SOMs)

Self-organising maps (SOMs) (Teuvo Kohonen 1990) uses an artificial neural network to visualise nonlinear relationships in multi-dimensional data on low dimensional grids (see Section 2.4.2). Given its functionality in detecting patterns in complex data and its successful use in other similar research (see Section 2.4.2), SOMs were chosen as the primary method for data mining via pipe burst modelling in this study. MATLAB (version R2019a) by MathWorks was the software used as the platform to carry out the SOMs analysis. The SOM Toolbox 2.1, a software library for implementing the algorithm for SOMs in MATLAB, developed by Alhoniemi et al. (2012) was used to facilitate the SOMs analysis. The implementation of SOMs was performed through the following steps:

- Identifying data operations for the SOM analysis.
- Carrying out a preliminary SOM analysis.
- Conducting the SOM analysis.
- Interpreting outputs from the SOMs analysis and summarising key findings.

In order to perform an analysis using SOMs, a set of required data operations using the SOMs Toolbox 2.1 in MATLAB were identified. These data operations were listed in order below:

- (1) Identify the variables that would be included in the analysis. For the SOM run, three comma-separated value (CSV) files were required as input into the model containing the following: (i) names of all variables used in the analysis (including pipe bursts); (ii) observation values for each variable; and (iii) names of units of measurement.
- (2) Specify the directory of the input files and activate the directory path.
- (3) Import and read the import data.
- (4) Run the model to compute the SOM.
- (5) Save and export the outputs of the SOMs analysis as figures.

In the analysis, only numeric (i.e., quantitative) variables drive the SOM. Non-numeric (i.e., qualitative) variables are used as categories to label the model. Therefore, if a SOMs analysis that only contained numeric variables was run, adding non-numeric variables and rerunning the model would not change the outputs of the component planes for the numeric variables and *U-matrix*.

A preliminary analysis was conducted with SOMs to check for noise in the data before conducting the stepwise analysis. For this analysis, the Complex Dataset which was considered the noisiest dataset was used. For the most part, the results showed that the data were not too noisy since expected correlations between variables, such as pipe material and pipe installation era, were observed. However, it was difficult to discern the categories of pipe material and land-use when used as non-numeric input variable in the SOM. As a result, the categories of these variables were regrouped to higher levels (see Sections 4.4.2 and 4.4.7).

After the preliminary analysis was carried out, the SOMs analysis was conducted according to the stepwise rules that were developed to guide data mining (see Section 5.3). Each SOM run followed the abovementioned set of data operations. The procedures used to interpret the SOMs outputs and summarise key findings were described below.

For each SOM analysis, the output is called a Kohonen SOM map which represents the feature space of the data being analysed. This low dimensional map is comprised of component planes for a *U-matrix* and each input variable. Component planes are made up of colour-coded hexagonal cells that represent neurons. These neurons were the processing units used for training the network to group or cluster similar high-dimensional data while preserving the original topological and metric properties of the input data. The colours of the cells, along with the scale bars alongside each component plane that show the range of values for each variable, are used to facilitate interpretation of the SOMs.

An example of a SOM output was shown in Figure 5.2. The *U-matrix* is used to represent how data are clustered over the entire feature space. Instances of clusters are represented by lighter colours (i.e., turquoise) while the space between clusters is represented by darker colours (i.e., blue). Thus, several small clusters were observed in the *U-matrix* component plane. For the detection of patterns among input variables, again, component planes were visually inspected. High and low values for variables were represented by red and blue, respectively, with intermediate values in between. In the *Burst* and *Pressure* component planes, instances of peak values were found in the same position of each plane. This can be interpreted as a correlation between peak bursts and peak pressures. Further to that relationship, by examining the *Diameter* component plane, it was observed that smaller values of diameter also coincided with that pattern. Another pattern that can be discerned is a relationship between installation era and material based on the similar distributions in those component planes. In the same way these component planes were examined, component planes for non-numeric variables used as categories to label a model can also be interpreted.

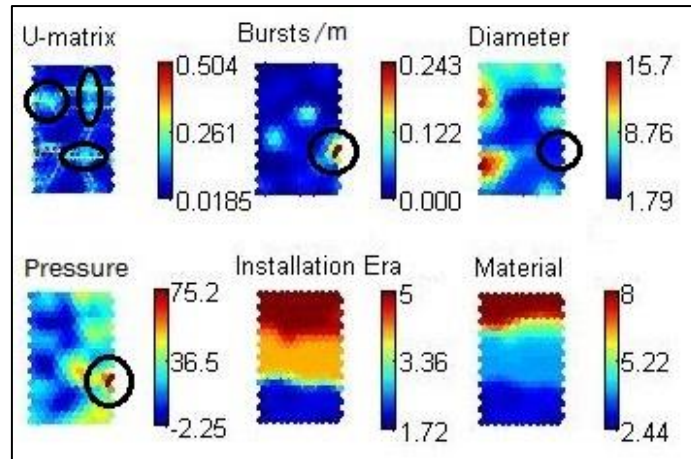


Figure 5.2 - An example of the output from a SOM analysis.

Patterns and trends were detected from visually inspecting SOM outputs. Interpreting findings required qualitative reasoning which could also be substantiated from established knowledge and observations (see Section 2.3). Ultimately, the patterns detected in each SOM run were compiled using the tabular format shown in Table 5.1. This compilation of key findings was used to facilitate a comparative analysis that informed a systematic synthesis of all findings (see Section 5.4.3).

Table 5.1 - Template used to summarise key findings from SOMS analyses.

SOM Run	Detected Patterns from Data Mining Analysis with SOMS		
	Low Bursts (Number of bursts)	Medium Bursts (Number of bursts)	High & Peak Bursts (Number of bursts)
1			
2	Summary of detected patterns		
⋮			

5.4.2 Evolutionary Polynomial Regression (EPR)

The second method used was evolutionary polynomial regression (EPR) (Giustolisi and Savic 2006) that combined numerical regression modelling with genetic algorithms to produce mathematical expressions (see Section 2.4.2). In 2009, Giustolisi and Savic expanded on previous work with the inclusion of a multi-objective function that developed multiple models of interpretable mathematical expressions which were simultaneously optimised for fitness of data and parsimony. The works of Giustolisi and Savic (2006 and 2009) were

packaged as a Microsoft Excel function named ‘EPR MOGA-XL’ which was programmed by Luigi Berardi, Daniele Laucelli, and Orazio Giustolisi in 2011. EPR MOGA-XL was used for this research. The following were sequentially performed in implementing the method:

- Identifying data operations for the EPR analysis.
- Conducting a preliminary analysis for selecting an expression structure for EPR modelling and checking for randomness in the data.
- Conducting the EPR analysis.
- Evaluating mathematical expressions and summarising key findings from the EPR analysis.

To perform an analysis with EPR, a set of required data operations were identified. The data operations were listed below in sequence of execution:

1. Identify the dependent and independent variables that would be included in the model run (based on the stepwise rules for data mining in Section 5.3) and extract the data from the associated dataset (i.e., Simple Dataset, Complex Dataset, or Complex Dataset).
2. Open the EPR MOGA-XL add-on function in Microsoft Excel and define tool settings as shown in Figure 5.3. The default settings were mostly kept with exception of the expression structure, inner function, and modelling type. The expression structure was determined through the preliminary analysis which was described later in this section. According to Mounce et al. (2014), inner functions do not necessarily increase model accuracy and can lead to longer model run times. Therefore, structures did not include inner functions. The modelling type was selected as statistical regression.
3. Import data for input variables.
4. Define labels for each input variable.
5. Set data.
6. Run EPR.
7. Save spreadsheet file with model outputs and performance metrics.
8. Save figures of mathematical expressions.

Evolutionary Polynomial Regression EPR MOGA-XL - Office Excel Add-In (1.0)										About EPR MOGA-XL		H E L P	Syntax	Settings	Output
													Data	EPR Run	EPRreg-XL
Filename	STATIC									Start EPR-MOGA Analysis		☐ Control ai ☐ Control zi ☐ PCS			
Plot Option	Cartesian plot											GA parameters			
Expression structure	Y=sum(ai*f(X1*X2))+ao											10	40		
Inner Function	No function											Bias (a0)			
Modelling Type	Statistical Regression											Regression Method			
[na nb nk]	0	0	0									LSaj>0 (Non-negative Least Squares)			
Scale Output	0	0										Optimization Strategy			
Scale Input	0	0										Min(Xi,SSE)			
Number of Terms	1	2	3	4	5	6	7	8	9						
Exponents	0	0.5	1	1.5	2	-0.5	-1	-1.5	-2						

Figure 5.3 - Main user interface of EPR MOGA-XL.

EPR MOGA-XL has seven built-in expression structures. Mounce et al. (2016) tested each structure to determine the best model structure to fit their data in an attempt to produce the highest performing models. Similarly, for this study, a preliminary EPR analysis was carried out to determine the best model structure. For this, two subsets of data from the Simple Dataset were used as input:

- Set 1: Pipe bursts, pipe diameter, and rainfall.
- Set 2: Pipe bursts, pipe diameter, rainfall, and month.

In total, 14 EPR runs were performed to test and find the best fit expression structure. The results were shown in Table 5.2. While Structures 1 and 2 had the highest and same COD, Structure 2 was chosen for simplicity.

Table 5.2 – Results of testing the built-in expression structures for EPR modelling.

No.	Expression Structure	Average COD of Models	
		Set 1	Set 2
1	$Y = \sum(a_i * X_1 * X_2 * f(X_1) * f(X_2)) + a_o$	49.2	49.0
2	$Y = \sum(a_i * f(X_1 * X_2)) + a_o$	49.2	49.0
3	$Y = \sum(a_i * X_1 * X_2 * f(X_1 * X_2)) + a_o$	48.3	49.1
4	$Y = \log(\sum(a_i * X_1 * X_2) + a_o)$	-7,237.0	-7,292.2
5	$Y = \exp(\sum(a_i * X_1 * X_2) + a_o)$	30.0	30.0
6	$Y = \sin(\sum(a_i * X_1 * X_2) + a_o)$	-20.8	-22.0
7	$Y = \tan(\sum(a_i * X_1 * X_2) + a_o)$	-9.5	-5.5

In the preliminary analysis, the check for randomness in the data were performed by partitioning another subset from the Simple Dataset (i.e., pipe bursts, diameter, and month) into training and testing data for EPR analyses. The model outputs and values of the COD exhibited no significant differences. Thus, data were not partitioned for training and testing in the EPR analysis. This was considered reasonable for several reasons:

- The purpose of the mathematical expressions produced from the EPR analysis was to assess the relationships between the variables and not for predicting pipe burst rates since non-numeric variables with more than two categories could not be included. Also, other data constraints that affected the precision and exactness of the data were not best suited for predictive modelling (see Section 4.7).
- Pipe burst modelling with EPR was used to verify findings from the SOMs analysis and not used as the lead method for detecting patterns.
- Discovering relationships between pipe burst and explanatory factors could be discerned from models with low COD values.
- The goal of utilising EPR analysis as a tool for data mining was knowledge discovery, and thus having more data available would support this effort.
- The data used in each EPR run was already being partitioned into subsets according to the factor selection of the stepwise rules. Furthermore, the EPR analysis generated multiple models that were compared and synthesised to determine a general consensus in pipe burst behaviour.

Before using either SOMs or EPR, the preliminary analyses carried out were used to inform how each method would be implemented. The EPR analysis was carried out according to the stepwise rules for data mining with consideration to the key findings from the SOMs analysis. Each EPR run followed the set of data operations given above. The mathematical expressions produced from the EPR run were ordered by parsimony. In evaluating and summarising key findings from EPR models, the following was considered: the COD, variables found in interaction terms which could indicate relationships between factors, the magnitude of coefficients, and statistical correlations.

5.4.3 Fuzzy Logic

The findings from SOMs and EPR were used to inform the development of a pipe burst risk model using fuzzy logic. Fuzzy Logic is a method that differs from traditional logic in that it allows a degree of uncertainty to be incorporated into reasoning based on computing degrees of truth. Since Fuzzy Logic was introduced by Zadeh (1965), it has been widely used in many applications including pipe failure studies (see Section 2.4.2). For this study, Fuzzy Logic was used as the method to create a single pipe burst risk model of the local WDN based on the findings from data mining. The Fuzzy Logic Designer application in Mathwork's MATLAB was used for the building of a fuzzy inference system (FIS) to produce the pipe burst risk model. The following steps were performed for implementing this method:

- Identifying data operations for the creating and running the FIS.
- Define fuzzy membership functions.
- Define the classification of risk scores.
- Determine the knowledge base that would be used to define the fuzzy rules of the FIS.
- Develop the FIS.
- Run the FIS to generate the risk scores to inform the pipe burst risk model.

To create and run the FIS, data operations were required. These data operations were listed below in the order of execution:

1. Open the Fuzzy Logic Designer application.
2. Add and name each input and output variable.
3. Define the fuzzy membership functions for each input and output variable.
4. For each linguistic variable (i.e., parameter) of each membership function, indicate the name, range, and type of fuzzy membership function.
5. For the FIS, define the fuzzy operators (i.e., 'and' and 'or'), and methods for implication, aggregation, and defuzzification.
6. Use the Rule Editor to input the fuzzy rules.
7. Run the model and compute the risk scores.
8. Export the risk scores.

In defining the membership functions for numeric (i.e., quantitative) variables, a template was developed to limit the subjectivity from user input. This template was adapted using the scales generated for component planes in the SOMs analysis. Using this template, data were partitioned into three classes: low, medium, and high. The range (i.e., difference between the highest and lowest value) was computed and divided by nine in order to produce fixed increments. These increments were used to consistently define the three classes (see Figure 5.4). For non-numeric (i.e., qualitative) variables, membership functions were defined as binary or defined in a way that most accurately represent the categories of a variable.

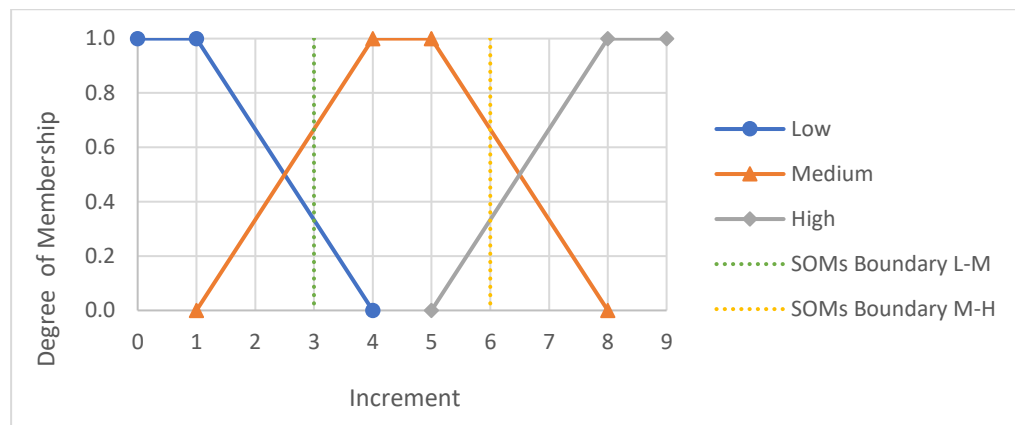


Figure 5.4 - Template for defining fuzzy membership functions.

The single output of the FIS was risk scores for pipes. The risk categories and ranges were shown in Table 5.3. However, since risk score was defined as a fuzzy set (see Figure 6.26), the boundaries in the fuzzy membership function were not fixed as in Table 5.3.

Table 5.3 - Defining risk scoring for the fuzzy inference system.

Risk Categories	Scoring
Low	Less than or equal to 1 (i.e., $\text{score} \leq 1$)
Medium	Greater than 1 and less than or equal to 2 (i.e., $1 < \text{score} \leq 2$)
High	Greater than 2 and less than or equal to 3 (i.e., $2 < \text{score} \leq 3$)
Peak	Greater than 3 and less than or equal to 4 (i.e., $3 < \text{score} \leq 4$)

The knowledge base that would be used to define the rules of the FIS was informed from the data mining analysis. Since SOMs was the primary method

for detecting findings, key findings from all of the models generated with SOMs were compiled and compared. In instances where boundaries of variables overlapped, these were synthesised. These findings were also corroborated with key observations from the EPR models. Based on these synthesised findings, fuzzy rules were developed.

The FIS was developed by following the data operations listed above. Built-in operators and methods within the FIS were defined as follows:

- ‘AND’ operator was based on the intersection operator used in classical set theory which was defined as ‘minimum’ in the FIS.
- ‘OR’ operator was based on the union operator used in classical set theory which was defined as ‘maximum’ in the FIS.
- The implication method determines how the output is presented. This was set to the default, ‘minimum’, which truncates the output fuzzy set.
- The aggregation method is the process in which rules are combined. This was set to ‘maximum’, to obtain the highest culminated value.
- Defuzzification, which defined how the single output value was derived, was set to centroid – returning the value in the center of the area under the output curve.

Having carried out all of the above, the FIS was run. The resulting output (i.e., risk scores) was used to inform the creation of a hotspot map (see Section 5.5) and tested with historical service repair data (see Section 0).

5.5 Procedures for Creating and Analysing Hotspot Maps

The FIS (see Section 5.4.3) assigned a score that represented the susceptibility of a pipe to frequent bursts, thus creating a pipe burst risk model. These scores were used to create a hotspot map by the following steps:

- (1) Export the risk scores produced from the FIS from MATLAB.
- (2) Import the risk scores into Microsoft Excel and append to the attribute data of the pipe network.
- (3) Classify the risk scores by category using an IF-statement function.

- (4) Export that file with the pipe network data and assigned risk scores and categories into ArcGIS as a CSV file.
- (5) In ArcGIS, linked the data from the CSV file to the pipe network using the Join function.
- (6) Edit the symbology of the pipe layer by risk category to depict (by colour) the distribution of risk in the pipe network.
- (7) Define map legend and export map.

For the hotspot map, a clustering analysis was carried out to determine if there were statistically significant clusters of higher risk pipes. This also allowed for the identification of spatial patterns that might not have been as clearly discernible on the hotspot maps. The first step in the cluster analysis was to assess if hotspot maps exhibited patterns of clustering, dispersion, or randomness – that is, determining the existence of spatial autocorrelation based on pipe location and risk score. The Spatial Autocorrelation (Moran's I) tool in ESRI's ArcMap was used calculate the Moran's I Index value to measure clustering and to compute a z-score (representing the standard deviation) and p-value (representing the probability that observed patterns were random) for evaluating statistical significance. If the map exhibited statistically significant clustering, then the Cluster and Outlier Analysis (Anselin Local Moran's I) tool in ESRI's ArcMap was used to geospatially model statistically significant clusters of hot spots, cold spots, and outliers. A pipe was classified: *High-High (HH)* if it was a higher risk pipe that was surrounded by other higher risk pipes; *Low-Low (LL)* if it was a lower risk pipe that was surrounded by other lower risk pipes; *High-Low (HL)* if was a higher risk pipe that was surrounded by lower risk pipes; and *Low-High (LH)* if was a lower risk pipe that was surrounded by higher risk pipes. Pipes that had a p-value that was not statistically significant were deemed as not showing any distinct clustering.

5.6 Validating Pipe Burst Risk Model

The pipe burst risk model, which was created using a FIS (see Section 5.4.3), was validated with records of service disruptions attributed to pipes with reported or suspected bursts. These records, which spanned from March 2019

to August 2020 (i.e., 1.5 years), were retrieved from public notices given by the BWA on its official account on Instagram. These records were used as the best available source of data since written requests to receive historical data on reported pipe bursts between 2013 and 2020 directly from the BWA were not fulfilled. Service disruptions notices displayed the date that the notice was given along with information on reported and suspected pipe bursts. Bursts were reported by diameter size, subdistrict, and parish (see Figure 5.5).

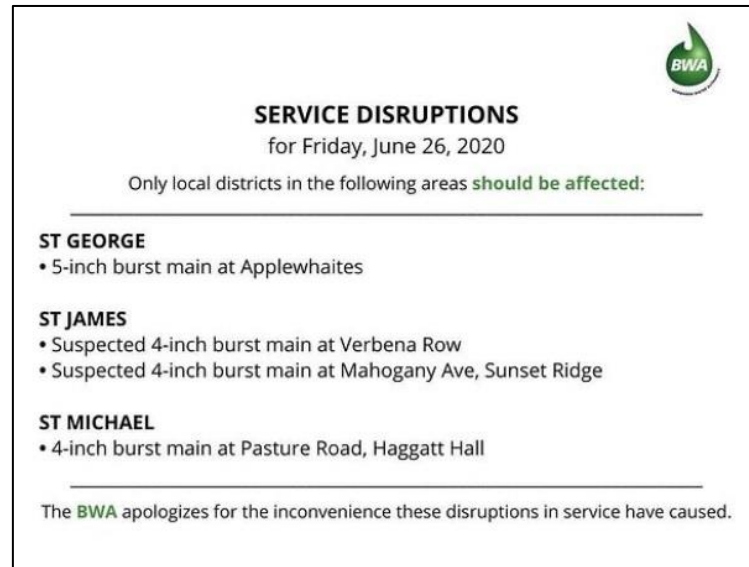


Figure 5.5 - A public notice shared by the BWA of service disruptions caused by pipe bursts.

Approximately 475 electronic notices were used to compile by hand 1,001 observations of reported and suspected bursts. From those observations, the description of bursts by subdistrict and by pipe diameter size had 709 matches with pipes in the Complex Dataset which were also recorded by subdistrict and by diameter. A map of these reported and suspected pipe bursts is shown in Figure 5.6.

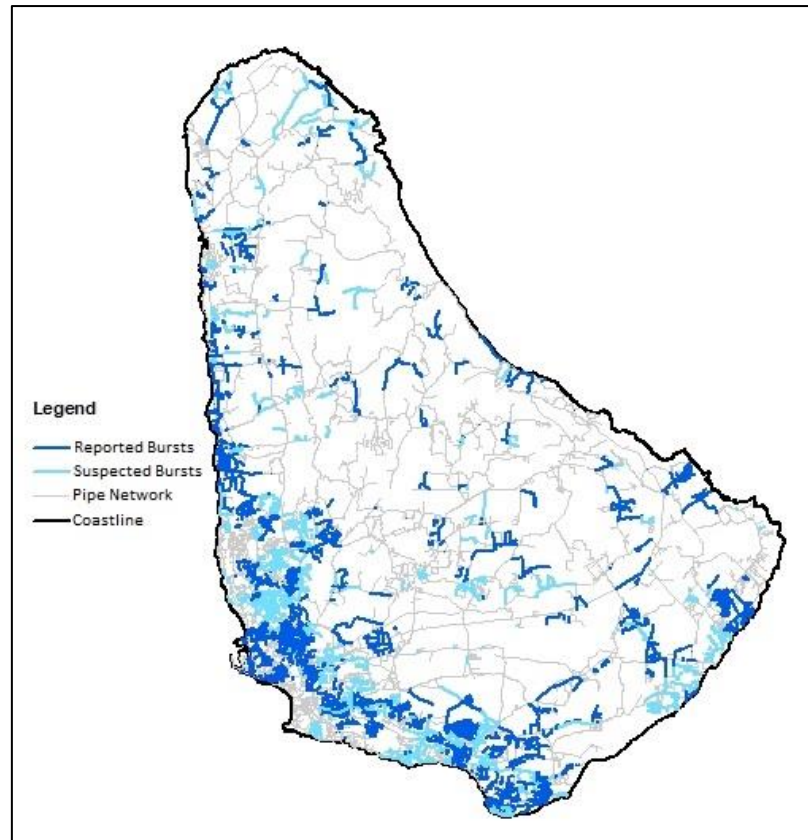


Figure 5.6 - Map of reported and suspected bursts between March 2019 and August 2020.

From the 709 bursts observations, 379 were unique. It was assumed that repeated observations indicated pipes that had multiple bursts. If bursts for a particular pipe group was recorded more than once outside of a 10-day period, then those instances of bursts were considered as separate incidents. This assumption was based on the development of standards service for the BWA which were determined in consultation with the Barbados Fair Trading Commission. The standards had set targets of having 90% of bursts on service pipes (i.e., smaller diameter pipes) be repaired within 5 days, and 95% of bursts on transmission pipes (i.e., larger diameter pipes) be repaired within 1 day (Barbados FTC 2019). Therefore, a 10-day period was used to determine multiple burst incidents with a doubling of the regulated time frame to act as a buffer for possible delays in repair.

Pipes identified in the service disruption and repair database by single or multiple burst incidents were matched and compared to the risk scores assigned

from the pipe burst risk model. This procedure provided a degree of validation of the pipe burst risk model despite data limitations.

5.7 Conducting the Assessment of Climate Change on Pipe Bursts

A part of the overarching aim was to inform the planning of local pipe replacement interventions by taking into consideration the implications of climate change on pipe bursts. Along that line, the research sought to explore how the effects of climate change, particularly on rainfall (i.e., the only climatic variable included in this study), could impact pipe burst occurrences. The procedures used in the assessment of the impact of rainfall and changes in rainfall on pipe bursts are described in this section.

For the first step, a model that depicted current pipe burst behaviour in the WDN was required. Thus, the pipe burst risk model which was developed using fuzzy logic (see Section 5.4.3) was also used as the base model for this assessment. A new sub-dataset of rainfall projections informed by climate change would be used as input into the model. It was preferred that the projections were in a geospatial format at a resolution greater than or equivalent to the scale at the subdistrict level which was between 2 km and 3 km (see Section 4.3.3) to match other model input data.

For the second step, climate change projections for rainfall had to be sourced since it was outside the scope of this research to produce such data. Hence, georeferenced climate change projections for rainfall in Barbados were sourced from Charlery and Nurse (2018). The projections were generated using direct area downscaling and the model boundary was derived from the Phase-6 of the Couple Model Intercomparison Project – a framework that was widely used for the evaluation of global climate models. Charlery and Nurse (2018) based the model on more than 30 years of historical data for Barbados. The downscaled resolution was 1 km and the data were processed at 0.2 km resolution. Projections for rainfall were generated using the latest climate change scenarios – Representative Concentration Pathways (RCPs) – from the IPCC. The lowest and highest emission greenhouse gas (GHG) scenarios (i.e., RCP2.6 and RCP 8.5, respectively) were chosen for the years 2035 and 2050. In this way, the

most extreme scenarios would be evaluated for medium (2035) and long-term horizons (2050).

For the third step, the projected rainfall data sourced from Charlery and Nurse (2018) had to be geospatially plotted and interpolated. The data were received as 48 CSV files – 1 file for each month in the year for the 4 climate change scenarios. The files contained monthly rainfall projections, latitude, and longitude for 2,442 data-points. The 48 files were merged into 4 files, each representing a scenario year, so that the average annual rainfall could be calculated for each data-point. The files were imported into a GIS and the data-points were plotted (see Figure 5.7). Annual average rainfall for each scenario was interpolated by taking the same approach used to interpolate rainfall in the creation of the pipe burst datasets (see Section 4.4.5). Thus, the co-kriging interpolation method was applied using the Geostatistical Analyst tool in ESRI's ArcGIS ArcMap to incorporate the data-points for projected rainfall and a digital elevation model of the island due to the orographic rainfall (see Section 7.2).

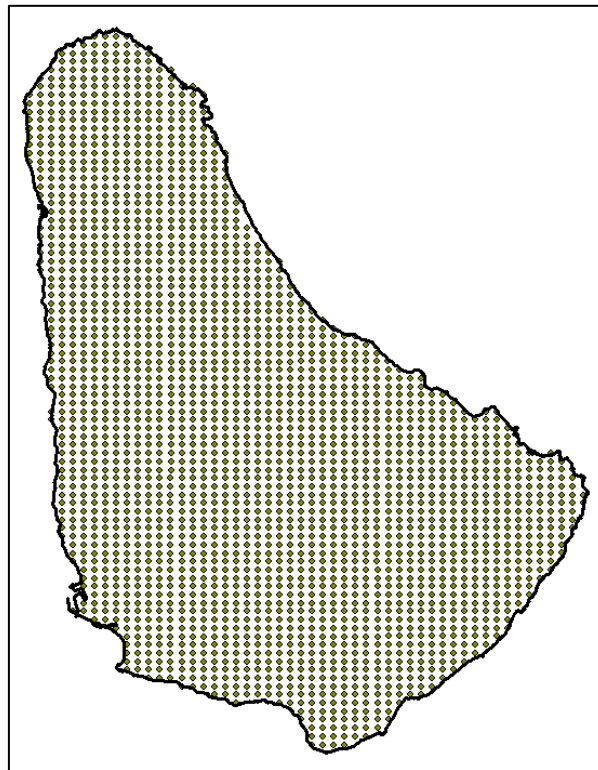


Figure 5.7 - Georeferenced data-points for climate change rainfall projections for Barbados.

For the fourth step, the data rainfall projections for each scenario were analysed by using a standard meteorological index called deciles. Deciles were used to compare changes from historical rainfall to future rainfall (see Section 7.3.2). The range of historical rainfall was ranked and divided into 10 equal scales (i.e., deciles) to establish a baseline about the average which was then used to compare future values. A decile map was produced for each scenario. In this way, a better understanding of the extent to which the amount and distribution of rain was likely to be altered by climate change was gained. In addition to deciles, the characteristics of the wet and dry seasons for each scenario were also analysed (see Section 7.3.1).

For the fifth step, the average value of projected rainfall had to be calculated for all subdistricts for each climate change scenario. The geospatial analysis methods used to calculate the average historical rainfall for subdistricts (see Section 4.4.5) were applied in the same manner. In this way, projected rainfall values for each scenario were linked to the Complex Dataset.

For the sixth step, multiple runs of the pipe burst risk model were carried out. The input data included significant factors from the Complex Dataset and projected rainfall which was swapped out for each climate change scenarios. In this way, the risk scores for the 2,442 pipe cohorts (grouped by subdistrict and by pipe diameter) were calculated for each climate change scenario.

For the seventh step, hotspot maps for each climate change scenario were produced and assessed. The risk scores that were generated from the pipe burst risk model for each scenario were exported from the FIS in MATLAB (version R2019a) and imported into a spreadsheet to determine the risk categories of each pipe cohort. The spreadsheet was imported into a GIS for linking the data to the geospatial representation of the pipe network. Schematic maps of the pipeline were created to depict risk using the techniques in Section 5.5. Histograms and summary tables for the four hotspot maps were generated to conduct a comparative analysis of how the distribution of pipe bursts varied under each climate change scenario.

For the final step, a cluster analysis was performed on the hotspot map produced for each climate change scenario. These maps were visually inspected to identify and compare patterns found in the hotspot maps. Histograms to show the frequency of the risk categories were also created. The procedures and methods used in the cluster analysis were given in Section 5.5.

5.8 Chapter Summary

In this chapter, the analytical process of the research was depicted (see Figure 5.1) and explained (see Section 5.2). For the data mining analysis, an iterative set of stepwise rules was developed to systematically guide how variables in each dataset would be analysed with each method (see Section 5.3). Details on the implementation of each method (i.e., SOMs, EPR, and Fuzzy Logic) that contributed to the development of the pipe burst risk model were provided (see Section 5.4). The risk scores assigned to pipes from the model were used to creating hotspot maps. Procedures for mapping and evaluating the spatial distribution of clusters on the hotspot maps were also detailed (see Section 5.5), along with the how the model was tested (see Section 0). Hotspot and cluster maps were also generated for the climate change assessment which incorporated downscaled projections of future rainfall. Eight action steps were carried out to conduct the assessment of climate change on pipe bursts (see Section 5.7). In the next chapter, the results of the data mining analysis were presented along with the final pipe burst risk model that was developed for the local WDN.

6 RESULTS: DATA MINING AND PIPE BURST MODELLING

6.1 Chapter Introduction

The previous chapter details how the analysis was carried out. The aim of this chapter is to present the results of the analysis – particularly, the outputs of the data mining analysis and pipe burst modelling. The following is contained in this chapter:

- (1) A data mining analysis based on pipe burst models generated with self-organising maps (SOMs) and evolutionary polynomial regression (EPR).
- (2) A comparison and synthesis of outputs from pipe burst modelling with SOMs and EPR.
- (3) The pipe burst risk model that was based on the knowledge that was derived and created using fuzzy logic.

This chapter is laid out in this way to present the results in the order of the flow of the analytical process (see Figure 5.1).

6.2 Data Mining with SOMs

SOMs, a type of ANN that allows for the detection of relationships among multivariate on low-dimensional grids (see Section 2.4.2), was the primary method used for this data mining analysis. Only the outputs for the SOMs analysis, which were presented by dataset, are shown in this section. An in-depth discussion of the results and implications is provided in Chapter 8.

6.2.1 SOMs Analysis – Simple Dataset

As described in Section 4.3.1, the Simple Dataset was based on monthly time-series data of reported pipe bursts between 2002 and 2012 recorded by pipe diameter size. The first objective (i.e., Step 1a) of the analysis was to identify any temporal trends in reported pipe bursts in the Simple Dataset using SOMs. Spatial trends could not be determined since variables in the Simple Dataset were not defined by boundaries smaller than an island-scale extent. For the first model run, input data were total monthly bursts, diameter, and month. The

output of the first run using SOM was shown as Run-A in Figure 6.1. Interpretations of the outputs followed the techniques detailed in Section 5.4.1.

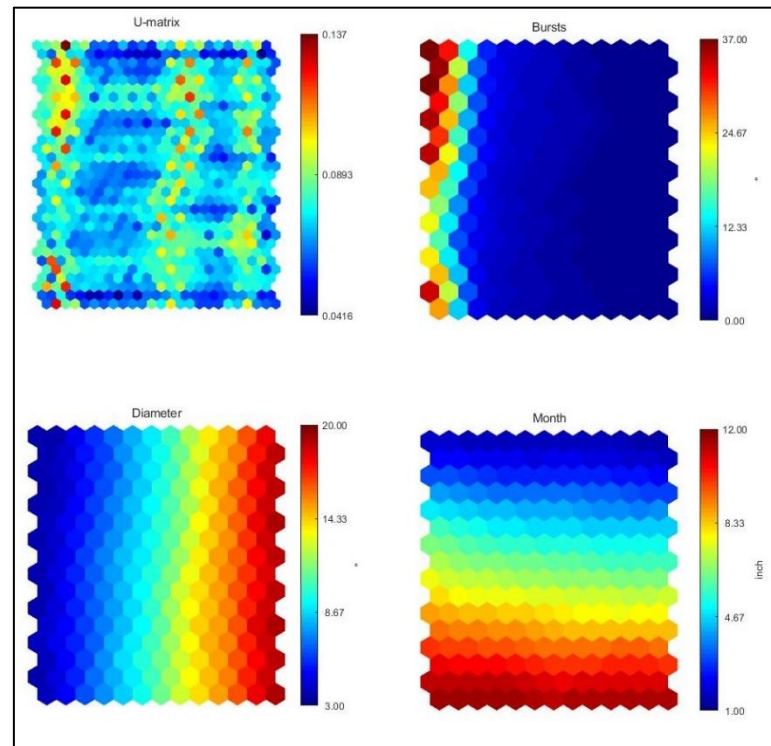


Figure 6.1 - SOMs result for monthly bursts analysis (Run-A) using Simple Dataset.

In Figure 6.1, examination of the component planes for *Bursts* and *Month* showed that pipe burst occurrences were highest (i.e., greater than 25 bursts per month – see scale of *Bursts*) between December and June. For the remaining months in the year, from July to November, pipe bursts occurred at a low (0 to 12 bursts per month) or medium (13 to 25 bursts per month) frequency. These results implied a temporal trend in that the upper third of pipe burst frequencies in the entire WDN took place between December and June.

Another observation from Run-A (see Figure 6.1) was that high and medium burst frequencies occurred in small diameter pipes that were between 3-inches and 6-inches. Furthermore, similar to the consensus in the literature (see Pipe Diameter in Section 2.4.2), pipe bursts and pipe diameter appeared inversely correlated (see *Bursts* and *Diameter* component planes). Ultimately, the *U-matrix* showed clustering of pipes (i.e., lighter coloured areas of the *U-matrix* component plane) with similar diameter sizes. Three clusters were

distinguished: (i) 3-inch to 5-inch pipes, (ii) 11-inch to 14-inch pipes, and (iii) 16-inch to 18-inch pipes. This likely denoted some similarities in the characteristics or burst behaviour of different pipe diameter groups.

The main finding from Run-A was the detection of the highest occurrence of pipe bursts during the months that also formed the dry season (December to May) while lower burst frequencies occurred in the wet season (June to November) and throughout the year. This temporal trend in pipe bursts appeared to align with the two distinct rain seasons in Barbados. To further investigate this finding, another run was carried out using total monthly bursts and diameter (same as in Run-A) as the quantitative variables driving the SOM model, while the type of rain season was used as a category to label the model. The use of variables as category labels in SOMs was explained in Section 5.4.1. The results were shown as Run-B in Figure 6.2.

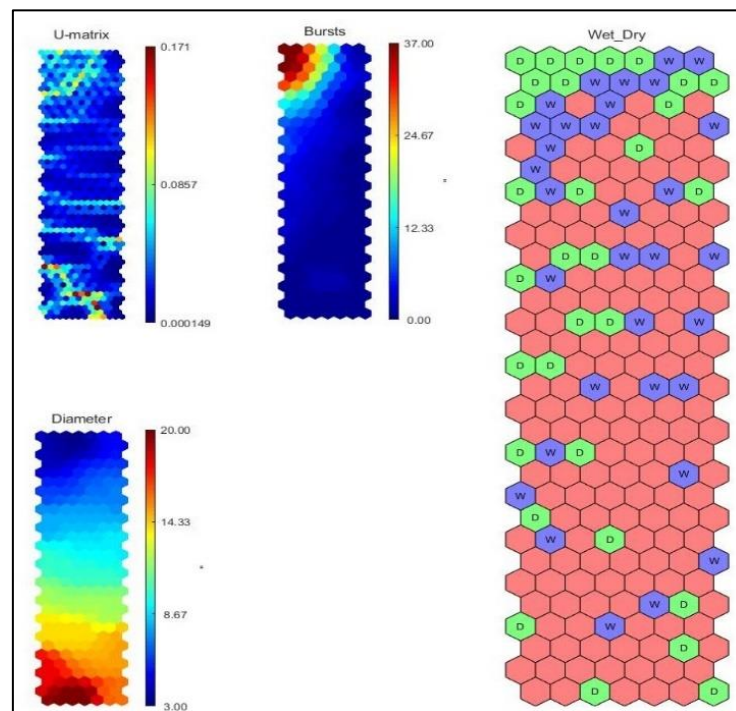


Figure 6.2 - SOMs result for seasonal temporal analysis (Run-B) using Simple Dataset.

For Run-B, visual inspection of the *Bursts* and *Diameter* component planes showed that peak bursts in small diameter pipes, particularly 3-inch and 4-inch, were distinctly associated with the dry season in the *Rain Season* labelled plane. Whereas, in the wet season it was prevalent that pipes with diameters

approximately between 4-inches and 6-inches exhibited a relatively medium burst rate that ranged from 13 to 25 bursts per month. The findings of Run-B were similar to that of Run-A. In the labelled plane, an empty cell implied that a winner could not be determined for the type of rain season associated with low pipe bursts. Overall, the distribution of the *U-matrix* of Run-B highlighted the following clusters in the dataset: (i) a cluster of small diameter pipes that had high and medium burst frequencies; and (ii) small clusters of larger diameter pipes that exhibited low burst frequencies.

The identification of a temporal trend in the occurrence of pipe bursts was the first objective of the analysis (i.e., Step 1a). This trend of frequent bursts being more prevalent in the dry season also aligned with the preliminary descriptive statistical analysis of the Simple Dataset (see Section 4.3.1). Having satisfied the first objective, the second objective (i.e., Step 1b) was to ascertain and assess other independent variables contributing to pipe bursts that also exhibited a similar trend to that which was discovered. This prompted the inclusion of amount of rainfall as a numerical input variable for analysis since its annual distribution defined the wet and dry seasons that the trend in peak pipe bursts closely followed. Though the previous analysis used type of rain season as a categorical label only, the inclusion of amount of rainfall as a numerical variable explicitly contributed in driving the SOM model. Thus, the input variables used by the SOM analysis to fulfil the second objective were total monthly bursts, diameter, and average monthly rainfall. The output of this analysis, Run-C, was shown in Figure 6.3.

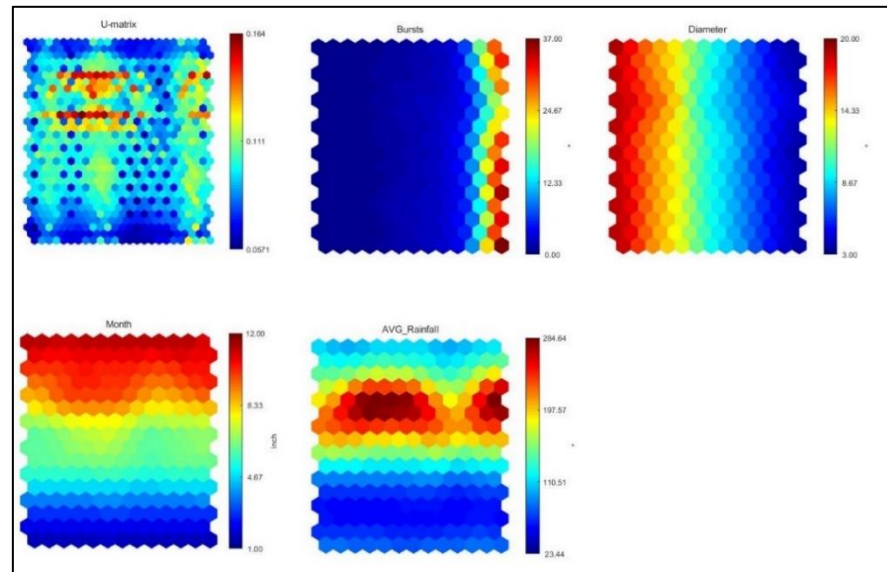


Figure 6.3 - SOMs results for average monthly rainfall analysis (Run-C) using Simple Dataset.

For Run-C, examination of the *Average Rainfall*, *Bursts*, and *Month* component planes showed that peak pipe bursts occurred between January and March when rainfall was low; that is, less than 90mm (see scale of *Average Rainfall* in Figure 6.3). Other high frequencies of pipe bursts occurred mostly during the dry season coinciding with low and medium rainfall. Conversely, in instances of high rainfall (i.e., over 197mm), pipe bursts were of low and medium frequencies. Regarding pipe diameter, high and medium pipe burst frequencies were found in small diameter pipes which was consistent with the findings of Run-A and Run-B. However, over the range of nearly all diameter sizes in the pipe network, bursts were low while the amount of rainfall varied. This finding implied that there were other factors besides rainfall that influenced pipe bursts. The *U-matrix* of Run-C showed two distinct clusters that closely resembled the instances of high rainfall in the *Average Rainfall* component plane.

To validate the inclusion of rainfall as a significant independent variable, another analysis was carried out with total monthly bursts, diameter, average monthly rainfall, and year as the numerical input variables driving the SOMs. The output of this analysis, Run-D, was shown in Figure 6.4. Inspection of the component planes showed two clusters of peak pipe bursts (see *Bursts* component plane) that aligned with the years 2002 to 2003, and 2008 to 2009.

Notably, the years 2003 and 2009 were predominantly dry and drought years, respectively (see Section 4.4.5). Overall, the *U-Matrix* showed a few clusters that coincided with clusters of high rainfall.

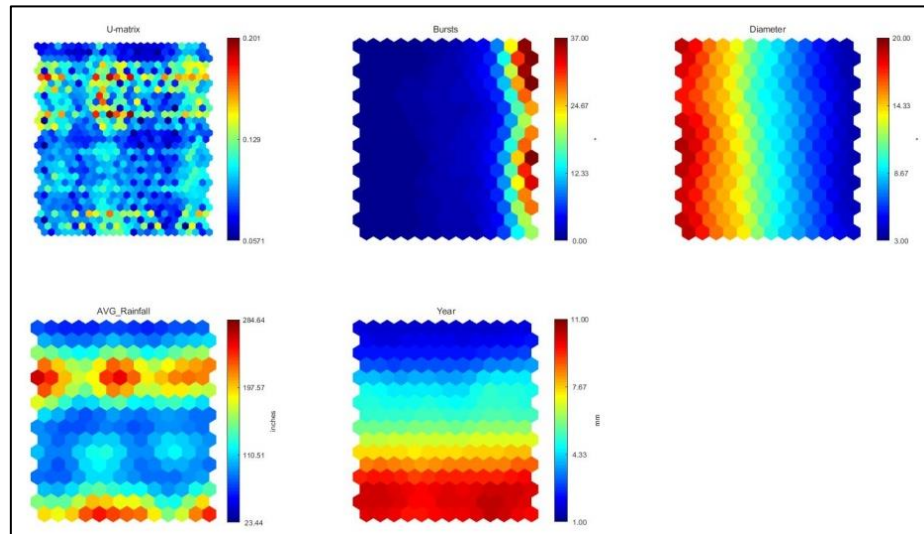


Figure 6.4 - SOMs results for average annual rainfall analysis (Run-D) using Simple Dataset.

The third and final objective of this analysis (i.e., Step 1c) was to investigate the effect of various statistics of the independent variable of interest, rainfall, on pipe bursts. Previous analyses (see Run-C and Run-D) used average as the descriptor of rainfall. Another useful descriptor of rainfall was the measurement of standard deviation since it represented the degree of change from the average. This descriptor was considered relevant since changes in rainfall could promote soil movement which was recognised as a significant factor of pipe failure in the literature (see Section 2.3). Therefore, previous analyses were repeated using the standard deviation in average monthly rainfall instead of average monthly rainfall. However, there was no significant difference in the outputs from the SOM analyses compared to using the average. The only notable observation was that the smallest and largest standard deviations occurred during the dry and wet seasons, respectively. Thus, average rainfall was chosen as the statistic for rainfall in the remainder of the data mining analysis. Key findings from the SOMs analyses conducted with the Simple Dataset were summarised in Table 6.1.

Table 6.1 - Summary of key findings from SOMs analysis using Simple Dataset.

SOM Run	Risk to Pipe Burst Occurrences		
	Low (0 – 12 monthly bursts)	Medium (13 – 25 monthly bursts)	High (>25 monthly bursts)
Run-A	Found mostly in medium and large diameter pipes throughout the year	Most pronounced between July and November in 3” to 6” pipes	Peak bursts between December and June in 3” and 4” pipes
Run-B	Found mostly in medium and large diameter pipes	Prevalent in pipes between 4” and 6” in the months of the wet season	- Peak bursts in 3” and 4” pipes - Peak bursts in the months of the dry season
Run-C	- Found mostly in medium and large diameter pipes - Other factors besides rainfall may be influencing bursts	Prevalent in small diameter pipes	- Peak bursts in 3” and 4” pipes - Peak bursts in driest months, January to March with rainfall less than 90mm - High pipe bursts coincided with low and medium average rainfall
Run-D	Likely different burst behaviour for pipes of varied diameters in dry season		- Peak bursts occurred in or around predominantly dry or drought years (2003 and 2009) - The type of year (i.e. dry or drought) may be considered a relevant predictor variable for pipe burst

The risk categories in Table 6.1 – low, medium, and high – were defined by dividing the scale associated with the component plane for bursts into three equal parts (see Figure 6.1). The highest burst rate values (depicted by darker red in a SOM output) were categorised as peak risk. This procedure used to categorise risk was used throughout the SOM analysis.

6.2.2 SOMs Analysis – Compound Dataset

The Compound Dataset was based on monthly reported pipe bursts from 2002 to 2012 which were recorded by 16 local districts delineated by the BWA (see Section 4.3.2). The first objective of the analysis (i.e., Step 2a) was to detect any trends in pipe burst occurrences in the spatiotemporal data of the Compound Dataset. The initial set of quantitative input variables driving the SOM model were total monthly pipe bursts in each district, month, and year, while location was used as a category to label the model. Location was an identifier for the general geographic location of each district – north, east, south, and west (see Table 4.3). The output for this initial analysis, Run-E, was shown in Figure 6.5.

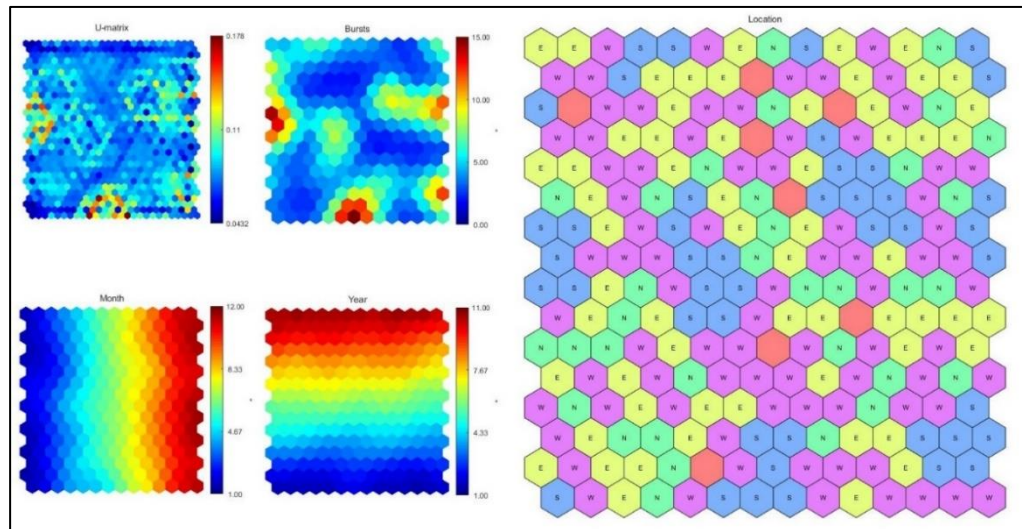


Figure 6.5 - SOMs results for initial spatial and temporal analysis (Run-E) using Compound Dataset.

From Run-E, examination of the *Bursts* and *Location* component planes showed a spatial pattern of peak pipe bursts predominantly occurring in southern districts. This pattern was also discerned from the preliminary descriptive statistical analysis of the Compound Dataset (see Section 4.3.2). These peak bursts in southern districts occurred during the dry season, particularly January and May, between 2002 to 2003 and 2007 to 2009. In other districts, pipe bursts of medium frequencies were prevalent in the west while lower frequencies were found mostly in the east and north. The *U-matrix* of Run-E showed clusters that coincided with clusters of peak and high bursts.

The second objective (i.e., Step 2b) was to assess the contribution of the significant independent variable identified from the analysis conducted with the Simple Dataset. As such, rainfall was added to the set of input variables for analysis and all data were rearranged by rain season since a temporal trend in pipe bursts was detected at that scale. The input variables were pipe bursts, rainfall, rain season, and location. Pipe bursts were totalled in each district for the dry and wet seasons over the years of available data. Rainfall was calculated as the average monthly rainfall in each rain season during that year. Rain season was included as an indicator variable to denote the dry and wet seasons using the digits 0 and 1, respectively. In this way, the use of rain season as a numeric variable, which drives the SOM, partitioned the data for a closer examination of

possible seasonal effects. The output from this SOM analysis, Run-F, was shown in Figure 6.6.

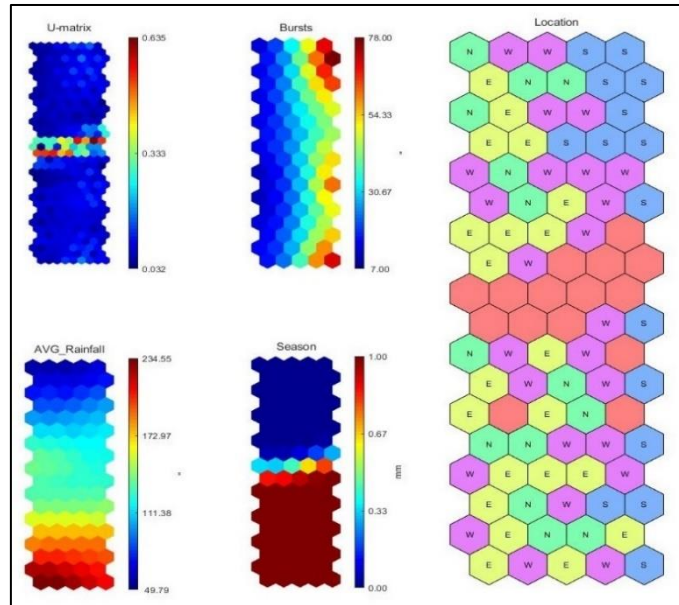


Figure 6.6 - SOMs results for rainfall analysis (Run-F) using Compound Dataset.

From Run-F, pipes with higher burst frequencies were predominantly found in southern districts. Pipes in western districts had medium and low burst frequencies. The eastern and northern districts were characterised by pipes with low burst frequencies. It was observed that more bursts occurred during the dry season than in the wet season. High bursts frequencies occurred in the southern districts during periods of low and high rainfall. This was an indication that additional factors were likely contributing to pipe bursts. Overall, the *U-Matrix* showed clustering at the rain season boundary.

For the third objective (i.e., Step 2c), other geospatial variables at the district level were added to the analysis. These additional non-numeric variables, soil type and land-use, were included as categories to label the model and not as numeric variables driving the SOM. The output of this SOM analysis, Run-G, was shown in Figure 6.7.

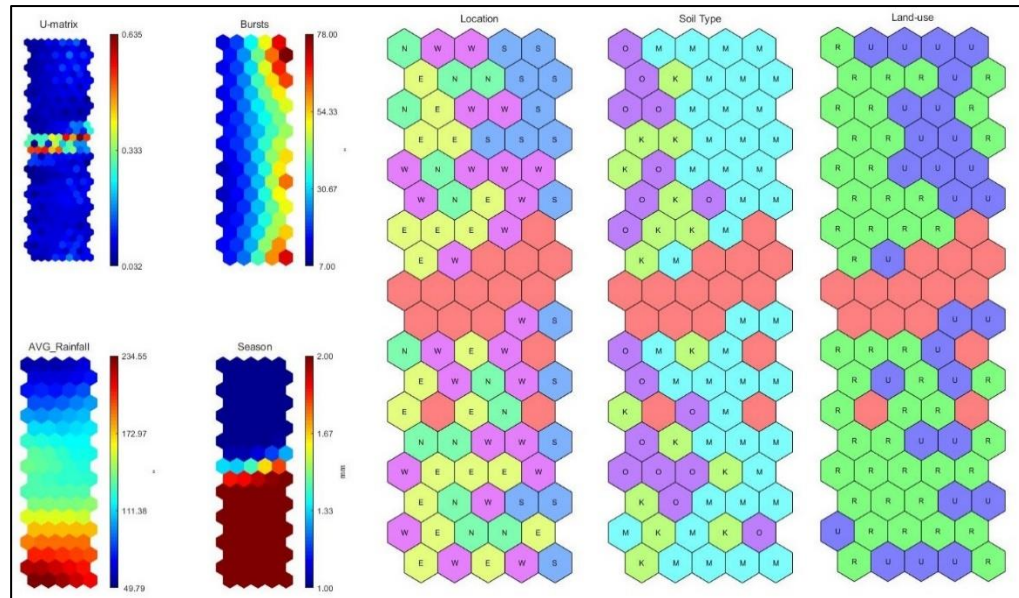


Figure 6.7 - SOM results for geospatial analysis (Run-G) using Compound Dataset.

Given that the quantitative variables in Run-F were unchanged in Run-G, the distribution of the *Bursts*, *Average Rainfall*, *Season*, and *U-Matrix* component planes remained the same (see Figure 6.6 and Figure 6.7). Inspection of the *Bursts* and *Soil Type* planes showed that high and medium pipe bursts were associated with Smectite clay, while pipes with low burst frequencies were mostly found in Kandoid clay and Scotland District clay. Regarding land-use, pipes with low burst frequencies were largely found in rural areas though some rural areas were associated with peak pipe bursts. Whereas pipes with medium burst frequencies mostly occurred in urban areas. Key findings from the SOMs analyses conducted with the Compound Dataset were summarised in Table 6.2.

Table 6.2 - Summary of key findings from SOMs analysis using Compound Dataset.

SOM Run	Risk to Pipe Burst Occurrences		
	Low (0 – 5 monthly bursts)	Medium (6 – 10 monthly bursts)	High (>11 monthly bursts)
Run-E	- Found mostly in eastern and northern districts - No discernible temporal pattern	- Prevalent in western districts - No discernible temporal pattern	- Peak and high bursts in southern districts - High bursts during dry season - Peak bursts in January and May
	Low (<30 monthly bursts)	Medium (30 – 55 monthly bursts)	High (>55 monthly bursts)
Run-F	Found mostly in eastern and northern districts, and some western districts	Prevalent in western districts	- Peak and high bursts in southern districts - More bursts occurred during the dry season than wet season

Run-G	• Low bursts found in Kandoid clay and Scotland District clay • Largely found in rural areas	• Medium bursts found in Smectite clay • Found mostly in urban areas	• High bursts found in Smectite clay • Peak bursts occurred in rural areas
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6.2.3 SOMs Analysis – Complex Dataset

The Complex Dataset was based on average annual reported pipe bursts between 2002 and 2012, which was normalised by pipe length. Pipe bursts were recorded by subdistrict and by pipe diameter size (see Section 4.3.3). The first objective (i.e., Step 3a) was to evaluate the key variables identified from previous analyses in Sections 6.2.1 and 6.2.2. Therefore, the SOMs analysis, Run-H, included average annual pipe burst rates (normalised by pipe length – see Section 4.6), diameter, average annual rainfall, location, and soil type. The output from Run-H was shown in Figure 6.8.

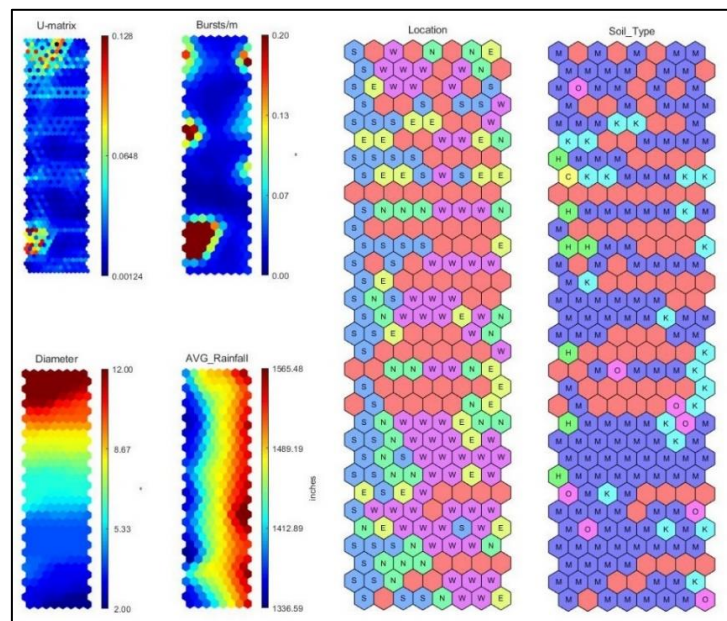


Figure 6.8 - SOMs results using key variables (Run-H) with Complex Dataset.

From Run-H, four instances of peak bursts were observed in the *Bursts* component plane. The largest of the clusters was associated with 3-inch and 4-inch diameter pipes that were found across all district locations in areas of Smectite clay and low to medium average rainfall. The two smallest clusters of peak bursts contained pipes with diameters between 10-inches and 12-inches in areas of predominantly Smectite clay. These areas were in western and southern

districts characterised by high and low average rainfall, respectively. The remaining cluster of peak bursts were associated with 6-inch and 7-inch pipes in southern districts with Smectite clay, shallow soils, and low rainfall. Across the entire dataset, the *U-matrix* showed two clusters of pipes – one of which was associated with the largest cluster of peak bursts.

In Run-H, the effect of rain season was not captured with the average annual rainfall. Hence, additional analyses were carried out using the average monthly rainfall in the dry and wet seasons over the period that bursts were recorded. The analysis that examined rainfall in the dry season, Run-I, showed that peak bursts were prevalent in small and medium diameter pipes in Smectite clay when rainfall was less than 75mm. For the analysis that included rainfall in the wet season, Run-J, peak bursts in small, medium, and large diameters occurred in Smectite and Kandoid clays, Smectite and Shallow Clays, and Smectite clay, respectively, with rainfall greater than 160mm.

The second objective (i.e., Step 3b) was to include the remaining explanatory variables in the analysis to detect additional geospatial trends. Again, analyses were arranged to examine the effects of the dry and wet seasons. In analysing the effects of the dry season, three model runs were conducted – each building on the previous. The first model, Run-K, included the same type of input variables in Run-I, however, land-use was added as a category label for the model as the remaining environmental variable. In the same way, the second model Run-L had all input variables of Run-K but also included the remaining physical variables – pipe material and installation era – as category labels for the model. Run-K and Run-L, which had the same quantitative variables driving the SOMs models, showed three clusters of peak bursts. Two of the clusters, one with small diameters and the other with medium diameters, were found in areas of low rainfall and Smectite clay. The third cluster of pipes with peak bursts were small diameter pipes found in areas with Kandoid, Scotland District and Smectite clays that experienced the highest rainfall in the dry season. The predominant pipe material across all three clusters was iron, whereas a predominant category of installation era and land-use could not be determined.

The third model, Run-M, included all input variables of Run-L with the addition of pressure and connection density, the remaining numeric operational variables. The SOMs output for Run-M was shown in Figure 6.9.

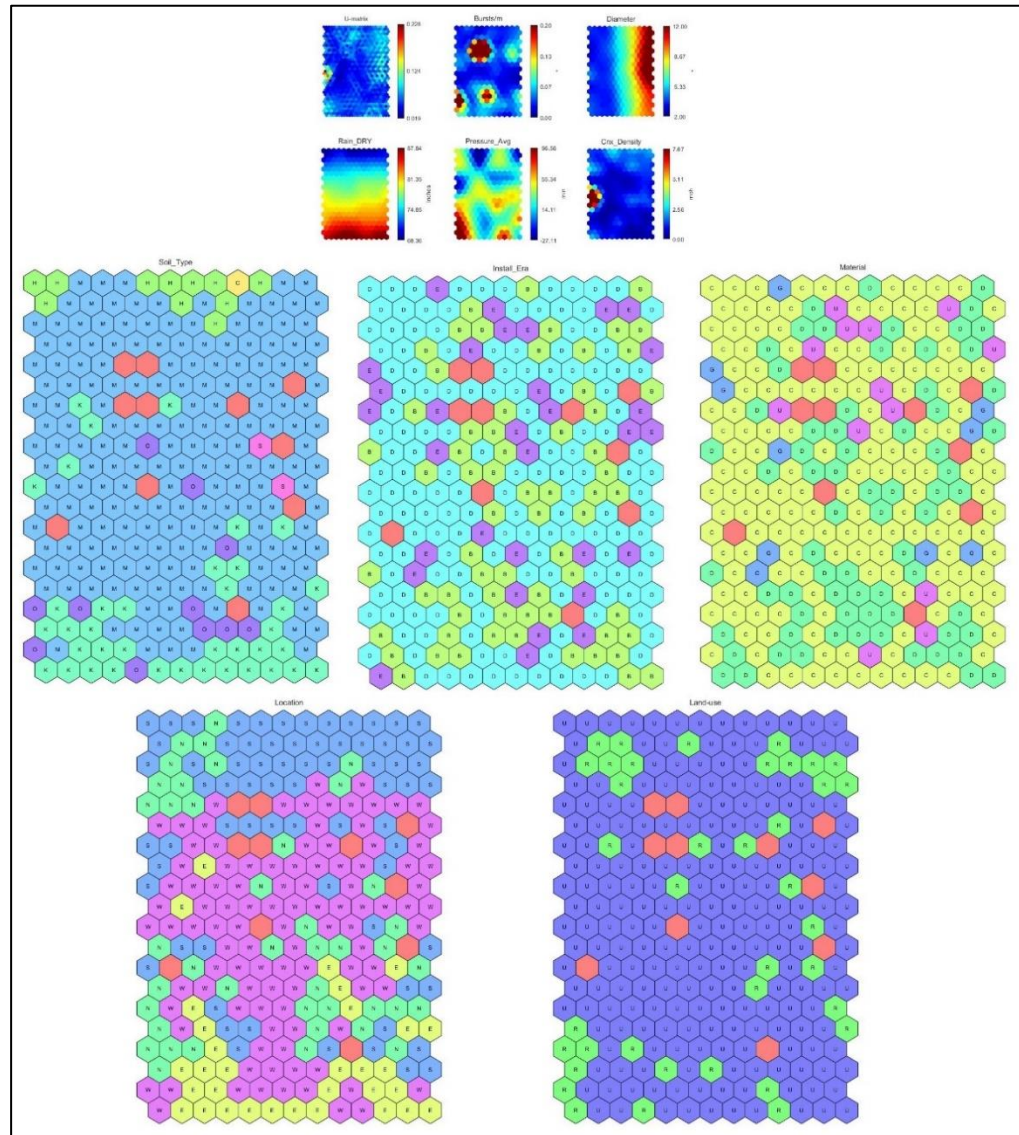


Figure 6.9 - SOM results for geospatial analysis (Run-M) in dry season using Complex Dataset.

Examination of the *Bursts* component plane in Run-M showed three clusters of peak bursts. The largest cluster consisted of pipes between 3-inches and 6-inches found mostly in urban areas in the southern districts. These pipes were mostly made up of iron material which were installed during various eras. The areas were characterised by low rainfall, Smectite clay, and low to average system pressures. For the other two smaller clusters of peak pipe bursts, one

cluster was evidently driven by peak pressures while the other appeared to be driven by a dynamic pattern between reduced rainfall and Smectite clay. Across the entire dataset, the *U-matrix* showed one dominant large cluster of pipes and a semblance of another clustered structure.

The effects of the wet season were similarly analysed as for the dry season. The first model, Run-N, was built on Run-J by including land-use (i.e., an environmental variable) as a category label for the model. The second model, Run-O, incorporated all input variables of Run-N and included pipe material and installation era (i.e., physical variables) as category labels. In Run-N and Run-O, which had the same quantitative variables driving the SOMs models, three clusters (two large and one small) of peak bursts were observed. The material of the pipes in all clusters was predominantly iron. Two of the clusters were primarily associated with southern districts while the other was linked to the remaining districts. No other discernible patterns were detected. For the third and final model, Run-P, pressure and connection density (i.e., operational variables) were included as numeric variables that contributed to driving the SOMs. The model outputs of Run-P were shown in Figure 6.10.

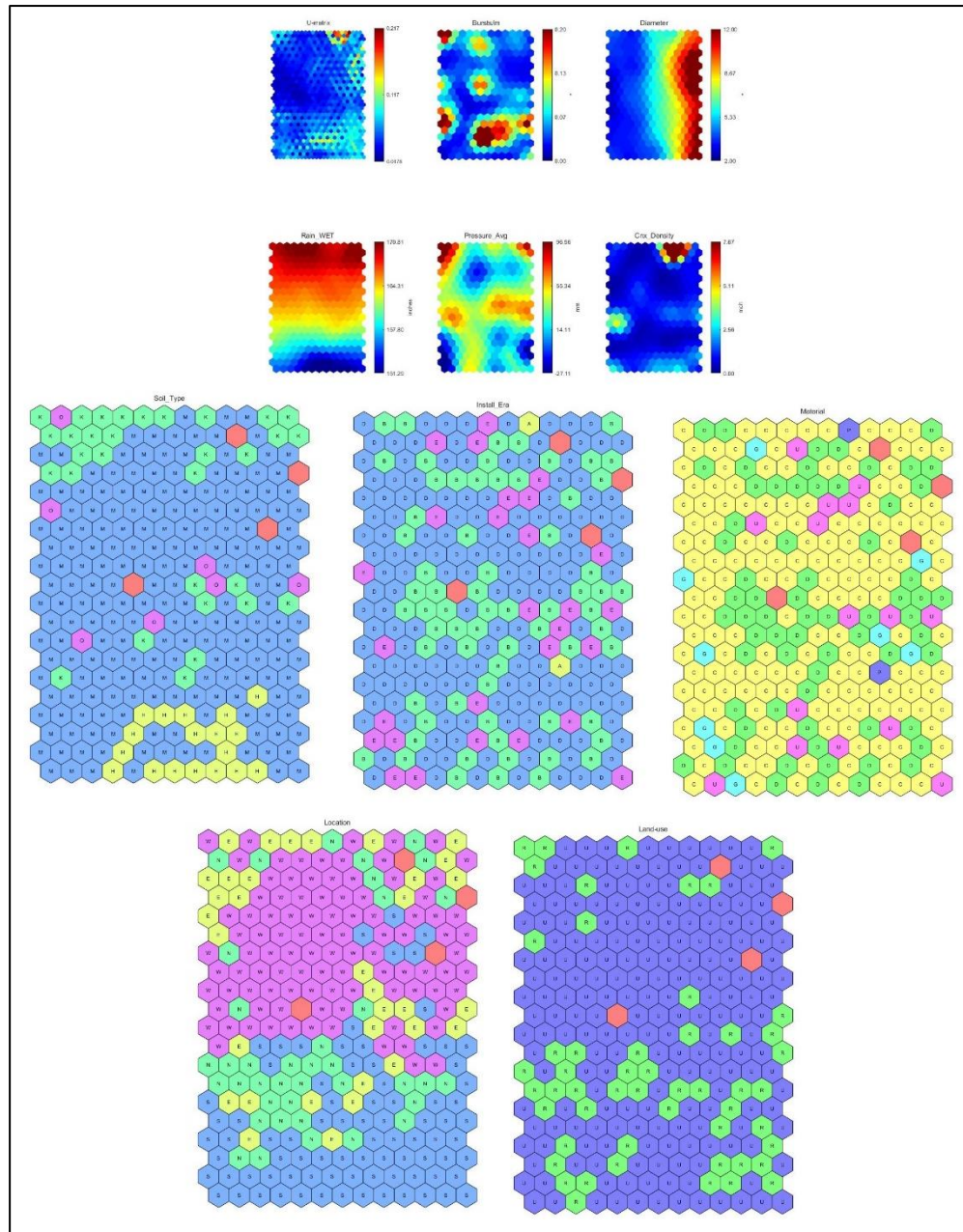


Figure 6.10 - SOMs results for geospatial analysis (Run-P) in wet season using Complex Dataset.

In Run-P, four clusters with peak pipe bursts were observed in the *Bursts* plane. A separate, tiny cluster of small diameter pipes in a rural area had peak bursts that were driven by high pressures. The other set of peak burst clusters formed a near horizontal band above the base of the *Bursts* component plane. These pipes were predominantly of iron material and laid within Smectite and Shallow clays in areas that received an average of 160 mm of rainfall in the wet season. Most of the pipes found in this set of peak clusters were laid pre-1970 in

southern and northern districts. Across the dataset, the *U-matrix* plane showed a distinct cluster which appeared to be associated with pipes that had a high connection density. Some other structures of weaker clustering were also apparent.

The third objective (i.e., Step 3c) was to assess the relative risk factor of qualitative variables by using numerical proxy values that would assist in driving the analysis instead of categorical labels that had no effect on the SOM. In the next SOM analysis, Run-Q, pipe material, installation era, and soil type were included in the analysis using numerical proxy values for risk to contribute bursts (see Section 4.5). Although land-use was considered as a possible explanatory variable, it was not coded as a numerical proxy for risk since it only had two parameters (see Section 4.5). Location functioned as an indicator for geographic location and data-partitioning; it was not considered an explanatory variable. The output for Run-Q was shown in Figure 6.11.

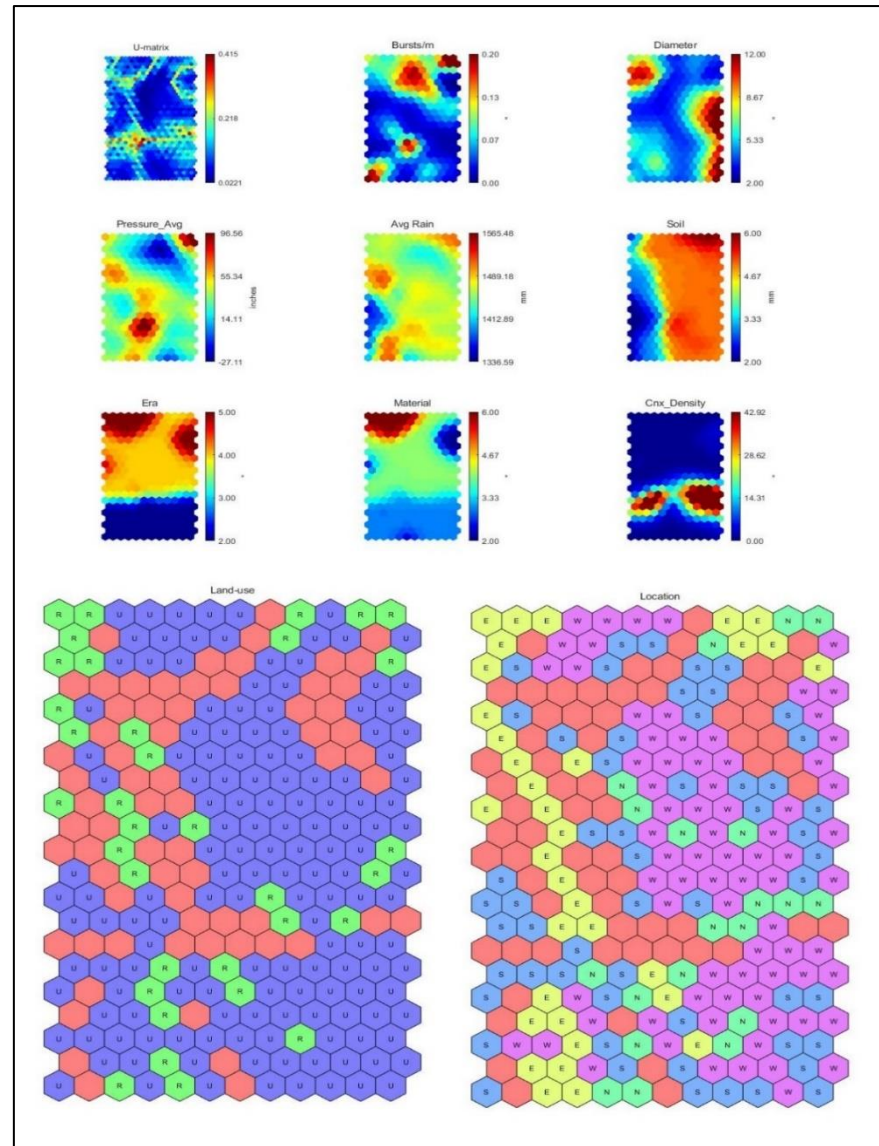


Figure 6.11 - SOM results for with proxy variables for risk (Run-Q) using Complex Dataset.

In Run-Q, four distinct clusters of peak pipe bursts were seen in the *Bursts* component plane. Two of the smaller clusters were clearly driven by high pressures above 115 PSI. Though the clusters differed in diameter, material, installation era, and soil type, pressure was peak. The other smaller cluster of peak bursts consisted of 5-inch and 6-inch ductile iron pipes (installed between 1970 and 1996) found in areas with variable rainfall and Shallow Soil. The largest cluster of pipe bursts was predominantly found in urban areas. The characteristics of those pipes were aged, small diameter, cast-iron pipes found in Smectoid clay with lower rainfall. The *U-matrix* showed several clusters

across the dataset. Key findings from the SOMs analyses conducted with the Complex Dataset were summarised in Table 6.3.

Table 6.3 – Summary of key findings from SOMs analysis using Complex Dataset.

SOM Run	Risk to Pipe Burst Occurrences		
	Low (0 – 0.07 bursts/m)	Medium (0.07 – 0.14 bursts/m)	High (0.14 – 0.20 bursts/m)
Run-H	-	-	• 3-inch and 4-inch pipes in Smectite clay in low and medium rainfall • 10-inch to 12-inch pipes in areas of Smectite clay, with low rainfall in southern districts or high rainfall in western districts
Run-I	-	-	• Small and medium diameter pipes in Smectite clay with less than 75mm rainfall
Run-J	-	-	• Small diameter pipes in Smectite and Kandoid clays with rainfall greater than 160mm • Medium diameter pipes in Smectite and Shallow clays with rainfall greater than 160mm • Large diameter pipes in Smectite clay with rainfall greater than 160mm
Run-K and Run-L	-	-	• Small diameter pipes of iron material in Smectite clay and low rainfall • Medium diameter pipes of iron material in Smectite clay and low rainfall • Small diameter pipes across the 3 clay soil types in low rainfall
Run-M	-	-	• Iron pipes between 3-inches and 6-inches, in Smectite clay, low rainfall, low pressure, and mostly in urban and southern districts • Small diameter iron pipes under peak pressure
Run-N and Run-O	-	-	• Iron pipes in southern districts
Run-P	-	-	• Small diameter pipes, in rural areas under high pressures • Iron pipes in areas of Smectite and Shallow clays that receive about 160 mm of rainfall in the wet season
Run-Q	-	-	• Peak pressures (above 115 psi) promote peak bursts despite other factors • 5-inch and 6-inch ductile iron pipes in Shallow soil and variable rainfall

			• Old, small diameter, cast iron pipes in Smectite clay and low rainfall
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6.3 Data Mining with EPR

Data mining through pipe failure modelling using SOMs analysis produced 17 models (i.e., from Run-A to Run-Q). Clustering patterns detected from the models revealed spatiotemporal trends and relationships between pipe bursts and explanatory factors. These findings provided domain knowledge, which was the goal of applying this method. Having gained insight into factors which appear to have an influence on the occurrence of pipe bursts, the stepwise process for data mining (see Section 5.3) was repeated using EPR. EPR is a method that integrated genetic algorithms with regression to derive numerical models (see Section 2.4.2). The results from the EPR analysis are presented in the following sections below. The mathematical expressions derived from the EPR analysis would be used in an attempt to substantiate the findings from the SOMs analysis and statistically quantify the strength and response of relationships among variables.

6.3.1 EPR Analysis – Simple Dataset

The first objective (i.e., Step 1a) of using EPR analysis was to generate mathematical expressions in order to detect trends and assess relationships among variables contained in the Simple Dataset. For the initial EPR analysis, monthly number of pipe bursts was the dependent variable (i.e., regression target) while diameter and month were included as independent (i.e., predictor) variables. The set of expressions generated, Set-A, was shown below in Figure 6.12.

EPR-MOGA Symbolic Expressions	COD (R ²)
(1) $Bu = +286.6115 \frac{1}{Di^2}$	0.485
(2) $Bu = +44.3909 \frac{1}{Di^2 Mo^{0.5}} + 265.8543 \frac{1}{Di^2}$	0.490
* Bu - Pipe bursts (monthly) Di - Pipe diameter (in inches) Mo - Month (with Dec = 1, Jan = 2, Feb = 3, etc.)	

Figure 6.12 - EPR expressions in Set-A for monthly pipe bursts using Simple Dataset.

In Equation 1 of Set-A, pipe diameter was the sole explanatory variable. A negative correlation was observed between pipe bursts and pipe diameter. Particularly, the occurrence of pipe bursts was inversely proportional to the square of the pipe diameter size. Equation 1 had a coefficient of determination (COD) of 0.485 which implied that pipe diameter size accounted for approximately 49% of the variation in the pipe burst model. Equation 2 was similar to Equation 1 except for an additional interaction term that included month as a factor. However, there was no significant improvement in the COD. Based on the coefficients for both terms in Equation 2, more weight was placed on the contribution of pipe diameter as a standalone independent variable than on the combined effect of the variables in the interaction term. Nonetheless, the interaction term showed that smaller pipe diameter sizes and months coded with lower digits (i.e., months of the dry season) promoted an increase in pipe burst occurrences. This observation of higher pipe bursts during the months of the dry season aligned with the findings of the SOM analyses that supported a seasonal trend in pipe burst occurrences.

To further investigate the seasonal effect of rainfall on pipe bursts, another EPR analysis was carried out. For this analysis, numerical indicators that represented month in the previous analysis (i.e., Set-B) were relabelled to indicate the type of rain season. Therefore, the input variables used for this analysis were monthly pipe bursts, pipe diameter, and type of rain season (as a binary variable). This EPR run only generated 1 expression that contained pipe diameter as the sole explanatory variable. Thus, it was not possible to examine the effect of rain season with that expression. For that reason, the dataset was divided by rain season and another series of EPR analyses were carried out – each with 924

observations. The equations produced from the analyses, Set-B, were shown in Figure 6.13.

EPR-MOGA Symbolic Expressions		COD (R ²)
(1) $Bu = +305.8105 \frac{1}{Di^2}$	(Dry season)	0.511
(2) $Bu = +267.4124 \frac{1}{Di^2}$	(Wet season)	0.448
*Bu - Pipe bursts (monthly) Di - Pipe diameter		

Figure 6.13 – EPR expressions in Set-B for monthly pipe bursts by rain season using Simple Dataset.

The negative correlation between pipe bursts and pipe diameter that was observed in Set-A also remained for the expressions in Set-B. Inspection of Equation 1 and Equation 2 showed high similarities with the only difference in the coefficient values. The higher coefficient in Equation 1 implied that pipe bursts were higher in the dry season than in the wet season for all diameter sizes. The COD for Equation 1 was also higher than the COD for Equation 2. This was an indication that pipe diameter was a better predictor of the variation in pipe bursts occurrences in the dry season than in the wet season.

The marginal improvement caused by splitting the dataset according to rain season prompted a regrouping of the dataset for further analysis. Monthly pipe bursts were totalled by annual rain season in each year. This reduced the number of observations from 924 to 154. The input variables for this analysis, by rain season, consisted of total pipe bursts in a rain season and pipe diameter. The expressions derived from this EPR analysis, Set-C, was shown in Figure 6.14. While the structure of the expressions in Set-C were similar to those in Set-B, the COD increased in both expressions – though more notably for the wet season. However, this slight improvement could result from the reduction in the size of the dataset.

EPR-MOGA Symbolic Expressions	COD (R ²)
(1) Bu = +1834.8632 $\frac{1}{Di^2}$ (Dry season)	0.562
(2) Bu = +1605.399 $\frac{1}{Di^2}$ (Wet season)	0.545
*Bu - Pipe bursts (rain season) Di - Pipe diameter	

Figure 6.14 – EPR expressions in Set-C for seasonal pipe bursts using Simple Dataset.

The second objective (i.e., Step 1b) was to generate and examine expressions that included independent variables that exhibited a trend similar to that which was observed for pipe bursts occurrences. In previous analyses, the temporal trend detected in pipe bursts followed closely to the patterns of the local rain seasons. For that reason, rainfall was added in the analysis. Accordingly, an EPR analysis was carried out with the following input variables: total bursts in a rain season, pipe diameter, dry season (as a binary variable), wet season (as a binary variable), and average monthly rainfall in a rain season. The models generated from the EPR run, Set-D, were shown in Figure 6.15.

EPR-MOGA Symbolic Expressions	COD (R ²)
(1) Bu = +1720.1311 $\frac{1}{Di^2}$	0.555
(2) Bu = +1605.399 $\frac{1}{Di^2}$ + 229.4642 $\frac{Dry}{Di^2}$	0.554
(3) Bu = +1603.6039 $\frac{1}{Di^2}$ + 10.4383 $\frac{Ra^{0.5}Dry^{0.5}}{Di^2}$	0.558
(4) Bu = +12396.3219 $\frac{1}{Di^2Ra^{0.5}}$ + 417.3501 $\frac{Dry^{0.5}}{Di^2}$ + 37.8842 $\frac{Ra^{0.5}}{Di^2}$	0.564
* Bu - Pipe bursts (rain season) Di - Pipe diameter Dry - Dry season (yes = 1 and no = 0) Ra - Average monthly rainfall for a rain season	

Figure 6.15 – EPR expressions in Set-D for seasonal pipe bursts with rainfall as interest variable using Simple Dataset.

Similar to the findings of Set-C, pipe bursts were higher in the dry season and negatively correlated with pipe diameter for the expressions in Set-D. Though rainfall was added as an input variable, the COD of the expressions in Set-D were essentially the same as in Set-C. Nonetheless, the expressions in Set-D were examined to discern any relationships or key observations.

Regarding the contribution of the type of rain season, there were no instances of wet season in any of the expressions (see Figure 6.15). The interactions term containing diameter and dry season in Equation 2 and Equation 4 indicated that smaller diameter pipes experienced higher burst frequencies in the dry season. Pertaining to rainfall, some terms had rainfall directly correlating to pipe bursts but the magnitude of the coefficients was very small when compared to other terms that showed an indirect correlation between rainfall and pipe bursts. This suggested that a significant portion of the total contribution of rainfall to higher pipe burst frequencies was attributed to lower values of rainfall rather than higher values. Notably, the weaker positive correlation between pipe bursts and rainfall was also detected in the SOMs analysis for pipes that had a medium risk to bursts incidents during wetter conditions (see Table 6.1).

The third objective (i.e. Step 1c) was to investigate the effect of other statistics of rainfall on pipe bursts. In Set-D, the average was the statistic used to depict monthly rainfall in a rain season. Therefore, the EPR analyses were repeated using total and standard deviation as the descriptors of rainfall. The mathematical expressions that resulted from the analyses were compared to the outputs of Set-D. However, there was no significant difference in the structure nor performance of the models. This was a similar finding in the SOM analysis (see Section 6.2.1).

6.3.2 EPR Analysis – Compound Dataset

The first objective (i.e., Step 2a) was to detect spatiotemporal trends in pipe burst occurrences using a basic set of variables in the Compound Dataset (see Table 4.14). Therefore, the input variables used for the initial EPR analysis of the Compound Dataset were pipe bursts, month, and location (represented as binary variables). However, the analysis resulted in expressions marked by a negative COD.

In an attempt to improve the model, the dataset was grouped by rain season. The regrouping of the data was prompted by the findings of previous analyses that detected a seasonal trend in pipe burst occurrences and due to the slight improvement to other EPR expressions when data were grouped by rain season

(see Figure 6.13 and Figure 6.14). Additionally, the binary variables for location were removed and the dataset was partitioned by district. In that way, it was still possible to examine relationships of pipes that were found in the same geographic area. Thus, a set of EPR analyses were carried out for each type of district location (i.e., north, south, east, and west) using total pipe bursts in a rain season as the dependent variable, and dry season and wet season as independent variables. The results showed a slight but not significant improvement from the previous run.

The second objective (i.e., Step 2b) was to assess the effect of including rainfall to the model as the variable of interest. Hence, EPR analyses were conducted for each location with total pipe bursts in a rain season, dry season (binary variable), wet season (binary variable), and average monthly rainfall in a rain season as input variables. The expressions produced from the analyses had either near-zero or negative CODs.

For the third objective (i.e., Step 2c), other geospatial variables remaining in the dataset were added as explanatory variables. Land-use was included as a binary variable in the analysis while soil-type was represented by risk values (see Section 4.5). Thus, the input variables for this EPR analysis were total pipe bursts in a rain season, dry season (binary variable), wet season (binary variable), average monthly rainfall in a rain season, land-use, and soil type. The models produced for the general location of each district, Set-E, were shown in Figure 6.16.

District	EPR-MOGA Symbolic Expressions	COD (R^2)
North	(1) $Bu = +9.507Soil^{0.5}$	0
	(2) $Bu = +141.1034\frac{1}{Soil^2} + 0.57027Soil^2$	0.178
	(3) $Bu = +94.0051\frac{1}{Soil^2} + 0.57137Soil^2 + 4.1358\frac{Rain^{0.5}}{Soil^2}$	0.187
East	(1) $Bu = +1.6596Rain^{0.5}$	-0.047
	(2) $Bu = +1.4678Rain^{0.5} + 5.4612Dry^{1.5}$	0.054
	(3) $Bu = +109.1417\frac{1}{Soil^2} + 0.007811RainSoil^{1.5} + 0.66671DrySoil^{1.5}$	0.059
South	(1) $Bu = +7.525e - 09\frac{1}{LU^{0.5}}$	0.097
	(2) $Bu = +1.4843e - 05\frac{1}{Rain^2LU^{0.5}} + 3.5841Rain^{0.5} + 7.8042Dry^{0.5}$	0.245
West	(1) $Bu = +2.6509Soil^{1.5}$	0.422
	(2) $Bu = +0.9579Soil^2 + 0.40203Rain^{0.5}$	0.439
	(3) $Bu = +26.5547LU^{0.5} + 0.068012Rain + 6.789Dry^{0.5}$	0.438
	(4) $Bu = +16.5105LU^{1.5} + 0.28454Rain^{0.5}Soil + 6.5067Dry^{1.5}$	0.453
* Bu - Pipe bursts (rain season) So - Soil risk (lowest = 1 and highest = 2) Ra - Average monthly rainfall for a rain season Dry - Dry season (yes = 1 and no = 0) LU - Land-use (urban = 1 and rural = 0)		

Figure 6.16 - EPR expressions in Set-E for seasonal pipe bursts for each district location using Compound Dataset.

Overall, Set-E was characterised by low model performance except for expressions generated for western districts which had the highest average COD, 0.44. Nonetheless, some correlations were discerned from the models to assist in corroborating the findings from the SOMs analysis. From the expressions in Set-E, it was observed that pipe bursts were higher in the dry season, particularly for districts in the east, west, and south. Also, rainfall and soil type appeared to be significant variables across the set of expressions. In fact, in the expressions with the highest COD, higher risk values of soil types implied an increase in pipe bursts. Additionally, urbanisation appeared to be a possible factor that contributed to pipe bursts in southern and western districts.

6.3.3 EPR Analysis – Complex Dataset

The first objective (i.e., Step 3a) of analysis of the Complex Dataset was to assess the contribution to pipe bursts by key variables previously identified. However, the type of rain season was not available. Also, given that the representation of variables as binary variables produced poor model performance in previous analyses (see Section 6.3.2) then soil type and location were not included in this analysis since these variables had more than two parameters. Therefore, an EPR analysis was conducted with pipe bursts as the dependent variable and pipe diameter and average annual rainfall (by subdistrict) as the independent variables. The expressions produced from this EPR run had near-zero and negative CODs.

For the second objective (i.e., Step 3b), other physical, operational, and environmental variables were included in the analysis. While there was an improvement in model performance as the average COD of the expressions from the model run, Set-F, was 0.21, only pipe diameter was shown as a key factor. Rainfall and land-use did not have a significant contribution to the models and pressure did not appear in any expression.

The third objective (i.e., Step 3c) was to assess the risk factor of qualitative variables that were excluded from previous analyses. Hence, the risk values of soil type, pipe material, and pipe installation era (see Section 4.5) were added to the input variables used for analysis of Set-F. The expressions generated from this analysis, Set-G, had an average COD of 0.22. The low CODs of both sets of expressions may be owed to the near zero values of the dependent variable, pipe bursts, having been normalised by pipe length. Nonetheless, the expressions across Set-G showed three interaction terms. For the first term, there was a consistent inverse correlation with pipe diameter and a positive correlation with the risk value of pipe material. Pipes with smaller diameters having a material of high-risk value contribute to pipe bursts. Also, soil type and installation era were factors for the first term. In the second term, pipe count and pipe length were key factors that had a positive correlation with pipe bursts. For the third term, pipe diameter was positively correlated with pipe bursts. That

is, pipes with larger diameters promoted pipe bursts with consideration given to pipe material and soil type. The key findings from the EPR analysis conducted across the three datasets were summarised in Table 6.4.

Table 6.4 – Summary of key findings from the EPR analysis.

Key Findings from EPR Analyses	
Observed Correlations	Model Set
Diameter is negatively correlated to pipe bursts	A, B, C, F
Bursts are higher in dry season than wet season	A, B, C, D, E
Lower rainfall in dry season promotes more bursts	D
Higher rainfall in wet season promotes more bursts	D
Soils with higher risk values promotes an increase in burst frequency	E
Larger diameter pipes promote bursts with consideration of material and soil	G
Factors in Combination Terms	Model Set
Diameter and rainfall	D
Rainfall and soil type	E
Diameter, material, and soil	G

6.4 Synthesising Key Findings from SOMs and EPR Analyses

Knowledge discovery about the factors that contribute to pipe bursts in the local WDN was the main goal of the data mining analysis presented in Section 6.2. Data mining was facilitated by modelling historical pipe bursts with possible explanatory factors using a stepwise, iterative, analytical approach (see Section 5.3) in which the methods of SOMs and EPR were separately applied. Employing this approach resulted in multiple pipe bursts models (see Section 6.2). The evaluation of each model contributed to the overall understanding of trends in pipe burst occurrences and correlations among input variables. These key findings from the data mining analysis were utilised for the identification of the main drivers of pipe bursts.

Identifying factors that significantly contributed to pipe bursts formed the first part of the research aim (see Section 1.4). The procedures developed for identifying and classifying factors by importance based on the findings from the data mining analysis were described in Section 5.4.3. SOMs were the primary method used for detecting patterns. The compilation of key findings from pipe burst models derived with SOMs (see Table 6.1, Table 6.2, and Table 6.3) was

used to classify factors by importance. A factor was categorised as significant if it exhibited a positive correlation with peak pipe bursts in at least two models. Additionally, a factor was categorised as relevant if it exhibited a positive correlation with pipe bursts (less than peak risk level) in at least two models. The result of this categorisation was shown in Figure 6.17.

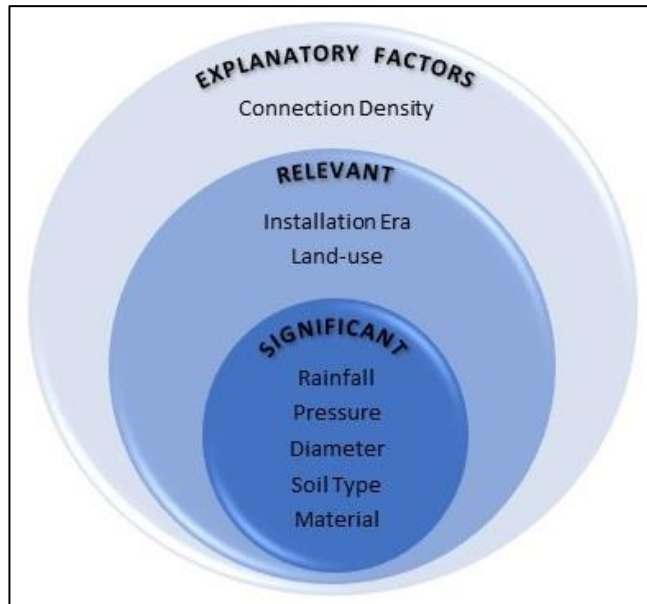


Figure 6.17 - Categorisation of explanatory factors by level of importance.

One of the purposes of the EPR analysis was used to corroborate the findings from the SOMs analysis. However, the correlations from the EPR models were generally weak for a small number of explanatory variables. This reduced confidence in models that performed poorly and thus the application of outputs was limited. Nonetheless, relationships among factors were discerned and the key findings from the EPR analysis (see Table 6.4) were compared to the categorisation of factors. EPR models with the highest performance metrics (i.e., R^2 greater than 0.50) contained one or more of the following explanatory variables: rainfall, diameter, and type of rain season. It was observed that these variables formed a subset of the significant variables identified in Figure 6.17. To an extent, this observation provided additional justification to the categorisation of factors by importance.

Besides categorising variables by importance, key findings from the SOMs analysis were compiled, compared, and synthesised in order to derive a

consensus about pipe burst behaviour across the entire network. Findings with comparable observations that contained the same variables were combined wherever bounds overlapped. From these synthesised findings, the set that pertained only to significant variables was listed below in Table 6.5.

Table 6.5 - A synthesis of pipe burst behaviour for significant variables from SOMs models.

Key Findings Synthesised from Pipe Failure Models		No. of Variables	Supporting Model Runs
Observed Correlations	Risk		
Diameter between 7" and 20"	Low	1	A, B, C, D
Diameter between 3" and 6" AND month is from July to November	Medium	2	A, B
Diameter between 5" and 6" AND month is from December to June	Medium	2	A
Diameter between 3" and 6" AND rainfall greater than 180mm	Medium	2	C
Diameter between 3" and 4" AND month is from April to June	High	2	A, B
Diameter between 3" and 5" AND rainfall less than 105mm AND Smectite clay AND iron material	High	4	C, D, G, H, K
Diameter between 3" and 4" AND month is from December to March	Peak	2	A, B
Diameter between 3" and 4" AND pressure above 85 meters of head AND material is iron	Peak	3	K, L, M
Diameter less than 10" AND monthly rainfall less than 80mm AND Smectite clay AND iron material	Peak	4	I

Despite the shortcomings in the performance of models derived using EPR, key findings from the EPR analysis (see Table 6.4) were compared and found to be consistent to the synthesised findings in Table 6.5.

6.5 Pipe Burst Risk Modelling with Fuzzy Logic

Multiple pipe failure models of the local WDN were produced from the data mining analysis using SOMs and EPR (see Sections 6.2 and 6.3). Each model contributed to the overall understanding of pipe burst occurrences, in particular: the level of importance of contributing factors (see Figure 6.17) and insight into the pipe burst behaviour of the network (see Table 6.5). These insights were used in the development of a single pipe burst risk model.

The pipe burst risk model was developed using fuzzy logic due to the shortcomings of the data mining analysis. Although patterns were detected from

the SOMs analysis, the performance of the EPR models implied that the correlations were weak. Thus, implementing a fuzzy analysis that could build on domain knowledge while accounting for the level of uncertainty in it was considered an appropriate approach given the nature of the data available. In this section, the details of the steps taken to create the pipe burst risk model using a FIS were presented.

6.5.1 Developing the Fuzzy Inference System (FIS)

Fuzzy inference is the process of assigning values of an input vector to an output vector with a set of user-defined rules. In this way, a FIS was designed to assign a set of pipe attributes and network conditions (input vector) to a relative risk score (output vector) using the knowledge derived from pipe failure data mining models (rules). Other applications of fuzzy logic in similar studies were discussed in Section 2.4.2.

The basic structure of the FIS consisted of two main components: fuzzy membership functions and fuzzy rules (see Figure 6.18). The FIS was developed using the Fuzzy Logic Toolbox in the MATLAB software from MathWorks that had built-in operators and methods for controlling the various processes applied to the components. The configuration of each fuzzy membership function and derivation of the set of fuzzy rules for the fuzzy inference process were described below. Details on the procedures used in the implementation of this method were given in Section 5.4.3.

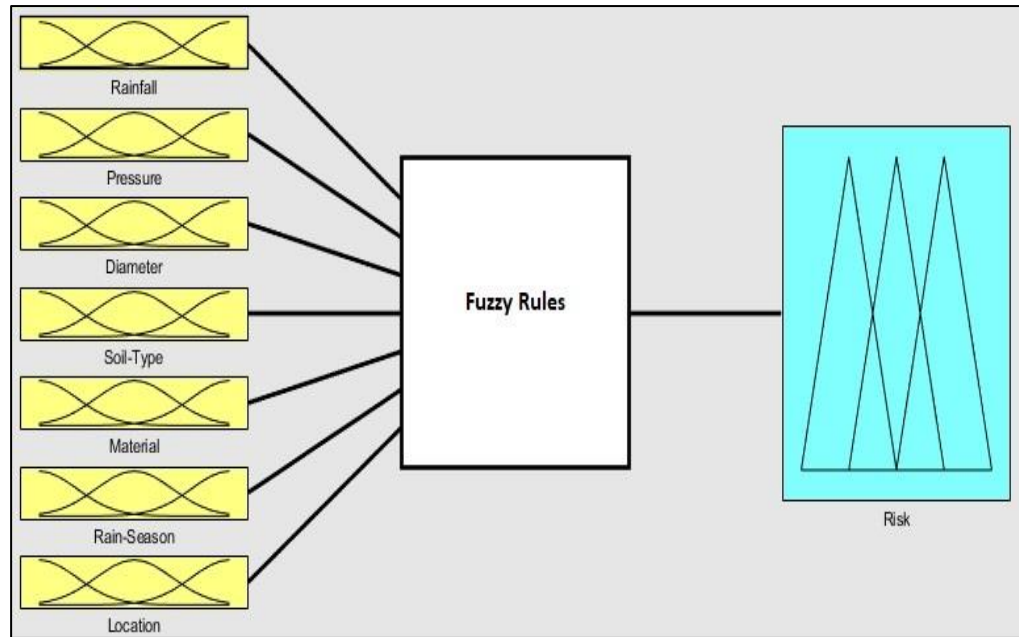


Figure 6.18 - Basic structure of the fuzzy inference system used in this study.

Fuzzy Membership Functions

For the FIS, fuzzy membership functions had to be created for all input and output variables. In developing fuzzy membership functions, the range of values (or parameters) of each variable were converted to linguistic variables. The extent to which values were associated with linguistic variables was defined by the degree of membership which used a continuous interval scale between 0 and 1. For example, 7-inch diameter pipes were defined as having a 0.68 degree of membership as ‘medium’ and 0.32 degree of membership as ‘small’ (see Figure 6.21). This meant that 7-inch diameter pipes were essentially considered as medium-sized pipes but to a lesser extent can also be thought of as small.

The fuzzy membership functions derived for each factor was guided by natural breaks in data distribution (see Figure 5.4) and the patterns discovered from data mining. The procedures for structuring fuzzy membership functions were given in Section 5.4.3. The fuzzy membership functions created for the pipe burst risk model were shown below.

Rainfall

The linguistic variables used to describe the members in the fuzzy set created for rainfall were dry, average, and wet. This fuzzy set characterised average

monthly rainfall based on the distribution and trends observed from historical rainfall between 2002 and 2012. The members of this fuzzy set were represented with a trapezoidal membership function (see Figure 6.19).

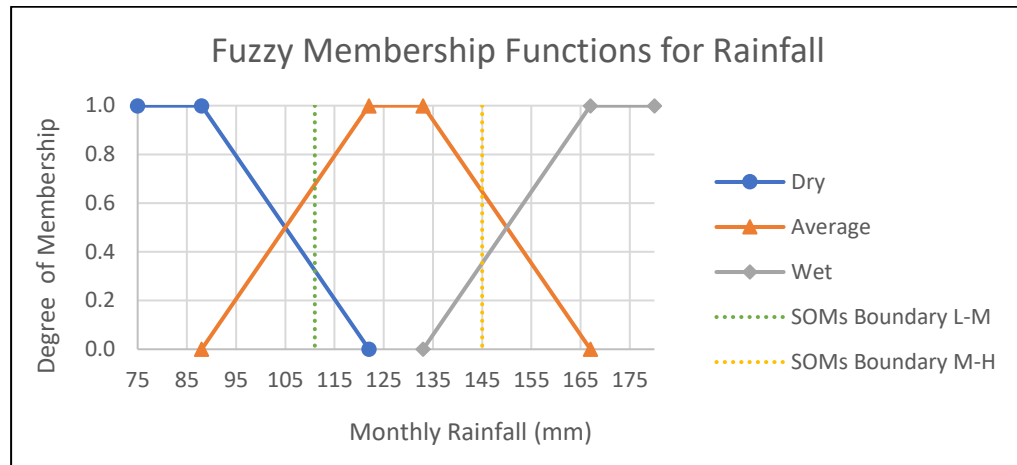


Figure 6.19 - Fuzzy membership functions for rainfall.

Pressure

The three linguistic variables used to define the members of the fuzzy set for average modelled pressure were: low, normal, and high. The trapezoidal and triangular membership functions were applied to the members in this fuzzy set. The fuzzy membership functions for pressure were shown in Figure 6.20.

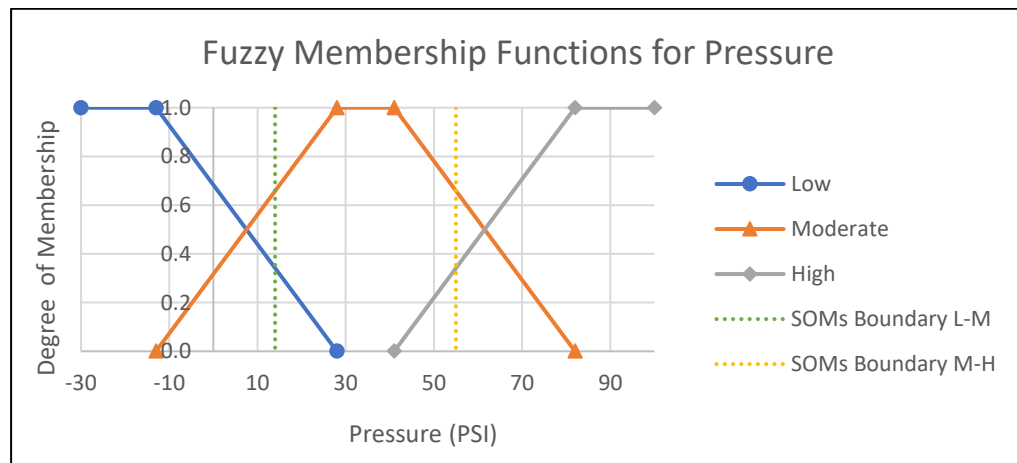


Figure 6.20 - Fuzzy membership functions for pressure.

Diameter

The linguistic variables used to characterise the members of the fuzzy set for pipe diameter were small, medium, and large. A trapezoidal membership

function was applied to the members in this fuzzy set. The fuzzy membership functions for diameter were shown in Figure 6.21.

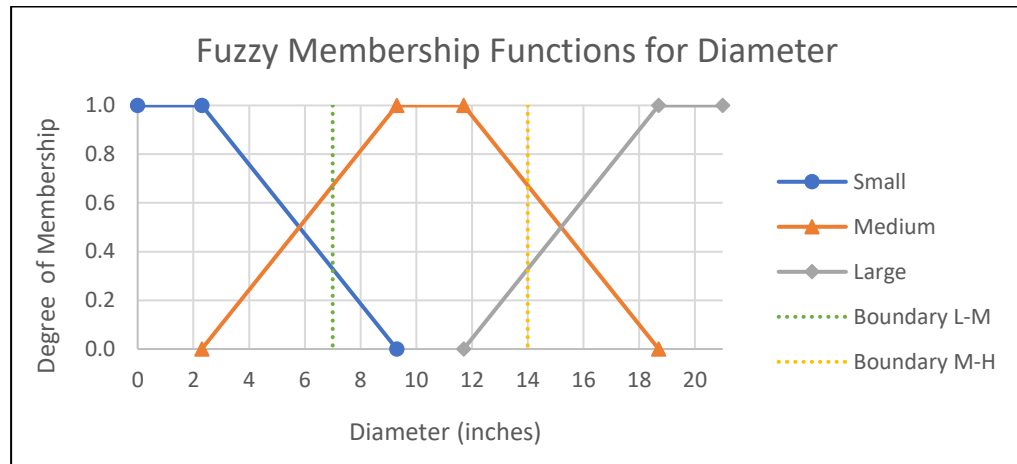


Figure 6.21 - Fuzzy membership functions for pipe diameter.

Soil Type

Two linguistic variables were used to define the members of the fuzzy set created for soil type: Smectite clay and other soil types. Given that Smectite clay was the predominant soil type that showed correlations with higher pipe burst occurrences (see Table 6.5), the variable was effectively represented as binary. A trapezoidal membership function was applied to the members in this fuzzy set (see Figure 6.22).

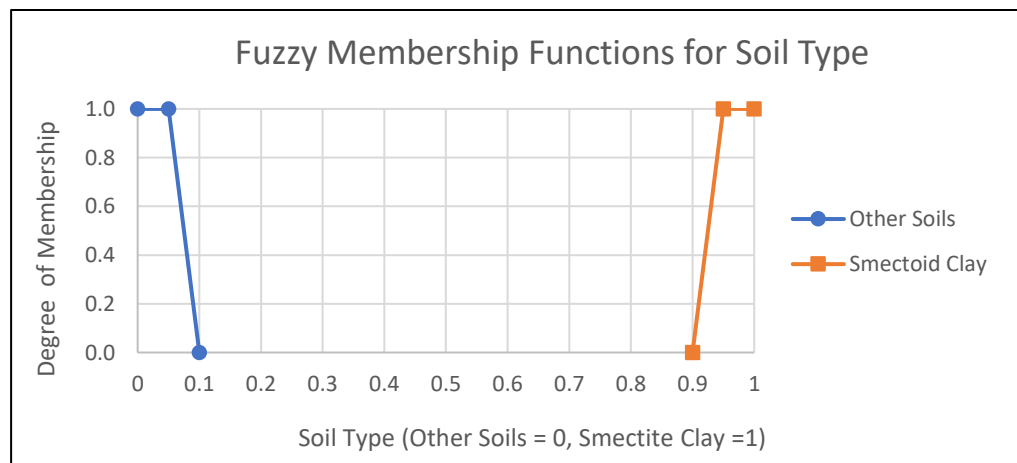


Figure 6.22 - Fuzzy membership functions for soil type.

Material

Two linguistic variables were used to characterise the fuzzy set created for pipe material: iron and non-iron. Since iron pipes exhibited a substantially higher correlation with pipe bursts than non-iron pipes (i.e., asbestos cement, HDPE, PVC, steel, and unknown material – see Table 4.6), the fuzzy set was arranged to show these categorical parameters as discrete variables. A trapezoidal membership function was applied to both members in this fuzzy set (see Figure 6.23).

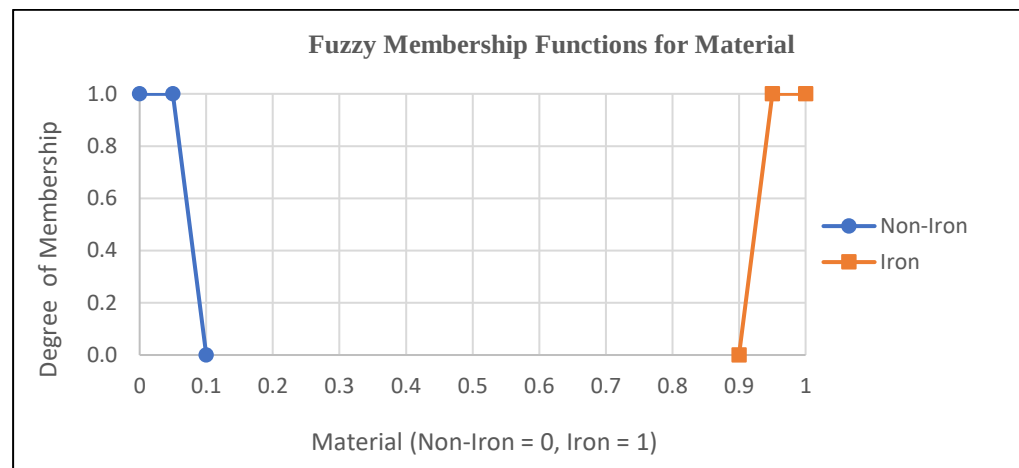


Figure 6.23 - Fuzzy membership functions for pipe material.

Rain Season (Month)

The linguistic variables used to define the members of the fuzzy set created for rain season were dry and wet although the local dry season was from December to May and the wet seasons was from June to November, the fuzzy set for rain season was created to define the degree to which each month can be considered as dry or wet. To inform how the degree of memberships were defined, historical rainfall data (see Section 4.4.5) were used to rank months from driest to wettest. The resulting fuzzy set for rain season, which used trapezoidal membership functions, was shown in Figure 6.24.

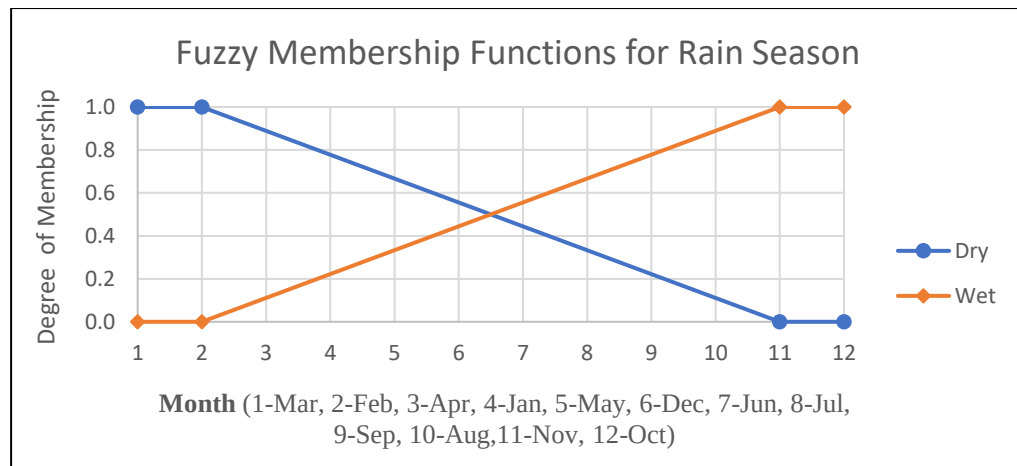


Figure 6.24 - Fuzzy membership functions for rain season.

Location

Location was variable used as an indicator for geographical location (see Section 4.6). From the key findings of the SOMs analysis, it was found that peak pipe bursts were prevalent in southern districts. Given that, location was included as an input variable in the FIS. The linguistic variables used to define the members of the fuzzy set created for location were ‘south’ and ‘other’ (i.e., all other districts) – reflecting the observations from the SOMs analysis. The use of location in this binary way also had the effect of partitioning the data. A trapezoidal membership function was applied to both members in this fuzzy set (see Figure 6.25).

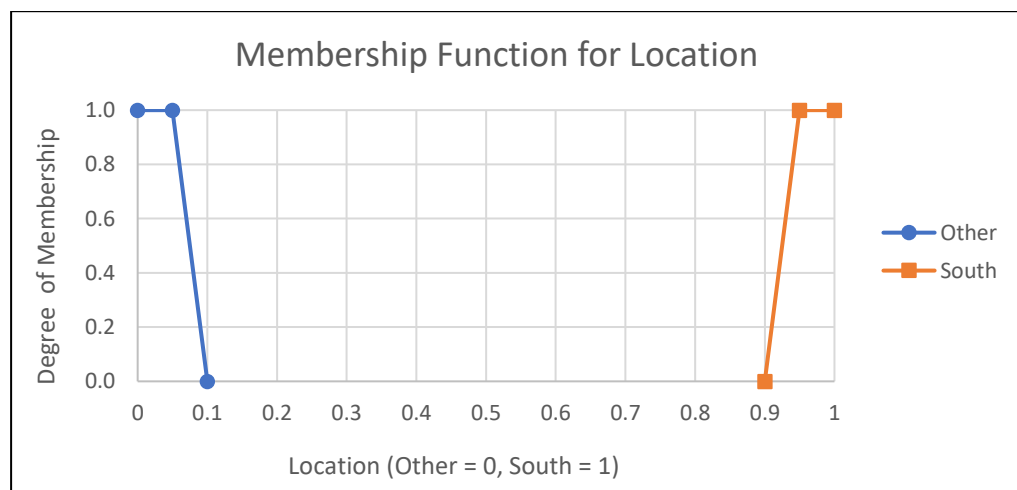


Figure 6.25 - Fuzzy membership functions for location.

Risk

Four linguistic variables were used to define risk: low, medium, high, and peak. The members in this fuzzy set matched the descriptions for contributions to pipe bursts in the SOM analyses. The trapezoidal and triangular membership functions were applied to the members in this fuzzy set (see Figure 6.26).

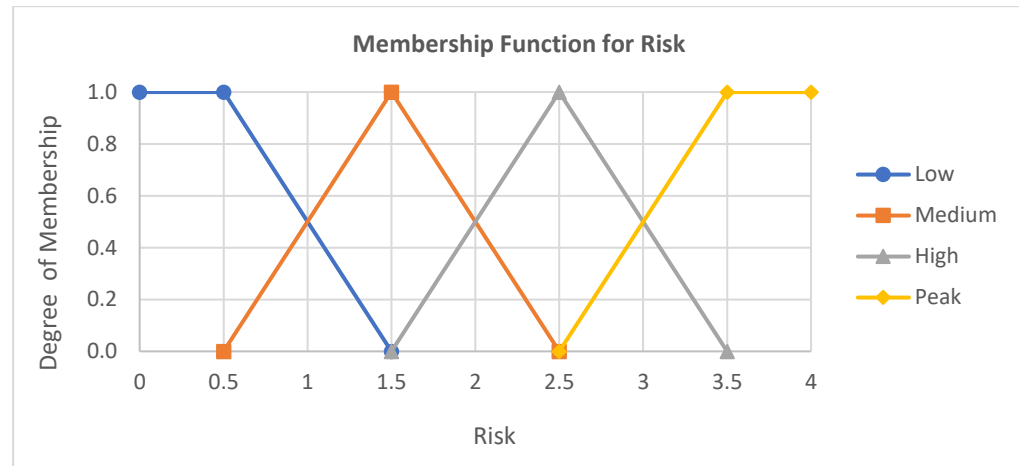


Figure 6.26 - Fuzzy membership functions for risk.

Fuzzy Rules and Fuzzy Inference Processing

Fuzzy rules are used to instruct the processing carried out in a FIS (MathWorks 2018; Salehi, Ghazizadeh, and Tabesh 2018; Ho, Lin, and Lo 2010; Dubois and Prade 1996). For this study, fuzzy rules were based on the set of synthesised key findings of pipe burst behaviour in the local WDN (see Table 6.5) and on the variants of those findings that also held true. Each rule was written as a conditional *if-then* statement that pointed to members of the fuzzy sets created for each significant variable. Compounded statements with multiple antecedents were joined with the *AND* operator (MathWorks 2018). The final compilation of fuzzy rules used in the FIS is listed below:

- If diameter is not small, then risk is low.
- If pressure is high, and diameter is small, and material is iron, then risk is peak.
- If rainfall is high, and diameter is small, and rain season is wet, then risk is low.

- If rainfall is low, and diameter is not small, and rain season is dry, then risk is low.
- If rainfall is low, and diameter is not small, and rain season is wet, then risk is low.
- If rainfall is low, and diameter is small, and soil-type is smectoid clay, and material is iron, and rain season is dry, and location south, then risk is peak.
- If rainfall is low, and diameter is small, and soil type is smectoid clay, and material is iron, and rain season is dry, then risk is high.

In the fuzzy inference process, fuzzy sets allowed for the fuzzification of input data. This fuzzy data were passed to the fuzzy rules that were given an equal weight of one and evaluated in parallel. The implication of each rule was calculated and the fuzzy outputs from all rules were aggregated into a single fuzzy set that was defuzzied to obtain a single value (i.e., a risk score). This defuzzification technique is known as the centroid calculation which gives the middle value of the area under the curve of the aggregated fuzzy outputs (MathWorks 2018). Overall, the fuzzy inference process contained 57 plots – a plot for each input and output variable (8) under each rule (7) and a single plot of the final defuzzied output (see Figure 6.27). The outputs of the pipe burst risk model were risk scores. Before using these risk scores to create a hotspot map of the network, the model was tested. The results of testing the pipe burst risk model were presented in the next section.

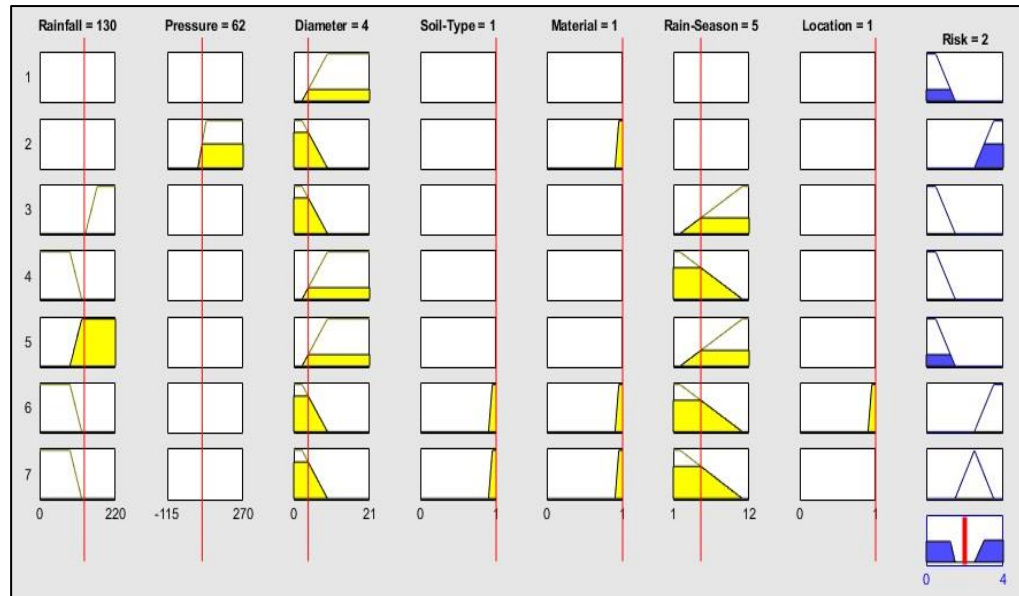


Figure 6.27 - Plots of the fuzzy inference system shown in the Rule Viewer in MATLAB.

6.5.2 Hotspot Map of the Pipe Network

The pipe burst risk model created in Section 6.5.1 was used to assign risk scores to each pipe in the local WDN. Input data for the model were extracted from the Complex Dataset (see Section 4.6) which more or less represented past network conditions. A total of 12 model runs, 1 run for every month, was performed. This resulted in a risk score for each pipe cohort (i.e., pipes grouped by subdistrict and by diameter – 2,442 groups) for each month. This allowed for a tally of the average risk scores across the year, dry season, and wet season. A hotspot map was produced to depict the average annual risk of the susceptibility of pipes to frequent bursts under historical conditions (see Figure 6.28). The procedures for classifying the risk scores and producing a hotspot map were detailed in Sections 5.4.3 and 5.5.

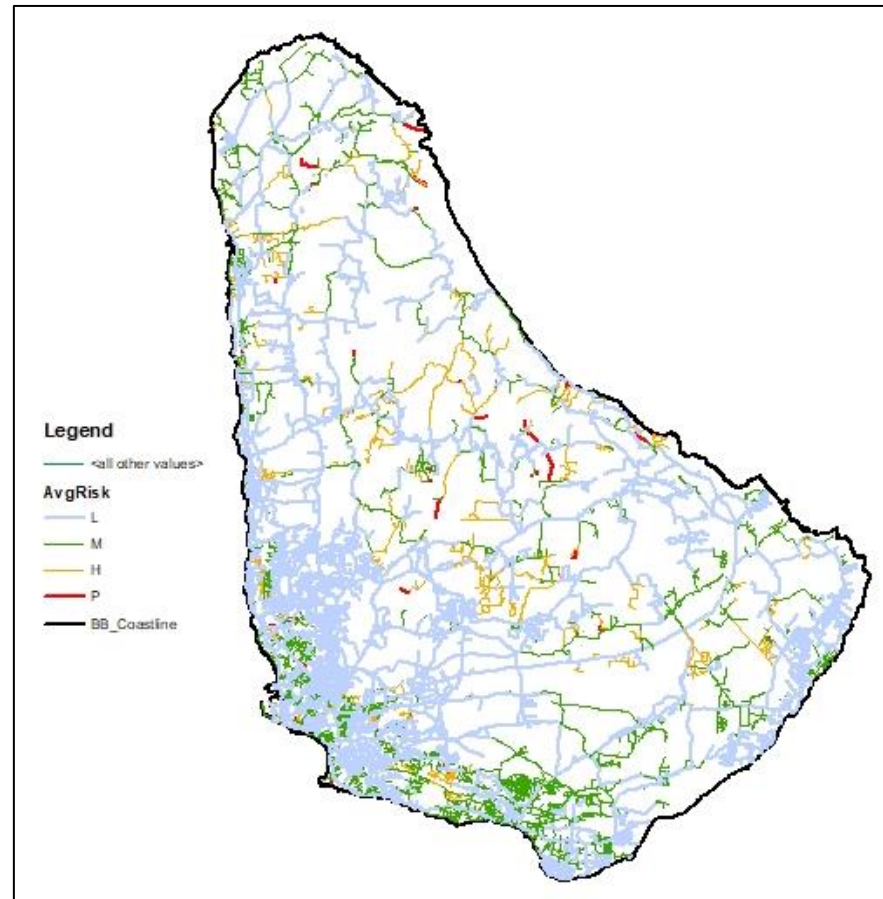


Figure 6.28 - Hotspot map of pipe network.

On the hotspot map in Figure 6.28, pipes classified as peak and high risk were seen mostly around the centre of the island (known for higher elevation) and in the north. The susceptibility of these pipes to peak bursts can be likely attributed to high pressure (see Table 6.6). Notably, these were also smaller diameter pipes that were laid in Smectite Clay. Medium risk pipes were spread across the island but seen predominantly in the south. Low risk pipes, which made up much of the pipeline (see Figure 6.29), were larger diameter pipes with lower average pressures.

Table 6.6 - Summary of pipes in hotspot map of pipe network.

Summary of Hotspot Model of the Network						
Risk Level	No. of Pipes	Rainfall ^a	Pressure ^a	Diameter ^a	Soil Type ^b	Material ^b
Low	11558	121.8	33.1	7.3	Smectite	CIP
Medium	5388	119.8	47.8	4.5	Smectite	CIP
High	1604	122.9	107.8	3.3	Smectite	CIP
Peak	67	124.2	110.5	1.6	Smectite	GI

^a Mean value^b Predominant value

A histogram of showing the distribution of risk by pipe cohort was shown in Figure 6.29). This distribution, with a larger number of low risk pipes, would allow for the identification of the most critical pipes which can assist in prioritisation pipes for replacement or rehabilitation interventions.

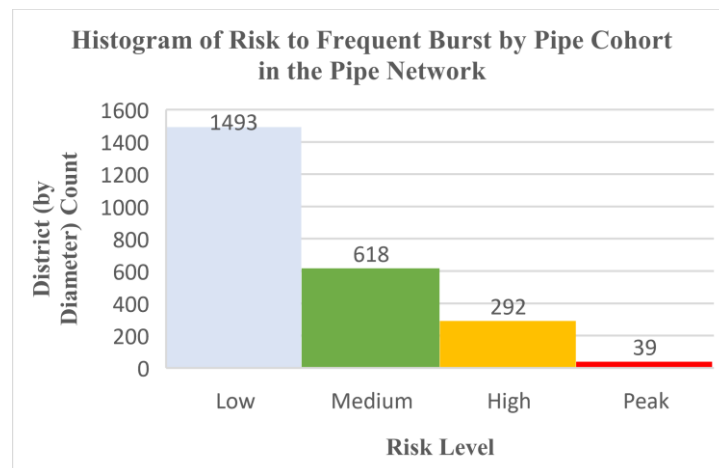


Figure 6.29 - Histogram of risk by pipe cohort for hotspot map of pipe network.

A breakdown of the distribution of risk by rain season was given in Table 6.7. While the figures were comparable, overall, the risk of pipes bursting was slightly higher in the dry season than in the wet season. It was observed that the number of pipes susceptible to peak risk was the same. This was likely because peak risk pipes were driven by high pressures which were not correlated or influence by rain season.

Table 6.7 - Distribution of risk to frequent pipe bursts by rain season for pipe network.

Distribution of Risk by Rain Season in Pipe Network		
Risk Level	Dry Season	Wet Season
	Total Number of Pipes	
Low	11,225 (60.3%)	12,150 (65.3%)
Medium	5,477 (29.4%)	5,017 (26.9%)
High	1,848 (9.9%)	1,383 (7.4%)
Peak	67 (0.4%)	67 (0.4%)

A cluster analysis of the hotspot map was performed to better identify and determine statistically significant clusters (see Figure 6.30). Using the Spatial Autocorrelation tool in ESRI's ArcMap, the Moran's I Index value was 0.193, z-score was 8.6, and p-value was less than 0.01. These values implied that the slight clustering observed in Figure 6.30 had less than 1% likelihood that the pattern could be the result of random chance, and that the clusters were statistically significant. Thus, the Cluster and Outlier Analysis tool in ESRI's ArcMap was used to model statistically significant clusters of hot spots, cold spots, and outliers (see Figure 6.30).

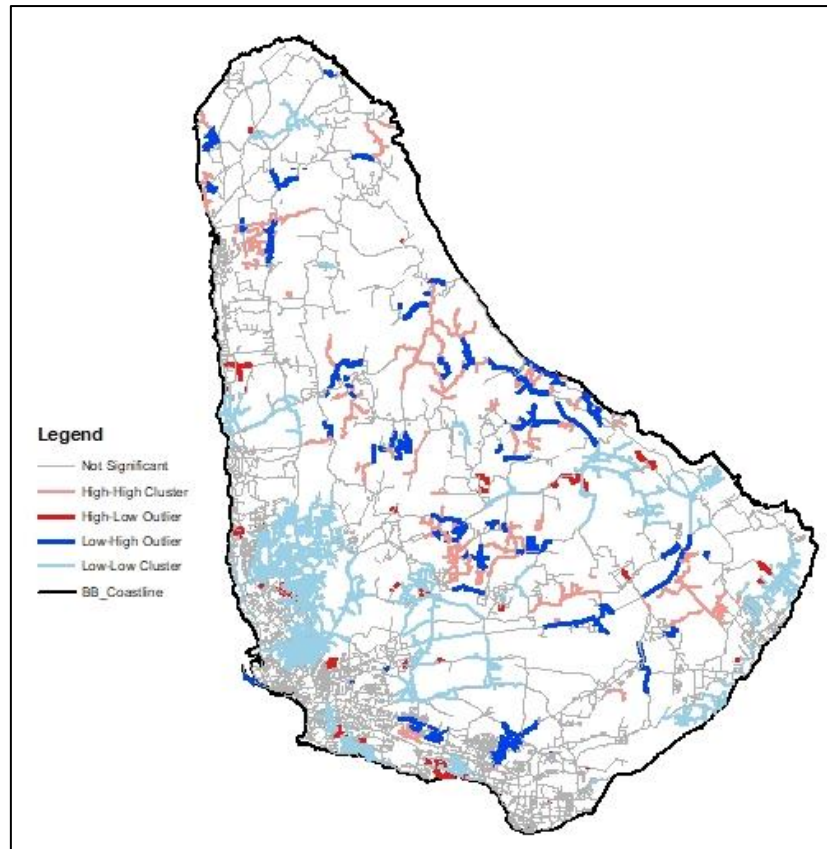


Figure 6.30 - Cluster analysis of hotspot map of pipe network.

In Figure 6.30, clusters of higher risk pipes are clearly concentrated in the central and eastern part of the island with some lower risk pipes, as outliers being found near high risk clusters, in between. Clusters of lower risk were prominent in the southwest. The distribution of pipes by cluster type was depicted in Figure 6.31.

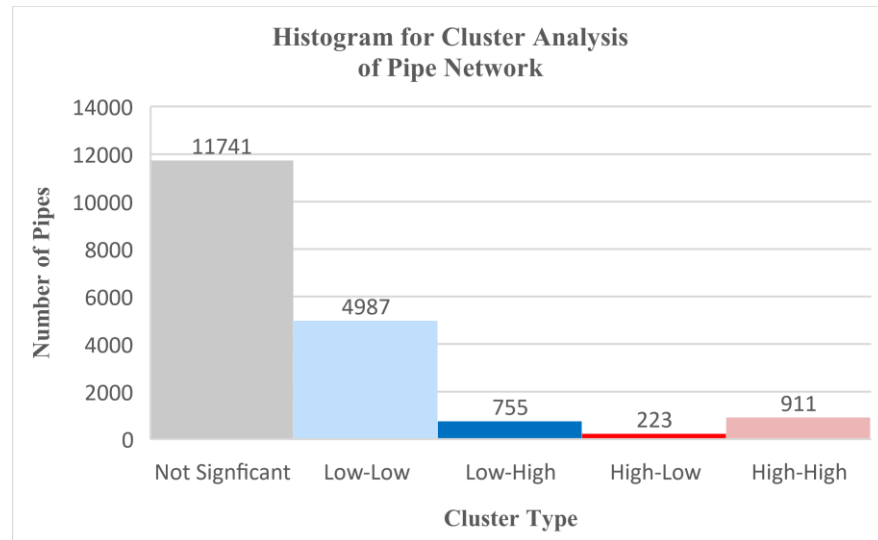


Figure 6.31 - Histogram for cluster analysis of hotspot map of pipe network.

6.5.3 Validating the Pipe Burst Risk Model

The pipe burst risk model (see Figure 6.28) was tested against historical records of service disruptions. This database of service disruptions was compiled from public service notices given by the BWA and comprised of 1,001 observations between March 2019 to August 2020 (see Section 0). Although this data likely represented only a subset of the bursts and had other limitations, no other validation data were able to be sourced as the request for pipe burst data between the period 2013 and 2020 was not met by the BWA.

From the 1,001 observations, 364 were unique. Assumptions were made to separate observations according to frequency – single bursts or multiple bursts (see Section 0). Based on this grouping, pipes from the service disruptions records (i.e., validation model) were matched to the pipe burst risk model according to pipe diameter size and subdistrict name. The results of this comparative analysis were shown below in Table 6.8.

Table 6.8 – Results from validating the pipe burst risk model.

Validating the Pipe Burst Risk Model			
Classification from Hotspot Model	Burst Frequency of Pipes in Validation Model		Total
	Single	Multiple	
Low Risk	104	76	180
Medium Risk	68	58	126
High Risk	33	25	58

From Table 6.8, it was shown that from the 180 pipes from the validation model that matched low risk pipes in the hotspot model, a majority were considered as having single burst incidents (104). Having more pipes with single bursts matching with low risk pipes in the hotspot model was an indication of relatively good model performance. Medium risk pipes in the hotspot model had a near-even matching to single (68) and multiple (58) burst pipes in the validation model. This was a fair result as medium risk can be considered as a moderate or average category. However, pipes with multiple bursts (25) in the validation model were slightly less than pipes with single bursts (33) that matched high risk pipes in the hotspot. Although this difference was small, a greater matching of high risk pipes in the hotspot model to multiple pipe bursts in the validation model would have proven better model performance. The validation model did not match any pipes that were classified as peak in the hotspot model but peak pipes were very few overall in hotspot model – less than 1% (see Table 6.6).

The distribution of risk of the validation model was also compared to the pipe burst risk model (see Table 6.9). The distributions were similar, with a descending number of pipes classified from low risk to peak risk. Notably, the validation model contained nearly double the percentage of high risk pipes (i.e., real bursts).

Table 6.9 - A comparison of the distribution of risk between the pipe burst risk model and validation model.

Comparing the Distribution of Risk		
Risk Levels	Pipe Burst Risk Model	Validation Model
Low Risk	11,558 (62.1%)	180 (49.5%)

Medium Risk	5,388 (28.9%)	126 (34.6%)
High Risk	1,604 (8.6%)	58 (15.9%)
Peak Risk	67 (0.4%)	0 (0.0%)

While this comparative analysis did provide some form of validation of the pipe burst risk model and insight into its accuracy of depicting the pipe burst behaviour of the network, there were several shortcomings in the validation data. Over the 1.5-year span, the validation data that were compiled appeared limited, containing only 1,001 observations although the General Manager of BWA had reported over 700 bursts per month in 2018 (Downes 2018). Additionally, the average number of pipe bursts per pipe in the validation data was less than two. This frequency was very low when compared to the higher rates which were used as the basis for the pipe risk hotspot model (see Table 6.1, Table 6.2, and Table 6.3). Furthermore, data in the validation model represented a subset of total pipe bursts (i.e., bursts that have been reported or suspected) and indicated where pipe repair efforts of BWA have been concentrated. Despite these shortcomings, there was still confidence in the pipe burst risk model which was based on the findings from data mining. Moreover, the clustering analysis of the hotspot map was carried out to identify statistically significant clusters in light of the deficiencies of validating the model.

6.6 Chapter Summary

In this chapter, multiple pipe bursts models which were generated from applying the iterative stepwise rules (see Section 5.3) that guided the data mining process were presented and evaluated (see Sections 6.2 and 6.3). A major trend discovered through data mining was that peak pipe bursts occurred during the dry season, and the set of significant factors that promoted such frequent bursts were rainfall, pressure, diameter, soil type, material, month (i.e., season), and location (see Section 6.4). With the knowledge and insight gained into pipe burst behaviour, a pipe burst risk model was developed to depict the dynamics of significant factors and to provide a relative risk score for the susceptibility of pipes to peak bursts (see Section 6.5). Considering the apparent role of rainfall in frequent pipe burst occurrences, a hotspot model of the pipe network under

historic conditions was created and validated. In that way, the effects of shifts in future rainfall due to climate change on pipe bursts could be assessed. This assessment on the implications of climate change on local pipe bursts is presented in the next chapter.

7 RESULTS: CLIMATE CHANGE ASSESSMENT ON PIPE BURSTS

7.1 Chapter Introduction

The aim of this chapter is to present the results of an assessment on the potential effects of future rainfall on the distribution of pipe bursts in the WDN in Barbados. This chapter includes the following:

- (1) A set of downscaled rainfall projections for Barbados that were produced using four climate change scenarios.
- (2) Statistical, seasonal, and geospatial analyses of the rainfall projections used for the assessment.
- (3) Hotspot maps for each climate change scenario which were derived from incorporating rainfall projections into the pipe burst risk model.

This chapter provides a depiction and evaluation of expected distributions of future rainfall before showing the results that changes in rainfall patterns can have on the location and likelihood of pipe burst occurrences. The implications of the results of this assessment are discussed in the next chapter.

7.2 Climate Change Projections of Rainfall for Barbados

In Chapter 6 – Results: Data Mining and Pipe Burst Modelling, rainfall was classified as a significant variable that contributed to peak pipe bursts (see Section 6.4). Rainfall was therefore used as a key input variable for the local pipe burst risk model presented in Section 6.5.2. This suggests that deviation from the historical average of rainfall which was used to configure bounds of the rainfall membership function in the model (see Section 6.5.1) might alter the allocation of risk representing the susceptibility of pipes to frequent bursts (see Section 6.5.2). Climate change is anticipated to affect the amount and distribution of rainfall. Hence, incorporating rainfall projections into the pipe burst risk model was potential means for investigating the possible medium to long-term impacts of climate change on burst hotspots.

Globally, the state of knowledge on climate change is assessed by the IPCC of the United Nations. Over the past two decades, the IPCC has presented a series of climate scenarios. The most recent set of scenarios, Representative

Concentration Pathways (RCPs), were used as the basis for climate predictions and projections in the IPCC Fifth Assessment Report. The RCP scenarios comprised of four discrete pathways (namely RCP2.6, RCP4.5, RCP6.0, and RCP8.5) that project how human activities could determine the concentration of GHGs and other atmospheric pollutants throughout the 21st century (IPCC 2014). The decimal number associated with each RCP scenario indicated the threshold of radiative forcing extending up to the year 2100. Radiative forcing is a measure of the effect of external drivers of climate change, such as GHGs, on the net radiative flux or heat trapped in the atmosphere. Scenarios were defined based on the level of radiative forcing: RCP2.6 was very low, RCP4.5 and RCP6.0 were intermediate, and RCP8.5 was very high.

For this study, the climate change scenarios RCP2.6 and RCP 8.5 were used. In this way, the two climatic extremes (i.e., the lowest and highest emission scenarios) could be compared. Rainfall projections were sourced from Charlery and Nurse (2018) who are climate modellers at the University of the West Indies in Barbados. Charlery and Nurse (2018) developed a climate model that generated projections for various meteorological variables under the RCP scenarios until 2100. From that model, the decadal averages of mean monthly rainfall were extracted for the medium and long-term horizons of 2035 and 2050, respectively (see Section 5.7).

The data received from Charlery and Nurse (2018) were projections for average monthly rainfall in the form of georeferenced data points at a downscaled resolution of 1 km (see Figure 5.7) – the highest resolution of climate change projections of rainfall for Barbados to-date. The data were prepared to represent the average annual rainfall at each point for each scenario year (see Section 5.7). Using the same techniques as in Section 4.4.5, the data points were imported into ESRI's ArcGIS ArcMap and interpolated with the co-kriging method in Geostatistical Analyst tool. This method allowed for the data points of projected rainfall and a digital elevation model to be incorporated into the interpolation. The results of the interpolation of future rainfall for each scenario (i.e., RCP2.6

in 2035, RCP2.6 in 2050, RCP 8.5 in 2035, and RCP8.5 in 2050) were shown in Figure 7.1.

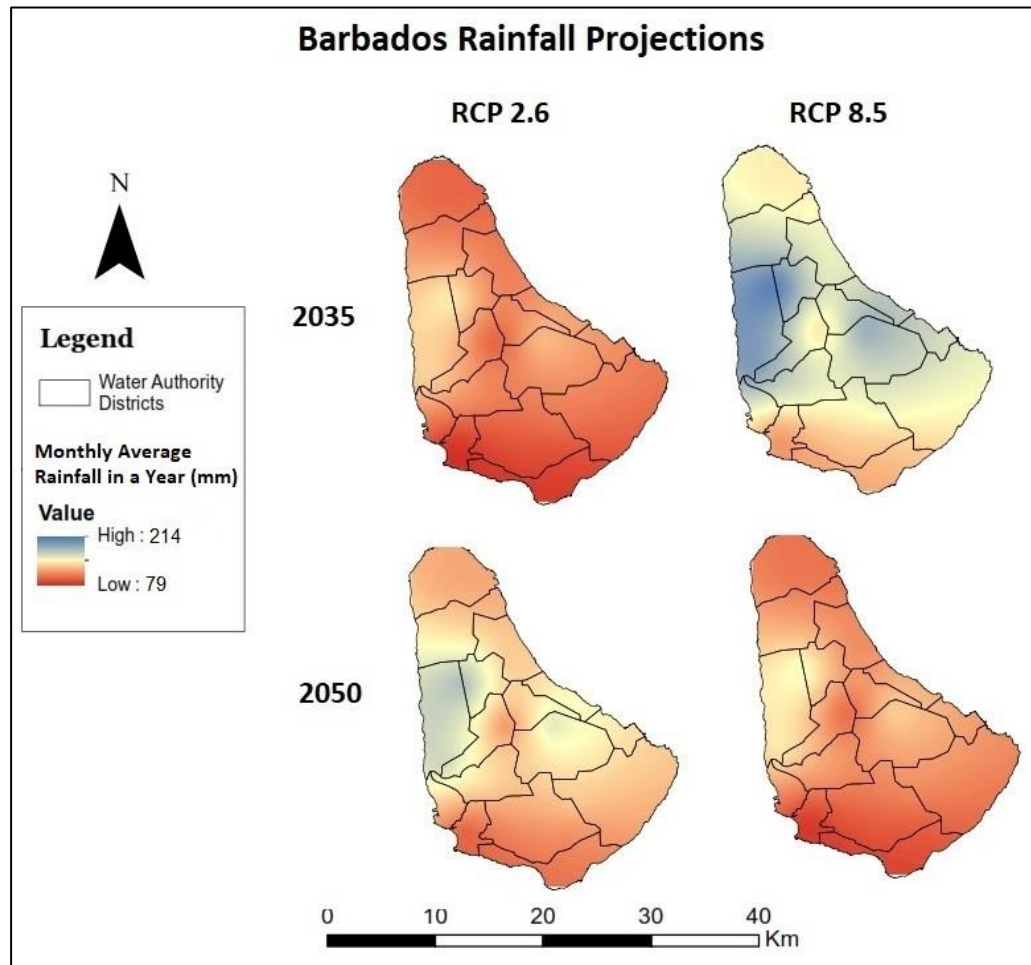


Figure 7.1 Barbados rainfall projections for RCP2.6 and RCP8.5 for the years 2035 and 2050.

In Figure 7.1, the amount of rainfall in each climate change scenario was depicted using choropleth maps that had the same scale and colour scheme to proportionally visualise and compare the variation in rainfall. For RCP2.6, the rainfall situation appears dry in 2035 and becomes more wet (particularly in the west and east) by 2050. For RCP 8.5, the situation appears very wet in 2035 and becomes very dry by 2050. Some of the patterns observed in these rainfall projections were also found by Campbell et al. (2021) who used an ensemble of simulations from a regional climate model. Campbell et al. (2021) projected that the largest drying occurred at the smallest long-term temperature goal of 1.5C (similar to RCP2.6 in 2035) and statistically significant drying over time as the

temperature goal increased (as in RCP8.5 from 2035 to 2050). However, the increase in wetting in RCP2.6 from 2035 to 2050 and the wetness in RCP8.5 in 2035 appear as anomalies. This may be explained by the averaging of inter-annual variability across the time slice. Additionally, Campbell et al. (2021) suggested that models at resolutions higher than 25km, as Charlery and Nurse (2018), can simulate local features that induce orographic rainfall that show differences in projections.

The projections from Charlery and Nurse (2018) were downscaled from a single global circulation model. Therefore, it represents only a possible set of projections for future rainfall and not an average from an ensemble of models. Nevertheless, these rainfall projections were sufficient for this exploratory study. The range of rainfall values for each scenario in Figure 7.1 was provided in Table 7.1, along with the total annual rainfall. Additional analysis on the rainfall projections were carried out in the next section.

Table 7.1 - Summary of projected rainfall for Barbados in climate change scenarios.

Characteristics of Projected Rainfall in Climate Change Scenarios		
Scenarios	Range of Interpolated Monthly Average Rainfall (mm)	Total Annual Rainfall (mm)
RCP2.6 2035	79 - 139	1,279
RCP2.6 2050	96 - 175	1,579
RCP8.5 2035	114 - 205	1,872
RCP8.5 2050	81 - 147	1,343

7.3 Analyses of Rainfall Projections

To gain a better understanding of the variations in the amount and distribution of rainfall in each scenario (see Figure 7.1), additional analyses were carried out. The results of these analyses were given in the sections below.

7.3.1 Statistical and Seasonal Analysis

For each scenario, the average monthly rainfall was computed as the mean of the projected rainfall value for all data-points (see Figure 5.7) in each month of a scenario year. The values for average monthly rainfall were plotted as line graphs in Figure 7.2 and Figure 7.3. Those line graphs were compared to the

averages over the 15-year average that spanned over the period of reported pipe bursts data that were used in this study. Rainfall amounts across all scenarios roughly followed the historical pattern, exhibiting a distinct dry season and wet season. For RCP2.6, average rainfall in each month was lower (i.e., drier) in 2035 than 2050 except in February and October. For RCP8.5, average rainfall in each month was higher (i.e., wetter) in 2035 than in 2050 except in March and September.

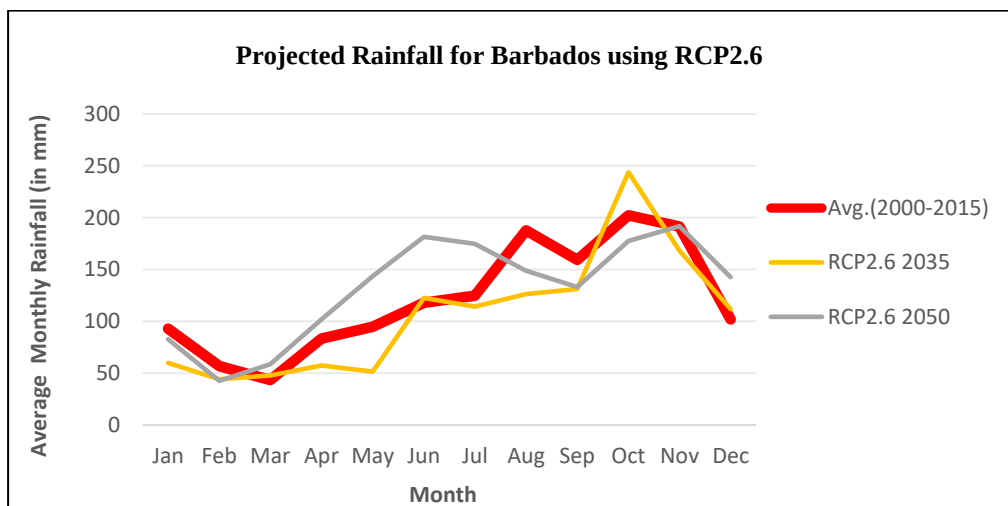


Figure 7.2 - Line chart of average monthly rainfall in RCP2.6 for 2035 and 2050.

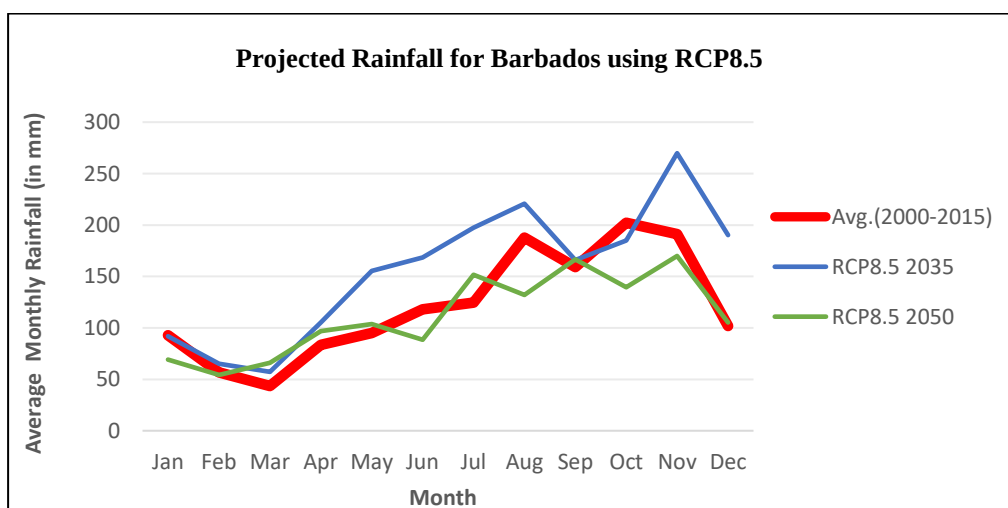


Figure 7.3 - Line chart of average monthly rainfall in RCP8.5 for 2035 and 2050

Using a similar statistical approach shown in the figures above, box plots of monthly rainfall average across each scenario year were produced (see Figure 7.4). These box plots allowed for an effective visualisation of the symmetry of

the data. RCP2.6 in 2035 and RCP 8.5 in 2035 exhibited a right-skewed data distribution. Whereas RCP2.6 in 2050 and RCP 8.5 in 2050 were more or less symmetrically distributed.

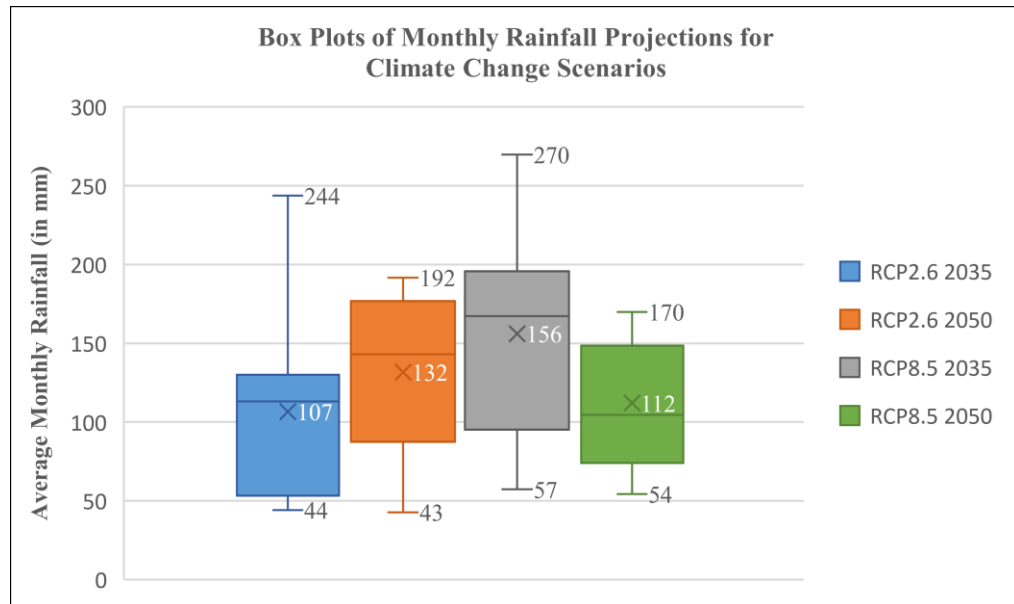


Figure 7.4 - Box plots using monthly rainfall in each scenario year.

In Figure 7.5, the average monthly values for rainfall from Figure 7.2 and Figure 7.3 were used to total the amount of rainfall by rain season for each scenario, as was done previously. Although RCP2.6 in 2035 had the lowest dry season, RCP8.5 in 2050 had the lowest wet season. RCP8.5 in 2035 had the highest dry season and wet season. In fact, it also had the highest range (i.e., the difference between the dry and wet seasons) – 106 mm. RCP8.5 in 2050 had the smallest range – 63 mm. The ranges for RCP2.6 in 2035 and RCP2.6 in 2050 were 99 mm and 82 mm, respectively.

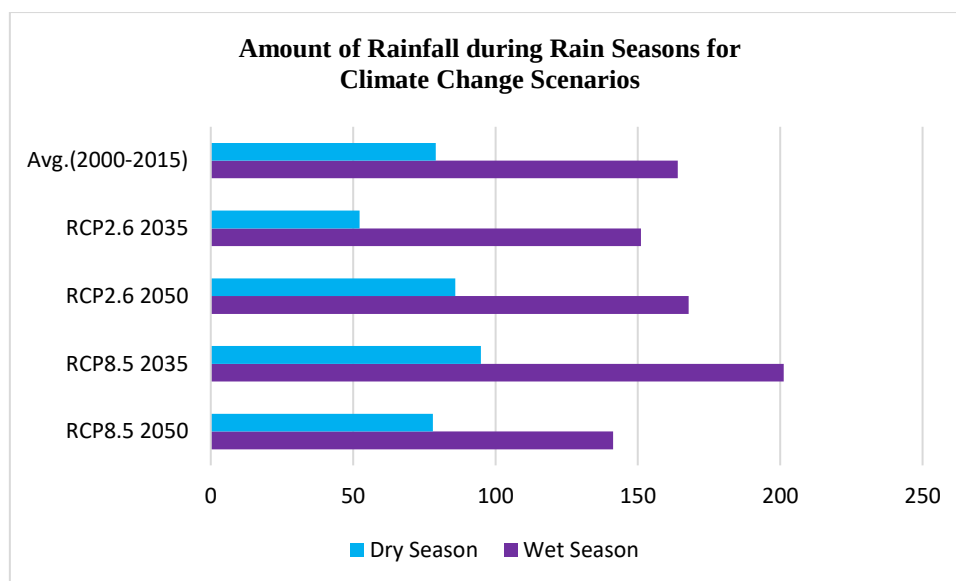


Figure 7.5 - Clustered bar chart showing rainfall amount by season for all scenarios.

7.3.2 Geospatial Analysis

A geospatial analysis of the climate change projections for rainfall was carried out to assess the relative changes in rainfall at every location across the island. Future rainfall was evaluated from the context of historical rainfall using a standardised index called deciles. Rainfall deciles, first described by Gibbs and Maher (1967), were used to rank rainfall data for a specific period and location into 10 equal divisions to determine a scale (WMO and GWP 2016). The established scale is then used to compare new rainfall data.

To establish rainfall deciles for this study, data was taken from the interpolated geospatial map of historical average monthly rainfall, between 2000 and 2015, in Barbados (reference). The values for rainfall ranged from 77 mm to 178 mm. The difference between the maximum and minimum values was calculated (i.e., 101 mm) and divided by 10. The resulting value (i.e., 10.1) was added to the minimum to define the first bin – 77 mm to 87 mm. This process was repeated until the tenth bin was defined. The bins were labelled according to Gibbs and Maher (1967). The final distribution for the deciles was shown in Table 7.2.

Table 7.2 - Decile bins for monthly rainfall from 2000 to 2015.

Decile Bins	Decile Label	Rainfall Range (in mm)
Below 1	Lowest projected	Less than 77.0
1	Much below average	77.0 – 87.0
2 – 3	Below average	87.1 – 107.0
4 – 7	Average	107.1 – 147.0
8 – 9	Above average	147.1 – 168.0
10	Much above average	168.1 – 178.0
Above 10	Highest projected	Greater than 178.0

The deciles in Table 7.2 were used to classify and map the values of projected rainfall to show, by location, the relative changes from the base period. The rainfall decile maps under RCP2.6 and RCP8.5 for the years 2035 and 2050 were shown in Figure 7.6.

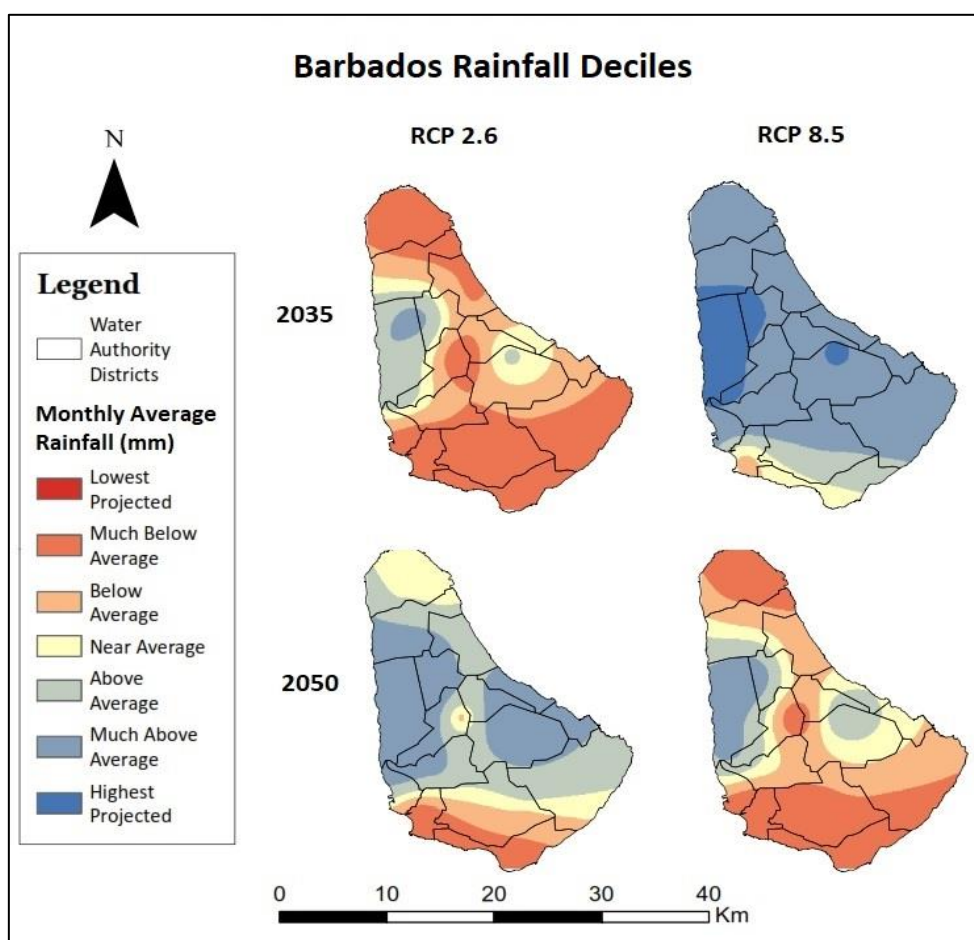


Figure 7.6 - Rainfall deciles from 2000 - 2015 applied to climate change scenarios.

In RCP2.6, extensive drying below the normal was seen in 2035 but by 2050, most areas were wetter ‘above’ and ‘much above’ the historical average (see Figure 7.6). Although in both scenarios (i.e., RCP2.6 in 2035 and RCP2.6 in 2050) there was considerable deviation from the norm, the general pattern was similar: drier areas in the north and south, and rainfall maxima in the west and east. For RCP8.5, most of the total land area exhibited extensive wetting much above the average in 2035. Moreover, RCP8.5 in 2035 was the only scenario that showed no significant drying and had the highest projected rainfall in areas primarily along the west coast. In RCP8.5 in 2050, the pattern of rainfall distribution was similar to RCP2.6 in 2035 though areas of local maxima were wetter and instances of drying were not as intense.

7.4 Pipe Burst Risk Modelling and Hotspot Mapping under Climate Change Scenarios

To assess the impacts of climate change on pipe bursts, projections for rainfall (see Section 7.2) were incorporated into the pipe burst risk model (see Section 6.5). Under each scenario, average rainfall for each subdistrict had to be computed since input data for the pipe burst risk model was on the scale of subdistricts as in the Complex Dataset (see Section 4.3.3). The same approach used to calculate rainfall for subdistricts in Section 4.4.5 was adopted. However, the difference was that instead of using historical rainfall, the projections for rainfall were used. Therefore, the procedures laid out in Section 4.4.5 were repeated for each scenario to produce specific values for average rainfall in all subdistricts.

Having computed average rainfall across all subdistricts under the four climate change scenarios, the data could be incorporated into the pipe burst risk model. The model required seven input variables – rainfall, pressure, diameter, soil type, material, month (i.e., a proxy for season), and location (see Section 6.5.1). The general structure of input data for the model was shown in Figure 7.7. For each climate change scenario, the model was run 12 times to generate risk scores for each month in that scenario year. Data for rainfall were the climate change projections for that scenario year while data for all other variables were

extracted from the Complex Dataset (see Section 4.6). In this way, any deviation in the frequency and distribution of pipe burst hotspots from past observations (see Section 6.5.2) were attributed to changes in future rainfall. The pipe burst risk model was run 48 times for this climate change assessment. The risk scores were averaged for each scenario year to generate hotspot maps which were presented in the sections below.

General Structure of Model Dataset for Climate Change Assessment								INPUT VARIABLES	
Pipe Cohorts	Rainfall	Pressure	Diameter	Soil Type	Material	Month	Location	Variables used to drive pipe burst risk model	
Pipe Group 1								FUTURE PROJECTIONS Rainfall values obtained from climate change projections for RCP2.6 and RCP 8.5 scenarios	
Pipe Group 2									
Pipe Group 3									
Pipe Group 4									
...								PARAMETER VALUES Besides rainfall and month, values for all other factors remain unchanged in scenarios	

Figure 7.7 - General structure of datasets used in the pipe burst risk model for the climate change assessment.

For each hotspot map, a spatial analysis was carried out to determine if clusters existed. If statistically significant clustering was detected, a cluster analysis was carried out to identify clusters of hotspots, coldspots, and outliers. The map produced from the cluster analysis was compared to the base hotspot map to inspect for any changes. Further details on the methods and implementation of the cluster analysis were given in Section 5.5.

7.4.1 Scenario 1 – RCP2.6, Year 2035

RCP2.6 in 2035 was the driest of the climate change scenarios used in this assessment (see Figure 7.1). The hotspot map of the pipe network generated for this scenario was shown in Figure 7.8. On the map, pipes of peak and high risk were predominantly found in the central and east parts of the island, with a few other clusters in the north and south. Medium risk pipes were concentrated in the south and north, while low risk pipes were found throughout the network. This spatial distribution of risk was similar to the hotspot map of the pipe network under historical conditions (see Figure 6.28). However, the main difference between the two hotspot maps was the increase in the number of higher risk pipes under this climate change scenario.

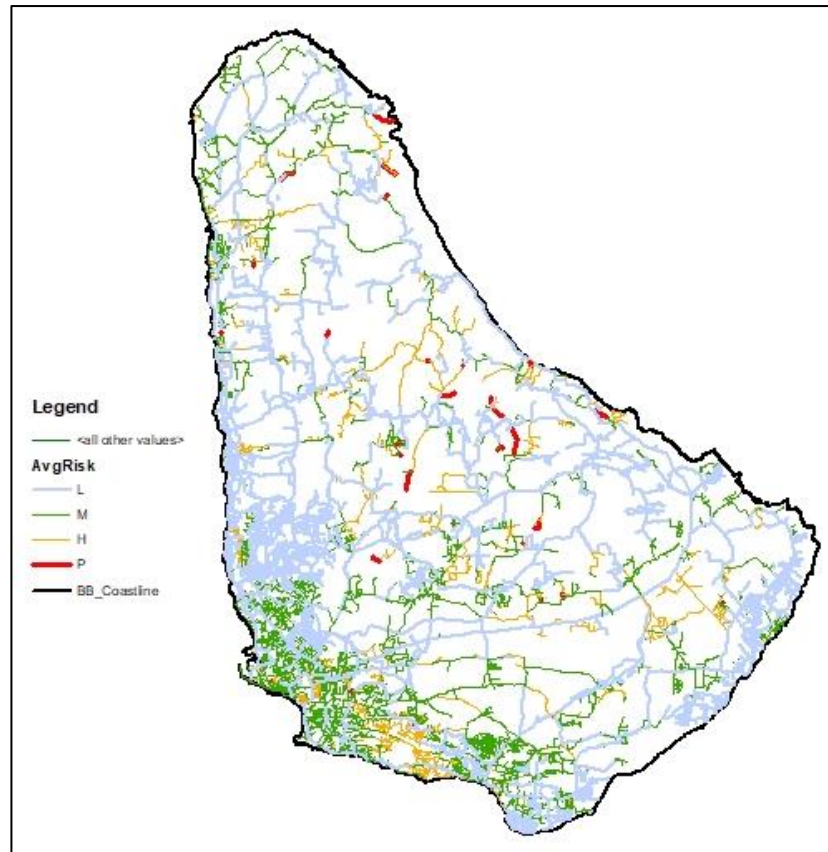


Figure 7.8 - Hotspot map for climate change scenario RCP2.6 2035.

This scenario had the largest number of higher risk pipes compared to the other climate change scenarios (see Table 7.3 and Figure 7.9). Pipes classified as high and peak risk were smaller diameter pipes that experienced higher pressures. When compared to the historical hotspot map (see Figure 6.28), the number of high risk pipes increased overall by 42% with its frequency nearly tripling in southern districts – likely owed to extensive reduction of rainfall in the south (see Figure 7.6). However, there was a decrease of 39% in the number of peak risk pipes. The average value of rainfall was nearly the same across all risk levels, including the predominant soil type and pipe material (see Table 7.3).

Table 7.3 - Summary of pipes in hotspot map of RCP2.6 2035.

Summary of Hotspot Model for RCP2.6 Year 2035						
Risk Level	No. of Pipes	Rainfall ^a	Pressure ^a	Diameter ^a	Soil Type ^b	Material ^b
Low	8194	109.2	35.5	7.9	Smectoid	CIP
Medium	8102	100.6	43.9	4.7	Smectoid	CIP
High	2280	101.8	85.3	3.2	Smectoid	CIP
Peak	41	109.1	119.2	1.6	Kandoid	GI

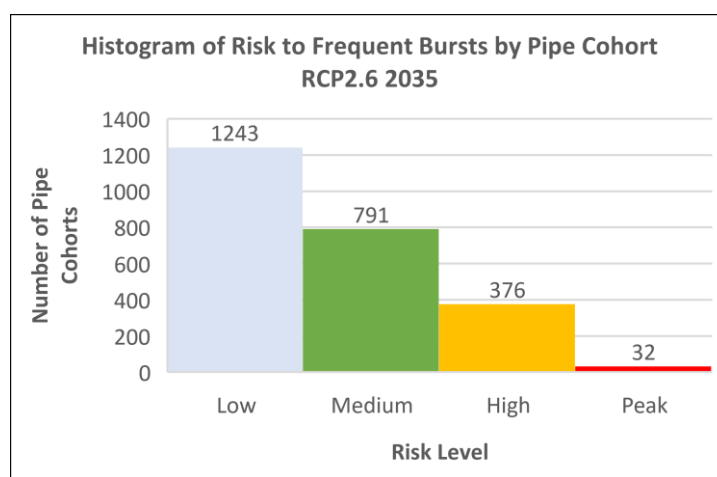
^a Mean value^b Predominant value

Figure 7.9 - Histogram of risk by pipe cohort for hotspot map of RCP2.6 2035.

The distribution of risk of pipes to frequent bursts was also shown by rain season (see Table 7.4). As expected, more pipes were at higher risk during the dry season than wet season. Most notably, the number of pipes classified as high risk during the wet season nearly tripled during the dry season.

Table 7.4 - Distribution of risk to frequent pipe bursts by rain season for RCP2.6 2035.

Distribution of Risk by Rain Season – RCP2.6 2035		
Risk Level	Dry Season	Wet Season
	Total Number of Pipes	
Low	8,153 (43.8%)	8,905 (47.8%)
Medium	5,859 (31.5%)	7,902 (42.4%)
High	4,566 (24.5%)	1,760 (9.5%)
Peak	39 (0.2%)	50 (0.3%)

Spatial and cluster analyses were carried out on the hotspot map in Figure 6.28. From the spatial analysis, the Moran's I Index value was determined as 0.21

which indicated slight clustering. The z-score and p-value were 9.31 and 0.00, respectively, which implied that clustering was most likely not random but statistically significant. Having verified the existence statistically significant clusters, a clustering analysis was carried out to identify map clusters of hot spots, cold spots, and outliers (see Figure 7.10).

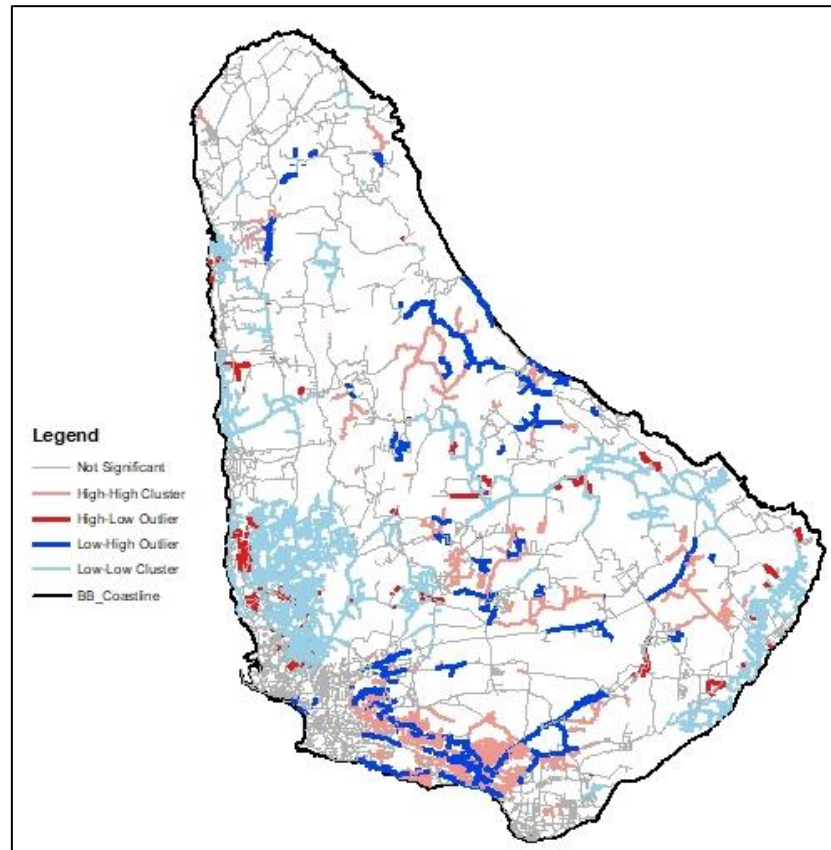


Figure 7.10 - Cluster analysis of RCP2.6 2035 hotspot map.

From the cluster map in Figure 7.10, clusters of higher risk pipes can be seen in the central, eastern, and southern parts of the island. Despite having the highest number of higher risk pipes across all scenarios, this scenario did not have the largest number of clusters of higher risk pipes (see Figure 7.11). That is, there was less clustering of higher risk pipes in this scenario. Lower risk pipes were clustered predominantly in the west and also in the east (see Figure 7.10).

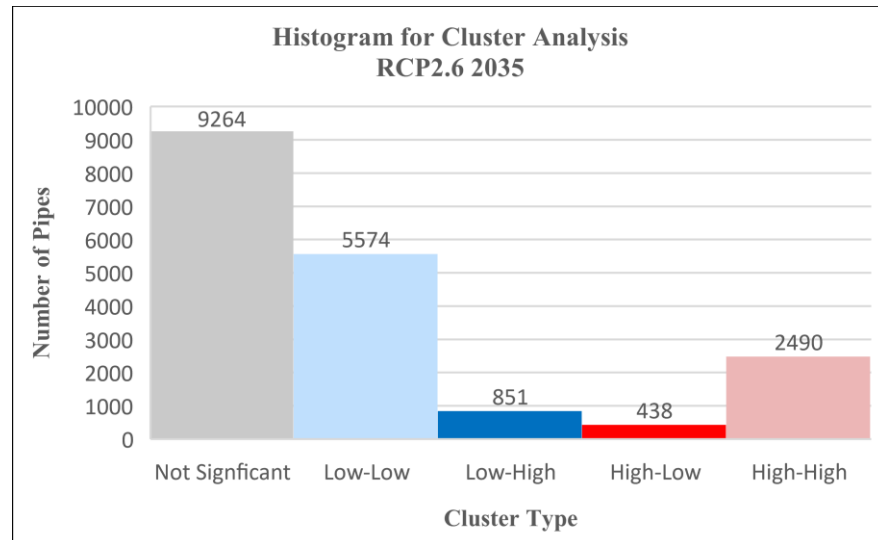


Figure 7.11 - Histogram for cluster analysis of RCP2.6 2035 hotspot map.

7.4.2 Scenario 2 – RCP2.6, Year 2050

RCP2.6 in 2050 was the second wettest of the climate change scenarios (see Figure 7.1). The hotspot map of the pipe network under this scenario was depicted in Figure 7.12. On the map, high risk pipes were mostly clustered in the central part of the island with very few instances of peak risk pipes. Medium risk pipes were largely found in the south, and low risk pipes were predominant throughout the network. Compared to the hotspot map of the network under historical conditions, there were high similarities in the distribution and frequency of risk to frequent pipe bursts.

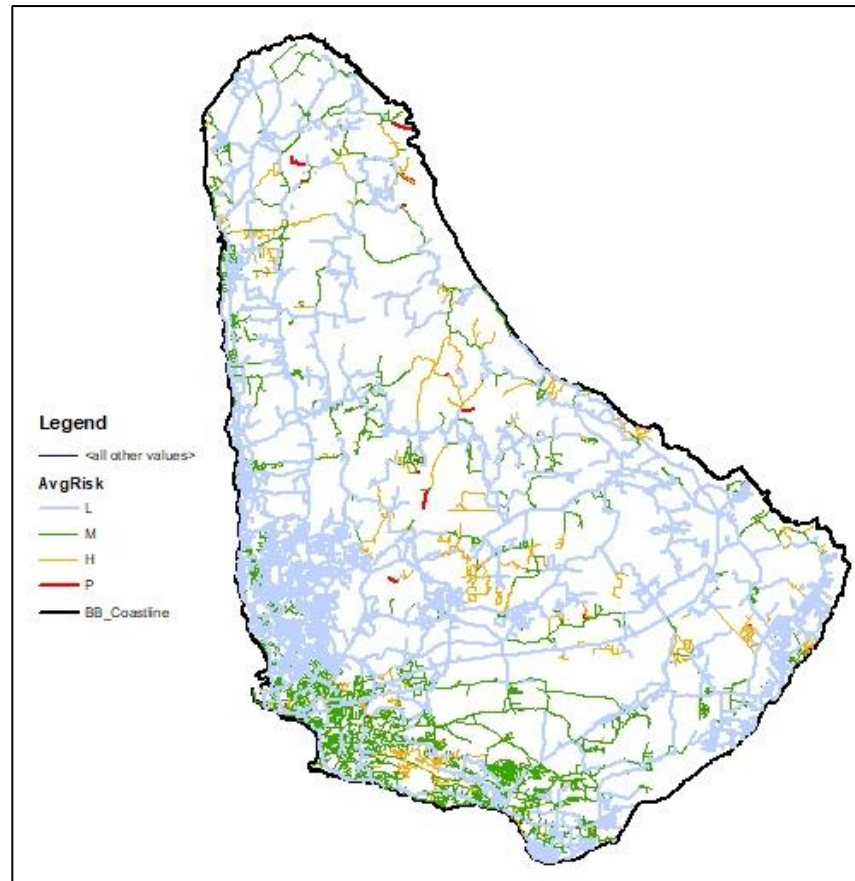


Figure 7.12 - Hotspot map for climate change scenario RCP2.6 2050.

The number of pipes classified in each risk level was shown in Table 7.5 and the frequency of risk distribution by pipe cohort was depicted with a histogram in Figure 7.13. Compared to the historical hotspot map in Figure 6.28, the number of peak pipes decreased by 61%, high risk pipes increased slightly by 2%, medium risk pipes increased by 24%, and low risk pipes decreased by 11%. In general, pipe burst behaviour remained the same with smaller diameter pipes under greater pressures being classified as higher risk pipes, and larger diameter pipes with lower pressures grouped as lower risk pipes. Additional patterns of other variables were not easily discerned given the similar distributions for rainfall, soil type, and pipe material.

Table 7.5 - Summary of pipes in hotspot map of RCP2.6 2050.

Summary of Hotspot Model for RCP2.6 Year 2050						
Risk Level	No. of Pipes	Rainfall ^a	Pressure ^a	Diameter ^a	Soil Type ^b	Material ^b
Low	10280	134.4	33.6	7.3	Smectoid	CIP

Medium	6679	121.4	51.5	4.7	Smectoid	CIP
High	1632	124.7	99.3	3.2	Smectoid	CIP
Peak	26	124.3	126.4	1.5	Smectoid	GI

^a Mean value

^b Predominant value

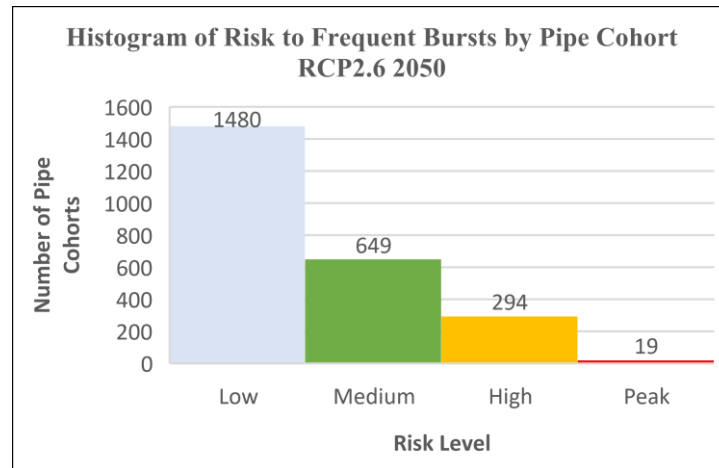


Figure 7.13 - Histogram of risk by pipe cohort for hotspot map of RCP2.6 2050.

The effect of rain season and by extension, rainfall, can be discerned from Table 7.6. During the dry season, pipes were more susceptible to frequent bursts. It was observed that while the wet season had a larger number of pipes classified as medium risk, the dry season had 3 times as many pipes classified as high risk.

Table 7.6 - Distribution of risk to frequent pipe bursts by rain season for RCP2.6 2050.

Distribution of Risk by Rain Season – RCP2.6 2050		
Risk Level	Dry Season	Wet Season
	Total Number of Pipes	
Low	9,535 (51.2%)	11,182 (60.1%)
Medium	5,460 (29.3%)	6,257 (33.6%)
High	3,591 (19.3%)	1,152 (6.2)
Peak	31 (0.2%)	26 (0.1%)

A spatial analysis was carried out on the hotspot map in Figure 7.12. From that spatial analysis, the Moran's I Index, z-score, and p-value were determined as 0.29, 13.11, and 0.00. These results indicated that clustering, which was more than slight, was statistically significant. Further to the spatial analysis, a

clustering analysis on the hotspot map in Figure 7.12 was carried out. The results of the clustering analysis were shown in Figure 7.14.

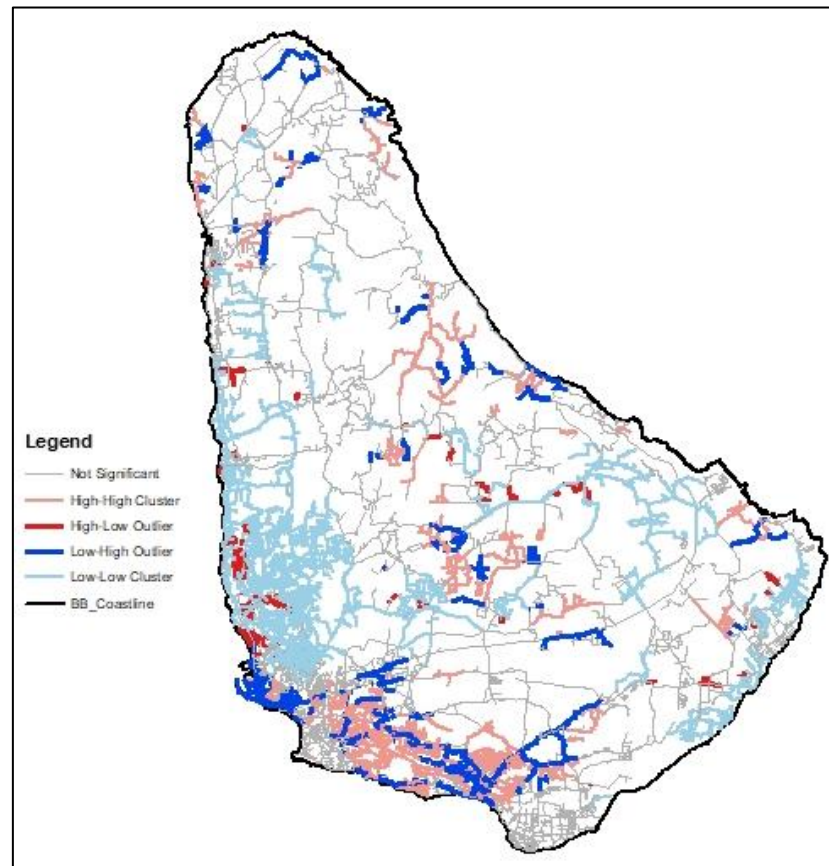


Figure 7.14 - Cluster analysis of RCP2.6 2050 hotspot map.

In Figure 7.14, clusters of higher risk pipes were mostly observed in the central and southern parts of the island. Clusters of lower risk pipes were prevalent on the west coast and also in the east – areas that were projected to received rainfall that was much above the historical average. The frequency of pipes by risk level in Figure 7.14 was depicted in the histogram in Figure 7.15.

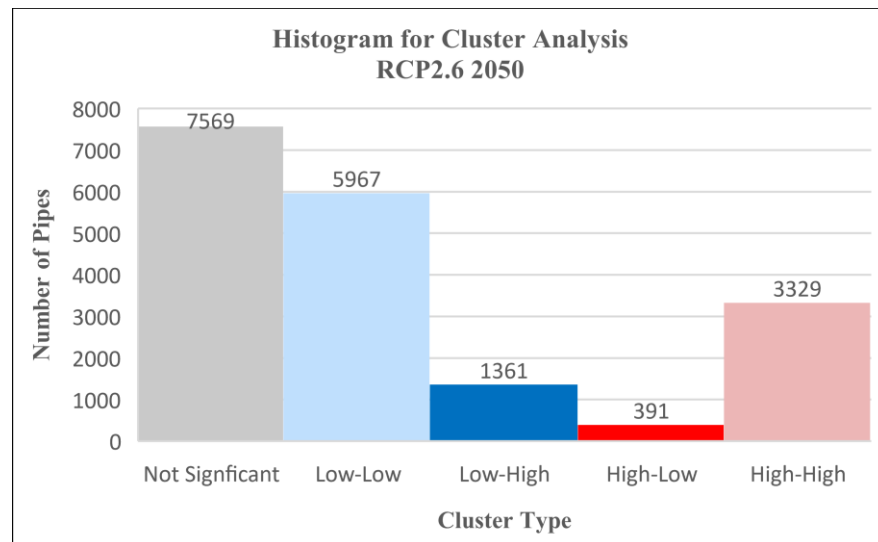


Figure 7.15 - Histogram for cluster analysis of RCP2.6 2050 hotspot map.

7.4.3 Scenario 3 – RCP8.5, Year 2035

RCP8.5 in 2035 was the second wettest of the climate change scenarios (see Figure 7.1). The hotspot map of the pipe network for this scenario was shown in Figure 7.16. On the map, low risk pipes were most prevalent. Some clusters of medium and high risk pipes were observed in the north, central, and south of

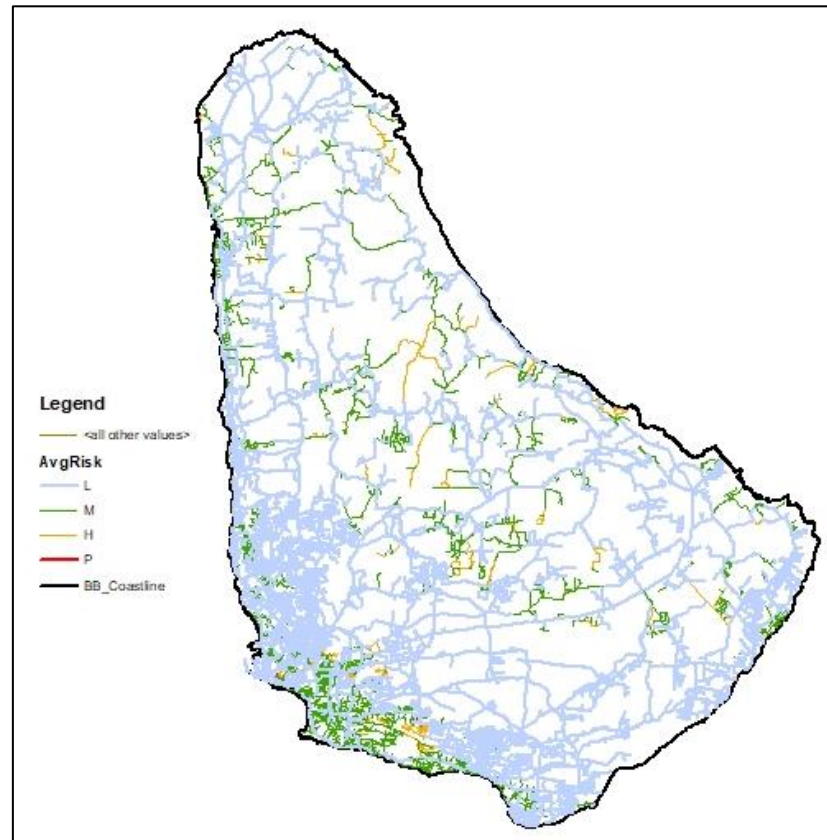


Figure 7.16 - Hotspot map for climate change scenario RCP8.5 2035.

the island. This distribution and frequency of risk was the most dissimilar of the climate change scenarios when compared to the hotspot map of the pipe network that captured past conditions (see Figure 6.28).

The frequency of pipes by risk level in the hotspot map in Figure 7.16 was given in Table 7.7. This frequency was also given by pipe cohort (see Figure 7.17). This scenario (i.e., RCP8.5 in 2035) had the highest number of low risk pipes of all climate change scenarios (see Table 7.7). It was also the only scenario that had an increase in the number of low risk pipes from the historical hotspot map.

Table 7.7 - Summary of pipes in hotspot map of RCP8.5 2035.

Summary of Hotspot Model for RCP8.5 Year 2035						
Risk Level	No. of Pipes	Rainfall ^a	Pressure ^a	Diameter ^a	Soil Type ^b	Material ^b
Low	13800	154.8	29.8	6.8	Smectoid	CIP
Medium	4211	151.5	87.0	4.3	Smectoid	CIP
High	602	151.0	129.7	2.9	Smectoid	CIP
Peak	4	127.5	90.5	1.8	Smectoid	GI

^a Mean value

^b Predominant value

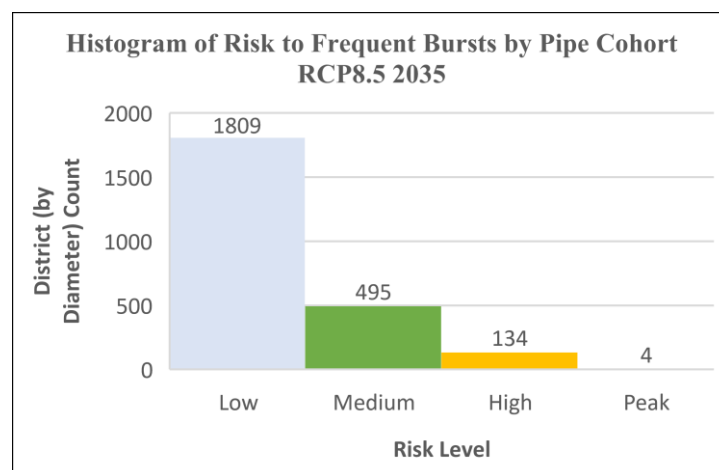


Figure 7.17 - Histogram of risk by pipe cohort for hotspot map of RCP8.5 2035.

The distribution of risk to frequent pipe bursts was shown according to rain season (see Table 7.8). During the wet season, less than 20% of the pipe network was classified as medium risk or higher. Even in the dry season, the number of pipes classified as low risk was higher than in the historical hotspot map overall.

Table 7.8 - Distribution of risk to frequent pipe bursts by rain season for RCP8.5 2035.

Distribution of Risk by Rain Season – RCP8.5 2035		
Risk Level	Dry Season	Wet Season
	Total Number of Pipes	
Low	12,946 (69.5%)	14,557 (78.2%)
Medium	4,091 (22.0%)	3,679 (19.8%)
High	1,575 (8.5%)	365 (2.0%)
Peak	5 (0.03%)	16 (0.1%)

A spatial analysis was performed on the hotspot map in Figure 7.16 to detect clusters in pipes according to risk value. From that analysis, the Moran's I Index value was calculated as 0.29, the z-score was 12.88, and the p-value was 0.00. These results were an indication that the statistically significant clustering had been detected. Following the spatial analysis, a clustering analysis was also carried out. The results of the clustering analysis that identified clusters of hotspots, coldspots, outliers were shown in Figure 7.18.

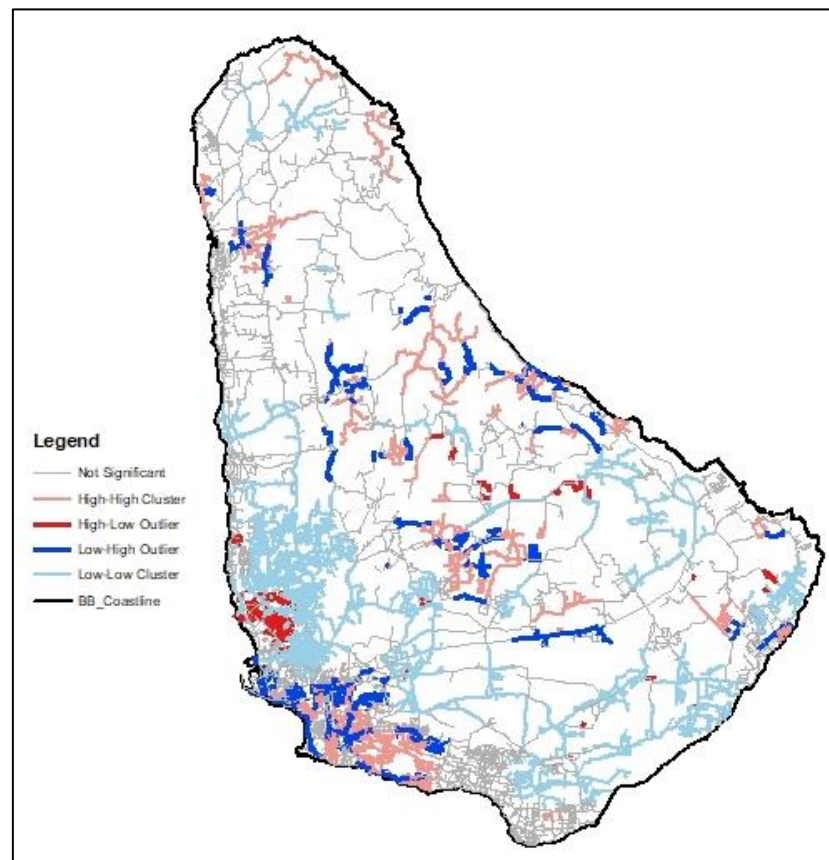


Figure 7.18 - Cluster analysis of RCP8.5 2035 hotspot map.

In Figure 7.18, clusters of higher risk pipes were observed in the north, central, and south of the island. However, clusters of lower risk pipes were predominant in the west, south, and east. The large numbers of clusters lower risk pipes observed in Figure 7.18 was reflected in the histogram shown in Figure 7.19.

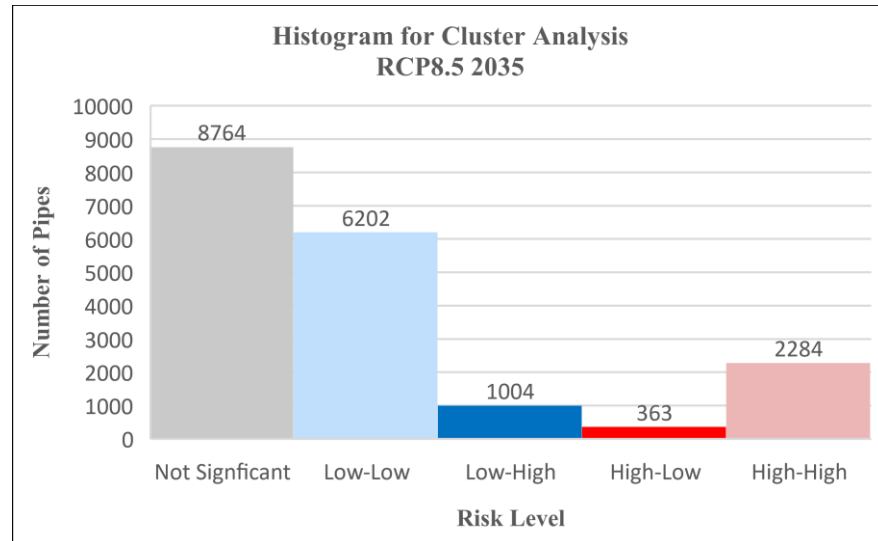


Figure 7.19 - Histogram for cluster analysis of RCP8.5 2035 hotspot map.

7.4.4 Scenario 4 – RCP8.5, Year 2050

RCP8.5 in 2035 was the second wettest of the climate change scenarios (see Figure 7.1). The hotspot map of the pipe network under this scenario was depicted in Figure 7.20. On the map, clusters of high risk pipes were seen in the north, central and south, with a few peak risk pipes scattered around the centre of the island. Medium risk pipes were predominant in the south and north of the island. Low risk pipes were found throughout the network but particularly clustered in the west and east. This general distribution of risk was similar to the spatial patterns observed in the hotspot map of the network under historical conditions although the frequencies of risk were projected to change.

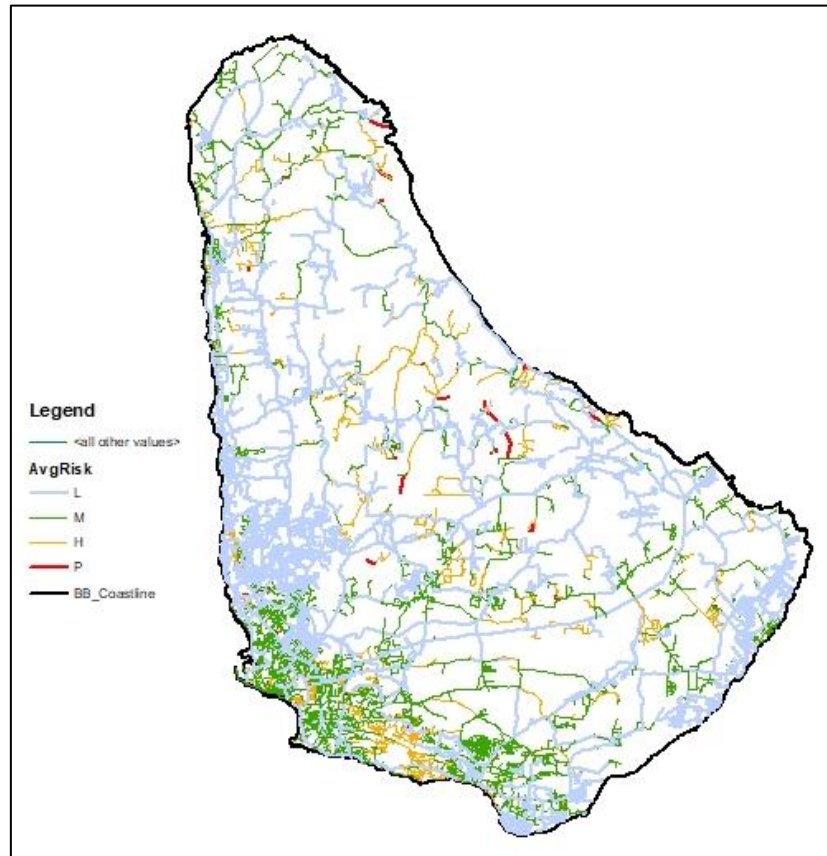


Figure 7.20 - Hotspot map for climate change scenario RCP8.5 2050.

In Figure 7.20, high and medium risk pipes were seen in the north, central, and south of the island. A few instances of peak burst pipes were discernible in the north and central parts while low risk pipes were prevalent across the network. The frequency of pipes for each risk level was shown in Table 7.9 and presented by pipe cohort in Figure 7.21. These frequencies were very similar to the hotspot model of RCP2.6 in 2035 which was likely due to these scenarios having comparative rainfall projections (see Figure 7.1). Pipes classified as peak and high risks were smaller diameter pipes under high pressures (see Table 7.9), while lower risk pipes were larger diameter pipes under low pressures.

Table 7.9 - Summary of pipes in hotspot map of RCP8.5 2050.

Summary of Hotspot Model for RCP8.5 Year 2050						
Risk Level	No. of Pipes	Rainfall ^a	Pressure ^a	Diameter ^a	Soil Type ^b	Material ^b
Low	8640	114.8	35.3	7.8	Smectoid	CIP
Medium	7674	104.5	44.7	4.7	Smectoid	CIP

High	2254	106.3	86.4	3.2	Smectoid	CIP
Peak	49	113.5	114.8	1.6	Kandoid/ Smectoid	GI

^a Mean value

^b Predominant value

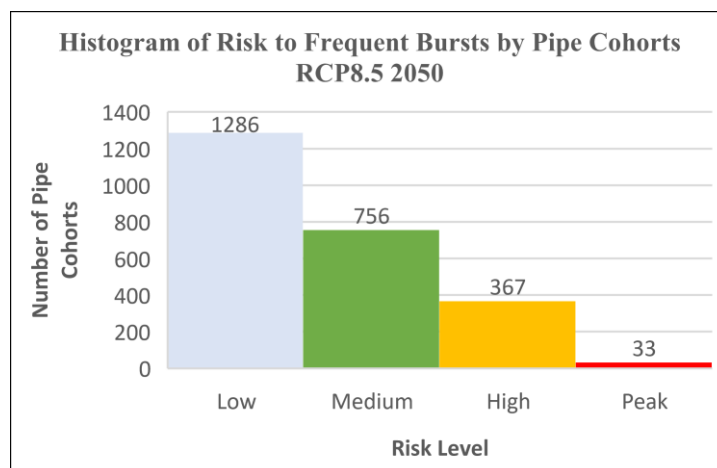


Figure 7.21 - Histogram of risk by pipe cohort for hotspot map of RCP8.5 2050.

The distribution of risk of pipes to frequent bursts was examined by rain season (see Table 7.10). The number of high risk pipes was 13% higher and the number of low risk pipes 10% lower during the dry season than in the wet season. However, the number of pipes classified as medium and peak risk in both rain seasons were comparable.

Table 7.10 - Distribution of risk to frequent pipe bursts by rain season for RCP8.5 2050.

Distribution of Risk by Rain Season – RCP8.5 2050		
Risk Level	Dry Season	Wet Season
	Total Number of Pipes	
Low	9,535 (51.2%)	11,182 (60.1%)
Medium	5,460 (29.3%)	6,257 (33.6%)
High	3,591 (19.3%)	1,152 (6.2)
Peak	31 (0.2%)	26 (0.1%)

A spatial analysis was carried out on the hotspot map in Figure 7.20. The analysis calculated the Moran's I Index, z-score, and p-value as 0.21, 9.54, and 0.00, respectively. The value of the Moran's I Index indicated that there was slight clustering. The values of the z-score and p-value determined that it was highly unlikely that the clustered pattern was random and that the observed

clustering was statistically significant. Subsequent to the spatial analysis, a clustering analysis was performed on the hotspot map in Figure 7.20. The output from the clustering analysis was shown in Figure 7.22.

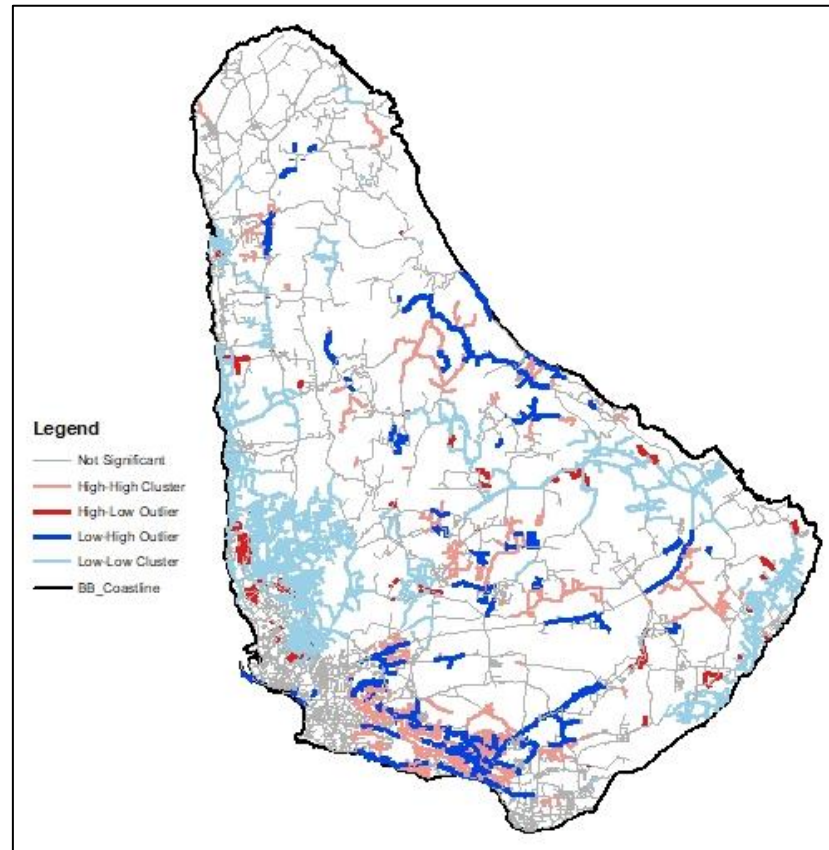


Figure 7.22 - Cluster analysis of RCP8.5 2050 hotspot map.

In Figure 7.22, clusters of higher risk pipes were seen in the central and southern parts of the island with a few outliers of lower risk pipes in between. The clusters of lower risk pipes were predominant on the west coast, where projected rainfall was ‘above’ or ‘much above’ the historical average (see Figure 7.1). The frequency of pipes in each cluster type was depicted in a histogram in Figure 7.23.

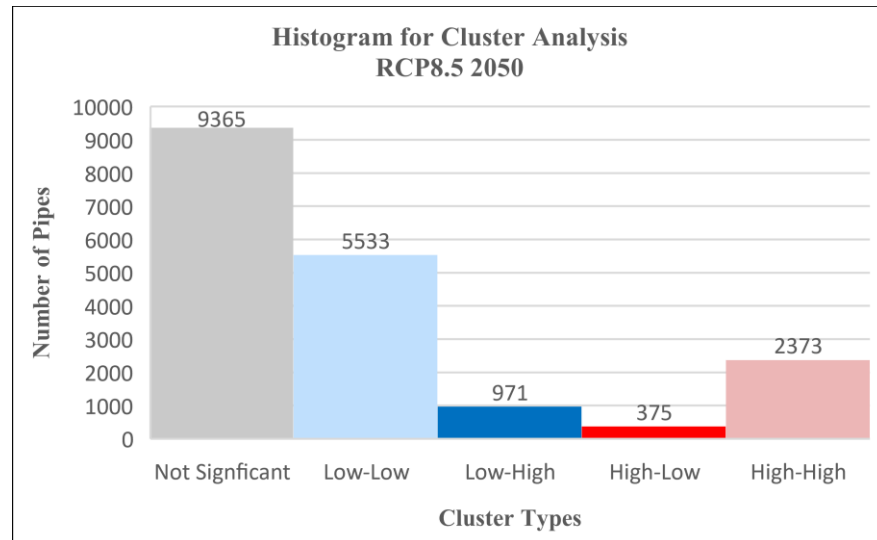


Figure 7.23 - Histogram for cluster analysis of RCP8.5 2050 hotspot map.

7.5 Chapter Summary

In this chapter, projections for future rainfall in Barbados under climate change scenarios RCP2.6 and RCP 8.5 for the years 2035 and 2050 were interpolated and analysed (see Section 7.2). Future rainfall was compared to historical rainfall to evaluate how the amount and distribution of rainfall varied in each climate change scenario (see Section 7.3). The rainfall analysis revealed that the drier scenarios were RCP2.6 in 2035 and RCP8.5 in 2050, and the wetter scenarios were RCP2.6 in 2050 and RCP8.5 in 2035. After completion of the rainfall analysis, the projections were used as input into the FIS (see Section 6.5.1) to generate hotspot maps for each scenario. It was observed that the frequency of pipe bursts increased in areas that experienced rainfall that was much below the historical average. Hence, more pipes were at a higher risk to peak bursts in the drier scenarios. A cluster analysis was carried out on each hotspot map to identify hotspots, cold spots, and outliers that were statistically significant. In the next chapter, the significance and implications of results of this assessment of climate change along with the results from data mining and pipe burst modelling in the previous chapter (see Chapter 6) is discussed.

8 DISCUSSION

8.1 Chapter Introduction

The aim of this chapter is to present the discussion of the results from the analyses and to examine the findings with respect to the existing body of knowledge. The following is contained in this chapter:

- (1) The interpretation and importance of the major findings determined from this research.
- (2) The theoretical, methodological, and practical implications of the findings that were presented.
- (3) Limitations of the study that impacted the findings.
- (4) The ways in which the validity and reliability of the research were enhanced.
- (5) Three sets of recommendations – (i) to address the limitations of the study, (ii) suggestions for further work, and (iii) ways in which the local water utility can manage the risks identified.

Key highlights in this chapter are the implications of the new knowledge gained about local pipe burst behaviour, and the contributions given to the emerging discourse of the impacts of climate change on pipe failure in a WDN.

8.2 Interpretation of Major Research Findings

This research was conducted to address the gap that although pipe age is a plausible factor that explains the high burst frequency in Barbados (see Section 1.2), no evidence-based research has been carried out to systematically determine the local drivers of pipe bursts and the broader implications of climate change. The interpretation of the four major findings from this research are presented in the sections below.

8.2.1 Categorisation of Local Pipe Burst Factors by Importance

The first major finding was the categorisation of factors that contributed to the occurrence of peak pipe bursts in Barbados. Each factor that was considered in the pipe failure analyses carried out in this study, which used several techniques

that were mostly data-driven, was ultimately categorised by level of importance (see Table 8.1).

Table 8.1 - Categorisation of explanatory factors by level of importance.

Categorisation of Factors by Importance		
Significant	Relevant	Not Classified
Rainfall Pressure Diameter Soil Type Pipe Material	Pipe installation era Land-use	Connection density

The categorisation of factors was derived from the synthesis of key findings of data mining analysis which was guided by a set of stepwise rules that was developed for this study (see Section 5.3). The data mining analysis was conducted via pipe burst modelling using self-organising maps (SOMs) and evolutionary polynomial regression (EPR) analyses (see Sections 6.2 and 6.3). Factors that had a positive correlation with peak pipe bursts in at least two models were considered significant while factors that showed a positive correlation with pipe bursts at risk levels lower than peak in at least two models were considered relevant (see Section 6.4). By this categorisation, pipe installation era (i.e., a proxy for pipe age) was considered relevant in the occurrence of peak pipe bursts in the network but it was not a significant factor.

The assertion that other variables were more significant than pipe age on the issue of '*what are the main drivers of pipe failure in Barbados*' was important for several reasons. Firstly, the classification was an outcome of a data-driven analysis that questioned the familiar notion and generalisation that pipe age would be the main driver of frequent pipe bursts. International studies such as Barton et al. (2019), Boxall et al. (2007), O'Day (1982) also supported that increases in failure in older pipes should not be automatically attributed as a function of pipe age alone.

Secondly, the categorisation provided a more complete understanding of the underlying aspects of pipe failure in the local WDN. While pipe age and pressure were known to promote pipe bursts (Husbands 2019), not much

attention had been given to investigating and addressing other possible contributing factors of pipe bursts (such as pipe material, rainfall, and soil type) in the network. Discerning important explanatory factors was a necessary step in building a robust knowledge base that could be used to make informed decisions.

Thirdly, the diverse set of significant and relevant factors highlighted the need for comprehensive data gathering for studies that can be used to inform decision-making in better management. Particularly for the BWA, there is a need to enhance its database and expand system modelling efforts. Considerable gaps were found in the datasets obtained from and being maintained by BWA (see Section 4.7). These internal datasets primarily contained physical and operational data about the pipe network. All other data had to be created or sourced from external entities (see Table 4.1). The fixing of data inconsistencies and inclusion of environmental data for system modelling in a geographical information system (GIS) were crucial in bringing about knowledge discovery in this study.

Though the quantity and quality of data are challenges faced by most water utilities (Wood and Lence 2009, Pelletier, Mailhot, and Villeneuve 2003), incorporating additional data in different formats from multiple sources can enhance pipe failure analysis (Wu and Liu 2017, Roman et al. 2014, Van Daal and Raterman 2013). This study used three discrete datasets that contained various spatiotemporal variables which were recorded at different spatial scales (see Table 4.14). This approach of incorporating and analysing multiple datasets was taken because of the deficiencies in the way that data were gathered and recorded. The research could have been carried out in a more straightforward manner had there been one comprehensive dataset which would have allowed for better control and validation. However, in spite of this, efficient use was made of available data from multiple sources (see Table 4.1) as some variables such as connection density and land-use were derived through innovative geospatial techniques (see Sections 4.4.7 and 4.4.8). The methodology

demonstrated that even where there are data challenges, if there is sufficient data then it is possible to derive meaningful outputs.

Lastly, the categorisation of rainfall and type of rain season (i.e., dry season or wet season) as significant factors had broader implications. Rainfall was a dynamic environmental factor that varied over time and space, and it was anticipated that climate change would alter its historical distribution. Although the impacts of climate change on rainfall patterns and water resources such as the local groundwater supply were examined and understood (Taylor et al. 2018; Charlery and Nurse 2018; Cashman, Nurse, and Charlery 2010), the potential effects of climate change on pipe failure were not known. In fact, not much research has been carried out on the impact of climate change on pipe failure in general (see Section 2.6.2). Thus, it was essential to not only understand how rainfall distribution influenced pipe burst behaviour but also to prudently assess the future implications of climate change on pipe failure.

8.2.2 Characterisation of the Behaviour of Peak Pipe Bursts

The second major finding was the detection of a seasonal trend in pipe burst occurrences which matched findings in other studies (Barton et al. 2019; Farmani et al. 2017; Wols and van Thienen 2016) and discovering the effects local explanatory factors have in influencing the location of frequent pipe failures in the WDN. This insight gained into the behaviour of pipe bursts was derived from modelling pipe failure with SOMs and EPR (see Sections 6.2 and 6.3). SOM was a technique that used an artificial neural network to detect and analyse non-linear relationships in complex data and EPR was a hybrid technique that combined numerical regression with genetic algorithms (see Section 2.4.2). These methods were systemically applied for data mining using an iterative stepwise methodology, as described in Section 5.3, to address the research question *‘are there any patterns or factors that can explain the high burst rate of the pipe network in Barbados?’*

Modelling pipe failure with time-series data of reported bursts revealed a temporal trend of higher burst occurrences during a specific period of the year. That period, from December to May, was the same interval as the local dry

season. It was also observed that instances of peak annual pipe bursts occurred around drought and predominantly dry years. Both trends suggested a link between frequent pipe burst occurrences and rainfall. These patterns were not easily discerned or conclusive during the preliminary stage of data analysis which used basic descriptive statistics to characterise the distribution of the data (see Section 4.3). Therefore, the application of more rigorous methods was critical in discerning and confirming this trend.

Increased pipe failure during dry periods was found in other works such as Barton et al. (2019), Wols and van Thienen (2016), and Gallage et al. (2012). Prior to this research, the pattern of higher occurrences of pipe bursts during the dry season had not been noted before in Barbados. Thus, the recognition of rainfall as a main factor in the occurrence of peak bursts was an important outcome of the research especially given the effect rainfall can have in certain pipe environments. Savic and Walters (1999) emphasised that rainfall had a direct impact on pipe failure remarking that it was the only factor other than leakage that affected soil moisture which in turn had several consequences. This concept by Savic and Walters (1999) was particularly relevant to this study since several pipe failure models showed that rainfall and soil type were among the set of other significant factors which influenced the location of peak bursts in the local network.

From the SOMs analysis, multiple pipe failure models were generated from the three datasets used for analysis (see Section 4.6). From the multiple models produced, it was found that pipes with the following attributes were prone to peak bursts: (1) small diameter, (2) iron material, (3) under extremely high water pressures, (4) laid in smectoid clay, and (5) within an area that experienced low rainfall (see Table 6.5). Furthermore, if total monthly rainfall fell below a threshold of 80mm then the susceptibility to peak bursts increased for small and medium diameter pipes fitting the criteria above. This was particularly observed in the southern districts. A few of the mathematical models generated from the EPR analysis also suggested that this combination of factors was significant in promoting bursts (Table 6.4). In this combined

effect, it was particularly the properties of the predominant soil type (i.e., Smectite Clay) swelling with increased moisture from rainfall and shrinking under extremely dry conditions (Abbey, Eyo, and Ng'ambi 2020) that likely promoted pipe failure (Gallage et al. 2012). The action of swelling caused an upward pressure to be applied on pipes while shrinking resulted in a loss of bedding support – prompting soil movement and instability (Gallage et al. 2012).

The expansion characteristics of clay depend on water, and thus, maximum swelling and shrinking action due to variations in soil moisture are experienced in climates with distinct wet and dry seasons (Krenz, Lee, and Owens 2006). In Barbados, though it is known that several types of clays exist (see Section 4.4.4) and land slippage events occasionally occur in Scotland District clay, it is the smectoid clay that has a significant shrink-swell capacity (Abbey, Eyo, and Ng'ambi 2020, Krenz, Lee, and Owens 2006). This swelling clay is found stretching inland from the north, west, and south coasts in Barbados (see Figure 4.5). In these areas, especially during the dry season, a bending stress would be induced on pipes by the smectoid clay. Pipes with small diameters or iron material would be most vulnerable to the bending stress and corrosion caused by cyclic wetting and drying. Understanding the dynamics of these explanatory factors gave reason for the trend detected in high burst occurrences during the dry season and provided insight into rationalising locations where frequent pipe failures would likely occur.

Another significant factor that had a role in the location of frequent pipe bursts was pressure. A direct correlation between peak bursts and high pressures was found in some of the pipe failure models generated from the SOMs analysis (see Section 6.2.3). The highest pressures in the network were observed particularly in the central part of the island (see Figure 4.13). In that area, there are pressure management issues because of the rise in elevation and spread of remote communities. Due to the decrease in water pressure with each unit increase in elevation, operational pressure was boosted in order to deliver water to customers. The increase in operational pressure subjected parts of the pipeline

to pressures that exceeded the maximum thresholds recognised by several water utilities (see Section 2.3.7), ultimately contributing to more frequent pipe bursts at those locations.

The only significant factor that indicated location was district area. This variable represented the geographic location (i.e., north, south, east, and west) of the district where each pipe was laid. The analysis of pipe failure suggested a correlation between peak bursts and southern districts. However, southern districts usually formed part of a combination of parameters which collectively promoted frequent bursts. Those other parameters included low rainfall, dry season, drought or predominantly dry years, and Smectite Clay. Due to the generalisation of district area, the southern district may have been a proxy for another factor. While Smectite Clay was predominant in southern districts, location was only used as a category to label the SOMs analysis to avoid autocorrelation and location was not included as variable in the EPR analysis with soil type. Nonetheless, it provided some insight into the areas where frequent pipe bursts occurred.

8.2.3 Pipe Burst Risk Model for the Local Pipe Network

The third major finding was the identification of the most critical pipes in the network that were prone to peak burst occurrences. A pipe burst risk model was developed using a FIS assigning a risk value to each pipe in the network (see Section 6.5.1). A FIS was used due to its capability to handle imprecise and vague data, flexibility of incorporating expert knowledge, and effectiveness in other similar applications (see Section 2.4.2). The basic structure of the FIS, which was founded on fuzzy logic, only considered the set of significant variables (i.e., rainfall, pressure, diameter, soil type, and material) along with the type of rain season and location (i.e., the general geographic area). For each input and output variable, a fuzzy membership function was created to define the range of fuzzy values for its parameters (see Section 6.5.1). The rules that governed the FIS, mapping inputs (i.e., pipe attributes) to outputs (i.e., risk), comprised of seven *if-then* statements based on key findings from pipe failure modelling (see Table 6.5). The pipe burst risk model was a tool that can be

useful in various applications such as pipe prioritisation for replacement, sensitivity analysis, and a basis for online simulations.

The basis of the pipe burst risk model consisted of a synthesis of multiple pipe failure models across different spatial and temporal scales. Fundamentally, the system represented one consolidated model of pipe failure. This was important because it took into account the new knowledge gained into the behaviour of peak pipe bursts in the network which also captured what was previously known. Prior to this study, with only a partial understanding of pipe burst behaviour, high risk was attributed to the oldest pipes in the network or pipes that experienced issues with pressure management. This approach has not proven successful since burst rates and water losses have continued to increase over the years. However, with a more complete model of pipe failure with the fuzzy ranking system, an evidence-based redistribution of risk was performed. This model was also validated in Section 6.5.3, showing relatively good and performances in identifying low and medium risk pipes. However, the model did not perform as well in matching pipes suspected of multiple bursts to pipes classified as high risk. The implications of the model validation were discussed in detail in Section 8.3.3.

The risk ranking system assisted in answering the research question: *which parts of the pipeline are most at risk to frequent bursts in Barbados?* The sole output of the fuzzy ranking system was a risk value (between 0 and 4) used to classify a pipe into one of the following risk levels: low, medium, high, and peak. This classification was used to identify critical pipes and inform the mapping of the spatial distribution of risk in the pipe network using a GIS. The maps provided a visualisation of a hotspot model of the pipeline (see Figure 6.28). The creation of a hotspot model of the pipe network was important because (1) it attempted to resolve one of the major challenges faced of pinpointing vulnerable parts of the pipeline; and (2) it was a tool that could then be used to inform and focus ongoing efforts of pipe replacement in the water distribution system. A cluster analysis of the hot spot model was carried out, showing statistically significant clusters of higher risk pipes around the central

and east parts of the island (see Figure 6.30). This cluster analysis also served to address the shortcomings in validating the hotspot model of the pipe network.

While the pipe burst risk model was built on the domain knowledge derived from multiple pipe burst models, the FIS also represented a standalone model. Given that, the FIS could also be used to explore how changes in any of the seven input factors would in turn alter the pattern of pipe failure in the network. In this study, the FIS was used to investigate the impact of climate change via IPCC scenarios on the distribution of risk to frequent pipe burst occurrences since rainfall was categorised as a significant factor.

8.2.4 Assessment of Climate Change on Pipe Failure

The fourth major finding was the extent to which the distribution of hotspots in the network can be influenced by the variation in future rainfall caused by climate change. This finding was established from the climate change assessment in which the pipe burst risk model of the network developed in Section 6.5 was used to explore and examine the effects of rainfall projections on the frequency and distribution of pipe burst occurrences in the future. The assessment incorporated downscaled geospatial rainfall projections that were derived from the most recent ensemble of climate change models for the lowest and highest emission scenarios from the IPCC – RCP2.6 and RCP8.5, respectively – for the years 2035 and 2050. This data represented the highest resolution of climate change projections of rainfall for Barbados to-date (Charlery and Nurse 2018). With consideration to changes in future rainfall compared to the historical average, RCP2.6 in 2035 (driest) and RCP8.5 in 2050 showed significant drying while RCP2.6 in 2050 and RCP8.5 in 2035 (wettest) exhibited substantial increases in wetness (see Figure 7.6). The final outputs of the assessment were hotspot models of the pipe network generated for each scenario (see Section 7.4).

The rainfall projections used in this study represented the specific outputs generated by Charlery and Nurse (2018) from a particular GCM. If a different GCM was used, then different results would have been generated. However, no one model would produce a right answer and evaluating various GCMs was not

an objective of this research. Rather, the goal was to present a proof of concept which can be used to evaluate how the results of the climate change assessment on pipe bursts could change with different GCMs.

A summary of the hotspot models that were produced under each climate change scenario was shown in Figure 8.1. Under RCP2.6, there was a considerable increase in the number of higher risk pipes in 2035 from the historical average of the network. By 2050 (in that same scenario), the frequency of risk was comparable to the network average. Under RCP8.5, there was a substantial decrease in the number of higher risk pipes in 2035 when compared to the historical network average. However, by 2050 the frequency of higher risk pipes was much above the network average. Regarding the frequency of higher risk pipes, overall, drier scenarios showed increases while wetter scenarios exhibited decreases. The spatial distribution of risk was similar to the historical across all scenarios except RCP8.5 in 2035 which had extreme wetting.

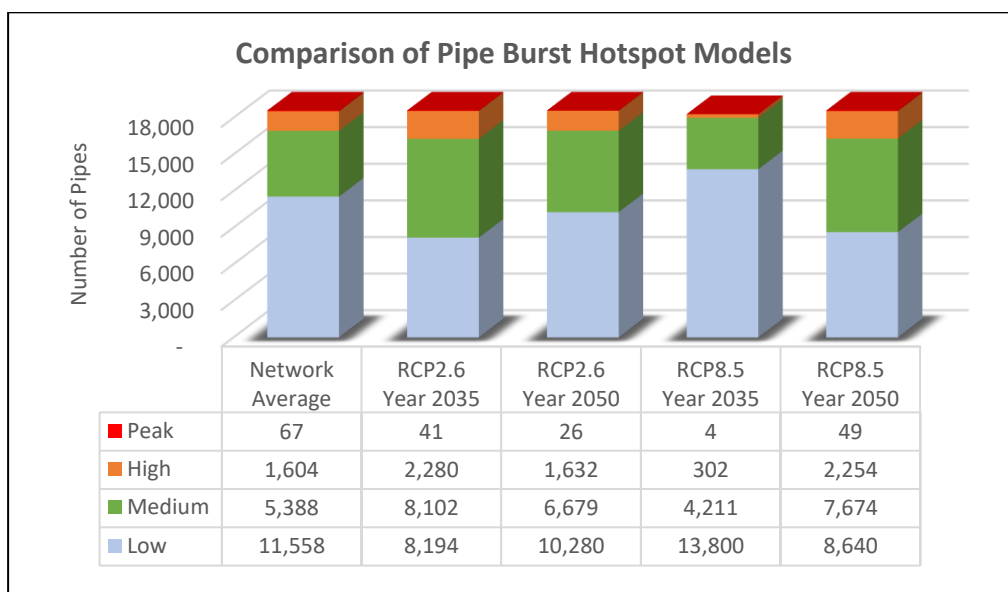


Figure 8.1 – A comparison of the hotspot models produced from the climate change assessment.

The hotspot model of the pipe network was validated as far as possible using service repair data (see Section 6.5.3) and considered to show a fair performance despite the limitations in the validation data. To address this shortcoming, a spatial and clustering analysis was conducted on the historical hotspot map of

the network and on the hotspot maps for each climate change scenario. From these analyses, it was determined that statistically significant clustering was observed on each hotspot map and these areas of hotspots, coldspots, and outliers were identified (see Figure 7.10, Figure 7.14, Figure 7.18, and Figure 7.22). That is, the clustering was not due to randomness or chance. These cluster maps increased the level of confidence in the findings from this climate change assessment.

The results from the climate change assessment were important for several reasons. Firstly, the findings gave foresight into the possible impacts that climate change could have on the frequency and distribution of pipe bursts in the future. The assessment showed possible outcomes of what could happen regarding pipe burst incidents if there were no interventions implemented to improve the state of the pipe network. This provided insight into the broader implications of climate change.

Secondly, the findings from the assessment added to the understanding of the possible impact of climate change on pipe burst risk. Climate change can pose a triple threat on a water supply system – *source, consumers, and infrastructure*. The effects of climate change on water sources and water demand had been widely investigated but not much research had been conducted on the impacts on the WDN itself. Therefore, this assessment aided filling this current gap in the literature (see Section 2.6.2).

Thirdly, building on the abovementioned, this study being conducted in Barbados was important for building knowledge at the local level and even more broadly for other SIDS which are also expected to experience the effects of climate change. Given that there was a limited number of pipe failure studies conducted in these places to begin with, then this assessment on the impacts of climate change on pipe bursts brought a degree of needed insight and a framework that can be adopted.

Lastly, based on the practical implications of this work and the appreciation that the effects of a changing climate are already being seen globally and locally, it would be in the interest of BWA to move from its current reactive approach of

addressing pipe bursts and more towards being proactive and giving forethought to planning. This was a gap in the approach to rehabilitation interventions in operations at the BWA and likely in most of the other water utilities in Caribbean SIDS (and similar regions). While works such as Mackey, Cashman, and Cumberbatch (2014) and Halcrow Inc. (2012) presented recommendations for pipe prioritisation and rehabilitation for the BWA, methods were subjective and climate change was not considered. It is important that future planning contain initiatives that seek to address current and anticipated issues, bridging the gap between research and application.

8.3 Implications of the Research

Here, the key implications of this study were discussed in the context of the four major research findings (see Section 8.2). The types of implications considered are theoretical, methodological, and practical.

8.3.1 Categorisation of Local Pipe Burst Factors by Importance

Theoretical Implications

Pipe bursts can be defined as a function of explanatory factors (Rezaei, Ryan, and Stoianov 2015, Skipworth et al. 2002). This concept was used as the basis in most pipe failure studies, including this research. From the various studies in the literature, an extensive compilation of factors was reported as contributing to burst occurrences (see Table 2.1). For this study, knowledge gained from data mining was used to identify and categorise pipe burst factors by importance – establishing that some factors were more significant than others. The factors found to be most significant (i.e., rainfall, pressure, diameter, soil type, and material) were congruent with other studies (see Section 2.3). However, pipe installation era (which was used as a proxy for pipe age) was not found to be a significant factor for frequent bursts though studies such as Berardi et al. (2008) consider the age of a pipe as the most important factor. Alternatively, the nature and constraints of the data (see Section 4.7) and the assumptions made (see Section 4.4.3) might have affected this finding. Nonetheless, other studies did not identify pipe age as the sole cause of peak bursts (Barton et al. 2019; Boxall

et al. 2007; O'Day 1982). There is an ongoing debate in the literature on the level of importance of factors.

The debate on pipe burst factors exists because burst events cannot always be explained by the set of physical and environmental factors usually attributed as the main causes of failure (Rezaei, Ryan, and Stoianov 2015). The variation of factors and conditions at a local level can lead to different findings about pipe burst behaviour (Rezaei, Ryan, and Stoianov 2015; Rajani and Tesfamariam 2007). Accordingly, the expression (see Equation 1) used to mathematically define pipe bursts by Skipworth et al. (2002) should be adjusted to include weights that would represent the relative importance of factors (see Equation 2).

Equation 2 - Revised burst equation (where x and a represent explanatory factors and weights, respectively).

$$\text{Pipe bursts} = f(a_1 \cdot x_1 + a_2 \cdot x_2 + \dots + a_n \cdot x_n)$$

In this study, factors were categorised as relevant, significant, or neither in relation to explaining peak pipe burst occurrences (see Section 6.4). In so doing, pipe age was categorised as a relevant factor. Although the vulnerability of a pipe generally increases with age, this study found that it was the combination of significant factors that contributed to frequent bursts. Factors categorised as relevant were essentially static and did not rapidly fluctuate on a short-term basis (i.e., hourly, daily, and weekly). Factors categorised as significant contained both static and dynamic variables. However, static factors which were classified as significant (i.e., soil type and pipe material) had a role in the dynamic effect of physical interactions with other variables. Taking this into consideration, it was determined that peak bursts are mostly driven by dynamic factors.

From the set of input variables used in the analyses, connection density was the only factor that was not categorised as relevant or significant since it did not exhibit any correlation with peak burst occurrences. While it is theorised that a higher density of connections can indicate potential weak points along the pipeline (Skipworth et al. 2002), connection density is rarely used as an

explanatory variable in the analysis of pipe failure. A probable reason for the omission of connection density from such studies is the lack of a standard measure for data modelling and analysis.

Methodological Implications

In generating the classification of factors, it was critical to carry out an extensive data gathering scheme for creating a database as robust as possible (see Chapter 4) and make efficient use of available data – an objective of this research (see Section 3.2). For the data mining analysis, three discrete pipe burst datasets at different temporal and spatial scales were sourced directly from the BWA. Additional data for explanatory factors were sourced in various formats from different sources which resulted in the inclusion of multiple variables across the datasets (see Table 4.14). For example, soil type was a newly digitised factor assembled for this work. Data were also extensively cleaned (see Sections 4.3 and 4.4) with many inconsistencies and gaps being addressed. This approach, along with the way in which some variables such as land-use and connection density were derived, was an important step in identifying possible factors and it also enhanced the data mining process given the structure of the data.

The GIS layer that was created to represent land-use was checked and validated against satellite images. However, no validation data existed for connection density. This may provide a rationale for connection density not being classified as relevant or significant. Connection density was derived by using spatial analysis tools in a GIS to divide the number of connections by the length of the associated pipe. However, there was a drawback as the number of connections was estimated from the geo-locations of water meters which may have contained some errors. No other instances of measuring and including connection density in pipe failure analysis were found in the literature besides Mackey, Cashman, and Cumberbatch (2014) and Cooper et al. (2000). While the contribution of connection density to pipe bursts has been recognised and theorised (Skipworth et al. 2002), there is still a gap in its use in pipe failure analysis.

The categorisation was based on the findings of the data mining analysis conducted with SOMs. The stepwise rules (see Section 5.3) that guided the data mining analysis created an iterative process that systematically informed factor selection and resulted in multiple pipe burst models. After the completion of the data mining analysis, it was observed that the least number of models that a factor (i.e., pressure and connection density) appeared in was three. Therefore, it was reasoned that if a factor aligned with peak bursts (which was easily discernible and the least subjective pattern to detect) in at least two models (i.e., a majority) establishing a correlation, then it was considered significant. If it aligned with pipe bursts a risk level less than peak in at least two models, then it was considered relevant. Thus, the categorisation was dependent on the number of models generated from the data mining analysis.

Practical Implications

In the classification, pipe age was not categorised as a significant but as relevant. That is, while pipe age had a role in burst incidents, it was not deemed as a key factor in the occurrence of peak bursts. Thus, it was implied that changing the oldest pipes in the network would not necessarily contribute to a reduction in water losses from bursts. While the effects of both significant and relevant factors increased the vulnerability of the pipe network to deterioration, strategic practical interventions should seek to first address significant factors. In doing so, in theory, it should reduce the occurrence of frequently bursting pipes and hence result in a considerable decrease in water losses.

8.3.2 Characterisation of the Behaviour of Peak Pipe Bursts

Theoretical Implications

The literature review provided sufficient background on the behaviour of pipe burst factors, identifying those that were commonly used in pipe failure studies (see Section 2.4.2). For this study, an inductive approach was used to identify which factors were most significant in contributing to peak pipe bursts in the water distribution system in Barbados. Data mining was used as the main form of analysis. The primary method chosen to detect patterns in local pipe burst

behaviour was SOMs. The use of SOMs for pipe burst modelling and investigating factors that influenced pipe failure in a WDN was novel. The combined use of SOMs to identify patterns and trends in the data and EPR to corroborate the findings from the SOMs with mathematical expressions proved effective.

The stepwise rules developed for data mining (see Section 5.3) promoted a parsimonious approach by beginning analysis with the most basic models and then sequentially adding and subtracting other variables. This was most effective for the SOMs since with EPR the law of parsimony was already incorporated in the process of producing mathematical expressions from regression. As variables were added in a stepwise manner, with consideration to the implications of outputs from previous analyses, the additional patterns detected enhanced the understanding behind observed trends. For example, the detection of a seasonal pattern in pipe bursts incidents led to the inclusion of rainfall and eventually other variables such as soil type and pipe material. Ultimately, the effect of the interactions between these factors was discerned and supported by the literature. EPR also showed some improvement when the Simple Dataset was partitioned by rain season, and the variables in combination terms matched the findings from SOMs (see Table 6.4). Models from both SOMs and EPR showed some level of prediction which was enhanced and improved by adding variables such as rainfall and soil type.

Trends in local pipe burst behaviour aligned with other observations from the literature. These findings were summarised below:

- Diameter: pipe burst frequency is inversely correlated to pipe diameter (Barton et al. 2019; Bruaset and Sægrov 2018; Shi, Zhang, and Ho 2013)
- Material: cast iron pipes are more susceptible to deterioration and bursts than ductile iron or plastic pipes (Barton et al. 2019; Wols et al. 2019; Bruaset and Sægrov 2018; Boxall et al. 2007)
- Soil Type: shrinking and swelling clays can exacerbate bursts in places with pronounced dry and wet seasons (Abbey, Eyo, and Ng'ambi 2020; Wood and Lence 2009; Krenz, Lee, and Owens 2006)

- Rainfall: higher pipe burst frequencies are typically observed during periods of lower rainfall (Barton et al. 2019; Wols and van Thienen 2016; Gallage, Chan, and Kodikara 2012)
- Pressure: higher pressures that exceed particular thresholds can promote pipe burst occurrences especially in smaller diameter pipes (Barton et al. 2019; Punurai and Davis 2017; Marzouk, Hamid, and El-Said 2015; Savic and Walters 1999)

Methodological Implications

The application of SOMs as the primary method for detecting patterns in the data proved effective since more trends and relationships were discerned from SOMs than EPR. In theory, the use of EPR as the secondary method was to corroborate and quantitatively assess the strength of the correlations found from the SOMs analysis. However, based on the COD of some the models derived with EPR, these correlations were weak.

In several of the EPR runs, there was a very small addition to the COD when more terms were added to expressions which reduced parsimony. Models with the highest CODs were generated using the time series data of the Simple Dataset although the Compound Dataset also had time series data, the inclusion of binary variables appeared to reduce model performance. For the Complex Dataset, which was not time series data, COD was low or near-zero. Generally, in the literature, EPR analyses was utilised for prediction with homogeneous time series data rather than for knowledge discovery with complex data. From this study, it appeared that SOMs was better able to handle noise in the data than EPR. Another reason that may explain the lower performance of EPR was that whole numbers likely made better regression targets (i.e., pipe bursts) with the Simple and Compound Datasets instead of the near-zero decimal values that resulted from normalising pipe bursts with pipe length in the Complex Dataset. Likewise, zero values in binary variables might have had a similar effect in lowering model performance.

Models produced with SOMs are not generally validated in the traditional way of splitting the data into training and testing. Thus, models derived with EPR

served to corroborate the correlations found from the SOMs analysis. Models generated with EPR can be split into training and testing data for validation. However, this was not carried out since in the preliminary EPR analysis using the Simple Dataset, which yielded the best performing EPR models, there was no change in model outputs or the COD when the data were split into training and testing data. Given that, and the fact that the purpose of the EPR models was not to predict but to be solely used to assess the relationships among variables, the sets of EPR models were not tested and validated (see Section 5.4.2).

Practical Implications

Previously, with only a partial understanding of pipe burst behaviour, the water utility would have been managing its network with a considerable degree of ambiguity. However, now having gained a fundamental understanding into pipe burst behaviour, a few practical implications were identified. The effect of rainfall, particularly its marked seasonality, on soil moisture influenced soil movement. The resulting movement likely placed bending stress on pipes and reduced soil stability which in turn could promote pipe burst incidents. Thus, water utilities should give consideration to soil type especially when native soils are used as pipe bedding. If pipes are not suited for such conditions, it is expected that bursts would continue in such hotspot areas. Furthermore, high water losses from WDNs through bursts and leaks would also affect soil moisture levels. This too can contribute to soil movement. Given the above, it is considered prudent to investigate the drivers of pipe failure prior to carrying out any interventions or instituting operational procedures regarding the pipe network.

Data mining was an effective approach for identifying the drivers of pipe bursts and providing insight into pipe burst behaviour. The methods were useful when methodologically applied. Patterns and relationships were found in multivariate data that contained both numeric and non-numeric variables at different scales – which was not common in pipe failure studies. Methods for data mining, particularly SOMs (as seen in this study), were capable of handling large,

complex datasets. The increased adoption of these methods in pipe failure studies could prove to provide more insight into variables, especially those that were not able to be included using traditional statistical analysis.

8.3.3 Pipe Burst Risk Model for the Local Pipe Network

Theoretical Implications

The methods used in the development of the pipe burst risk model, particularly SOMs and Fuzzy Logic, were not purely quantitative, data-driven methods. There were qualitative aspects in the implementation of these methods and interpretation of the outputs (see Section 5.4). The trade-offs between the approach being data-driven and knowledge-based were appropriate for this study. The patterns detected from data mining across the multivariate, non-linear datasets were discerned using the unsupervised machine learning capabilities of SOMs. The knowledge derived which was logically and qualitatively synthesised with Fuzzy Logic, was ideal for the creation of a single pipe burst model.

In the FIS, rain season was added as an input variable. While the linguistic variables for rain season were ‘dry’ and ‘wet’ (see Figure 6.24 - Fuzzy membership functions for rain season.), the input was a numeric value that represented each month. Thus, in theory, this pipe burst risk model included a time aspect. The implication of this was that the changes in the risk of pipes to frequent bursts can be evaluated across each month of the year.

Conceptually, the pipe burst risk model depicted the pipe burst behaviour of the local WDN. This depiction was comparable to a Likelihood of Failure (LOF) which is an aspect considered in the standardised industry approach for the prioritisation of pipes for replacement. However, the Consequence of Failure (COF) is not captured in the pipe burst risk model. That is, for example, the model does not take into consideration the implications of bursts to a larger diameter pipe (or trunk main) than a smaller diameter pipe (or service pipe).

Methodological Implications

The method of fuzzy logic was used to develop the pipe burst risk model. Fuzzy membership functions were created for all input and output model variables (see Section 6.5.1). These user-defined functions increased flexibility since models based on fuzzy logic have the capacity to represent vagueness, interpret imprecise data, and handle uncertainty. In the fuzzy membership function for diameter (see Figure 6.21), a 7-inch diameter pipe had a 0.68 degree of membership as “medium” and a 0.32 degree of membership as “small”. Interpreted plainly, a 7-inch diameter pipe was considered a medium-sized pipe for the most part but it was still somewhat small. In the same manner, model rules pertaining to burst behaviour of small and medium pipes were applied to all 7-inch pipes according to the degree of membership. This type and handling of ambiguity in data was particular to fuzzy logic, whereas EPR required exactness. Considering that the data for this study were gathered in different formats from several sources, varied on temporal and spatial scales, and consisted of quantitative and qualitative variables, then the choosing a fuzzy logic was an appropriate method – especially for capturing the findings that were synthesised from the data mining analysis.

SOMs and EPR proved to be useful tools for data mining, detecting patterns in pipe burst behaviour. However, there were shortcomings in applying these methods. For example, the highest performance metrics for the COD of models derived with EPR ranged between 49% and 56%. These models contained two explanatory variables – diameter and rainfall – and indicated the overall effect of the type of rain season. However, other key significant variables such as pressure were not included in the models. Additionally, EPR models did not clearly distinguish the effects of the various parameters of factors (i.e., the types of soil or types of pipe material). SOMs and EPR, were primarily driven by data and carried out processes in a black or grey box mode without intervention from the user. Therefore, by encoding rules for the FIS, the resulting pipe burst risk model improved on previous model gaps by building on domain knowledge. Procedures for encoding rules and defining fuzzy membership functions were

set to limit the subjectivity of user input (see Section 5.4.3). Thus, these sets of methods – data-driven (SOMs and EPR) and expert-driven (fuzzy logic) – were carried out to complement each other.

The pipe burst risk model was tested with historical data of service repairs. The results of the validation (see Table 6.8) implied that the model performed relatively well in classifying low risk pipes, fair in classifying medium risk pipes, and could be improved in classifying high risk pipes. However, the validation data had several limitations (see Section 6.5.3) and assumptions about that data were also made to differentiate pipes that had single or multiple bursts. Additionally, the validation data appeared to underestimate the number of pipe bursts and was well under the thresholds used to define high and peak pipe bursts in the pipe burst models (see Section 6.5.3). By definition, the service repair data identified where efforts of the BWA were concentrated and only represented a subset of total pipe burst incidents. Thus, in the instances where the validation data did not show a greater match of pipes with multiple bursts to pipes classified as high risk in the hotspot model might not necessarily indicate that the pipe burst risk model was inaccurate as a whole. It was more likely that those pipes indicated areas where BWA efforts should be placed. Ultimately, higher quality and quantity of validation data was needed.

Validating models that depict or predict pipe failure is usually difficult. These models can only be validated with data that are based on the repair work and asset management information maintained by a local water utility. Such data can be heavily biased and dependent on many factors:

- If and when a burst becomes visible and is reported.
- Institutional policy and procedures for collecting and reporting pipe bursts and repairs.
- Quality of record keeping (i.e., missing or erroneous data entries or delays in recording).
- Technology that is used for burst detection.

Therefore, expectations of the performance of such models should be managed and take into account the levels of uncertainty and limitations in validation data.

Practical Implications

In developing the pipe burst risk model, the behaviour of pipe burst occurrences in the local WDN was characterised. This was fundamental for understanding, assessing, and addressing pipe failure. In this study, this pipe burst risk model was used to assess the impacts of climate change. Thus, the model and associated hotspot map could also be useful and practical for the BWA.

The pipe burst risk model can support decision-making in relation to planning interventions for pipe replacement and rehabilitation. Since the pipe burst risk model only took into account significant factors, then hotspot map of the pipe network that was produced from using the model focused on areas of high and peak bursts. Therefore, targeting higher risk pipes for repair or replacement should help to reduce water losses substantially. If hotspots in the pipeline are not addressed, observed rates of frequent pipe bursts should be expected to continue and possibly worsen, and not necessarily in a linear fashion. Hence using a hotspot approach to target interventions would be a useful tool to maximise the use of scarce resources.

8.3.4 Assessment of Climate Change on Pipe Failure

Theoretical Implications

The main conclusion drawn from the assessment was that there is some evidence to suggest that climate change will have an effect on pipe burst behaviour in coming years. This study contributes not only to the global number of pipe failure studies that included a climate change assessment but also to the limited few conducted in tropical and sub-tropical areas. The results suggest that climate phenomena such as temperature and frost are not the only variables of interest, noting that the studies in which these featured were in works about WDNs found in higher latitudes (see Section 2.6.2).

It is crucial to re-emphasise that the climate change assessment, in and of itself, did not classify rainfall as a significant factor. The classification of significant factors was derived through data mining and pipe failure modelling (see Sections 6.2 and 6.4). The climate change assessment did however examine how

different distributions of rainfall in various scenarios would influence pipe bursts occurrences. In theory, based on the findings of the assessment, if future rainfall was below or much below the historical average, smaller diameter pipes of iron material would be more susceptible to frequent bursts in areas such areas that contained Smectite Clay. This was based on the proposition that the shrinking and swelling properties of Smectite Clay can promote soil movement especially in places with defined dry and wet seasons (Krenz, Lee, and Owens 2006). Conversely, while wetter conditions can promote swelling movements in Smectite Clay, saturated soils do provide more soil stability in theory (Barton et al. 2019). Hence, there was a decrease in the number of higher risk pipes in RCP8.5 in 2035 (the wettest scenario with rainfall much above the historical average). Thus, it was the influence of climate change on rainfall and the dynamic interactions of soil type, pipe diameter, pipe material and soil moisture that characterised the threat of climate change on a WDN.

The hotspot maps generated for each climate change scenario only provided insight into possible outcomes for the frequency and distribution of risk of pipes to peak bursts. Since these maps represented plausible conditions in the future, data to validate these maps do not yet exist. Nevertheless, these maps provided useful insight (as described in the above paragraph) and the spatial and clustering analyses that were conducted identified statistically significant clusters on the hotspot maps.

The data for the rainfall projections used in this study represented possible distributions under IPCC climate change scenarios. While there was confidence in the reliability this data that were sourced from Charlery and Nurse (2018), it was outside the scope of this work to validate these downscaled projections but rather to use the data to demonstrate potential impacts on local pipe burst behaviour if the scenarios were to manifest.

Methodological Implications

The methodology developed and used in this research was novel. It adopted a geospatial approach to incorporate downscaled climate change projections of rainfall at a 1km x 1km resolution into a quantitative pipe failure model. The

results showed that better spatial coverage of climate data is highly desirable. To carry out the analysis, GIS was used as a data linking tool to associate projections with the input data for the fuzzy risk ranking model that generated outputs at the sub-district level. Thus, high-resolution geospatial outputs from climate models can be fitted into pipe failure models. The general framework of this methodology can be used by other pipe failure studies to facilitate the assessment of climate change projections especially given the lack of such applications in the literature. The framework can be widely applied despite differences in the formats of climate data and configurations of pipe failure models. Furthermore, all inputs require geospatial properties in order to support data linkage and maintenance of data resolution.

The means of complementing a geospatial analysis platform with a pipe failure model also proved effective for scenario analysis. The outputs generated from the scenario analysis were sufficient to visualise and allow comparative insight into the changes of hotspots across time and space. However, as inherent with any predictions of the future, there was no data to contrast with or validate hotspot maps for climate change scenarios. Nonetheless, these hotspot maps along with associated cluster maps, were still useful in depicting and anticipating future pipe burst behaviour (see above *Theoretical Implications*) and had insight for current and future applications (see below *Practical Implications*). This type of analysis was critical in contributing and expanding the body knowledge and for supporting decision-making in the operations and planning in a water utility company.

Practical Implications

In characterising peak pipe burst behaviour, it was determined that the cycle of wetting and drying (caused by rainfall particularly during the dry season) increased the vulnerability of iron pipes to bending stress and corrosion in clay soils. It thus followed that pipe burst rates were exacerbated in areas which were predicted to experience drying below the historical average because of climate change. Based on the GCM used by Charlery and Nurse (2018), the distribution of rainfall appears to shift considerably over the 15-year span in each scenario.

In RCP2.6, conditions in certain areas were drier than normal in 2035 then became wetter in 2050; and the converse was projected for RCP8.5. Despite the order in which it was predicted to occur, substantial drying was anticipated in both RCP scenarios. Based on the findings in this study, a reduction in rainfall likely promoted soil movement that had the effect of promoting frequent burst occurrences in the aged and fragile pipe network. Therefore, it would be prudent for the local water authority to anticipate this future shock which scenario analysis revealed.

The forecast horizon of the projections in the climate change assessments were medium and long-term. Nonetheless, in the analysis, rainfall was examined at a monthly interval. Thus, it was implied that short-term climate phenomena such as droughts may also have an impact on pipe bursts.

8.4 Limitations of the Research

Some of the major findings from this research were broadly applicable to other pipe networks in similar conditions – in particular, how significant factors influenced peak pipe bursts and the anticipated behaviour of pipe bursts in the future due to climate change projections of rainfall. Despite the degree of generalisability of the research, there were limitations specific to this study that affected the results. Most of these limitations stemmed from data deficiencies (see Section 4.7) and in validating the pipe burst risk model (see Sections 6.5.3 and 8.3.3). Other limitations that impacted the research were discussed below.

In the literature, most pipe failure studies used pipe material to cohort pipes into homogeneous groups for analysis to improve model performance. This technique was also adopted for pipe failure studies that used EPR analysis (Giraldo-González and Rodríguez 2020; Farmani et al. 2017; Berardi et al. 2008). Out of the three datasets used for pipe failure modelling in this study (see Section 4.5), only the Complex Dataset included pipe material. A preliminary EPR analysis was carried out to test if model performance was enhanced if pipes were cohorted by pipe material. However, model performance had not improved (see Section 5.4.2). Given that, splitting pipes into cohorts

according to material was not carried out. Furthermore, it was imperative that procedures for data mining were consistent for analysis across each dataset.

Although pipes were not split into cohorts according to pipe material, partitioning input into training and testing data for EPR analysis was a common technique to check for generalisability. Thus, a preliminary EPR analysis was carried out with the splitting of the Simple Dataset into training and testing data to test for changes in model outputs and performance due to randomness in the data. However, there was not much change in model accuracy. Thus, partitioning of the data into sets for training and testing was not carried out for analyses. This was justified since the stepwise iterative rules that guided the data mining analysis (see Section 5.3) resulted in splitting of the data by factors across all datasets and created multiple models (see Sections 6.2 and 6.3). Besides the metric used to indicate the performance of each model (i.e., COD), the findings from all models were compared and synthesised as a means of validation (see Section 6.4). Furthermore, since the goal of implementing the EPR analysis was knowledge discovery and not predicting pipe bursts, observational data was used in its entirety rather than being partitioned.

8.5 Validity and Reliability of the Research

The rigour of a research is typically evaluated on two dimensions: validity and reliability. Validity refers to how accurately a research measured what it had set out to measure while reliability refers to how consistently the methodology and methods of a research would be if repeated (Heale and Twycross 2015). This research aimed to identify the significant drivers of pipe bursts in the WDN in Barbados and to evaluate the implications of climate change. The approach of modelling pipe bursts was taken in an attempt to achieve both parts of the research aim. The techniques used to improve the rigour of this research, and provide a workaround for some of the limitations, were presented in this section.

The validity of the research was enhanced by:

- Adopting the widely accepted concept that pipe bursts can be modelled as a function of its explanatory factors as the basis of this study.

- Incorporating and analysing three distinct datasets that reported pipe bursts occurrences at different temporal and spatial scales.
- Measurably improving data quality before analysis were conducted.
- Applying multiple methods for data mining and comparing findings in to derive a consensus on insight about pipe burst behaviour.
- The use of a FIS for creating a single pipe burst risk model accounted for, to a certain extent, inexactness, randomness in data, and shortcomings in the synthesis of findings.
- The use of service repair incidents to compare with the hotspot model of the network provided a limited form of validation.
- The clustering analysis conducted on the hotspot maps was a means of increasing the confidence in the results of the pipe burst risk model for the network despite limited validation data.
- Using rainfall projections from four scenarios to assess the implications of climate change.

The reliability and rigour of this research was enhanced by:

- Replicable techniques used to build the geo-spatial datasets.
- The methodological modelling framework and analytical process developed for this study.
- Applying stepwise iterative rules that guided how each factor in each dataset would be analysed with each method. This increased the likelihood that if the data mining analysis were repeated, the similar results and findings would be made.
- The creation of a template (see Figure 5.4) to guide how fuzzy membership functions were defined made increased the consistency and reliability of this process.
- A set of procedures created to classify factors by importance and identify significant factors.

8.6 Recommendations

Most of the limitations identified in this research pertained to data (see Sections 4.7 and 8.4). The issues mainly consisted of data gaps, deficiencies in data

quality, and challenges in data not being available or collected. Addressing these issues was outside of the scope and control of this study. Therefore, recommended activities that can be carried out in an attempt to resolve these limitations were given in this section. Despite the shortcomings, the research can be used to facilitate further work. Thus, recommendations for further work were also given. Additionally, ways for the local water utility to address the risks to frequent pipe bursts based on the findings of this study were proposed.

8.6.1 Recommendations to Address Data Limitations of the Research

The research outputs also support good practices for improving water utility data. Despite difficulties with data, this research exhibited how data can be used to provide fundamental insights and the need to enhance data generation. Suggestions for good practices that can be adopted by the BWA were described below.

- (1) Activities for collecting, cleaning, compiling, linking, and updating internal databases relating to pipe bursts, pipe burst factors, and the actual pipe network should be carried out routinely. There were many challenges with the data, especially having to fix inconsistencies, that were experienced during this which can be resolved if proper maintenance of data is routinely exercised. If this is not addressed, the poor state of data will continue. Furthermore, analyses such as data mining are dependent on data quality. The BWA should also establish and maintain relationships with other local entities that host relevant external data for the purpose of facilitating regular data exchanges.
- (2) The deficiencies of the hydraulic model should be addressed in order to improve its accuracy, and hence, allow for better use of such an extensive and expensive tool. The database created in this study can help in filling some of the gaps in input data for the hydraulic model which usually needs to be updated and recalibrated regularly. With a more accurate model, the BWA can pinpoint areas under high pressure. This would also be instrumental in developing a pressure management strategy and assist in facilitating any internal water audits.

- (3) Pipe bursts observed by the general public can assist the BWA in identifying and addressing bursts as quickly as possible. The BWA has made provisions for bursts to be reported via telephone, fax, postal mail, electronic mail, and a help desk located in the main office building. However, the BWA should seek to develop an application for mobile devices (such as cellular phones and tablets) that would allow real-time reporting of leaks. Information such as a global positioning system (GPS) location, address, pictures, videos, date, time stamp, etc. can be received instantly. United Utilities, a water utility registered in England and Wales, reported good success of a mobile application that allows customers to report leaks (United Utilities 2021).

8.6.2 Recommendations for Further Work

Data mining was an effective approach for identifying the main drivers of pipe failure and understanding pipe burst behaviour. Further work can seek to optimise some of the processes within the analytical framework through automation. For example, once input data has been prepared, the process of applying multiple data mining techniques with machine learning capabilities to assist in finding the best-fit model can be automated. Automating processes as far as possible can help to shift the model from static to dynamic. There may be potential for creating software or online tool that can assist in this process of automation.

If the automation process as described above increases the performance of pipe burst risk modelling, then the applications of the model can be expanded with further work. The approach used to derive the pipe burst risk model was comparable to the likelihood of failure (LOF) since the physical, operational, and environmental factors that promoted pipe bursts were assessed. A next step could be the development of the consequence of failure (COF), which considers the social and economic implications of pipe failure. It is a common industry practice to combine the LOF and COF for the generation of a pipe prioritisation plan. In some studies, a limitation in deriving the LOF and COF was the reliance on input and methods that were mainly qualitative and subjective (El-Abassy et al. 2019, Li et al. 2015, Mackey, Cashman, and Cumberbatch 2014). However,

this study used an inductive approach with mostly quantitative methods to minimise subjectivity. Thus, further work can produce the COF to complement the risk ranking (i.e., LOF). The work can also consider the financial costs associated with the civil works involved with pipe replacement (i.e., costs of materials, labour, and other expenses) and carry out a cost benefit analysis.

Another pertinent means of expanding the research stems from rainfall being categorised as a significant factor of pipe burst occurrences and the implications from the assessment of climate change on bursts. The research evaluated the medium and long-term impacts of climate change on the distribution of pipe bursts using downscaled rainfall projections. Further research can consider short-term impacts of other climatic phenomena such as droughts or floods. Over the past two decades, Barbados has been negatively affected by several instances of droughts and flash flooding from intensive short-term weather events. Such events that have considerable influence on fluctuations in soil moisture (which can also be incorporated as a possible explanatory factor in the model) can be studied given the relationship between pipe bursts and the shrink-swell characteristics of the predominant local soil - Smectite Clay.

Tools that generate predictions for short-term rainfall events can be coupled with an improved pipe burst risk model. This coupling can provide possible short-term indication of burst behaviour which could be used to prepare response programmes. Examples of such tools are given below:

- Stochastic weather generators can produce weather data for a particular location in the format of time-series. A continuous time-series of short-term weather data can also allow for the incorporation of variability.
- Climate Predictability Tool is a software package that produces downscaled seasonal climate forecast models using model output statistics and general circulation models (GCMs), sea-surface temperatures, or other appropriate predictors (Mason and Tippet 2016). This tool was developed at the International Research Institute for Climate and Society at Columbia University.

- The Caribbean Climate Outlook Forum (CariCOF) Drought Outlook is an overview and guide of the drought concerns in the Caribbean Region which serves as an early warning tool. This tool was developed in partnership by CariCOF and the Caribbean Institute for Meteorology and Hydrology.

8.6.3 Recommendations for the Local Water Utility

The Netherlands boasts one of the lowest leakage rates among water utilities across the globe – approximately 6% (EurEau 2017). The low level of leakage was credited to cyclically repeating the actions involved in research and maintenance (Rosenthal 2016). This real-world example exemplifies some of the benefits of practically applying lessons learnt from research and fostering. One of the objectives of this study was to show how the approaches adopted for this research have the potential to inform the work of water utilities such as the planning of pipe replacement interventions by the BWA. In this way, several recommendations were given.

- (1) A strategic pressure management plan should be developed to ensure that certain parts of the pipe network (particularly smaller diameter pipes made of iron materials) are not subjected to extremely high pressures since it was observed peak bursts correlated with pressure levels which exceeded industry standards. Proper pressure management should be written and adopted into the operations of the water utility. These actions would assist in decreasing the high levels of water loss.
- (2) Procedures for the selection and laying of pipes should be revised since the effect of soil movement (due to the changes in moisture in Smectite Clay) seems to explain the higher frequencies of bursts occurring in smaller diameter pipes of iron material found in Smectite Clay especially during the dry season (see Section 4.4.4). Actions such as providing supportive pipe bedding, using less expansive soils for a more neutral and stable back-fill, and selecting a more flexible pipe material can assist in reducing the high number of pipe bursts.
- (3) In areas that are projected to experience rainfall that is much below the historical average (see Figure 7.6), it is expected that soil movement will

increase due to the shrink-swell characteristic of Smectite Clay. If pipes that are laid in that soil type in those areas are to be replaced, then consideration should be given to using pipe materials that are more flexible and tolerant to bending – such as plastic-based materials (which were less than 3% of pipe materials identified in BWA’s pipe network – see Table 4.6). This action would make the pipe network more resilient to external shocks such as climate change. In the industry at large, PVC pipes are being widely used for new pipe installations in WDNs (Duffy 2019). Therefore, it is probable that pipes that have been or will be replaced BWA will be PVC or plastic-based material.

- (4) The hotspot map of the pipe network which identified pipes that were most vulnerable to frequent bursts can be used to inform and complement on-going and future pipe replacement efforts. Currently, the BWA typically responds to pipe bursts in a reactive manner – addressing bursts only after it has been seen and reported. The hotspot map can not only inform the prioritisation of pipe replacement but also assist in proactively direct efforts to higher risk areas.

8.7 Chapter Summary

In this chapter, the following were considered as major research findings:

- (1) The identification of significant factors (i.e., rainfall, pressure, diameter, soil type, and material) that contribute to frequent pipe bursts in the aged WDN in Barbados.
- (2) The seasonal trend of higher burst incidents during the dry season and insight into understanding the dynamics between peak pipe bursts and significant factors.
- (3) The pipe burst risk model that was used to characterise pipe burst behaviour and identify potential hotspots in the local WDN.
- (4) The impact of climate change on altering the amount and distribution of future rainfall can influence the location of hotspots.

Most of these findings were congruent with observations from other related works in the literature. The implications from this research were used to

contribute to ongoing discussions in the literature, particularly for SIDS, especially on the impacts of climate change on pipe burst as this was an existing gap. Recommendations were given to address some of the limitations identified and for further work. In the next chapter, which is the final chapter in the main body of the thesis, conclusions were drawn and contributions of the research were given.

9 CONCLUSION

9.1 Chapter Introduction

The aim of this chapter is to reflect on the research as a whole, drawing out the main conclusions and highlighting the significance of the study. Here, the following is presented:

- (1) Conclusions coming out from the research which addresses the research aim and research questions.
- (2) Contributions and relevance of the research especially to the wider body of knowledge.
- (3) A final reflection.

This chapter is laid out in this way to facilitate a link between what had been initially set out to be accomplished and what was actually achieved.

9.2 Revisiting the Research Aim and Research Questions

The research aim was *to identify and understand which factors contribute to pipe bursts in the WDN in Barbados, and thus better inform the planning of pipe replacement interventions that considers the implications of climate change* (see Section 1.4). A total of eight possible explanatory factors of pipe burst occurrences, identified from the literature, were investigated in this study through a data mining analysis. The patterns and trends detected from the analyses carried out, contributed to the development of the conclusions pertaining to identifying the most important factors (RQ-1, see Section 9.2.1) and understanding local pipe burst behaviour (RQ-2 and RQ-3, see Section 9.2.2 and 9.2.3). An assessment that was conducted found that the likely impacts of climate change on pipe bursts may be significant (RQ-4, see Section 9.2.4). The implications of these findings were used as the basis for considerations and recommendations to the local water utility (RQ-5, see Section 9.2.5).

9.2.1 Research Question 1

RQ-1: What are the main drivers of pipe burst occurrences in the WDN in Barbados?

Based on the results of this research work, it was concluded that the main drivers of pipe burst occurrences in the WDN in Barbados are rainfall, pressure, pipe diameter, soil type, and pipe material. These factors were categorised as significant – determined as having the highest contributions to peak pipe burst occurrences (see Section 6.4). These findings are based on the results of a data mining analysis that was guided by iterative stepwise rules (see Section 5.3) and conducted via pipe burst modelling using SOMs and EPR (see Sections 6.2 and 6.3). There is a high degree of confidence that the categorisation of significant factors is accurate since it was derived by comparing multiple pipe burst models generated from applying two different methods across three discrete datasets.

9.2.2 Research Question 2

RQ-2: 2. What patterns can be identified in the behaviour of pipe bursts in the WDN in Barbados?

Several patterns of the pipe burst behaviour in the WDN were identified:

- (1) Generally, across the network, the rate of pipe bursts was considerably higher in smaller diameter pipes than larger diameter pipes. Particularly, high pressures (above 115 PSI) contributed to the occurrence of frequent pipe bursts in smaller diameter pipes made of iron material (see Table 6.5).
- (2) Pipe bursts were highest during the dry season (i.e., between December and May). This was considered to be a very critical pattern as it prompted the inclusion of rainfall into analysis which allowed for a closer examination of the relationships between the amount of rainfall and frequency of burst occurrences.
- (3) Smaller diameter pipes made of iron material were more prone to bursts likely due to the shrinking and swelling action caused by changes in the soil moisture content of Smectoid Clay (a predominant soil type in Barbados) particularly during the dry season. The movement of the soil and decreased

support caused by drying conditions would place additional bending stress on aged pipes that have already surpassed its expected life cycle.

While pipe age certainly accounts for deterioration, this study found that dynamic effects such as high internal water pressure (and possibly water hammer) and soil movements appear to account for frequent pipe bursts. Furthermore, it was concluded that small diameter pipes in swelling and shrinking clays increases risk to peak bursts. In conjunction with rainfall, the effect of dry and wet seasons would contribute to marked shrinking and swelling. Therefore, these factors should be considered in combination.

9.2.3 Research Question 3

RQ-3: In Barbados, which areas of the pipeline in the WDN are most susceptible to frequent pipe bursts?

The areas of the local pipeline that are most susceptible to frequent bursts were identified in the hotspot map of the network (see Figure 6.28). The conclusion was that these areas of the pipeline had a combination of characteristics associated with the main drivers of pipe bursts. Areas contained pipes with the following attributes: a small or medium-sized diameter (less than 10-inches), subjected to high pressures particularly over 115 PSI, or made of iron material. Areas also had the following environmental conditions and geographic: low rainfall of a monthly average less than 105mm, native soil was predominantly Smectite Clay (see Section 4.4.4), or located in southern districts (see Table 4.3).

9.2.4 Research Question 4

RQ-4: How can the effect of climate change impact the occurrences of pipe bursts in the WDN in Barbados?

One of the factors identified as contributing to local pipe bursts is the presence of clays that shrink and swell. The actions of this clay are linked to soil moisture. Hence, changes in rainfall patterns can have an impact on pipe bursts. By gaining this insight into pipe burst behaviour, a climate change assessment was conducted to indicate the extent to which rainfall may influence pipe bursts and

to understand the risks associated. A set of downscaled rainfall projections for Barbados were generated by Charlery and Nurse (2018) from a particular GCM under the RCP climate change scenarios by the IPCC. While the projections represented only one plausible outcome for future rainfall, these allowed for exploration of the possible implications of climate change on pipe bursts.

From the assessment, it was showed that when rainfall was projected to decrease much below the historical average (see Figure 7.6), a pattern of higher pipe bursts in smaller diameter pipes made of iron material laid in Smectite Clay was observed particularly during the dry season (see Section 6.4). In the driest scenario, pipes classified as high risk increased by 42% (see Section 7.4.1). Conversely, areas that were projected to experience rainfall much above the historical average showed a decrease in the frequency of pipe bursts as it was likely that soil saturation improved soil stability (see Sections 7.4.2 and 7.4.4). Thus, it was concluded that a reduction in rainfall and increased drying caused by climate change has the potential effect of increasing pipe burst rates due to the inclination clays to shrink and swell because of fluctuations in soil moisture which can place bending stress on pipes (see Section 4.4.4).

9.2.5 Research Question 5

RQ-5: What considerations should be given, if any, by the Barbados Water Authority in planning pipe replacement interventions with respect to climate change?

Based on the findings of this research, it was concluded that the BWA should consider climate change in the planning of pipe replacement interventions. In the assessment, climate change was shown to possibly have a significant impact on pipe burst behaviour (see Section 7.4). In two of the climate change scenarios (i.e., RCP2.6 in 2035 and RCP8.5 in 2050), the number of high risk and medium risk pipes increased by 41% and 46% from the average distribution, respectively. These are considerable increases which can indicate higher losses of water and energy, and larger costs for more frequent repairs and replacements of burst pipes. Therefore, the BWA should consider implementing procedures in pipe rehabilitation initiatives that can enhance the resiliency of the network

to the impacts of climate change. The following recommendations are given: (1) adding bedding support for all newly laid pipes, (2) increasing bedding support for already-laid pipes wherever possible and feasible, (3) using less expansive soils as a back-fill instead of the native soil type, and (4) switching to more flexible pipe materials if iron pipes laid in Smectoid Clay are being replaced.

9.3 Contributions of the Research

The novel aspects of this work were the research scope and case study, the modelling and analytical frameworks, and the approach used to assess the impact of climate change on pipe bursts. These aspects allowed for contributions to be made not only to the wider body of knowledge but also at the local level. The set of contributions are given below.

Contributions to the Literature

(1) Scope and Case Study

This study added to the limited number of pipe failure studies conducted in subtropical and tropical regions which is considered as a gap in the literature by Shi, Zhang, and Ho (2013).

(2) Quantification of Explanatory Factors

Geospatial techniques for quantifying connection density, land-use, and traffic loading were developed since these factors are not usually included in analysis of pipe failure (see Section 4.4).

(3) Frameworks

A modelling framework (see Figure 3.2), analytical process (see Figure 5.1), and stepwise rules (see Section 5.3) were developed for this study. These were used to guide the data mining analysis which led to the identification of the local drivers of pipe bursts and to direct how the scenario-based assessment of climate change on pipe bursts was to be conducted. These tools can be adopted by other studies with similar objectives.

(4) Combined Use of Methods

SOMs and EPR were used as tools for data mining to promote knowledge discovery by modelling pipe bursts with historical data. The insight gained from data mining was used to form a knowledge domain for a pipe burst risk model that assigned scores to pipes based on susceptibility to frequent burst occurrences. The pipe burst risk model was developed with Fuzzy Logic. Implementation of these methods not only exhibited the effectiveness and usefulness of data mining in pipe failure analysis but also how methods that are mostly driven by data (SOMs and EPR) can work in complement with expert systems (Fuzzy Logic).

(5) Key Findings

Key findings from the data mining analysis contributed to the existing understanding of the factors that contribute to pipe burst behaviour in WDNs. These findings are listed below:

- Pipe age is not always the sole or main factor that promotes pipe bursts (Barton et al. 2019, Boxall et al. 2007, O'Day 1982).
- Smaller diameter pipes are more susceptible to bursts than larger diameter pipes (Barton et al. 2019; Bruaset and Sægrov 2018; Shi, Zhang, and Ho 2013).
- High pressures promote bursts in smaller diameter pipes (Barton et al. 2019; Punurai and Davis 2017; Marzouk, Hamid, and El-Said 2015; Savic and Walters 1999).
- Higher bursts are observed during the dry season than wet season (Barton et al. 2019; Wols and van Thienen 2016; Gallage, Chan, and Kodikara 2012).
- Soils that shrink and swell can induce a bending stress on pipes that lead to increased bursts especially with seasonality (Abbey, Eyo, and Ng'ambi 2020; Wood and Lence 2009; Krenz, Lee, and Owens 2006).

(6) Climate Change Assessment on Pipe Bursts

The approach of incorporating downscaled geospatial climate change projections for rainfall at a resolution of 1km x 1km into a pipe burst model based on fuzzy logic is novel. The scale of the resolution of the rainfall projections is comparable to the historical data used for pipe burst modelling. In the literature, there are a limited number of pipe failure studies that considered climate change – most of which examined the effects of temperature in high latitude areas (see Section 2.5). Furthermore, climate projection data were not high resolution.

Contributions at the Local Level

- (1) Identification of the main drivers of peak pipe bursts in the WDN and insight into pipe burst behaviour and the dynamic relationships among factors. An evidence-based disproving of the perpetuated opinion that pipe age was the main driver of pipe bursts in the local WDN.
- (2) A pipe burst risk model to depict local pipe burst behaviour was developed which can be continually updated and used for other applications (see Section 6.4).
- (3) A ranking of all pipes in the WDN indicating the level of susceptibility to frequent burst occurrences and a hotspot map for visualisation which can be used to inform planning of pipe replacement interventions (see Section 6.5.2).
- (4) A set of recommendations that can be used to help water utilities prudently address pipe bursts, climate change impacts, water losses, and other shortcomings (see Section 8.6).

9.4 Final Reflection

Most water utilities are challenged with data deficiencies that seemingly prevent analyses from being carried out to support evidence-based decision-making and planning. However, this research demonstrated that despite challenges in data, the strategic implementation of methods that are not purely data-driven but also allow for expert knowledge to be incorporated can facilitate the detection of

critical patterns and creation of useful tools. The research also showed the importance of fundamentally understanding the local factors that affect the performance of WDNs. Having identified rainfall as one of the main drivers of pipe bursts in Barbados, further work was carried out to investigate the possible impacts of climate change and to evaluate how potential risks, such as increased water losses, can be addressed.

Day-Zero in Cape Town brought global attention to the possibility that even major cities could run out of water because of a combination of factors including: inadequate planning, operation and maintenance practices, and climate change. Therefore, all water utilities – especially in countries that are on the frontline of climate change such as the Caribbean and SIDS – should seek to understand and manage WDNs efficiently, sustainably, and resiliently. While this research was novel for the Caribbean Region, it is hoped that it will be the catalyst of many other related works.

“Where cities do run out [or experience a shortage] of water, the difficulty in most instances, up to now has been not so much a shortage of water as a shortage of vision...what is required in most instances is not more water as such but more forethought as to future needs and possibilities and the willingness to finance preparation of plans and the provision of the additional waterworks needed to bring the water to the people.” (Grima 1972)

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